# CRITICAL THINKING FOR LIFE

MICHAEL GANNOTTI

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Michael Gannotti


Figure 1: Cover

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This illustrated edition merges the documented final draft with its Chicago bibliography. Endnotes run in one series. Chapter-opening images are original still-lifes, not portraits of persons or children.

# Contents

# Preface

## Not a Bicycle

Two documents, decades apart.

The first is a public sentence. Robert Ennis, in a working definition already in print in *Educational Leadership* in 1985 and restated in his later outlines, wrote: "Critical thinking is reasonable reflective thinking focused on deciding what to believe or do."[1] The sentence includes action. It is the best one-line definition a *life* book can use, because a life is not only appraisal of other people's arguments. It is sharing, consenting, hiring, voting, refusing. Ennis does not settle how to teach toward that sentence. He names the activity.

The second is a constraint. Daniel Willingham, in the Summer 2007 *American Educator*, answered a question the 1980s had spent exhorting:

> People who have sought to teach critical thinking have assumed that it is a skill, like riding a bicycle, and that, like other skills, once you learn it, you can apply it in any situation. Research from cognitive science shows that thinking is not that sort of skill. The processes of thinking are intertwined with the content of thought (that is, domain knowledge).[2]

You can teach maxims. Without background knowledge and practice, they will not implement. Domain-specific critical thinking can be taught. There is no proven way to teach a content-free general skill directly. The bicycle passage is quoted at length in Chapter 3; the preface needs only the bicycle. Two facts, decades apart: the best life definition includes action; the pedagogical constraint says you cannot implement the definition without domain knowledge and practice.

[1] Robert H. Ennis, "A Logical Basis for Measuring Critical Thinking Skills," *Educational Leadership* 43, no. 2 (October 1985): 44–48, already as the "working definition"; restated in Ennis, "The Nature of Critical Thinking: Outlines of General Critical Thinking Dispositions and Abilities," last revised September 2015, https://criticalthinking.net/wp-content/uploads/2024/04/The-Nature-of-Critical-Thinking.pdf. Prefer Ennis's own later restatements over a Foundation reprint. The public sentence used here is the later restatement and includes *do*. Variants drop or add a comma and sometimes "and"; a 1990 ERIC reprint (ED320015) quotes the 1985 p. 45 wording as "reflective and reasonable thinking that is focused on deciding what to believe or do." The 1985 EL full PDF was paywalled this pass.

[2] Daniel T. Willingham, "Critical Thinking: Why Is It So Hard to Teach?," *American Educator* 31, no. 2 (Summer 2007). Reading Rockets authorized reprint, https://www.readingrockets.org/topics/comprehension/articles/critical-thinking-why-it-so-hard-teach, fetched 30 August 2026. Bicycle passage quoted at length in Chapter 3, not the full essay; preface uses the constraint only. Prefer the article body over the AFT deck pull-quote "There's no such thing as critical thinking 'skills.' " Willingham did not say critical thinking cannot be taught.

This book is not a skills poster, not a test-prep course, and not a sequel to *Autonomous AI and Education*. The popular story collapses slogan, skill, test, political compliment, and life into one transferable faculty, already promised by every decent school. No serious account in the research this manuscript stands on endorses all of those conjuncts. The preface names the collapse. The introduction takes it apart. The chapters show which conjuncts survive.

What this book is not, in one page. Not a Paul–Elder franchise: steal the 1981 warning that training on "neutral" cases can make students "more sophistic rather than less so"; leave criticalthinking.org. Not a CCTST-prep course: the test is not the thing; Delphi is expert consensus for assessment, not a discovery of nature. Not a P21, OECD, or UNESCO trial: those are slogan-machines and demand-signals. Not five AI chapters: Willingham, Bastani, and Fan already live in the education book; this book needs one chapter on the life condition that a machine will finish the sentence. Not a What Works Clearinghouse-grade claim where none exists; as of 30 August 2026 there is no WWC intervention report for P4C, argument mapping, Paul–Elder, or named critical-thinking programs.

The coalition, in miniature, is Ennis plus Dewey plus Willingham plus a Paul-ish sophistry warning plus Lipman for childhood judgment, with McPeck's *about what?* kept as a constraint. That is a coalition. Coalitions hide contradictions. The manuscript promises to say, on the page, where they veto each other. The introduction maps the ten chapters. Chapter 2 chooses in public. This preface does not dump the table.

The contract with the reader is documentary. Chicago notes; load-bearing numbers cited on the page; longer source fights — UNVERIFIED flags, vendor-claim labels, preprint labels, journalism labels — in the notes. Items the research marked unverified stay flagged until a primary is read. Kant as a critical-thinking source, named-state media-literacy bill text, and Nyhan 2020's full *PNAS* PDF remain among them. Dewey 1910 pages 74 and 82 have now been opened for the English phrase; Hitchcock's "educational goal-name" framing is still his, and those pages are not quoted here. Kosmyna is a preprint, treated in Chapter 9, not here. Composite hypotheticals, if a later chapter needs one to show a move, are labelled *hypothetical* in the sentence that introduces them. No portraits of living children. No invented dedication. No invented classroom as reported fact. No recruiter "#1 skill" treated as a finding. Open questions are left standing rather than filled with analogy.

The pair of documents is the book's spine, not its bibliography. Ennis without Willingham becomes a poster of the activity with no account of why maxims fail. Willingham without Ennis becomes a veto on generic courses with no public sentence for a life that includes *doing*. Dewey's 1910 definition of reflective thought — active, persistent, careful consideration of belief in light of grounds and further conclusions — sits behind both, and his lantern sentence already vetoes the ready-made apparatus.[3] Socrates, Bacon, and Kant did not say "critical thinking"; this preface will not open as

[3]John Dewey, *How We Think* (Boston: D. C. Heath, 1910), Project Gutenberg text, https://gutenberg.org/files/37423/37423-h/37423-h.htm, chap. 1 (reflective-thought definition, wording verified) and chap. 3 (lantern/specificity sentence, spent in Chapter 2). Dewey uses the English phrase at 1910: 74 — "The essence of critical thinking is suspended judgment; and the essence of this suspense is inquiry to determine the nature of the problem before proceeding to attempts at its solution" — and at 1910: 82 — "So far as we conduct each of these processes in the light of the other, we get valid discovery or verified critical thinking." Pages opened 30 August 2026 against Gutenberg and the Mead Project (chaps. 6–7). Hitchcock's claim that those pages introduce the term *as the name of an educational goal* remains Hitchcock's framing, not a sentence on those pages. Not coinage.

if they did. Coinage is unverified and unnecessary. The English educational goal-name is modern. The work is not.

A third document is always waiting to be treated as the pair: a recruiter survey, a 96 percent miss rate on a climate `.org`, a practice score that rose while an unaided exam fell. Those belong in the chapters that can carry their designs. NACE's 4.49 is not number one and is not a skill assessment. Breakstone's 96 percent is a civic baseline, not a proof that schools never taught thinking. Bastani's −17 percent is a school dissociation, already treated at length in the education book, cited here as the life condition rather than as a product tour. The preface holds to two documents so the collapse can be named before the numbers arrive.

If the popular story is a collapse, the reader needs the corrections before any program is explained at length. That is the introduction's job.

# Introduction

## The Conjunction That Does Not Survive

The popular story is a conjunction, and it is how the phrase functions in the wild. Critical thinking is a single, general, content-free skill, teachable in a standalone course or a poster of maxims, measurable by a short test, possessed as a personal trait that licenses political authority, and already promised by every decent school.[4] Every school, ministry, and HR department now says the words. P21 unpacks them as one of the 4Cs. OECD's Learning Compass 2030, after a one-paragraph survey of Sternberg, Facione, Ennis, and Bailin, defines critical thinking as "questioning and evaluating ideas and solutions (OECD, 2016)." UNESCO bins "critical and innovative thinking" with creativity, entrepreneurship, and resourcefulness. DeAngelo and colleagues found more than 99 percent of faculty calling the development of critical thinking very important or essential. Tsui found 92 percent of students believing they had made gains.[5] Self-report is not a test. Frameworks are not trials. The conjunction does not care. It is one faculty, already delivered, certifiable, and politically licensing.

A reader has met this story in a mission sentence, a competency framework, a hiring rubric, a political compliment, and a timed test, often in the same week. The story is congratulatory when the speaker is "we" and catastrophic when the speaker is "they" or "kids today." Both tones need the same object: a single transferable faculty that a decent school should already have installed. Chapter 1 names the three folk uses — slogan, compliment, product — and the collapse that treats them as one. This introduction does the prior job. It puts eight corrections in the way, so that a reader who still wants the popular story has to go through the documents. Almost none of the popular story is how the record works.

Almost none of that is how the record works. The phrase has a history and six non-equivalent conceptions. Transfer is rare. Implicit expectation fails. Tests operationalize constructs; they do not name a mental organ. Civic and professional life are domains. Machines finish sentences. This introduction's job is to make the reader unable to treat a slogan, a CCTST score, and a life of judgment as the same object. Eight corrections, then a map, then the design problem the chapters have to solve. Documents do the work. No invented classroom. Composite hypotheticals, if a later chapter needed one to show a move, would be labelled *hypothetical* in the sentence that introduces

[4] Conjunction as stated in the 01-definitions research memo, §9, and Chapter 1 of this book.

[5] P21 *Framework Definitions* (2009), ERIC ED519462; OECD Learning Compass 2030 construct note; UNESCO *Assessment of Transversal Competencies* (2016); Linda DeAngelo et al., HERI faculty survey 2009, >99 percent; Lisa Tsui 1998, 92 percent self-reported gains. Frameworks are not trials. Self-report is not a test.

them. This introduction needs none. The popular story is already public. The corrections are already sourced. The map is already the book.

## Correction 1 — No coinage myth

Do not write that anyone coined *critical thinking*. David Hitchcock, in the *Stanford Encyclopedia of Philosophy*, reports that Dewey (1910) *used* the term as the name of an educational goal, more commonly calling the same goal reflective thinking. Pages 74 and 82 of the 1910 edition have now been opened (Gutenberg and Mead Project). Dewey writes, at 1910: 74, “The essence of critical thinking is suspended judgment”; at 1910: 82, that conducting induction and deduction in the light of each other yields “valid discovery or verified critical thinking.” Those sentences use the English phrase. They do not say that the phrase is the name of an educational goal; that framing is Hitchcock’s.[6] Edward Glaser, in 1941, stabilized the phrase as a research heading. The 1980s industrialized it: California GE, Paul’s conference, College Board, *A Nation at Risk*, Delphi. The Foundation for Critical Thinking’s “roots in the mid-late 20th century” blurb is inconsistent with Dewey 1910 and Glaser 1941 and should not be followed. Socrates did not say “critical thinking.” Bacon did not. Kant did not. Kant as a critical-thinking source is unverified here. Coinage myths are a way of skipping the work of saying what the words have meant.

The honest ancestry is practices without the English phrase: elenchus as accountability to reasons in public; Bacon’s idols as a classification of standing error, which Dewey already translated into educational language; Locke and Mill as Dewey actually quotes them. Chapter 2 does that work. The introduction only needs the refusal: a 2,500-year brand-narrative is not historiography, and an Enlightenment “think for yourself” is atmosphere, not Ennis. Freire’s “critical” is a live competitor in folk use and Hitchcock’s non-example for the mainstream concept. Mixing them in one origin-story is how slogans are born. Glaser both educationalized the phrase and, in the same dissertation, limited what general training can do. The 1980s did not discover the activity; they scaled a name into requirements, conferences, textbooks, and tests. Facione’s report exists because assessment and instruction at system scale needed a shared construct. A shared construct is not a discovery of nature. That distinction is Correction 5’s, and it starts in the history.

Dewey’s 1910 definition of the *activity* is worth keeping almost verbatim, and it is not a skills list:

> Active, persistent, and careful consideration of any belief or supposed form of knowledge in the light of the grounds that support it, and the further conclusions to which it tends, constitutes reflective thought.[7]

The same book refused the apparatus the later slogan would need: thinking “is specific, not a

[6] David Hitchcock, “Critical Thinking,” *Stanford Encyclopedia of Philosophy*, first published 21 July 2018, substantive revision 12 October 2022, https://plato.stanford.edu/entries/critical-thinking/; History supplement, https://plato.stanford.edu/entries/critical-thinking/history.html. Dewey 1910: 74, 82 opened 30 August 2026 against Project Gutenberg (https://gutenberg.org/files/37423/37423-h/37423-h.htm) and the Mead Project transcription of chaps. 6–7. Page 74: “The essence of critical thinking is suspended judgment; and the essence of this suspense is inquiry to determine the nature of the problem before proceeding to attempts at its solution.” Page 82: “So far as we conduct each of these processes in the light of the other, we get valid discovery or verified critical thinking.” Hitchcock’s claim that those pages introduce the term *as the name of an educational goal* remains Hitchcock’s framing, not a sentence on those pages. Not coinage.

[7] John Dewey, *How We Think* (Boston: D. C. Heath, 1910), Gutenberg text, ch. 1. Wording verified.

machine-like, ready-made apparatus to be turned indifferently and at will upon all subjects, as a lantern may throw its light as it happens upon horses, streets, gardens, trees, or river."[8] One sentence names an activity. The other already vetoes the bicycle. Anyone who treats Dewey as the grandfather of a content-free skills course has not read both. Hitchcock reports that the 1933 restatement dropped most uses of "critical" and "uncritical," settling on reflection; "critical" occurs once as a *component* of reflection, not the whole. The 1933 full text was not independently opened for this book; the phases and the diction claim stay marked as Hitchcock's report.[9] The Eight-Year Study treated the two phrases as synonyms. Dewey, on that reading of 1933, did not. The title-phrase is held anyway, because Ennis's public sentence still does more work for a life book than a market-punished rename to "reflective thinking," and because the folk uses have to be refused under the name the public actually uses. Open question 18 stands.

## Correction 2 — Six accounts cannot design the same book

Hitchcock's shared *concept* — careful, goal-directed thinking, after Bailin and colleagues — is not a license to collapse Ennis, Delphi/Facione, Paul–Elder, Willingham, McPeck, and Lipman. They cannot all design the same book. Chapter 2 prints the table. The introduction prints the coalition and the vetoes in one paragraph.

Ennis: "Critical thinking is reasonable reflective thinking focused on deciding what to believe or do."[10] Best public sentence for a life book, because it includes action. It does not settle pedagogy. Delphi / Facione 1990, ERIC ED315423: expert consensus *for purposes of educational assessment and instruction*; 46 panelists; six rounds, 11 February 1988–25 September 1989; not "the APA's official definition." Table 1, quoted, not paraphrased:

> We understand critical thinking to be purposeful, self-regulatory judgment which results in interpretation, analysis, evaluation, and inference, as well as explanation of the evidential, conceptual, methodological, criteriological, or contextual considerations upon which that judgment is based. CT is essential as a tool of inquiry. As such, CT is a liberating force in education and a powerful resource in one's personal and civic life. While not synonymous with good thinking, CT is a pervasive and self-rectifying human phenomenon. The ideal critical thinker is habitually inquisitive, well-informed, trustful of reason, open-minded, flexible, fair-minded in evaluation, honest in facing personal biases, prudent in making judgments, willing to reconsider, clear about issues, orderly in complex matters, diligent in seeking relevant information, reasonable in the

[8] Dewey 1910, lantern sentence, spent in Chapter 2.

[9] Hitchcock, SEP History supplement, on Dewey 1933 dropping "critical"/"uncritical." 1933 full text not independently fetched; phases and diction as Hitchcock's report.

[10] Robert H. Ennis, "A Logical Basis for Measuring Critical Thinking Skills," *Educational Leadership* 43, no. 2 (October 1985): 44–48, already as the "working definition"; restated in Ennis, "The Nature of Critical Thinking: Outlines of General Critical Thinking Dispositions and Abilities," last revised September 2015, https://criticalthinking.net/wp-content/uploads/2024/04/The-Nature-of-Critical-Thinking.pdf. Prefer Ennis's own later restatements over a Foundation reprint. The public sentence used in this book is the later restatement ("reasonable reflective thinking focused on deciding what to believe or do"). Variants in his publications drop or add a comma and sometimes "and"; a 1990 ERIC reprint of the 1985 article (ED320015) quotes p. 45 as "reflective and reasonable thinking that is focused on deciding what to believe or do." The 1985 EL full PDF was paywalled this pass. The *do* is in every variant used here.

> selection of criteria, focused in inquiry, and persistent in seeking results which are as precise as the subject and the circumstances of inquiry permit.[11]

The panel split about 61/30 on whether dispositions are *in* the meaning; 52 percent rejected an ethical component in the meaning. CCTST is one operationalization of that construct, not the meaning. Paul–Elder is a curriculum brand; steal the 1981 strong-sense warning; leave the franchise. Willingham is the cognitive-science constraint. McPeck: thinking is always about X. Lipman: judgment, community of inquiry, childhood. The coalition this book can honestly run is Ennis plus Dewey plus Willingham plus a Paul-ish sophistry warning plus Lipman for childhood. Where Willingham vetoes Ennis-style general courses, the book says so. Where Ennis's outline still earns pages, it earns them as aim, not as a reason to ignore knowledge. Delphi cannot be the meaning. McPeck's ban on general courses is rejected; his *about what?* is kept.

Adjacent accounts exist and are not promoted by silence: Siegel's critical spirit as an educational ideal; Bailin against generic discrete skills; Halpern's teach-for-transfer optimism; Fisher and Scriven; Stanovich's rationality; Kuhn's dialogic practice; Freire/hooks. They constrain later chapters. They are not this book's six. A writer who collapsed the six plus the adjacent into one operational definition would have written the popular story in scholarly clothes. The introduction's correction is the impossibility proof: they cannot design the same book. The coalition is a choice, made in public in Chapter 2, with vetoes left standing in Chapter 10. Open question 1 is that choice's remainder.

## Correction 3 — Not a bicycle-skill

Willingham 2007, quoted in Chapter 3, not paraphrased away here:

> People who have sought to teach critical thinking have assumed that it is a skill, like riding a bicycle, and that, like other skills, once you learn it, you can apply it in any situation. Research from cognitive science shows that thinking is not that sort of skill.[12]

He did not say critical thinking cannot be taught. Domain-specific critical thinking can be taught. Generic add-ons have limited success. 2019: "We are not even sure the general skills exist, but we are quite sure there is no proven way to teach them directly."[13] Three-point boil-down, second point not dropped: it is not a skill deployable regardless of context; there *are* metacognitive strategies that make it more likely; doing what the strategies call for depends on domain knowledge and practice. Glaser 1941:175 already: disposition transfers better than skill; skill limited by knowledge. McPeck 1981: "I teach thinking" simpliciter is empty. The book that promises a content-free general power is making the claim McPeck called conceptually empty.

[11] Peter A. Facione, *Critical Thinking: A Statement of Expert Consensus for Purposes of Educational Assessment and Instruction* (The Delphi Report), ERIC ED315423 (1990), Table 1. Quoted, not paraphrased. Not "the APA's official definition." Forty-six panelists, six rounds, 11 February 1988–25 September 1989. Round 5B: about 61 percent dispositions-in-the-meaning versus about 30 percent strict proceduralism — a split, not a count of 61 experts. 52 percent reject ethics-in-the-meaning. Full PDF fetched 30 August 2026 from https://files.eric.ed.gov/fulltext/ED315423.pdf.

[12] Daniel T. Willingham, "Critical Thinking: Why Is It So Hard to Teach?," *American Educator* 31, no. 2 (Summer 2007). Full bicycle passage in Chapter 3. Prefer the article body over the AFT deck line. He did not say CT cannot be taught.

[13] Willingham, *How to Teach Critical Thinking* (NSW Department of Education, 2019).

Even three-year-olds can think critically in a domain they know; even trained scientists fail outside theirs. Stage is domain knowledge, not a birthday. Surface structure captures attention; deep structure is what would transfer; novices do not see it. Willingham's band/garden pair: 19 percent saw the analog; a hint raised solution only to 35 percent. Expertise is encapsulated: neurologists do not diagnose cardiac cases well; technical writers do not automatically write newspaper articles; professional philosophers remain susceptible to irrelevant problem features. Chapter 3 spends those findings. The introduction only needs the veto on the bicycle, and the second point the popular story drops: metacognitive strategies *do* help; they do not supply the knowledge the strategies call for. A life-skill book that is only maxims is the thing Willingham is diagnosing. A life-skill book that is only "just teach history and the thinking will come" ignores the second point. Willingham 2019's four-step plan — identify what critical thinking *means in each domain*; identify the content those tasks need; sequence; plan three to five years of revisiting — is a design constraint, not this book's teaching chapter. Choosing content expresses values; not choosing is still a choice. Everyday life is not a school subject and not "everything in general." The architectural decision Chapter 3 names, and Chapters 7–10 run, is domains of a life as the subjects: money, health, news, work, friendship, citizenship, and the fluent sentence a machine will finish. How general is general enough remains Open question 2.

## Correction 4 — Transfer is rare; implicit expectation fails

Thorndike and Woodworth (1901): spread of practice only where identical elements are concerned; formal discipline fails. Detterman (1993): the default prediction is failure; likelihood tracks similarity. Gick and Holyoak: Duncker radiation about 10 percent spontaneous; analog without hint about 30 percent; analog plus hint about 75 percent. The gap is *noticing*, not application skill.[14] Near is not far. Chess, music, and working-memory "thinking gyms" shrink toward zero with active controls (Sala and Gobet).

Philip Abrami and colleagues, 2008: improvement "cannot be a matter of implicit expectation."[15] Immersion — critical thinking as a hoped-for by-product — was the weakest design. Abrami 2015: 341 *effect sizes*, not 341 studies; g+ = 0.30 on standardized critical-thinking tests; Ennis-type mixed / infusion / general / immersion differences **not statistically significant**; dialogue plus authentic problems plus mentoring g = 0.57 from 19 studies. Do not cite 2015 as proof that mixed won a horse race. Design mixed anyway, for transfer reasons. Huber and Kuncel 2016: college about 0.5–0.6 SD over four years on generic tests; curriculum-wide critical-thinking campaigns and nursing's mandated curriculum did not add a reliable increment.[16] Stamping the phrase on every syllabus is immersion-by-committee. Do not mash 0.30 with 0.57 with Gorard +0.12 with EEF 2021's 0 months with Roozenbeek's field *h* of 0.09. They are different objects.

What teaching actually moves, previewed: Philosophy for Children has two large independent RCTs with different stories — Gorard 2015 small attainment effects, more for FSM; EEF 2021,

[14] Thorndike and Woodworth 1901; Detterman 1993; Gick and Holyoak 1980/1983. Spent in Chapter 4. Sala and Gobet 2016/2017 on thinking gyms.

[15] Philip C. Abrami et al., "Instructional Interventions Affecting Critical Thinking Skills and Dispositions," *Review of Educational Research* 78, no. 4 (2008): 1102–1134. Headline quoted: cannot be a matter of implicit expectation.

[16] Abrami et al. 2015, 341 effect sizes, g+ = 0.30; combination g = 0.57, 19 studies; Ennis-type NS. Huber and Kuncel 2016, college ~0.5–0.6 SD. Do not mash effect sizes.

198 schools, five padlocks, **0 months**. Argument mapping is a candidate method for the argument-analysis slice; van Gelder 2015 is explicit that no large RCT exists. Domain-embedded is the load-bearing beam: CVS meta g = 0.61; Reisman 2012 Reading Like a Historian, a quasi-experiment, not an RCT, not a generic-CT result. Classroom talk: more student talk is not more critical thinking (Murphy et al. 2009). First-year seminars move GPA $\delta = 0.02$ and retention $\delta = 0.11$ — wrong outcome. WWC has no named-CT intervention report as of 30 August 2026.[17] Chapter 5 is that record. The introduction's correction is smaller: you cannot get a bicycle out of these numbers, and you cannot get the numbers by hoping. Dual process is the doorway Chapter 4 will walk through, not a System 1 / System 2 poster. Type 1 is not the villain; Type 2 can rationalize; myside is largely independent of cognitive ability; Kahan 2012's U.S. identity pattern is not a human universal (Pröpper, Geiger, Blanken, and Brick 2022). Debiasing: consider-the-opposite has the best classical support; Morewedge-style games move lab bias scores, not civic behavior. Kuhn: most people are weak at genuine evidence, alternatives, and counterarguments on everyday social issues; explanation is not argument. Reflection is theater when uninformed, fluency-fooled, identity-serving, post-hoc, or unscored. Those are Chapter 4's. The introduction only needs them as further reasons the conjunction cannot survive: a general faculty that was a bicycle would not have this much machinery underneath, and this much failure to deploy.

## Correction 5 — The test is not the thing

Delphi is expert consensus for assessment, not a discovery of nature. Independent CCTST subscale alphas as low as 0.21. Watson–Glaser often one factor. CLA+ constructed-response reliability 0.43. HCTA retired. No HALO test; halo is a rater effect; HCTSR is a holistic rubric. Kuncel: little discriminant validity of common critical-thinking tests against *g*; little evidence they predict grades or job performance better than IQ.[18] Arum and Roksa's 45 percent "no gain" is a CLA finding, partly motivation — not "critical thinking cannot be taught." Liu, Bridgeman, and Adler: conclusions about college learning can flip depending on whether students are induced to try. A book that treats a CCTST score as what critical thinking is has confused an instrument with a concept. Chapter 6 is that refusal at length. AIAS is language for assessment redesign, not an RCT that oral defense improves the activity; Mike Perkins, not David Perkins. "Used in more than 350 institutions" is a dissemination claim, not an effect size. A take-home essay is no longer, by itself, evidence of the student's judgment; that sentence is older than generative models (Frederiksen 1984); the models made it unskippable. Redesign when the construct is judgment: unseen cases, process evidence, oral defense. Those have rater problems. They are still better matched to the construct than a badge. CART is a research prototype, not a school-ready instrument. Vendor subscores are not instructional steering. A semester of practice-item gains is not a skill gain.

[17] Gorard, Siddiqui, and See 2015; EEF 2021 P4C, 0 months, 5/5 padlock; van Gelder 2015, no large RCT of argument mapping; Schwichow et al. 2016, CVS g = 0.61; Reisman 2012 QES; Murphy et al. 2009; Permzadian and Crede 2016; WWC intervention-reports index 30 August 2026, negative search is the finding. Spent in Chapter 5.

[18] Liu, Frankel, and Roohr, ETS RR-14-10 (2014); Kuncel 2011 via Huber and Kuncel; Arum and Roksa 2011; Liu, Bridgeman, and Adler 2012. HCTA retired. No HALO test.

## Correction 6 — Recruiter "#1 skill" is folklore unless sourced

NACE *Job Outlook 2025*, 237 respondents, 162 members = 19.2 percent of eligible: communication **4.57**, critical thinking **4.49**, teamwork 4.43 on a five-point importance scale. 96.1 percent rate critical thinking very or extremely important; 53.5 percent rate graduates very or extremely proficient. What they test, when they test, is a situated interview story — "describe a time when you anticipated a problem" — not a Facione skill and not a Watson–Glaser.[19] WEF *Future of Jobs 2025*: *analytical thinking*, seven of ten firms, a survey, and the phrase is not "critical thinking." Treat unsourced "critical thinking is the #1 skill employers want" as folklore. Sourced to NACE or WEF it is still a recruiter survey, not a skill assessment, and not evidence that generic courses produce job performance. NACE 237 is not a probability sample of all employers.

Chapter 8 will restate this with the workplace detail and add the clinic and the court as domains, not as proof of a transferable faculty. Casscells, Schoenberger, and Graboys: 11 of 60 in a Harvard Medical School sample gave the correct ~2 percent on a base-rate item; the modal answer was 95 percent. Natural frequencies restore insight. NAM 2015: most people will experience at least one diagnostic error. Newman-Toker 2023: 795,000 is a modeling estimate, range 598,000–1,023,000, not a census. Thompson and Schumann's prosecutor's fallacy is a teaching object. The cab problem's "correct" 41 percent is a live academic dispute. Superforecasting hygiene is closer to thinking-for-life than Wason-card performance, and it is still not a workplace RCT of the phrase. The introduction's correction is the labor-market sentence. The later chapter's correction is that stakes have knowledge and error-costs, not a faculty.

## Correction 7 — Civic life is a domain; backfire is rare

CRAAP trains fakeable signals. Wineburg, Breakstone, Ziv, and Smith's *Educating for Misunderstanding*: college students used strategies recommended by library pages; those "best practices" can make students more susceptible, because they train attention onto features adversaries can fake. Breakstone and colleagues, 2021: 3,446 high-school students; **96 percent** never uncovered a climate site's fossil-fuel ties; two thirds failed to distinguish news from ads; more than half treated an anonymously posted Facebook video shot in Russia as "strong evidence" of U.S. voter fraud.[20] Lateral reading — leave the page; who's behind it, what's the evidence, what do other sources say — is a different object of practice, with trial evidence (Wineburg and McGrew 2019; Wineburg et al. 2022). Six COR lessons moved credibility-task scores and still left about half the points on the table. That is not civic-outcome transformation.

Wood and Porter, 2019: five experiments, more than 10,000 participants, 52 issues, no item triggered a worldview backfire. Nyhan 2020: backfire effects are "extremely rare in practice"; interpretations of Nyhan and Reifler 2010 outstripped the findings. *The Debunking Handbook 2020*: "Do not refrain from attempting to debunk or correct misinformation out of fear that doing so will backfire."[21] Continued influence is the real problem, not ironic strengthening. Do not write a "never

[19] NACE *Job Outlook 2025*; WEF *Future of Jobs 2025*. Recruiter surveys, not skill assessments.

[20] Breakstone et al. 2021, *Educational Researcher*. Wineburg and McGrew 2019: 10 fact checkers, 10 Ph.D. historians, 25 Stanford undergraduates; Wineburg et al. 2022. *Educating for Misunderstanding* (2020) is a SHEG working paper; PDF fetch timed out; do not invent extra percentages.

[21] Wood and Porter 2019; Nyhan 2020 *PNAS* (full PDF UNVERIFIED; phrasing from snippets plus handbook);

correct — it backfires” book. Accuracy nudges change sharing discernment under inattention, not belief; partisan moderation is a live dispute. Inoculation works on technique recognition in the lab ($d$ 0.28–0.68), more weakly on YouTube ($h = 0.09$), and decays without boosters. Twenty-five state media-literacy laws are laws; Media Literacy Now has not evaluated classrooms. A law that says “teach critical thinking / media literacy” is not COR, not inoculation, and not an accuracy prompt. Guess et al. 2020: Facebook-style tips +26.5 percent discernment in a U.S. sample, +17.5 percent in a highly educated Indian online sample, decay, **null** in rural northern India. Checklist tips help a little, fade, and fail where the information ecology is different. Civic *outcomes* — turnout, vote quality, scam victimization, vaccine uptake, wild sharing over a year — are not shown in the trials fetched. That is Open question 6, standing.

## Correction 8 — One AI chapter, not five

Willingham, Bastani, and Fan already live in *Autonomous AI and Education*. This book needs the *life* condition: a machine will finish the sentence. Bastani 2025: unguarded GPT +48 percent on practice, **−17 percent** versus never-AI on a closed-book isomorphic exam; GPT Tutor +127 percent practice, unaided exam approximately at control; students did not perceive the loss. Fan 2025: better essays, no significant knowledge gain or transfer, fewer evaluation and orientation moves — “metacognitive laziness.” Stadler: ease without depth. Liang: 61.22 percent of human TOEFL essays flagged as AI. Kosmyna 2025 is a **preprint**, n = 54 / 18, not population cognitive decline.[22] When a machine finishes the sentence, evaluation is the remaining human work. Two lanes — formation versus performance — is a design hypothesis, not a fitted life-span schedule, not existing district policy. No RCT shows that “evaluate the chatbot” transfers to non-AI thinking. Willingham predicts it will not, without domain knowledge. One chapter. Not a sequel.

The evaluative loop is this book’s title under machine conditions: interpret; check laterally, numerically, or against a source; refuse a fluent wrong answer; orally defend with the model closed. That loop is Ennis plus Willingham plus Wineburg plus Gigerenzer. It has no transfer RCT. Detectors fail and fail unfairly; collapsing performance into a detector is Liang’s false positives. Offloading is old (Risko and Gilbert; Sparrow, Liu, and Wegner); the new risk is offloading *evaluation*, with fluency as the false cue that evaluation occurred. Illusory truth (Fazio et al. 2015) plus machine fluency is the civic mechanism. Chapter 9 is short on purpose. If it ran past a sequel’s length, it would have been the wrong chapter.

## Map of the book

Ten chapters, one paragraph, then the design problem.

Chapter 1 names the folk collapse: slogan, political compliment, test-prep product; P21/OECD/UNESCO

*Debunking Handbook 2020*, fetched in full 30 August 2026. Roozenbeek et al. 2022; Maertens et al. 2021. MLN January 2026: 25 states; implementation not evaluated.

[22] Bastani et al. 2025, *PNAS*; Fan et al. 2025, *BJET*; Stadler et al. 2024; Liang et al. 2023, 61.22 percent average false-positive rate on 91 human TOEFL essays (arXiv HTML; *Patterns* HTML rounds to 61.3 percent; 18/91 flagged by all seven detectors; 89/91 = 97.8 percent flagged by at least one); Kosmyna et al. 2025, arXiv:2506.08872, **preprint**, n = 54 / 18 (Ye Tong Yuan; Ashly Vivian Beresnitzky). Division of labor with *Autonomous AI and Education*: one chapter, not five.

as slogan-machines; self-report is not a test; recruiter "#1" is folklore unless sourced; one concept, rival conceptions; thesis and map, without dumping the six-account table and without quoting the bicycle. Chapter 2 is the history and the choice: Dewey to Delphi to six accounts; the coalition and the vetoes; dispositions in or out of the meaning. Chapter 3 quotes the bicycle passage and installs the knowledge constraint; expertise is encapsulated; domains of a life as the subjects. Chapter 4 is why maxims fail: transfer, dual process, myside. Chapter 5 is what teaching actually moves: Abrami's explicitness; P4C's two RCTs, different stories; argument mapping as a candidate without a large RCT; domain-embedded practice as the beam. Chapter 6: the test is not the thing. Chapter 7: leave the page; civic and media life. Chapter 8: clinic, workplace, court as domains with named knowledge and error-costs; 795,000 as a modeling estimate. Chapter 9: machines that finish the sentence; one chapter, not five. Chapter 10: a life of deeds, not a badge; stages as implication, not a developmental RCT; Guess ~7× sharing, not decline; equity older than the chatbot; five deeds with sources. Coda: what an honest book can promise versus what it cannot. The coda will not say the future is bright. Brightness is not in the documents. What can be promised is a definition, a constraint, a family of accounts, a transfer default, a teaching combination that is not a horse race, a civic habit with a remaining miss rate, domains rather than a faculty, one AI chapter, and five deeds. What cannot is listed there so it cannot be filled here.

The design problem is not which list of thinking skills to teach. It is which knowledge, which practiced recognitions, which reflective habits, and which social conditions make good judgment more likely in the situations a life actually contains — a vote, a share, a test result, a hiring story, a sentence a machine will finish. Implicit expectation cannot be the design. A bicycle cannot be the object. A short test cannot be the meaning. A badge cannot be the life. The chapters that follow are argument, then evidence, then deeds. Open questions are named as a list the book will not fill: coalition contradictions; year-scale far transfer; dispositions in the meaning; civic outcomes; incremental validity after *g*; twenty-five-state implementation; "evaluate the chatbot" transfer; standpoint of 1988–89 expert panels; whether the title-phrase is still the right one. Hold the title unless a better public sentence than Ennis's appears. It has not.

The remaining open questions the outline owned, restated as a list this introduction will not fill: how general is general enough for a life skill; dose of knowledge versus maxims versus incentives; debiasing under identity load; what P4C does when attainment is zero at scale; argument mapping with a real control; why nursing did not beat other majors; motivation on low-stakes tests; older adults in 2026's short-video ecology; knowledge-rich as equity without a fetched U.S. COR-gap RCT; Dual Process 2.0 as trainable mindware; Schraw's domain-generality of metacognition as the author's preference, not a settled finding. A 2030 classroom, a twelve-habit bicycle, and a fitted two-lane life-span schedule would fill several of those at once. They are not in the documents. The coda copies the cannot-list so a later reader cannot treat a named gap as a solved chapter.

The thesis the sources do not contradict, restated as the design problem's answer: critical thinking is not a transferable bicycle-skill. It is domain knowledge plus evaluation habits practiced on real stakes, for life, including when a machine will finish the sentence. Ennis's sentence for the definition. Willingham for pedagogy. Transfer rare. Implicit expectation failed. The test not the thing. Civic and professional life as domains. Evaluation as the remaining human work. Deeds, not a badge. That is the payload of the title. It is also the constraint on every chapter that would rather sell a slogans-and-posters curriculum. The popular story will try to come back as a summary of this introduction: "so critical thinking is important, but you need knowledge too." That sentence is the conjunction with a knowledge clause appended, which is what P21 already had and what slogan-use already drops. The corrections are stricter. Not a bicycle. Not six accounts in one. Not implicit expectation. Not the test. Not recruiter folklore. Not a civic vaccine. Not five AI chapters. Not a badge. If a later page reinstalls any of those, it has left the record. The chapters that follow were written to make that reinstalling harder, not to make the title easier to sell. A serious general reader, and a practitioner who will steal a table, both need the same thing from an introduction: the conjunction named, the corrections sourced, the map of ten chapters, and the design problem stated without a program attached. That is this file. The record, already on disk in the ten chapters, does the rest. Begin with the slogan named. Refuse it with documents. Then the life, on stakes that do not grade you and will not wait for a course to finish.

Chapter 1 does not tour programs. It names the folk uses so later chapters can refuse them without sounding pedantic.

# Chapter 1

## The Folk Collapse: Slogan, Skill, and Life

Figure 2: A fallen picture frame and a poster of silhouetted figures on a wooden floor before a dark podium and curtain.

Every school, ministry, and HR department now says the words. A Partnership for 21st Century Skills framework unpacks "critical thinking and problem solving" as one of the 4Cs, then adds, in a clause routinely dropped, that those who can think critically "must build on a base of core academic subject knowledge."[23] A National Association of Colleges and Employers survey ranks

[23]Partnership for 21st Century Skills, *P21 Framework Definitions* (December 2009), ERIC ED519462; later Battelle for Kids reprint. Advocacy coalition, not a research synthesis. Knowledge clause in the same document: "Those who

critical thinking 4.49 on a five-point importance scale — just behind communication at 4.57 — and then, when it tests, asks a situated interview story: describe a time when you anticipated a problem.[24] A school mission sentence says "we teach critical thinking," and means, often, that we are not *merely* about facts, that we have Bloom's top rungs, that we assigned a debate. Three artifacts. One collapse.

The collapse this book exists to refuse is the conjunction: critical thinking is a single, general, content-free skill, teachable in a standalone course or a poster of maxims, measurable by a short test, possessed as a personal trait that licenses political authority, and already promised by every decent school.[25] That sentence is how the phrase functions in the wild. No serious account in the research this book stands on endorses all of its conjuncts. Dewey, Glaser, McPeck, and Willingham deny content-freeness. Delphi denies "not synonymous with good thinking" even while listing skills. Ennis is general in *aim* but his conception is a long outline, not a bicycle. Paul and Elder deny that skills without intellectual traits are enough. Frameworks assert demand. Tests assert measurability. Slogans assert possession.

This chapter names the slogan so later chapters can refuse it without sounding pedantic. It does not tour programs. It does not dump six accounts; that table is Chapter 2. It does not quote Willingham's bicycle passage; that quotation is Chapter 3. It previews the constraint, states the thesis, and makes the folk uses visible. The rest of the book will show which conjuncts survive contact with the record.

A book that opened on a reconstructed classroom — a third-grade "critical thinking hour," a seminar in which a gifted teacher asked the right question — would have done the folk conjunction's work for it. Scenes persuade by fluency. Documents persuade by being checkable. The research rule this manuscript runs on is the same rule Chapter 7 will recommend for a feed: leave the page of the popular story and open the named paper. Composite hypotheticals, if a later chapter needs one to show a move, will be labelled *hypothetical* in the sentence that introduces them. This chapter needs none. The artifacts are public. P21's PDF, NACE's Likert table, a mission sentence you can find on a thousand .edu pages, and the conjunction as the 01-memo printed it are enough.

Socrates did not invent critical thinking. Bacon did not. Kant did not. The English educational goal-name is modern; the practices are not. Chapter 2 will do the ancestry without a coinage myth. This chapter's only job on that front is to keep the origin-story from walking in as a fourth folk use: "we teach what Socrates taught." We do not. We teach, when we teach anything this book will defend, evaluation habits in domains people actually know, on stakes that can hurt. The slogan says something easier.

The popular story that this chapter is taking apart has a tone as well as a content. It is congratulatory. Schools that say the words are serious. Students who say they have gained the skill are successful. Recruiters who rate it important are demanding. Citizens who call themselves critical thinkers are

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can think critically and communicate effectively must build on a base of core academic subject knowledge."

[24] National Association of Colleges and Employers, *Job Outlook 2025*. 237 respondents; 162 members = 19.2 percent of eligible. Communication 4.57, critical thinking 4.49. Recruiter survey, not a skill assessment; not a probability sample of all employers. Sample interview as given in NACE's Career Readiness Competencies: Critical Thinking PDF (2025).

[25] Conjunction as stated in the 01-definitions research memo, §9, 30 August 2026. No serious account in that memo's §§1–7 endorses all conjuncts.

the adults in the room. The tone is why the conjunction is hard to refuse in public. Refusing it sounds like refusing seriousness, success, demand, and adulthood. The documents do a different job. They show that the congratulatory sentence is several sentences, that they do not agree, and that the one that survives contact with cognitive science, teaching trials, tests, civic baselines, and fluent machines is narrower and more expensive than a mission stamp. Narrower: domain knowledge plus evaluation habits. More expensive: practiced on real stakes, for years, including when a machine will finish the sentence. Congratulation is cheap. The activity is not.

## 1.1 Three folk uses, none identical to the scholarly family

Three live folk uses, all parasitic on the scholarly family, none identical to it.

The first is the school-mission slogan. "We teach critical thinking." Often the sentence means: we are not *merely* about facts; we have Bloom's top rungs — analysis, synthesis, evaluation — as if Bloom had been a theory of the activity rather than a vocabulary for educational objectives; we assigned a debate.[26] Compatible with no shared conception, no domain practice, and no assessment that could fail. Ontario's official gloss, which Hitchcock quotes from a 2013 social-sciences guideline, is a respectable ministerial version of the same: a process of thinking about ideas or situations "in order to understand them fully, identify their implications, make a judgement, and/or guide decision making," plus a skills laundry list — questioning, predicting, analysing, synthesizing, examining opinions, identifying values, detecting bias.[27] It is Ennis-flavored policy language. It is not Ennis's conception. A mission stamp that cannot fail is not a practice. It is a claim of possession, which is one of the conjuncts.

Bloom is the vocabulary the mission stamp borrows, and Bloom is not a critical-thinking theory. *Taxonomy of Educational Objectives* (1956) gave schools a ladder. "Higher-order" became a compliment. Analysis, synthesis, evaluation became the rungs a school could point at when a parent asked whether the children were being asked to think. Hitchcock notes Bloom's page 38 among the educationalization documents; this book treats the taxonomy as a language, not as a finding that the top rungs are a transferable skill.[28] The debate-as-evidence version of the slogan is the same substitution in a classroom register. A debate can be a practice of reasons in public, which is something the pre-history of this phrase actually contains. A debate can also be weak-sense combat with a rubric attached. The mission sentence does not distinguish. It cannot. Distinguishing would require a conception, and the slogan is what you say when you have not chosen one.

The second is the political compliment, and its inverse insult. "We are the critical thinkers; they are the sheeple." The phrase, used this way, is not a description of an activity. It is a badge of faction. Paulo Freire's sense of "critical" as conscientization-plus-action — which Hitchcock's *Stanford Encyclopedia* entry treats as a non-example of the mainstream concept — lives here.[29] So does the

[26] Bloom's taxonomy as vocabulary, not a CT theory. Benjamin Bloom et al., *Taxonomy of Educational Objectives* (1956), p. 38 via Hitchcock. Do not treat Bloom as a critical-thinking account.

[27] Ontario ministerial gloss via David Hitchcock, "Critical Thinking," *Stanford Encyclopedia of Philosophy*, first published 21 July 2018, substantive revision 12 October 2022. 2013 social-sciences guideline. Ennis-flavored policy language, not Ennis's conception.

[28] Bloom's taxonomy as vocabulary, not a CT theory. Benjamin Bloom et al., *Taxonomy of Educational Objectives* (1956), p. 38 via Hitchcock. Do not treat Bloom as a critical-thinking account.

[29] Hitchcock, SEP entry: Freire's *Pedagogy of the Oppressed* (1970 trans.) as a non-example of the mainstream

inverse partisan capture of "critical" as licensed suspicion of the other side. Delphi's majority was right to refuse building a moral-political program into the *meaning* of the term; the folk use does it anyway. Richard Paul's "weak sense" — skilled defense of one's side — is exactly this compliment, given a name by someone who thought it was a problem rather than a brand.[30] A person who uses "critical thinking" as a way of saying "my side is the rational one" has not described a skill. They have paid themselves a compliment and invoiced the other side for a deficit. Pew is not needed to see the use; the use is public language. This book will not invent a poll. It will treat the compliment as what the 01-memo named: a folk use, parasitic on the scholarly family, identical to none of it.

The compliment has a scholarly cousin that it misquotes. Paul's strong-sense / weak-sense distinction was a diagnosis: training on "neutral" cases can make students "more sophistic rather than less so." The folk use heard "critical" and kept the combat. Freire heard "critical" and meant liberation. Hitchcock is right to keep Freire as a non-example of the mainstream *concept*; he is also right that the folk *use* does not care. A life book that pretended the compliment was a misunderstanding that a definition could cure would be naive about what the phrase is for in public. It is for ranking people. Ennis's sentence is for deciding what to believe or do. Those are not the same speech act. Chapter 10 will keep correlative care and refuse constitutive partisan ethics for that reason. This chapter only needs the folk fact: the phrase is used as a compliment, and the compliment is doing political work no expert panel authorized.

The third is the test-prep product. Watson–Glaser for hiring; CCTST and CCTDI for accreditation; LSAT "logical reasoning"; CLA+ performance tasks; ACT CAAP; Cornell tests; the Halpern Critical Thinking Assessment, now retired.[31] Each operationalizes *a* conception under psychometric constraint, usually multiple-choice argument analysis. Possin, Rear, and others have argued that several of these fail to measure what their names promise. The product category does a different job in public: it teaches that critical thinking is whatever the timed items are. A person who has "passed the critical thinking test" has a score. Chapter 6 will refuse to treat the score as the meaning. This chapter only needs the folk fact. The market already collapsed measurement with possession.

Watson–Glaser is a test lineage, not a definition; Bernard and colleagues found a single-component structure across sixty published studies; independent subscale alphas have been reported as low and wide. CCTST operationalizes Delphi; independent subscale alphas as low as 0.21; Huber and Kuncel's nursing result is the outcome-level critique — despite the National League for Nursing's requirement and heavy CCTST use, nursing students did not show larger long-term generic-CT gains than other majors. CLA+ is a writing-plus-reasoning performance task; Arum and Roksa's

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concept. A live competitor in folk use.

[30] Richard W. Paul, "Teaching Critical Thinking in the 'Strong' Sense: A Focus on Self-Deception, World Views, and a Dialectical Mode of Analysis," *Informal Logic* 4, no. 2 (1981): 2–7. PDF fetched 30 August 2026 from https://informallogic.ca/index.php/informal_logic/article/download/2766/2207. Exact wording: students trained to recognize "bad" reasoning in " 'egocentrically' neutral" cases "become more sophistic rather than less so, more skilled in 'rationalizing' and 'intellectualizing' the biases they already have." Hitchcock's later gloss "more able sophists" is a paraphrase, not Paul's 1981 sentence. Steal the warning; do not outsource theory to the Foundation for Critical Thinking.

[31] Instrument map from Ou Lydia Liu, Lois Frankel, and Katrina Crotts Roohr, "Assessing Critical Thinking in Higher Education: Current State and Directions for Next-Generation Assessment," ETS RR-14-10 (2014). HCTA retired as of a 2024 *Journal of Intelligence* review (PMC10890380). No instrument named HALO was found; halo is a rater effect; HCTSR is a holistic rubric. Vendor: Insight Assessment (CCTST/CCTDI); Pearson / TalentLens (Watson–Glaser); CAE (CLA/CLA+).

45 percent is a CLA finding. HCTA had everyday scenarios and a life-event correlation that did not control for *g*, and it is retired. No HALO test exists; holistic rubrics have rater-agreement and halo problems.[32] The folk use does not know these distinctions. It knows a product with the words in the name. That is enough for the conjunction's "measurable by a short test." A book that treated the product category as the scholarly family would have let a vendor write the definition.

The three uses share a family resemblance. Careful thinking, directed at a goal, is somewhere in the background of all three — Hitchcock's concept, after Bailin and colleagues.[33] The danger is not that the resemblance is fake. The danger is the conjunction that treats slogan, compliment, and test as one transferable faculty, already delivered. The scholarly family can survive a family resemblance. It cannot survive that collapse.

A reader who wants a fourth folk use can have "21st-century skill" as a variant of the first. It is the mission stamp with a date attached, as if the century had discovered a new mental organ. P21's 4Cs, WEF's skills outlook, UNESCO's TVCs, and a thousand strategic plans use the dating as urgency. Urgency is not a construct. The activity Ennis named — deciding what to believe or do, reasonably, reflectively — is not younger than 2000 and not older than Dewey in its educational goal-name. Dating it to the century is how a framework sells demand. This book will use "twenty-first century" for dates. It will not use it as a kind of thinking.

## 1.2 The conjunction this book refuses

Print the collapse as a single sentence the rest of the book can point at.

> Critical thinking is a single, general, content-free skill (or small skill-set), teachable in a standalone course or a poster of maxims, measurable by a short test, possessed as a personal trait that licenses political authority, and already promised by every decent school.

Walk the conjuncts.

*Single.* One thing. Not a family of conceptions. Not Ennis's outline plus Delphi's assessment construct plus Paul's strong-sense warning plus Willingham's knowledge constraint plus McPeck's *about what?* plus Lipman's judgment. One mass noun. Chapter 2 will show that the six cannot design the same book. This chapter only needs the folk assumption: that they can, because they share a name.

*General.* Deployable across domains, the way a bicycle, once learned, is deployable whenever a bicycle is present. The preview, not the quotation: Willingham's 2007 argument is that thinking is not that sort of skill; the processes of thinking are intertwined with the content of thought; maxims

[32] Liu et al. 2014 as n. 41. Huber and Kuncel 2016 nursing result: despite NLN's CT requirement and heavy CCTST use, nursing students did not show larger long-term generic-CT gains than other majors. Bernard, Zhang, Abrami, et al. 2008: Watson–Glaser one-component structure, 60 published studies. Butler 2012 HCTA r = −.38 with negative real-world outcomes, no control for *g*; instrument retired.

[33] Hitchcock, SEP, following Sharon Bailin, Roland Case, Jerrold R. Coombs, and Leroi B. Daniels, "Conceptualizing Critical Thinking," *Journal of Curriculum Studies* 31, no. 3 (1999): 285–302 (1999b). Careful, goal-directed thinking; three-feature core.

without knowledge and practice will not implement.[34] The full bicycle passage is Chapter 3. Glaser, in 1941, already split the same way: the attitude of wanting evidence transfers better than skill in applying logical methods, which is "limited by" pertinent knowledge.[35] McPeck: thinking is always about X; "I teach thinking" simpliciter is empty.[36] Dewey's lantern sentence, already on the page in Chapter 2, said thinking "is specific, not a machine-like, ready-made apparatus to be turned indifferently and at will upon all subjects."[37] Four denials of content-freeness, from 1910 to 2007, and the folk conjunction still needs the content-free skill or the slogan does not work. A skill you must rebuild in every domain is not what a mission stamp promises. Transfer is the empirical name of this conjunct, and Chapter 4 will spend it: Thorndike and Woodworth (1901), spread of practice only where identical elements are concerned; Detterman (1993), the default prediction is failure and likelihood tracks similarity; Gick and Holyoak, analog plus hint far above analog without, the gap is *noticing*, not application skill.[38] Near is not far. A book that treats a same-semester gain on a similar test as evidence of unaided workplace and civic judgment a year later, identity on the line, will overclaim. The folk conjunction needs that overclaim. Without it, "general" is only Ennis's aim, and aim is not a bicycle.

*Content-free.* The same denial, from the other side. A poster of maxims — "consider both sides"; "look for deep structure"; "evaluate the source" — is the pedagogical expression of content-freeness. You can teach the maxims. Without background knowledge and practice the student cannot implement them. Even three-year-olds can think critically in a domain they know; even trained scientists fail outside theirs. That is Willingham again, previewed, not quoted. The conjunction needs content-freeness because a content-bound skill cannot be promised by every decent school as a single offering.

*Teachable in a standalone course or a poster of maxims.* The 1980s industrialized this conjunct: California GE, Paul's conference, College Board reasoning competency, *A Nation at Risk*'s "higher-

[34] Daniel T. Willingham, "Critical Thinking: Why Is It So Hard to Teach?," *American Educator* 31, no. 2 (Summer 2007). Bicycle passage quoted in full in Chapter 3, not here. Three-point boil-down previewed: (1) CT is not a skill deployable regardless of context; (2) there *are* metacognitive strategies that make CT more likely; (3) doing what the strategies call for depends on domain knowledge and practice. Prefer the article body over the AFT deck pull-quote "There's no such thing as critical thinking 'skills.' " Willingham did not say "CT cannot be taught."

[35] Edward M. Glaser, *An Experiment in the Development of Critical Thinking* (New York: Teachers College, 1941), p. 175, via Hitchcock. Disposition transfers better than skill; skill limited by pertinent knowledge. Sixty-six years before Willingham.

[36] John E. McPeck, *Critical Thinking and Education* (New York: St. Martin's Press, 1981), p. 8. "The propensity and skill to engage in an activity with reflective scepticism." Thinking always about X; X is never "everything in general." Do not cite McPeck as "there are no general thinking abilities of any kind" without the writing/speaking replies and his exception for some dispositions.

[37] John Dewey, *How We Think* (Boston: D. C. Heath, 1910), Gutenberg text. Lantern sentence quoted and spent in Chapter 2. Reflective-thought definition: "Active, persistent, and careful consideration of any belief or supposed form of knowledge in the light of the grounds that support it, and the further conclusions to which it tends, constitutes reflective thought."

[38] Edward L. Thorndike and Robert S. Woodworth, "The Influence of Improvement in One Mental Function upon the Efficiency of Other Functions," *Psychological Review* 8 (1901). Douglas K. Detterman, "The Case for the Prosecution: Transfer as an Epiphenomenon," in *Transfer on Trial*, ed. Detterman and Robert J. Sternberg (Norwood, NJ: Ablex, 1993). Mary L. Gick and Keith J. Holyoak, analogical transfer: Duncker radiation ~10 percent spontaneous; analog without hint ~30 percent; analog plus hint ~75 percent (1980, 1983). Spent in Chapter 4; previewed here as the empirical name of "general."

order intellectual skills," Delphi.[39] A course you can schedule on Tuesdays, or a framework page you can hang, is what institutions can buy. Domain-embedded practice — control-of-variables, Reading Like a Historian, diagnosis as knowledge-plus-cases — is what Chapter 5 will treat as the load-bearing beam. It is slower. It does not look like a critical-thinking offering. The conjunction prefers the course and the poster because those are the products. Abrami 2015's Ennis-type differences — mixed, infusion, general, immersion — were not statistically significant; design mixed anyway, for transfer reasons, not because 2015 proved mixed won a horse race. The pedagogical combination that actually showed up was dialogue plus authentic problems plus mentoring, g = 0.57 from 19 studies, against an overall g+ of 0.30 on standardized tests.[40] Those numbers are Chapter 5's. This chapter only needs the folk preference: the standalone course and the poster are what the conjunction can sell. Domain-embedded practice is what the record can defend. They are not the same offering, and the slogan-machines do not distinguish them.

*Measurable by a short test.* Watson–Glaser, CCTST, CLA+. Insight Assessment sells the Delphi operationalization. Pearson / TalentLens sells the Watson–Glaser lineage. CAE sells CLA+. Vendor, in each case.[41] Independent CCTST subscale alphas have been reported as low as 0.21; CLA+ constructed-response reliability 0.43; HCTA retired; no HALO test.[42] Nathan Kuncel: little discriminant validity of common critical-thinking tests against *g*.[43] The conjunction needs the short test because possession has to be certifiable. Chapter 6's rule — the test is not the thing — is the refusal of this conjunct. This chapter only needs to name it.

*Possessed as a personal trait that licenses political authority.* The compliment. A trait you have, they lack; therefore your side may speak and theirs may not. Delphi's 52 percent who rejected ethics-in-the-meaning were refusing this conjunct as a matter of definition.[44] Folk use installs it anyway. Paul's weak-sense is the trait in action: skilled, and selfish. The book's ethics close, in Chapter 10, will keep Ennis's correlative care and refuse constitutive partisan ethics. The folk conjunction cannot survive that refusal, which is why the compliment is so durable. It does political work the scholarly family did not authorize.

*Already promised by every decent school.* DeAngelo and colleagues, HERI, 2009: more than 99

[39]Industrialization: California GE / Dumke 1980, via Hitchcock; Richard Paul's annual conference from 1980; College Board 1983 reasoning competency, via Hitchcock; National Commission on Excellence in Education, *A Nation at Risk* (1983), "higher-order intellectual skills"; Facione Delphi commission 1987–90, ERIC ED315423.

[40]Abrami et al. 2015, *Review of Educational Research* 85(2): 341 effect sizes; g+ = 0.30; dialogue + authentic + mentoring g = 0.57 (19 studies). Ennis-type differences NS. Abrami et al. 2008: improvement "cannot be a matter of implicit expectation." Spent in Chapters 4–5.

[41]Vendor label as required by the brief's bibliography group D. Frameworks and tests are not interchangeable with trials.

[42]Liu et al. 2014: independent CCTST subscale alphas 0.21–0.51 (Leppa 1997) vs. author-reported ~0.68–0.70; Watson–Glaser subscale alphas 0.17–0.74 (Loo and Thorpe 1999). Bernard et al. 2008: Watson–Glaser one-component structure. Zahner 2013: CLA+ constructed-response reliability 0.43, test-level 0.87 largely driven by the multiple-choice section.

[43]Nathan R. Kuncel, 2011, via Christopher R. Huber and Nathan R. Kuncel, "Does College Teach Critical Thinking? A Meta-Analysis," *Review of Educational Research* 86, no. 2 (2016): 431–468. Little discriminant validity of common CT tests against *g*; little evidence they predict grades or job performance *better than IQ*.

[44]Peter A. Facione, *Critical Thinking: A Statement of Expert Consensus for Purposes of Educational Assessment and Instruction* (The Delphi Report), ERIC ED315423 (1990). 46 panelists; 52 percent reject ethics-in-the-meaning; ~61/30 split on dispositions-in-the-meaning. Not "the APA's official definition."

percent of faculty say developing critical thinking is very important or essential.[45] Tsui, 1998: 92 percent of students believed they had made critical-thinking gains.[46] Self-report is not a test; that is the next section but one. The conjunct is the promise, not the measurement. A school that did not say the words would be confessing to being *merely* about facts. The slogan-machines in the next section are how the promise became mandatory.

No serious account endorses all of those conjuncts together. That is not a curiosity. It is why a book called *Critical Thinking for Life* has to open on a refusal rather than on a program. A program that tried to keep the whole conjunction would be a slogan. A program that dropped the conjunction and kept the activity would have to say, in public, which conjuncts die. They die at different rates. Demand is real. Measurability of *something* is real. Generality of *aim* is Ennis's point. Content-freeness dies. The bicycle dies. The trait-as-political-license dies. The already-promised dies as soon as Abrami's "cannot be a matter of implicit expectation" is allowed on the page. The rest of the book is that sorting.

A note on what "refuse" does not mean. Refusing the conjunction is not refusing the activity. Dewey's reflective thought — active, persistent, careful consideration of belief in light of grounds and further conclusions — is worth keeping almost verbatim, and Chapter 2 will keep it. Ennis's sentence is the public definition this life book can actually use. Willingham's constraint is how you teach toward that definition without lying about transfer. Lateral reading is a civic habit with trial evidence. Natural frequencies restore insight on a class of number-problems physicians fail. Accuracy prompts move sharing at the margin. None of those survive as a bicycle. All of them survive as work. The folk collapse is dangerous not because it uses the words, but because it uses them to promise a faculty that would make the work unnecessary. If you already *have* critical thinking, as a trait, as a course completed, as a test passed, you do not need to leave the page, recode the number, or notice that you do not know. The conjunction is a permission structure for skipping the deeds. Naming it is how the permission gets revoked.

## 1.3 Slogan-machines: P21, OECD, UNESCO

Why every school now says the words: not because a research construct was settled, but because frameworks and ministries needed a vocabulary for demand.

P21 — the Partnership for 21st Century Skills, later Learning — published *Framework Definitions* in December 2009 (ERIC ED519462). "Learning and innovation skills" equal the 4Cs: creativity, **critical thinking**, communication, collaboration. Critical thinking and problem solving are unpacked as: reason effectively (inductive, deductive, "as appropriate to the situation"); use systems thinking; make judgments and decisions (analyze and evaluate evidence, arguments, claims, beliefs; alternative points of view; synthesize; interpret; "reflect critically on learning experiences and processes"); solve problems, including "non-familiar" problems, and ask significant questions.[47] P21

[45] Linda DeAngelo, Sylvia Hurtado, John H. Pryor, et al., *The American College Teacher: National Norms for the 2007–2008 HERI Faculty Survey* (Los Angeles: HERI, UCLA, 2009). >99 percent very important or essential.

[46] Lisa Tsui, "Fostering Critical Thinking through Effective Pedagogy," *Journal of Higher Education* 70, no. 4 (1999) / Tsui 1998 as cited in the teaching-stream memo: 92 percent of students believed they had made CT gains. Self-report, not a test. (Tsui 1998 as given in the brief's Chapter 1 fact 4.)

[47] P21 2009, ERIC ED519462. See n. 23.

*does* say, in the same document: “Those who can think critically and communicate effectively must build on a base of core academic subject knowledge.” That clause is routinely dropped in slogan use. Dropping it is how a framework that contained a knowledge constraint became a slogan-machine for content-free skill. P21 was an advocacy coalition, business plus education, not a research synthesis. A 2008 companion, *21st Century Skills, Education and Competitiveness*, reports employer-survey ranking of critical thinking / problem solving and proposes a U.S. Department of Education “Office of 21st Century Skills.” Uptake is not evidence. “21st-century skill” does not name a new mental organ.

The OECD Learning Compass 2030 construct note *Critical Thinking*, after surveying Sternberg, Facione, Ennis, and Bailin, defines critical thinking as “questioning and evaluating ideas and solutions (OECD, 2016).”[48] It is then plugged into the Anticipation-Action-Reflection cycle. WEF *Future of Jobs* (2016) is cited as labor-market atmosphere. This is a policy construct note, one paragraph wide, citing the scholarly family and then choosing a short operational phrase for mapping curricula. It is not a rival theory. It is not a trial. It is how a ministry can say the words and point at a PDF.

UNESCO Bangkok / ERI-Net (2013–2015, refined later) puts “transversal competencies” in six domains, the first of which is “critical and innovative thinking” — examples listed: creativity, entrepreneurship, resourcefulness, application skills, reflective thinking, reasoned decision-making (UNESCO, *Assessment of transversal competencies*, 2016).[49] SDG 4.7 / Education 2030 is the policy envelope. UNESCO documents treat TVCs as overlapping with “21st century skills.” Critical thinking sits in a bin with entrepreneurship. That fact is diagnostic of slogan-collapse. A grouping for curriculum reform and teacher training is not a validated construct. A book that cited UNESCO TVC as evidence that a generic skill had been specified, taught, or measured would have mistaken a ministerial bin for a finding.

Standpoint, planted. P21 is employer-facing: a business-plus-education advocacy coalition that wanted a U.S. ED office and used employer surveys as atmosphere. OECD’s construct note is a mapping tool for curricula after a one-paragraph survey of the scholarly family. UNESCO TVC is Asia-Pacific ministerial, SDG-enveloped, and bins critical thinking with entrepreneurship. Delphi, which the frameworks sometimes cite by way of Facione, is North American, college-experienced, philosophy-heavy, 1988–89. A global life book that treated any of these as “what critical thinking is” would have a standpoint problem even if the documents had been trials. They are not trials. The standpoint problem is therefore not that a representative expert panel was misread. It is that demand-vocabularies from particular rooms were generalized as a human faculty. Chapter 10 returns to who is left out. This section only needs the diagnostic: when critical thinking sits in a bin with resourcefulness, the phrase is doing slogan-work.

The U.S. policy stack is a stack of uses, not a research construct. No single U.S. Department

[48] OECD, Learning Compass 2030 construct note, *Critical Thinking*. “The OECD defines critical thinking as questioning and evaluating ideas and solutions (OECD, 2016).” Policy construct note, one paragraph wide. Not a rival theory; not a trial.

[49] UNESCO Bangkok / ERI-Net, *Assessment of Transversal Competencies* (2016). “Critical and innovative thinking” binned with creativity, entrepreneurship, resourcefulness. Diagnostic of slogan-collapse. Asia-Pacific ministerial standpoint.

of Education definition of critical thinking was found that functions as one.[50] *A Nation at Risk* (1983) — "higher-order intellectual skills," cited by Willingham 2007. College Board 1983 — reasoning as a basic competency, via Hitchcock. California GE requirement (Dumke 1980) — the most operational U.S. *system* definition of that era. Goals 2000-era NCES / National Center for Postsecondary Teaching, Learning, and Assessment survey (Jones and colleagues, 1994/1995) used Delphi / CCTDI language to ask educators and employers what college graduates should be able to do; Facione (2000) reports "strong national consensus" on those outcomes, which is a consensus-on-a-survey, not a finding that Delphi-CT is being achieved. Common Core and later state standards more often say "reason abstractly," "evaluate arguments," "use evidence" than "critical thinking" as a standalone standard — unverified as a systematic census; this book will not assert a full standards count it did not do.

What a framework can do for this book is modest and real. It can show that the words are now mandatory in rooms that allocate money, write standards, and hire. It can show vocabulary drift: "critical thinking and problem solving," "analytical thinking," "critical and innovative thinking," "higher-order intellectual skills," "reasoning competency," each doing slightly different work, all available to be collapsed. It cannot show that a generic skill has been specified. It cannot show that the skill has been taught. It cannot show that it has been measured. Rule for the rest of the pages: cite frameworks as evidence of demand and of vocabulary drift. Never as evidence that a generic skill has been specified, taught, or measured. P21, OECD, UNESCO TVC, NAMLE, NACE rubrics, and WEF are slogan-machines and demand-signals. They are not trials. A law that says "teach critical thinking / media literacy" is, Chapter 7 will add, not Civic Online Reasoning, not inoculation, and not an accuracy prompt. Twenty-five state media-literacy laws, on Media Literacy Now's January 2026 count, are laws; MLN has not evaluated classroom implementation.[51] The slogan-machine's last product is a statute that says the words.

NAMLE's 2023 Core Principles are the professional-association cousin of the same machinery. Media literacy education is defined as habits of inquiry and skills of expression necessary for people to be "critical thinkers, thoughtful and effective communicators, and informed and responsible members of society." Principle 3: curious, open-minded, self-reflective inquiry, emphasizing reason, logic, and evidence. Principle 4: active inquiry, reflection, and critical thinking. Principle 10.4: identify quality, reliable, accurate information. The 2023 revision committee lists SHEG among sources drawn on.[52] This is a framework. NAMLE does not report an RCT on that page. The contrast the later civic chapter can print is already visible: a principle that says "critical thinking" is not Wineburg's three questions and not a trial. P21, OECD, UNESCO, NAMLE, and the U.S. stack are how the words became mandatory. They are not how the activity became teachable.

[50] No single US ED research-construct definition found in the 01-memo fetch, 30 August 2026. Stack of uses, not a construct. A systematic census of US state standards' use of the phrase is **UNVERIFIED**; not asserted here.

[51] Media Literacy Now, policy map and *U.S. Media Literacy Policy & Impact Report*, looking through 30 October 2025, page last updated January 2026: 25 states have media-literacy laws; eleven took new steps since January 2024. MLN states it has **not evaluated** classroom implementation. Illinois 2021, New Jersey 2022, Delaware 2022 statutory details **UNVERIFIED** pending bill text.

[52] National Association for Media Literacy Education, Core Principles, 2023 revision, fetched namle.org/resources/core-principles, 30 August 2026. Framework, not a trial. SHEG listed among sources drawn on by the revision committee.

## 1.4 Self-report is not a test

Faculty and students already believe the slogan is being kept.

Linda DeAngelo, Sylvia Hurtado, John Pryor, and colleagues, in the 2007–2008 HERI faculty survey published 2009: more than 99 percent of faculty say developing critical thinking is very important or essential.[53] That is a mission consensus, not a measured change. A faculty that did not check the box would be announcing that development of thinking was optional. The number is a ceiling. Ceilings of this kind do not move when teaching moves. They are how the promise became unanimous.

Lisa Tsui, 1998: 92 percent of students believed they had made critical-thinking gains.[54] Self-report is not a test. A student who has been told, on every syllabus, that this course develops critical thinking will, at a high rate, report that it did. Huber and Kuncel's later result, which Chapter 4 and Chapter 5 will spend, is the measurement the self-report is not: ordinary college already moves generic tests about half a standard deviation in four years, and stamping "critical thinking" on every syllabus — curriculum-wide campaigns, nursing's mandated curriculum — did not add a reliable increment.[55] Immersion-by-committee is what the mission stamp becomes when it is operationalized as a hoped-for by-product. Abrami 2008: improvement cannot be a matter of implicit expectation. The self-report literature is implicit expectation, surveyed.

Plant, not spend: later chapters will show that ordinary college already moves generic tests about half a standard deviation in four years, and that the movement is not the same object as a life of judgment. Huber and Kuncel's publication-year moderator — model-predicted four-year gain about 1.22 SD for 1963 versus about 0.33 for 2011 — is multiply interpretable, not a clean "kids today" claim, and is not used here as one.[56] Niu, Behar-Horenstein, and Garvan (2013): college interventions, 40 effect sizes, mean 0.195.[57] Small, on the tests used, not far. Faculty importance at 99 percent and student self-reported gains at 92 percent are not in tension with those numbers once self-report is refused as a test. They are in tension with the folk conjunction, which needs the 99 percent and the 92 percent to mean possession. Possession is what a mission stamp asserts. Measurement, when it is honest, is smaller, nearer, and not a civic outcome.

Arum and Roksa's 45 percent "no gain" on the CLA in the first two years of college is the other pole, and it is not "critical thinking cannot be taught." It is a CLA finding, partly motivation; Huber and Kuncel treat CLA as criterion-contaminated, not a pure critical-thinking test; Liu, Bridgeman, and Adler showed that conclusions about college learning can *flip* depending on whether students are induced to try.[58] This chapter plants the warning. Chapter 6 spends it. The folk conjunction

[53] DeAngelo et al. 2009. See n. 44.

[54] Tsui 1998. See n. 45.

[55] Huber and Kuncel 2016. Mixed-design 4-year CT-skill gain 0.59 SD; purely longitudinal 0.46 SD. Curriculum-wide CT campaigns not necessarily incremental; nursing mandate did not beat other majors on domain-general tests. Planted here; spent in Chapters 4–5.

[56] Huber and Kuncel publication-year moderator: model-predicted 4-year gain ~1.22 SD for 1963 vs. ~0.33 for 2011 — multiply interpretable, not a clean "kids today" claim.

[57] Lian Niu, Linda S. Behar-Horenstein, and Cyndi W. Garvan, "Do Instructional Interventions Influence College Students' Critical Thinking Skills? A Meta-Analysis," *Educational Research Review* 9 (2013): 114–128. 40 ES, mean 0.195.

[58] Richard Arum and Josipa Roksa, *Academically Adrift* (Chicago: University of Chicago Press, 2011). 24 institu-

uses DeAngelo and Tsui as proof of possession and Arum and Roksa, when it uses them, as proof of crisis. Neither is a definition. One is a mission survey. One is a low-stakes performance task. Self-report is not a test. A test is not the thing.

The crisis version of the folk story is worth a paragraph because it is the conjunction with the sign flipped. If 45 percent of students show no CLA gain, and if CLA is "critical thinking," then college is failing to deliver the bicycle. Huber and Kuncel's critique — criterion contamination, CLA not a pure CT test — and Liu, Bridgeman, and Adler's motivation result together say the 45 percent cannot bear that weight. Abrami's 0.30 says something can be moved on standardized CT tests, small, not far, not by implicit expectation. EEF 2021's 0 months says P4C as an attainment program at scale did not move KS2 reading. Breakstone's 96 percent says the civic baseline, before ChatGPT, was not a population of critical thinkers the schools had already made. The honest picture is not crisis-as-missing-bicycle. It is a set of small, near, incomplete, heterogeneous results on different objects, against a public language that needs a single faculty either possessed or catastrophically absent. Heterogeneity is the story. The conjunction cannot tell it.

First-year seminars are the higher-education version of the mission stamp, and they are the wrong outcome for a critical-thinking claim. Permzadian and Crede (2016): first-year seminars move GPA by $\delta = 0.02$ and one-year retention by $\delta = 0.11$.[59] Weinstein on Montclair: the *word* on syllabi without a unifying theoretical basis. K–12 standards already distributed some of the right practices — Common Core MP.3, NGSS Practice 7, C3 Dimensions 3–4 — and did not create a critical-thinking course. Naming is not teaching. A school that stamps the words on every syllabus and then points at DeAngelo's 99 percent has produced a survey result. Abrami's 0.30 and Huber and Kuncel's college slope are what teaching, and ordinary college, actually move on the tests used. They are not far. They are not civic. They are not a reason to keep the conjunction. They are a reason to stop treating self-report as the evaluation.

## 1.5 Recruiter "#1 skill" is folklore unless sourced

Employers say the words too. What they measure is a story.

NACE *Job Outlook 2025*, data 5 August–16 September 2024, 237 respondents, 162 of them members (19.2 percent of eligible members), plus 75 nonmembers:

| Competency | Importance (1–5) | Rated proficiency of recent grads (1–5) |
|---|---|---|
| Communication | 4.57 | 3.62 |
| **Critical thinking** | **4.49** | **3.67** |

tions, >2,300 students; 45 percent no statistically significant CLA gain in the first two years. Ou Lydia Liu, Brent Bridgeman, and Rachel Adler, "Measuring Learning Outcomes in Higher Education: Motivation Matters," *Educational Researcher* 41, no. 9 (2012): 352–362. Motivation on low-stakes tests can flip college-gain conclusions. Do not cite Arum and Roksa as "critical thinking cannot be taught."

[59] Vahe Permzadian and Marcus Crede, "Do First-Year Seminars Improve College Grades and Retention? A Systematic Review and Meta-Analysis of Academic and Persistence Outcomes," *Review of Educational Research* 86, no. 1 (2016). GPA $\delta = 0.02$, one-year retention $\delta = 0.11$ — wrong outcome for a CT claim. CCSS MP.3, NGSS Practice 7, C3 Framework 2013: standards distributed practices; they did not create a CT course.

| Competency | Importance (1–5) | Rated proficiency of recent grads (1–5) |
|---|---|---|
| Teamwork | 4.43 | 4.00 |
| Professionalism | 4.31 | 3.50 |

96.1 percent call critical thinking very or extremely important; 53.5 percent call graduates very or extremely proficient. Resume attribute “problem-solving skills”: 88.3 percent. Skills-based hiring: 64.8 percent. GPA screening: 46.4 percent.[60] NACE’s own definition of the competency: “Identify and respond to needs based upon an understanding of situational context and logical analysis of relevant information.” Sample interview item: “Can you describe a time when you anticipated a problem in advance…?” What they test, when they test, is a situated story, not a Facione skill and not a Watson–Glaser. NACE 2026’s skills-based hiring page: 70 percent of Job Outlook 2026 employers report skills-based hiring; GPA screening 42 percent “this year.”

WEF *Future of Jobs Report 2025*: **analytical thinking** remains the top core skill, “seven out of 10 companies considering it as essential.”[61] That is a WEF employer survey, not a validated critical-thinking test, and the phrase is *analytical thinking*, not “critical thinking.” AI and big data are the *fastest-growing* skills, not the current number one. Reading, writing, and mathematics show a small net *decline* in expected importance — a survey forecast, not a finding that literacy stopped mattering.

Treat unsourced “critical thinking is the #1 skill employers want” as folklore. Sourced to NACE 2025 it is still not number one; communication edges it. Sourced to WEF 2025 it is a different phrase. In both cases it is a recruiter survey, not a skill assessment, and not evidence that teaching isolated critical-thinking skills produces job performance. NACE 237 is not a probability sample of all employers. Importance is a Likert rating. A behavioral-interview story is closer to Willingham — knowledge plus practice in a domain — than to a generic course. Chapter 8 will restate this with the workplace detail. This chapter kills the unsourced labor-market sentence before it can walk through the rest of the book as a finding.

The folklore has a function. If employers want a general skill, schools can promise a general course. If what employers actually ask is a story about a problem in a situation, the honest offering is domain practice plus the habit of noticing. The slogan-machine prefers the first. The record is the second.

Beware survey folklore, restated as a checklist because the sentence will try to return. These are convenience and member surveys, not probability samples of all employers. “Importance” is a Likert rating, not a work-sample test of Facione skills or Watson–Glaser items. NACE 2025 does **not** make critical thinking number one; communication does. WEF’s word is *analytical thinking*. What gets tested, when skills-based hiring is real, is a situated story (STAR, behavior) or a work sample — which is closer to Willingham than to a generic course. “Critical thinking is the most important skill of the twenty-first century,” without a citation, is folklore. With a WEF or NACE citation

[60]NACE *Job Outlook 2025*, as n. 24. Resume, skills-based hiring, GPA figures from the same report. NACE 2026 skills-based hiring page: 70 percent report skills-based hiring; GPA screening 42 percent.

[61]World Economic Forum, *Future of Jobs Report 2025*, skills outlook. Analytical thinking, 7/10 companies, survey. Not “critical thinking.” Not a skill assessment.

it is still a recruiter survey, not evidence that teaching isolated skills produces job performance. Open question 14 of this book is whether anyone hiring at scale uses a validated critical-thinking instrument that predicts job performance after controlling for general cognitive ability and domain knowledge. It was not in the NACE or WEF pages fetched. Chapter 8 will not invent it.

NACE's sample behaviors for the competency — gather diverse sources, anticipate needs, summarize data with bias awareness, communicate rationale — are closer to Ennis's outline than to a bicycle, and they are still a rubric, not a trial. Skills-based hiring, on the 2025 report, is used most at interviewing (89.7 percent of those who use it) and screening (67.5 percent). Practices: competency-based job descriptions 73.7 percent, interview rubrics 52.6 percent, internally created assessments 38.6 percent, game-based assessments 2.6 percent, digital badges 3.5 percent.[62] The badge is almost unused. The interview story is the modal test. A life book that treated the unused badge as the employer's demand would have inverted the record. The folklore sentence — "#1 skill" — inverts it in a different direction: it takes a Likert rating, strips communication's edge, swaps WEF's "analytical thinking" back into "critical thinking," and prints a labor-market law. This section is the strip-back.

## 1.6 One concept, rival conceptions

David Hitchcock, in the *Stanford Encyclopedia of Philosophy*, treats the many definitions as rival *conceptions* of one *concept*: careful thinking directed to a goal.[63] After Bailin and colleagues (1999b): done to make up one's mind about what to believe or do; the thinker tries to meet standards of adequacy and accuracy; the thinking meets them to some threshold. That is a useful writer's move. It is not a license to collapse Ennis, Facione's Delphi, Paul–Elder, Willingham, McPeck, and Lipman into one operational definition. They cannot all design the same book.

Chapter 2 will print the table. This chapter needs only the preview. Ennis: reasonable reflective thinking focused on deciding what to believe or do; a long outline of dispositions and abilities; Cornell / Ennis–Weir; the best public sentence for a *life* book because it includes action. Delphi / Facione: expert consensus for assessment and instruction; six skills; a 61/30 split on whether dispositions are in the meaning; 52 percent reject ethics-in-the-meaning; CCTST as *one* operationalization of *one* consensus statement, not a discovery of nature. Paul–Elder: a curriculum brand; strong sense versus weak sense; classroom vocabulary; not a meta-analysis. Willingham: the cognitive-science constraint; domain-specific critical thinking can be taught; no proven way to teach a content-free general skill directly. McPeck: thinking is always about X; the *about what?* demand. Lipman: judgment, community of inquiry, childhood, criteria, self-correction, context-sensitivity; pedagogy, not EEF months. Adjacent accounts exist and are not the six: Siegel, Bailin, Halpern, Fisher and Scriven, Stanovich, Kuhn, Freire/hooks. The six cannot be collapsed. The adjacent cannot be promoted by silence.

[62] Same NACE 2025 report: among those who use skills-based hiring, interviewing 89.7 percent, screening 67.5 percent. Competency-based job descriptions 73.7 percent; interview rubrics 52.6 percent; internally created assessments 38.6 percent; game-based assessments 2.6 percent; digital badges 3.5 percent. Badge almost unused; interview story modal.

[63] Hitchcock, SEP. Rival conceptions of one concept. Chapter 2 prints the six-account table; this chapter does not dump it.

The live debate this section has to own is whether "critical thinking" is still the right title-phrase. Dewey preferred reflective thinking. Some cognitive scientists prefer scientific thinking, historical thinking, rationality, judgment. The market will punish a title change; the argument might not.[64] This book holds the title. The work of this chapter is to make the folk uses visible so later chapters can refuse them without sounding pedantic. A title change that left the conjunction intact would be cosmetics. A title held, with the conjunction named and refused, is an argument.

The coalition this manuscript will run, named here so Chapter 2 can choose it in public rather than smuggle it, is Ennis (believe *or do*) plus Dewey (inquiry, not only appraisal) plus Willingham (knowledge and practice) plus a Paul-ish warning about weak-sense sophistry plus Lipman for childhood judgment, with McPeck's *about what?* kept as a constraint. That is a coalition. Coalitions hide contradictions. Where Willingham vetoes Ennis-style general courses, this book will say so. Where Ennis's outline still earns pages, it earns them as a specification of aim. Delphi remains the source of every "six skills" list, named as such — expert consensus for assessment and instruction, not a discovery of nature. Paul–Elder remains a brand from which this book steals a warning and not a theory. The folk conjunction cannot tell those differences because it does not need them. A slogan that had to say "Ennis's aim, Willingham's constraint, not Delphi-as-nature, not Paul's franchise" would not be a slogan.

Standpoint is the other live constraint, planted here, spent in Chapters 2 and 10. Delphi is North American, college-experienced, philosophy-heavy, 1988–89. P21 is employer-facing. UNESCO TVC is Asia-Pacific ministerial. A global *life* book that treats these as "what critical thinking is" has a standpoint problem. Who is left out of expert panels remains Open question 21. Naming it in Chapter 1 is how the slogan-machines are kept from pretending to be universal.

## 1.7 Thesis, constraint, and the map

Critical thinking is not a transferable bicycle-skill. It is domain knowledge plus evaluation habits practiced on real stakes, for life, including when a machine will finish the sentence. Ennis's public sentence — "reasonable reflective thinking focused on deciding what to believe or do" — is the best *life*-book definition because it includes action, but it does not settle pedagogy.[65] Willingham 2007/2019 is the pedagogical constraint: thinking processes are intertwined with the content of thought; maxims without knowledge and practice will not implement; domain-specific critical thinking *can* be taught; there is no proven way to teach a content-free general skill directly. Transfer is rare and similarity-bound. Implicit expectation fails. The test is not the thing. Civic and professional life are domains, not proof of a transferable faculty. When a machine finishes the sentence, evaluation is the remaining human work. A life of critical thinking is a life of deeds, not

[64] Open question 18 of the research brief. Hold the title unless a better public sentence than Ennis's appears. Standpoint: Open question 21.

[65] Robert H. Ennis, "A Logical Basis for Measuring Critical Thinking Skills," *Educational Leadership* 43, no. 2 (October 1985): 44–48, already as the "working definition"; restated in Ennis, "The Nature of Critical Thinking: Outlines of General Critical Thinking Dispositions and Abilities," last revised September 2015, https://criticalthinking.net/wp-content/uploads/2024/04/The-Nature-of-Critical-Thinking.pdf: "Critical thinking is reasonable reflective thinking focused on deciding what to believe or do." Includes creative activities (hypotheses, alternatives, plans). Best *life*-book definition because it includes action; does not settle pedagogy. Variants drop or add a comma and sometimes "and"; the *do* is in every variant used here. Willingham 2007/2019 is the pedagogical constraint.

a badge.

Which conjuncts survive. Demand survives: schools, ministries, and recruiters want a word for careful, goal-directed thinking that includes action. Generality of *aim* survives: Ennis's believe-*or-do* is why this can be a life book and not only an argument-appraisal book. Something measurable survives, with a warning: tests operationalize constructs; they do not name a mental organ. Content-freeness does not survive. The bicycle does not survive. The standalone course as the engine does not survive Abrami's explicitness finding or Willingham's domain maps, though Ennis's mixed *proposal* will earn a design paragraph, not an RCT claim. The short test as the meaning does not survive Chapter 6. The trait that licenses political authority does not survive Delphi's majority, Ennis's correlative-not-constitutive care, or Paul's own warning about sophistry. The already-promised does not survive self-report, Huber and Kuncel, or a 96 percent miss rate on a climate `.org`.

The map, one paragraph, no effect sizes. Chapter 2: what the phrase has meant, Dewey to Delphi to six accounts, the coalition and the vetoes. Chapter 3: you cannot think about what you do not know; the bicycle passage, quoted; expertise is encapsulated. Chapter 4: why maxims fail — transfer, dual process, myside. Chapter 5: what teaching actually moves; implicit expectation fails; domain-embedded practice as the beam; P4C's two RCTs, different stories. Chapter 6: the test is not the thing. Chapter 7: leave the page; civic and media life; lateral reading versus CRAAP; backfire rare, continued influence real. Chapter 8: clinic, workplace, court as domains with named knowledge and error-costs. Chapter 9: machines that finish the sentence; one chapter, not five; evaluation as the remaining human work. Chapter 10: a life of deeds, not a badge; stages as implication; equity older than the chatbot. Coda: what an honest book can promise versus what it cannot.

Before choosing a conception, show what the phrase has meant. That is Chapter 2. The folk uses will still be on the page. They are supposed to be. A book that made them invisible would have to refuse them later, and would sound as if it were correcting the reader's vocabulary rather than the public conjunction. This chapter did the naming. The record does the rest.

Two open questions this chapter leaves standing, on purpose. Whether "critical thinking" is still the right title-phrase — Open question 18 — is held, not filled. No better public sentence than Ennis's has been offered by the folk uses, and the folk uses are not in the business of offering one. Standpoint of 1988–89 expert panels, P21, and UNESCO TVC — Open question 21 — is named, not solved. A Chapter 1 that filled either with a coinage story or with a claim of representativeness would have rejoined the popular story it exists to take apart. The corrections begin in the introduction. The conceptions begin in Chapter 2. The bicycle is quoted in Chapter 3. The slogan has been named. That is the work.

One more preview, because the title has a payload the folk uses do not. "For life" is not a K–12 unit and not a NACE PDF. It is the claim that the activity Ennis named is practiced on stakes that do not grade you — a vote, a share, a test result, a hiring story, a sentence a machine will finish — and that the practice requires knowledge in those domains. Chapter 7 will be civic. Chapter 8 will be clinic, work, court. Chapter 9 will be machines, one chapter not five, because Willingham and Bastani and Fan already live in *Autonomous AI and Education* and this book needs the life condition, not a sequel. Chapter 10 will be deeds. None of those chapters can do their jobs if Chapter 1 has left the conjunction unnamed. A reader who still thinks critical thinking is a bicycle, a test, a compliment, and a promise will read Wineburg as a citizenship vaccine, Bastani as an anti-AI brief, and Ennis as a poster. This chapter is the refusal that makes those misreadings harder. It is not yet the argument that replaces them.

The thesis, restated without a program attached: not a bicycle-skill; domain knowledge plus evaluation habits on real stakes, including when a machine finishes the sentence. Ennis for the public definition; Willingham for pedagogy. The coalition named; the vetoes promised. Open questions 18 and 21 standing. No invented classroom. No invented US ED definition. No recruiter "#1" without NACE's 4.57/4.49 and WEF's different phrase. No Pew poll that was not fetched. The phrase as compliment is on the page as a folk use. The bicycle is previewed, not quoted. The six accounts are previewed, not dumped. What a Chapter 1 of this book is allowed to do is name the slogan, kill the folklore, stamp the mission sentence as a stamp, and hand the reader to history. Chapter 2 is the history. A reader who came for a skills poster has already been told it will not be in this book. A reader who came for a life of judgment has been told what the popular story will try to sell instead, and which documents refuse the sale. That is enough for a first chapter. The bicycle is next but one. The record is already in the building.

# Chapter 2

## What the Phrase Has Meant: Dewey to Delphi to Six Accounts

Figure 3: A lantern, a stack of bound books, a tied folio, and a spiral notebook on a dark desk.

Two sentences from the same 1910 book already contain the argument this chapter has to keep in view. John Dewey, writing *How We Think* for teachers, defined reflective thought as “active, persistent, and careful consideration of any belief or supposed form of knowledge in the light of the grounds that support it, and the further conclusions to which it tends.”[66] A few chapters later

[66]John Dewey, *How We Think* (Boston: D. C. Heath, 1910), chap. 1. Wording checked against the Project Gutenberg transcription, https://gutenberg.org/files/37423/37423-h/37423-h.htm (accessed August 30, 2026). David Hitch-

he refused the picture that would make that definition easy to industrialize: thinking, he wrote, "is specific, not a machine-like, ready-made apparatus to be turned indifferently and at will upon all subjects, as a lantern may throw its light as it happens upon horses, streets, gardens, trees, or river."[67] One sentence names an activity worth keeping almost verbatim. The other already vetoes the bicycle. Anyone who treats Dewey as the grandfather of a content-free skills course has not read both.

This chapter is the history and the choice. The English educational goal-name is modern. The practices are not. Socrates did not say "critical thinking." Bacon did not say it. Kant did not say it. Dewey *used* the phrase as a name for an educational goal; Edward Glaser *stabilized* it as a research heading; the 1980s *industrialized* it into requirements, conferences, textbooks, and tests.[68] Nobody in this record needs to have coined the words. Coinage myths are a way of skipping the work of saying what the words have meant.

David Hitchcock, in the *Stanford Encyclopedia of Philosophy*, treats the many definitions as rival *conceptions* of one *concept*: careful thinking directed to a goal.[69] That is a useful writer's move. It is not a license to collapse Robert Ennis, Peter Facione's Delphi panel, Richard Paul and Linda Elder, Daniel Willingham, John McPeck, and Matthew Lipman into one operational definition. They cannot all design the same book. The work of this chapter is to show the family, print a table of what each account can and cannot design, and then choose in public: Ennis's sentence for a *life* book, because it includes action; Dewey for inquiry rather than mere appraisal; Willingham for the knowledge constraint; a Paul-ish warning about weak-sense sophistry; Lipman for childhood judgment; McPeck for the demand that thinking always be *about* something. Delphi is the source of every "six skills" list that will appear later in this book, named as such — expert consensus for assessment and instruction, not a discovery of nature.

That choice is a coalition. Coalitions hide contradictions. The vetoes belong on the page, not in a later draft.

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cock cites the same sentence to Dewey 1910: 6 and to the 1933 restatement at p. 9; see Hitchcock, "Critical Thinking," *Stanford Encyclopedia of Philosophy*, first published July 21, 2018, substantive revision October 12, 2022, https://plato.stanford.edu/entries/critical-thinking/.

[67]Dewey, *How We Think* (1910), chap. 3, "Natural Resources in the Training of Thought" (Gutenberg text, Heath 1910 pagination near pp. 38–39). The lantern/specificity sentence is not in chap. 1 with the reflective-thought definition; it is a few chapters later in the same volume.

[68]On Dewey's *use* of the English phrase, see Hitchcock, "Critical Thinking: History," *Stanford Encyclopedia of Philosophy*, https://plato.stanford.edu/entries/critical-thinking/history.html (live page re-fetched 30 August 2026). Pages 74 and 82 of Dewey 1910 have now been opened; see n. 80. Hitchcock's claim that those pages introduce the term *as the name of an educational goal* remains Hitchcock's framing. On Glaser's 1941 stabilization of the phrase as a research heading, see Edward M. Glaser, *An Experiment in the Development of Critical Thinking*, Teachers College Contributions to Education no. 843 (New York: Teachers College, Columbia University, 1941), quoted via Hitchcock, "History." On the 1980s industrialization (California GE, Paul's conference, College Board, *A Nation at Risk*, Delphi), see §2.5 and nn. 100–105. This chapter does not claim that anyone *coined* the English phrase.

[69]Hitchcock, "Critical Thinking," following Robert H. Ennis following John Rawls on concept versus conception; the three-feature core is from Sharon Bailin, Roland Case, Jerrold R. Coombs, and Leroi B. Daniels, "Conceptualizing Critical Thinking," *Journal of Curriculum Studies* 31, no. 3 (1999): 285–302 (1999b), as reported by Hitchcock. The companion 1999a paper in the same issue is "Common Misconceptions of Critical Thinking," 269–83 — against generic discrete skills, not the three-feature core.

## 2.1 Ancestry without coinage

The examined life is older than the English phrase. Plato's dialogues give the book a picture of thinking as accountability to reasons in public: *elenchus*, cross-examination that tests a claim by drawing out its implications until inconsistency appears. The moral frame is the *Apology*; the method is adversarial, interpersonal, and typically destructive of unearned confidence rather than productive of a positive doctrine. Meno's paradox, the slave-boy geometry episode, and the *Euthyphro* hunt for a definition are the teaching set-pieces because they show the cost. What elenchus cannot license is the sentence "Socrates invented critical thinking." He did not use the English phrase. "Critical thinking" as an educational goal-name is modern. Freire's politically freighted use of "critical," which Hitchcock's encyclopedia entry treats as a non-example of the mainstream concept, is a different descendant, not elenchus under another name.[70] The standing risk is the one any life book has to keep: skilled questioning can become a blood sport.

Francis Bacon, in 1620, named obstacles rather than a competency. *Novum Organum*, Book I, aphorisms 39–44, lists four "idols" that "beset the human mind": idols of the Tribe (species-level distortions — the senses as a false mirror, the over-reading of order into nature); of the Cave or Den (causes peculiar to a person — temperament, education, private hobby-horses); of the Market or Marketplace (those that come from intercourse and language, words forcing the understanding); and of the Theatre (received systems as stage-plays, dogmatic philosophies and perverted rules of demonstration).[71] Dewey's second chapter in *How We Think* walks the reader through those four and treats them as a classification of standing sources of error. His own moral, which is closer to a life-skill brief than most twenty-first-century competency frameworks, is that "only systematic regulation of the conditions under which observations are made and severe discipline of the habits of entertaining suggestions can secure a decision that one type of belief is vicious and the other sound."[72] Bacon is diagnosing obstacles to natural-philosophical inquiry, not writing a general-education syllabus. The idols are ancestors of bias-talk. They are not a taxonomy this book can relabel "critical-thinking skills" and be done.

Dewey also quotes Locke at length on four "wrong measures of probability": implicit faith in others; passion in place of reason; "want of… large, sound, roundabout sense"; and "giving up our assent to the common received opinions." He quotes Mill: "To draw inferences has been said to be the great business of life."[73] Hitchcock notes that those lengthy quotations of Bacon, Locke, and Mill indicate that Dewey "was not the first person to propose development of a scientific attitude of mind as an educational goal."[74] That is the honest ancestry: practices and diagnoses that later educational writers, especially Dewey, explicitly harvested. It is not a pedigree in which every famous name said the English words.

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[70]On elenchus and on Paulo Freire, *Pedagogy of the Oppressed* (1970 trans.), as a non-example of the mainstream concept, see Hitchcock, "Critical Thinking." Freire and bell hooks remain live competitors in folk use of "critical"; they are not treated here as a seventh of the six accounts in §2.6.

[71]Francis Bacon, *Novum Organum* (1620), bk. I, aphorisms 39–44. Gutenberg/Spedding-style numbering; https://www.gutenberg.org/files/45988/45988-h/45988-h.htm. Dewey's educational uptake is *How We Think* (1910), chap. 2, not Bacon's own vocabulary.

[72]Dewey, *How We Think* (1910), chap. 2 (Gutenberg text).

[73]Dewey, *How We Think* (1910), chap. 2, quoting John Locke on "wrong measures of probability" and John Stuart Mill, "To draw inferences has been said to be the great business of life."

[74]Hitchcock, "History."

Kant's *sapere aude* is often recruited into origin-stories of this kind. Treat Enlightenment "think for yourself" as atmosphere, not as a definition this book can cite as if it were Ennis. Kant as a source for the English educational slogan is unverified here; this chapter did not open a primary Kant text that uses anything equivalent.[75] The same discipline applies to the Foundation for Critical Thinking's website history: the claim that "the term 'critical thinking' has its roots in the mid-late 20th century" is inconsistent with Dewey 1910 and Glaser 1941 and should not be followed, and the companion claim of a 2,500-year developing tradition is brand-narrative, not historiography.[76]

What the pre-history can do for a life book is modest and real. It gives a picture of thinking as public accountability to reasons; a classification of standing sources of error that Dewey already translated into educational language; and a warning, from elenchus itself, that skill without a governing aim produces more able combatants. What it cannot do is furnish a coinage, a skills list, or a license to say that the Enlightenment named a K–12 competency. The load-bearing ancestor is not Socrates. It is Dewey.

## 2.2 Dewey 1910 / 1933: inquiry, not a skills list

*How We Think* (1910) is the document this book can actually stand on. The reflective-thought definition, already quoted, is Dewey's account of the *activity*. The surrounding marks, all from the Gutenberg text of the 1910 edition, fill in what that activity costs and what it is not.

Reflective thought "alone is truly educative in value." It is "a conscious and voluntary effort to establish belief upon a firm basis of reasons." Immediate acceptance of a suggestion is "uncritical thinking, the minimum of reflection." "Reflective thinking, in short, means judgment suspended during further inquiry; and suspense is likely to be somewhat painful." "To maintain the state of doubt and to carry on systematic and protracted inquiry—these are the essentials of thinking."[77] Those sentences already do work later chapters will need. The activity is educative, not ornamental. It is effort, not a trait you either have or lack. Uncritical thinking is named as the *minimum* of reflection, not as the opposite of a branded skill set. Suspended judgment is affectively expensive. The essentials are doubt maintained and inquiry carried on — a process picture, not a fallacy list.

The lantern sentence is the knowledge constraint in 1910 prose. Thinking is not a ready-made apparatus. It is not a light you swing onto horses, streets, gardens, trees, or river as you happen to pass them. A book that treats Dewey as the grandfather of generic critical-thinking *skills courses* is misreading him. The same man who gave the century its best short definition of the activity also said the activity is specific to subject-matter. That specificity is proto-Willingham and proto-McPeck. It is not an embarrassment to be cleaned up by a later skills taxonomy.

[75] **UNVERIFIED as a CT source.** Kant's *sapere aude* is often recruited into origin-stories. The research fetch of August 30, 2026, did not open a primary Kant text that uses anything equivalent to the English educational slogan. Treat Enlightenment "think for yourself" as atmosphere, not as a definition this book can cite as if it were Ennis.

[76] Foundation for Critical Thinking, "Defining Critical Thinking," https://www.criticalthinking.org/pages/defining-critical-thinking/766 (fetched August 30, 2026). Curriculum-brand site, not a meta-analysis. The history blurb ("roots in the mid-late 20th century"; "developing throughout the past 2,500 years") is brand-narrative, not historiography, and is inconsistent with Dewey 1910 and Glaser 1941.

[77] Dewey, *How We Think* (1910), chap. 1 (Gutenberg text). All five marks in this paragraph were checked against that transcription.

Dewey's process picture was restated in 1933, when he replaced "steps" with "phases" to kill the implication of an invariant sequence. The 1910 edition had used "steps" at page 72. The 1933 restatement, as Hitchcock quotes it from pages 106–107, lists five phases: suggestions, in which the mind leaps forward to a possible solution; an intellectualization of the difficulty or perplexity into a problem to be solved; use of one suggestion after another as a leading idea, or hypothesis, to initiate and guide observation; mental elaboration of the idea (reasoning as *part* of inference, not the whole); and testing the hypothesis by overt or imaginative action.[78] The sequence is preceded by a perplexed situation and followed by a cleared-up one. I have not independently opened a 1933 full text; the phases and the claim about dropping "critical" are Hitchcock's report of that edition, and they stay marked as such.[79]

The English phrase itself is a use-claim, not a coinage-claim. Hitchcock's History supplement says Dewey (1910: 74, 82) "introduced the term 'critical thinking' as the name of an educational goal, which he identified with a scientific attitude of mind. More commonly, he called the goal 'reflective thought,' 'reflective thinking,' 'reflection,' or just 'thought' or 'thinking.' "[80] Those pages have now been opened. At 1910: 74 Dewey writes: "The essence of critical thinking is suspended judgment; and the essence of this suspense is inquiry to determine the nature of the problem before proceeding to attempts at its solution." At 1910: 82: "So far as we conduct each of these processes in the light of the other, we get valid discovery or verified critical thinking."[81] The sentences use the English phrase. They do not say that the phrase is the name of an educational goal; that framing is Hitchcock's. Trust Hitchcock for the use-claim. Do not upgrade it to "Dewey coined the term." Coinage is unverified and unnecessary.

Hitchcock also reports that the 1933 restatement, without explaining why, "eliminated the previous edition's uses of the words 'critical' and 'uncritical,' thus settling firmly on 'reflection' or 'reflective thinking.' " In 1933 "critical" occurs once: "a person may not be sufficiently critical about the ideas that occur to him" (1933: 16) — being critical is a *component* of reflection, not the whole.[82] The Eight-Year Study of the Progressive Education Association, by contrast, treated "critical thinking" and "reflective thinking" as synonyms. Dewey himself, on Hitchcock's reading of 1933, did not.

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[78]The five phases are Hitchcock's quotation of Dewey, *How We Think: A Restatement of the Relation of Reflective Thinking to the Educative Process* (Boston: D. C. Heath, 1933), 106–7. The 1910 edition had used "steps" at p. 72. Dewey replaced "steps" with "phases" to kill the implication of an invariant sequence.

[79]**UNVERIFIED: Dewey 1933 full text.** This chapter did not independently open a 1933 edition. The phases, the claim that 1933 dropped most uses of "critical" and "uncritical," and the single remaining use of "critical" at 1933: 16 ("a person may not be sufficiently critical about the ideas that occur to him") are from Hitchcock, "History."

[80]Hitchcock, "History."

[81]Dewey 1910, pp. 74 and 82, opened 30 August 2026 against Project Gutenberg (https://gutenberg.org/files/37423/37423-h/37423-h.htm; chap. 6 begins p. 68, chap. 7 p. 79) and the Mead Project transcriptions (https://brocku.ca/MeadProject/Dewey/Dewey_1910a/Dewey_1910_f.html; https://brocku.ca/MeadProject/Dewey/Dewey_1910a/Dewey_1910_g.html). Page 74: "The essence of critical thinking is suspended judgment; and the essence of this suspense is inquiry to determine the nature of the problem before proceeding to attempts at its solution." Page 82: "So far as we conduct each of these processes in the light of the other, we get valid discovery or verified critical thinking." Hitchcock reports that those pages introduce the term as the name of an educational goal, identified with a scientific attitude of mind. That goal-name claim is Hitchcock's framing, not a sentence on those pages. Not coinage.

[82]Hitchcock, "History," reporting Dewey 1933: 16. See n. 78 on the unopened 1933 text. The Eight-Year Study's synonymy of "critical thinking" and "reflective thinking" is also Hitchcock, "History."

What Dewey gives this book is therefore four things, not a franchise. He gives a definition of the activity worth keeping almost verbatim. He gives a process picture that is inquiry rather than fallacy-spotting: perplexity, hypothesis, testing, a cleared-up situation. He gives an insistence that thinking is specific to subject-matter — the lantern, not the bicycle. And he gives a civic and individual value claim, happiness and the reduction of social waste, that a life book is entitled to inherit without pretending he wrote a test. He does not give a skills list. He does not give a transferable apparatus. He does not give the 1980s movement permission to treat "critical thinking" as a course you can schedule on Tuesdays.

The 1933 drop of the word "critical" is a fact about Dewey's diction, not a reason to abandon the title-phrase. Dewey preferred reflective thinking. Some cognitive scientists prefer scientific thinking, historical thinking, rationality, judgment. The market will punish a title change; the argument might not. This book holds the title and spends its pages making the folk uses visible so later chapters can refuse them. Dewey's 1910 definition of the *activity* still does more work than most of the later one-liners. His lantern sentence still vetoes more of the popular story than most of the later critiques. The job is to keep both.

## 2.3 Educationalization: Eight-Year Study, Glaser, Bloom

Between Dewey and the 1980s movement, three things happen to the phrase. It becomes a school-and-college *goal-name*. It is operationalized into tests. It is folded into Bloom's taxonomy and then exploded into a reform vocabulary that every mission statement can use without specifying a conception. None of those three is a settled construct. All three are why later decades could industrialize a family resemblance and call it a skill.

The Progressive Education Association's Eight-Year Study is the first large American institutionalization of the phrase as a measurable educational goal. Schools in the study adopted critical or reflective thinking as an aim. The University School of Ohio State University, as quoted by Hitchcock from the Commission on the Relation of School and College (1943: 745–746), put the synonymy and the civic claim in one paragraph:

> Critical or reflective thinking originates with the sensing of a problem. It is a quality of thought operating in an effort to solve the problem and to reach a tentative conclusion which is supported by all available data. … The success of democracy depends to a large extent on the disposition and ability of citizens to think critically and reflectively about the problems which must of necessity confront them.[83]

The evaluation staff classified many school objectives under "clear thinking" or "critical thinking" and built tests of interpreting data, the nature of proof, and applying principles (Smith, Tyler, and the Evaluation Staff, 1942).[84] I have not independently fetched the primary Eight-Year Study reports; the quotations and the testing program are Hitchcock's. What matters for this book is the

[83]Commission on the Relation of School and College, as quoted by Hitchcock, "History," from the University School of Ohio State University report (1943: 745–46). **Primary Eight-Year Study reports not independently fetched;** rely on Hitchcock.

[84]Eugene R. Smith, Ralph W. Tyler, and the Evaluation Staff, *Appraising and Recording Student Progress* (New York: Harper, 1942), as reported by Hitchcock, "History." Primary report not independently fetched.

institutional fact. By the early 1940s, an American school-reform project could treat "critical thinking" and "reflective thinking" as interchangeable names for a democratic disposition-plus-ability, and could try to measure it. That is educationalization. It is not yet a philosophical construct, and it is not yet a commercial test industry.

Edward M. Glaser's 1941 Teachers College dissertation, *An Experiment in the Development of Critical Thinking*, does both of the next jobs at once. It educationalizes the phrase as a research heading, and in the same volume it limits what general training can do. The experiment, in the fall of 1938, ran eight lesson units with four grade-12 classes against a control, pre- and post-tested with an Otis mental-ability test and the Watson–Glaser Tests of Critical Thinking, built with dissertation sponsor Goodwin Watson.[85]

Glaser defines critical thinking by rewriting Dewey:

> Critical thinking calls for a persistent effort to examine any belief or supposed form of knowledge in the light of the evidence that supports it and the further conclusions to which it tends.[86]

The rewrite is honest about its parent. Dewey's "active, persistent, and careful consideration" becomes Glaser's "persistent effort to examine"; "grounds that support it" becomes "evidence that supports it." The activity is still inquiry in light of grounds and further conclusions. What Glaser adds, and what the later test industry will treat as the whole story, is a three-part construct: "(1) an attitude of being disposed to consider in a thoughtful way the problems and subjects that come within the range of one's experience; (2) knowledge of the methods of logical inquiry and reasoning; and (3) some skill in applying those methods."[87] Attitude, knowledge of methods, skill in applying them. Already, in 1941, the construct is not a single skill.

Glaser's conclusion, which still constrains this book, is the 1941 version of Willingham 2007. Hitchcock quotes it from page 175:

> The aspect of critical thinking which appears most susceptible to general improvement is the attitude of being disposed to consider in a thoughtful way the problems and subjects that come within the range of one's experience. An attitude of wanting evidence for beliefs is more subject to general transfer. Development of skill in applying the methods of logical inquiry and reasoning, however, appears to be specifically related to, and in fact limited by, the acquisition of pertinent knowledge and facts concerning the problem or subject matter toward which the thinking is to be directed.[88]

Disposition of wanting evidence transfers better than skill in applying logical methods. Skill is limited by pertinent knowledge. A book that cites Glaser only as the father of the Watson–Glaser test, and not as an early knowledge-constraint theorist, is cherry-picking. The Watson–Glaser Critical Thinking Appraisal became the commercial descendant — editions from 1942 through 1980

[85] Glaser, *An Experiment in the Development of Critical Thinking* (1941). Full 1941 volume not independently fetched. Experimental details (fall 1938, eight lesson units, four grade-12 classes vs. control, Otis and Watson–Glaser) are from Hitchcock, "History."

[86] Glaser 1941: 6, as quoted by Hitchcock, "History"; cf. Dewey 1910: 6 and Dewey 1933: 9.

[87] Quoted on the Foundation for Critical Thinking "Defining Critical Thinking" page (n. 75), citing Glaser 1941; also in Hitchcock and in later secondary texts. Page sometimes given as 5. Full 1941 volume not independently fetched.

[88] Glaser 1941: 175, as quoted by Hitchcock, "History." Full volume not independently fetched.

and later; listed in the Delphi report's Appendix A.[89] It is an instrument, not a definition. Later psychometric critiques, including Kevin Possin's argument that the instrument has a serious validity problem ("the more you know, the lower your score"), belong to the assessment chapter. The writer's obligation here is simpler: do not treat Watson–Glaser scores as the meaning of critical thinking.

Bloom is why every school mission statement can say "we teach higher-order thinking" without specifying a conception. Bloom and colleagues, *Taxonomy of Educational Objectives,* Handbook I: Cognitive Domain (1956: 38), as quoted by Hitchcock:

> This has been labeled "critical thinking" by some, "reflective thinking" by Dewey and others, and "problem solving" by still others. In the taxonomy, we have used the term "intellectual abilities and skills."[90]

Analysis, synthesis, and evaluation became "higher-order thinking skills." Ennis later objected that Bloom's categories lack criteria usable across domains: "analysis" in chemistry is not "analysis" in literature.[91] The revised Anderson and Krathwohl taxonomy (2001) still treats critical thinking as cutting across remember / understand / apply / analyze / evaluate / create rather than as one cell. Bloom is a vocabulary, not a critical-thinking theory. A life book that inherits "higher-order" as if it named the same activity Dewey defined has already collapsed the family.

The sequence from Eight-Year Study to Glaser to Bloom is therefore not a progress toward one construct. It is a fork. One branch treats critical or reflective thinking as a democratic educational goal and tries to test interpreting data, the nature of proof, applying principles. One branch rewrites Dewey, splits attitude from skill, and finds that skill is limited by knowledge — then, commercially, becomes a hiring-and-admissions instrument. One branch files the whole family under "intellectual abilities and skills" and gives the schools a ladder. The first tight philosophical construct in the educational literature is still ahead. It is Ennis in 1962. The sentence this book uses is the 1980s revision of that construct, not Bloom's ladder and not a Watson–Glaser score.

## 2.4 Ennis: from "correct assessing" to believe *or do*

Robert H. Ennis's 1962 paper in the *Harvard Educational Review*, "A Concept of Critical Thinking," is the first tight philosophical construct in this lineage. The starting point is B. Othanel Smith (1953): thinking as examining statements to determine meaning and whether to accept or reject them. Ennis adds a normative component. His own later recollection, in a 2011 *Inquiry* essay, is the wording this chapter uses, because the 1962 PDF was not independently fetched:

[89] Watson–Glaser editions 1942–80 and later; listed in Peter A. Facione, *Critical Thinking: A Statement of Expert Consensus for Purposes of Educational Assessment and Instruction*, ERIC ED315423 (1990), Appendix A. On later validity critique, see Kevin Possin, discussed in the assessment chapter; do not treat Watson–Glaser scores as the meaning of critical thinking.

[90] Benjamin S. Bloom, Max D. Engelhart, Edward J. Furst, Walter H. Hill, and David R. Krathwohl, *Taxonomy of Educational Objectives*, Handbook I: *Cognitive Domain* (New York: Longmans, Green, 1956), 38, as quoted by Hitchcock, "History." Not independently fetched.

[91] Robert H. Ennis, "A Logical Basis for Measuring Critical Thinking Skills," *Educational Leadership* 43, no. 2 (1985): 44–48, objecting that Bloom's categories lack criteria usable across domains. The revised taxonomy is Lorin W. Anderson and David R. Krathwohl, eds., *A Taxonomy for Learning, Teaching, and Assessing* (New York: Longman, 2001).

> For my first definition, I amended Smith's definition by holding that critical thinking is "the correct assessing of statements" (Ennis, 1962, p. 83). … I added "correct" because I believed and still do believe … that "critical thinking," as used in the critical thinking movement, was not merely a descriptive term, but also a term of approbation.[92]

Twelve "aspects," three dimensions (logical, criterial, pragmatic). Explicitly, the 1962 paper did *not* include judging value statements. This is the paper John McPeck later treats as "the prevailing view." It is a construct for assessing *statements*. It is not yet a life-book definition, because a life is not only a set of statements to accept or reject. Action is not in the sentence.

Ennis developed his second, still-current definition in the 1980s. He says so himself: "I developed my second (and current) definition of 'critical thinking' in the 1980's."[93] The 1985 *Educational Leadership* piece already has it as the "working definition":

> Critical thinking is reasonable reflective thinking focused on deciding what to believe or do.[94]

Variants in his own publications drop or add a comma and sometimes an "and" ("reasonable *and* reflective"). The substance does not change. So defined, Ennis notes, critical thinking includes creative activities — formulating hypotheses, questions, alternatives, plans — and is practical because deciding what to believe or do is practical. The *or do* is why this can be a life book and not only an argument-appraisal book. Appraisal of statements is inside the sentence. Construction of alternatives is inside it. Action is inside it.

Ennis treats the one-sentence formula as the *concept* — ordinary use among critical-thinking supporters — and the long outline of dispositions and abilities as a *conception* usable as curriculum goals and assessment specifications. Usefulness, "not elegance or mutual exclusiveness," is the design criterion.[95] The 2011/2015 outline at criticalthinking.net, Ennis's own site, not the Foundation's, summarizes the ideal thinker as disposed to try to "get it right," to present a position honestly and clearly, and to care about others; and able to clarify, seek and judge the basis for a view, infer wisely, imaginatively suppose and integrate, and do these with dispatch, sensitivity, and rhetorical skill. The last disposition, care for others, is marked "auxiliary, not constitutive."[96]

That marking is load-bearing for this book's ethics and for Chapter 10. In 1996, in *Informal Logic*,

[92] Robert H. Ennis, "Critical Thinking: Reflection and Perspective, Part I," *Inquiry: Critical Thinking Across the Disciplines* 26, no. 1 (2011): 4–18. The 1962 paper is Ennis, "A Concept of Critical Thinking: A Proposed Basis for Research on the Teaching and Evaluation of Critical Thinking Ability," *Harvard Educational Review* 32, no. 1 (1962): 81–111. Full 1962 PDF **not independently fetched**; the definition "the correct assessing of statements" (1962: 83) is from Ennis's own 2011 recollection. B. Othanel Smith, 1953: 130, is the starting point Ennis names.

[93] Ennis, "Reflection and Perspective, Part I" (2011).

[94] Ennis, "A Logical Basis for Measuring Critical Thinking Skills" (1985), already as the "working definition." Confirmed across Ennis 1991, 1996, 2011, and the 2015 PDF at criticalthinking.net (n. 94). Variants drop or add a comma and sometimes "and" ("reasonable *and* reflective").

[95] Robert H. Ennis, "The Nature of Critical Thinking: Outlines of General Critical Thinking Dispositions and Abilities," last revised September 2015, https://criticalthinking.net/wp-content/uploads/2024/04/The-Nature-of-Critical-Thinking.pdf. Author's own site; not the Foundation for Critical Thinking. Usefulness, "not elegance or mutual exclusiveness," is the design criterion Ennis states there.

[96] Ennis, "The Nature of Critical Thinking" (2015); the 2011 *Inquiry* outline marks care for others as "auxiliary, not constitutive."

Ennis splits the difference the Delphi panel would not settle. Care for persons' dignity is a "*correlative*" disposition — without it a critical-thinking program "would be deficient and perhaps dangerous" — but it is not constitutive of critical thinking.[97] The distinction lets a life book keep danger in view without smuggling a moral-political program into the *meaning* of the term. Delphi's majority, as the next section will show, refused to build ethics into the meaning. Folk use does it anyway. Ennis's correlative is the honest way to name the danger and still have a definition that can travel.

What Ennis can design follows from the concept/conception split. He can design a public sentence that includes action. He can design a comprehensive goals list for a course or an across-the-curriculum program. He can design the Cornell tests and the Ennis–Weir essay test. He can design a vision of critical thinking across the curriculum — CTAC, 2013 and 2018 — as a *proposal*, not as a completed multi-institution randomized trial.[98] What he cannot design, by the sentence alone, is a theory of transfer; a cognitive-science account of why maxims fail; a reason to treat critical thinking as independent of domain knowledge. Ennis *argues against* strong McPeckian subject-specificity, most carefully in the 1989 *Educational Researcher* paper that distinguishes conceptual, epistemological, and empirical subject-specificity and argues that McPeck overreaches on the first two.[99] His definition is silent on how a life-skill book should handle knowledge. That silence is why Willingham still constrains this book even if Ennis is the epigraph.

Hitchcock notes that Ennis, in 2016, lists fourteen philosophically oriented scholarly definitions plus three dictionary definitions, and treats them as rival conceptions of one concept.[100] That is the right bibliographic posture for this book: show the family, then choose. The choice of Ennis's sentence as the public definition is a choice about *aim* — belief or action — not a claim that Ennis settled pedagogy. Pedagogy is Willingham's constraint, Glaser's 1941 split, McPeck's *about what?*, Dewey's lantern. The sentence earns the title page. It does not earn a content-free semester.

The 1962 construct and the 1980s sentence are not the same object. "Correct assessing of statements" is a philosopher's tightening of Smith. "Reasonable reflective thinking focused on deciding what to believe or do" is a public formula that can govern a life, a course, or a chapter outline. A book that quotes only the later sentence, and pretends Ennis never excluded value judgments, never needed a correlative for care, and never faced McPeck, is using Ennis as a slogan. This book uses him as a definitional spine and then says, in the coalition section, where another account vetoes a course designed from the outline alone.

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[97] Robert H. Ennis, "Critical Thinking Dispositions: Their Nature and Assessability," *Informal Logic* 18, nos. 2–3 (1996). Care for persons' dignity is a "correlative" disposition; a CT program without it "would be deficient and perhaps dangerous."

[98] Robert H. Ennis, "Critical Thinking Across the Curriculum: A Vision," *Topoi* 37 (2018): 165–84, developing the 2013 proposal. A *proposal*, not a completed multi-institution randomized trial.

[99] Robert H. Ennis, "Critical Thinking and Subject Specificity: Clarification and Needed Research," *Educational Researcher* 18, no. 3 (1989): 4–10. Conceptual versus epistemological versus empirical subject-specificity. Cited via Hitchcock; PDF not independently fetched.

[100] Hitchcock, "Critical Thinking," reporting Ennis (2016).

## 2.5 The 1980s: California, Paul, *A Nation at Risk*, Delphi

The 1980s did not discover critical thinking. They scaled it into requirements, conferences, textbooks, and tests. Facione's 1990 report is a document of that scale-up: written *because* assessment and instruction at system scale needed a shared construct. The movement's self-portrait, in Facione's opening, is from "cottage industry" to "growth industries" — textbooks, staff development, university requirements, state frameworks and tests — and then, with success, the "vexing" questions: what exactly are those skills and dispositions? How teach? How assess at campus, district, and state scale?[101]

The stabilizing events are a short list. In 1980, Richard Paul begins the annual International Conference on Critical Thinking and Educational Reform in California; Hitchcock reports tens of thousands of educators over the years of that conference.[102] The same year, California State University trustees approve a general-education critical-thinking requirement. Chancellor Glenn Dumke's 1980 directive, as quoted by Hitchcock, asks for instruction designed to yield "the ability to analyze, criticize, and advocate ideas, to reason inductively and deductively," with "minimal competence" including distinguishing fact from judgment, belief from knowledge, and "an understanding of the formal and informal fallacies of language and thought."[103] That is the most operational U.S. *system* definition of the era. It is a requirement language, not a research construct, and it already leans toward informal logic and fallacy.

In 1983, the College Board's *Academic Preparation for College* names reasoning as one of six basic academic competencies. The same year, *A Nation at Risk* laments missing " 'higher-order' intellectual skills" — the phrase Willingham 2007 will cite as the lamentation that produced two decades of exhortation and little improvement.[104] From 1983, the Association for Informal Logic and Critical Thinking holds sessions at American Philosophical Association divisional meetings. Textbooks had already begun the retitling: Howard Kahane's *Logic and Contemporary Rhetoric* (1971) pioneered applied informal logic; by the 1980s, introductory logic courses are being renamed "critical thinking."[105] Ennis's 1985 *Educational Leadership* piece, already discussed, puts the working definition into the practitioner literature. In 1987, the APA Committee on Pre-College Philosophy asks Facione to inquire. The Delphi process begins in February 1988.

Peter A. Facione, *Critical Thinking: A Statement of Expert Consensus for Purposes of Educational Assessment and Instruction*, ERIC ED315423, is the document. Last revisions November 1989; copyright 1990 P. A. Facione; prepared *for* the Committee on Pre-College Philosophy of the Amer-

[101] Facione, *Critical Thinking* (ERIC ED315423, 1990), opening. Full text fetched August 30, 2026, from https://files.eric.ed.gov/fulltext/ED315423.pdf. Last revisions November 1989; © 1990 P. A. Facione. Prepared *for* the APA Committee on Pre-College Philosophy; not an APA official doctrine.

[102] Hitchcock, "History," on the annual International Conference on Critical Thinking and Educational Reform (California, from 1980; Richard Paul's orbit) and "tens of thousands of educators."

[103] Glenn S. Dumke, 1980 California State University chancellor's directive, as quoted by Hitchcock, "History."

[104] College Board, *Academic Preparation for College* (1983), naming reasoning as one of six basic academic competencies, via Hitchcock, "History." National Commission on Excellence in Education, *A Nation at Risk* (Washington, DC: U.S. Government Printing Office, 1983), on missing " 'higher-order' intellectual skills," as cited by Daniel T. Willingham, "Critical Thinking: Why Is It So Hard to Teach?," *American Educator* 31, no. 2 (Summer 2007).

[105] Howard Kahane, *Logic and Contemporary Rhetoric* (Belmont, CA: Wadsworth, 1971), via Hitchcock, "History." Association for Informal Logic and Critical Thinking (AILACT) sessions at APA divisional meetings from 1983: Hitchcock, "History."

ican Philosophical Association.[106] It is Facione's report of a Delphi panel. It is not "the APA's official definition of critical thinking" in the sense of a professional-association doctrine. The method: Delphi, six rounds, 11 February 1988 to 25 September 1989. Forty-six panelists. Facione reports roughly 52 percent philosophy, 22 percent education, 20 percent social sciences, 6 percent physical sciences. Names in Table 7 include Ennis, Lipman, Paul, Norris, Johnson, Browne, Keeley, Beyer, Costa, Hitchcock. Harvey Siegel is *not* listed. John McPeck is *not* listed. Consensus is not a vote-count; dissenting opinions remain in the mix. One of the forty-six asked to be listed as participant only, not as supporting the document.[107] The purpose, stated in the document, is a conceptualization "of sufficient clarity, accuracy and richness" to guide *educational assessment and instruction*, aimed at what a generally educated lower-division college thinker should be able to do — and then avowedly an *ideal*, not a description of typical sophomores.

Quote Table 1. Do not paraphrase it.

> We understand critical thinking to be purposeful, self-regulatory judgment which results in interpretation, analysis, evaluation, and inference, as well as explanation of the evidential, conceptual, methodological, criteriological, or contextual considerations upon which that judgment is based. CT is essential as a tool of inquiry. As such, CT is a liberating force in education and a powerful resource in one's personal and civic life. While not synonymous with good thinking, CT is a pervasive and self-rectifying human phenomenon. The ideal critical thinker is habitually inquisitive, well-informed, trustful of reason, open-minded, flexible, fair-minded in evaluation, honest in facing personal biases, prudent in making judgments, willing to reconsider, clear about issues, orderly in complex matters, diligent in seeking relevant information, reasonable in the selection of criteria, focused in inquiry, and persistent in seeking results which are as precise as the subject and the circumstances of inquiry permit.[108]

Two dimensions: cognitive skills and affective dispositions. The six skills, Table 3: interpretation (categorization, decoding significance, clarifying meaning); analysis (examining ideas, detecting arguments, analyzing arguments); evaluation (assessing claims, assessing arguments); inference (querying evidence, conjecturing alternatives, drawing conclusions); explanation (stating results, justifying procedures, presenting arguments); self-regulation (self-examination, self-correction).[109] Virtual unanimity (N > 95 percent) on analysis, evaluation, and inference as central. Strong consensus (N > 87 percent) on interpretation, explanation, and self-regulation. Residual objections live in the report itself: one assessment expert wanted interpretation treated as communication, not as critical thinking; another noted that self-regulation is meta-level and hard to test with paper and pencil. Any later list of "the six skills" is speaking Delphi. This book will say so.

An important finding in the report, not in the one-sentence definition, is the knowledge constraint

[106] Facione, ERIC ED315423 (1990). See n. 100.

[107] Facione, ERIC ED315423 (1990), on method, dates (February 11, 1988–September 25, 1989), forty-six panelists, disciplinary mix, Table 7 names, and the one participant who asked not to be listed as supporting the document. Siegel is not listed. McPeck is not listed.

[108] Facione, ERIC ED315423 (1990), Table 1. Quoted in full; not paraphrased.

[109] Facione, ERIC ED315423 (1990), Table 3, with the consensus thresholds (N > 95 percent on analysis, evaluation, inference; N > 87 percent on interpretation, explanation, self-regulation) and the residual objections named in the report.

the panel did not vote out: "while CT skills themselves transcend specific subjects or disciplines, exercising them successfully in certain contexts demands domain-specific knowledge, some of which may concern specific methods and techniques used to make reasonable judgments in those specific contexts." Recommendation 3: "one cannot overemphasize the value of a solid liberal education to supplement the honing of one's CT skills and the cultivating of one's CT dispositions."[110] The panel also says, bluntly, that not every useful cognitive process is critical thinking; it sits in a family with problem-solving, decision-making, and creative thinking, "the conceptual overlaps … have yet to be examined satisfactorily."

The dispositions split is the fact this book must not flatten. The panel *agreed* on a list of affective dispositions of the "good critical thinker": inquisitiveness, concern to be well-informed, alertness to opportunities to use critical thinking, trust in reasoned inquiry, self-confidence in reasoning, open-mindedness, flexibility, fair-mindedness, honesty about biases, prudence in judging, willingness to reconsider; plus issue-specific clarity, orderliness, diligence, reasonableness, care, persistence, precision. They *did not agree* that those dispositions are part of the *meaning* of "critical thinking." About 61 percent, in Round 5B, held that dispositions are part of the meaning; a skilled but lazy closer of inquiry is *not* a critical thinker. About 30 percent held that "CT" is strictly a judgmental *process*; the sophist is a critical thinker, just not a *good* one.[111] On the normative and ethical twist: 52 percent reject building ethics into the meaning of "CT"; only 17 percent want to deny the title to, for example, a defense attorney who uses the skills to acquit a guilty client. The finding, in the report's own words: "The mistaken notion that CT has a normative component is rejected. … What 'CT' means, why it is of value, and the ethics of its use are best regarded as three distinct concerns."[112]

From that construct Facione, and Noreen Facione, built instruments. The California Critical Thinking Skills Test (CCTST), 1990/1992, California Academic Press, is a multiple-choice skill measure (analysis, evaluation, inference, induction, deduction; later numeracy), now sold by Insight Assessment. The California Critical Thinking Disposition Inventory (CCTDI), Facione and Facione 1992, is a Likert inventory. Seven empirically factored constructs emerged: truth-seeking, open-mindedness, analyticity, systematicity, confidence in reasoning, inquisitiveness, maturity of judgment. Facione's 2000 *Informal Logic* paper is explicit that these seven *emerged from factor analysis* of items written to the Delphi ideal; they are not identical to Table 5.[113] This book will not treat a CCTST score, a CCTDI profile, or Insight Assessment marketing copy as "what critical thinking is." They are one operationalization of one consensus statement produced for assessment-

[110] Facione, ERIC ED315423 (1990), on domain-specific knowledge required to exercise the skills, Recommendation 3 on liberal education, and the family of problem-solving, decision-making, and creative thinking.

[111] Facione, ERIC ED315423 (1990), Round 5B: about 61 percent dispositions-in-the-meaning versus about 30 percent strict proceduralism. Table 5 lists the affective dispositions of the "good critical thinker."

[112] Facione, ERIC ED315423 (1990): 52 percent reject building ethics into the meaning of "CT"; 17 percent would deny the title to, e.g., a defense attorney who uses CT skills to acquit a guilty client. Finding: "The mistaken notion that CT has a normative component is rejected. … What 'CT' means, why it is of value, and the ethics of its use are best regarded as three distinct concerns."

[113] Peter A. Facione, "The Disposition Toward Critical Thinking: Its Character, Measurement, and Relationship to Critical Thinking Skill," *Informal Logic* 20, no. 1 (2000). CCTST: California Academic Press, 1990/1992; later sold by Insight Assessment (vendor). CCTDI: Facione and Facione 1992; seven empirically factored constructs (truth-seeking, open-mindedness, analyticity, systematicity, confidence in reasoning, inquisitiveness, maturity of judgment) are *not* identical to Delphi Table 5.

and-instruction purposes by a self-selected expert panel that did not include McPeck and did not settle the dispositional-meaning dispute.

Jones and colleagues (1994/1995), a U.S. Department of Education–initiated survey of educators, employers, and policymakers on Goals 2000, used Delphi skills and CCTDI factors as the *expressions* of critical thinking for the survey. Facione later reported "strong national consensus" on those outcomes.[114] That is uptake. It is consensus-on-a-survey. It is not independent validation of the construct, and it is not a finding that Delphi-critical-thinking is being achieved.

Standpoint belongs on this page, not only in a later ethics chapter. Delphi is North American, college-experienced, philosophy-heavy, 1988–89. McPeck, the principal critic of general-skills courses, is absent. Siegel, whose *Educating Reason* (1988) is the other major philosophical statement of the decade, is not listed. A global *life* book that treats this panel as "what critical thinking is" has a standpoint problem. The panel did real work. Table 1 is citable. The six skills are citable as Delphi's list. The 61/30 split and the majority rejection of an ethical component in the meaning are citable as the panel's own unfinished business. None of that makes Delphi a discovery of nature.

## 2.6 Six accounts: what each can and cannot design

Hitchcock's shared core, after Sharon Bailin and colleagues (1999b), is three features: the thinking is done to make up one's mind about what to believe or do; the thinker tries to meet standards of adequacy and accuracy; the thinking meets them to some threshold. "Careful goal-directed thinking."[115] Everything below is a *conception* of that concept. They are not interchangeable modules. Collapse them into one operational definition and the rest of this book will fight itself.

**Ennis** (1962, then 1985 onward). Core formula: reasonable reflective thinking focused on deciding what to believe or do, plus a long outline of dispositions and abilities. What it *is*: a philosophical-educational conception, a curriculum-and-assessment specification. What it can design: a critical-thinking course or a CTAC goals list; Cornell and Ennis–Weir style assessment; a public one-sentence definition that includes action. What it cannot: a theory of transfer or of knowledge; a reason to ignore McPeck and Willingham. The constraint it still exerts: this book needs a definition this clean. Ennis's "believe *or do*" is why the title can say *for life* without quietly meaning *for argument appraisal*.

**Facione / APA Delphi 1990.** Core formula: purposeful, self-regulatory judgment resulting in interpretation, analysis, evaluation, inference, plus explanation of the grounds; skills plus dispositions of an *ideal*. What it *is*: expert-consensus construct *for assessment and lower-division instruction*. What it can design: skill taxonomies; outcome language for accreditation; CCTST and CCTDI as *one* operationalization. What it cannot: a settled meaning of "critical thinking" (the panel split on

[114] Elizabeth A. Jones et al., National Center for Postsecondary Teaching, Learning, and Assessment survey (1994/1995), U.S. Department of Education–initiated, Goals 2000 era, as reported by Facione, "Disposition Toward Critical Thinking" (2000). Uptake of Delphi/CCTDI language, not independent validation of the construct.

[115] Hitchcock, "Critical Thinking," after Bailin et al., "Conceptualizing Critical Thinking" (1999b). The six-account table in this section follows the book brief and the stream memo `01-definitions.md` §10. Do not add a seventh account as if it were one of the six. 1999a in the same issue is "Common Misconceptions of Critical Thinking"; that paper is the adjacent "against generic discrete skills" piece, not the three-feature core.

dispositions-in-the-meaning; the majority rejected a normative component); proof of generality; a K–12 life curriculum. The constraint: any book that lists "the six skills" is speaking Delphi. It must say so, and must not confuse the test with the construct. Willingham cannot be collapsed into Facione. Delphi lists skills that "transcend" subjects and then, in the next breath, demands domain-specific knowledge to exercise them. Willingham's claim is stronger and different: the processes of thinking are intertwined with the content of thought; critical thinking is not a skill with the usual properties of skills. Those are not two wordings of one finding.

**Paul and Elder / Foundation for Critical Thinking.** Treat this as a curriculum brand with a large popular footprint, not as a meta-analysis and not as the scholarly state of the art. Institutional facts: Richard Paul, annual International Conference from 1980 at Sonoma State; Foundation for Critical Thinking; Linda Elder as co-author of the *Miniature Guide* series. The 1987 Scriven and Paul draft statement for the National Council for Excellence in Critical Thinking Instruction, still posted as "Defining Critical Thinking" on the Foundation site:

> Critical thinking is the intellectually disciplined process of actively and skillfully conceptualizing, applying, analyzing, synthesizing, and/or evaluating information gathered from, or generated by, observation, experience, reflection, reasoning, or communication, as a guide to belief and action. In its exemplary form, it is based on universal intellectual values that transcend subject matter divisions: clarity, accuracy, precision, consistency, relevance, sound evidence, good reasons, depth, breadth, and fairness.[116]

Two components in the same statement: information-and-belief generating and processing *skills*, and the *habit*, based on intellectual commitment, of using those skills to guide behavior. Contrast with mere information, mere possession of skills, and mere exercise of skills "without acceptance of their results." A distinction between selfish or sophistical use and fairminded use — Paul's "weak sense" versus "strong sense." In 1981, in *Informal Logic*, Paul had already named the worry this book should steal: teaching critical thinking in the strong sense as a focus on self-deception, world-views, and a dialectical mode of analysis, because training to recognize "bad" reasoning in " 'egocentrically' neutral" cases makes students "more sophistic rather than less so, more skilled in 'rationalizing' and 'intellectualizing' the biases they already have."[117] Strong-sense critical thinking is a moral-psychological ambition, not a skill list.

The three-layer schema, in the *Miniature Guide* (4th ed., 2006) and later reprints: eight elements of thought (purpose; question at issue; information; interpretation and inference; concepts; assumptions; implications and consequences; point of view); intellectual standards applied to those elements; intellectual traits or virtues said to grow from repeated application of standards to elements (intellectual humility, courage, empathy, integrity, perseverance, confidence in reason, autonomy,

[116] Michael Scriven and Richard W. Paul, 1987 draft statement for the National Council for Excellence in Critical Thinking Instruction, posted as "Defining Critical Thinking" at https://www.criticalthinking.org/pages/defining-critical-thinking/766 (fetched August 30, 2026). Foundation site: curriculum brand, not a systematic review.

[117] Richard W. Paul, "Teaching Critical Thinking in the 'Strong' Sense: A Focus on Self-Deception, World Views, and a Dialectical Mode of Analysis," *Informal Logic* 4, no. 2 (1981): 2–7. PDF fetched 30 August 2026 from https://informallogic.ca/index.php/informal_logic/article/download/2766/2207. Exact: students trained on " 'egocentrically' neutral" cases "become more sophistic rather than less so, more skilled in 'rationalizing' and 'intellectualizing' the biases they already have." Hitchcock's SEP gloss "more able sophists" / "even more entrenched in the egocentric and sociocentric biases with which they began" is a paraphrase of that 1981 worry, not a quotation from the article. Steal the warning; do not outsource theory to the Foundation.

fairmindedness).[118] Lists of standards vary slightly across Foundation publications: commonly clarity, accuracy, precision, relevance, depth, breadth, logic, significance, fairness; sometimes completeness. The Foundation's own elements page lists seven "virtually universal" standards in one passage (clarity, precision, accuracy, relevance, depth, breadth, logic) and fuller lists elsewhere.[119] Do not pretend there is one canonical nine. A later Elder one-liner (September 2007, on the same defining page): "Critical thinking is self-guided, self-disciplined thinking which attempts to reason at the highest level of quality in a fair-minded way."[120]

What Paul–Elder can design: classroom routines, a shared vocabulary, checklists for taking apart an essay or a plan, a virtue-talk that students can actually hear. University-wide adoptions exist; Payette and Ross (2016) describe Louisville. Moseley and colleagues (2005) judged Paul's model applicable across subjects and levels — a *framework review*, not a randomized trial. Sheffield (2018) notes many U.S. Quality Enhancement Plans written by or with Paul and Elder, typically on a five-year clock he judges too short.[121] What it cannot design: a cognitive-science account of transfer; an evidence claim that the *Miniature Guide* produces general critical thinking; a substitute for domain knowledge; historiography. The website's history blurb is brand-narrative. This book can borrow classroom vocabulary and steal the 1981 warning. It will not outsource theory to criticalthinking.org.

**Willingham** (2007, 2019). Core formula, in the 2019 commonsensical definition: thinking is critical if it is *novel* (not just recalling a prior conclusion), *self-directed* (not merely executing instructions), and *effective* (respects conventions that make useful conclusions more likely) — and what counts as effective varies by domain.[122] The 2007 *American Educator* essay is the load-bearing constraint; its bicycle passage belongs in Chapter 3, quoted at length, not paraphrased here. The three-point boil-down, which this chapter needs in order not to mis-file him under Facione, is already enough: critical thinking is not a skill that can be acquired and deployed regardless of context; there *are* metacognitive strategies that, once learned, make critical thinking more likely; the ability to do what the strategies call for depends on domain knowledge and practice.[123] What Willingham

[118]Richard W. Paul and Linda Elder, *The Miniature Guide to Critical Thinking Concepts and Tools*, 4th ed. (Dillon Beach, CA: Foundation for Critical Thinking Press, 2006). Eight elements; intellectual standards; intellectual traits. Curriculum brand; large popular footprint; not a research synthesis.

[119]Foundation for Critical Thinking, "The Elements of Reasoning and the Intellectual Standards," https://www.criticalthinking.org/pages/the-elements-of-reasoning-and-the-intellectual-standards/480. This page lists seven "virtually universal" standards in one passage (clarity, precision, accuracy, relevance, depth, breadth, logic); other Foundation publications add significance, fairness, completeness. Internal list-variation is a fact; do not pretend there is one canonical nine.

[120]Linda Elder, September 2007, on the Foundation "Defining Critical Thinking" page (n. 75).

[121]David Moseley et al., framework review (2005), judged Paul's model applicable across subjects and levels — a *framework review*, not an RCT. Paul Payette and Diane Ross, on the University of Louisville adoption (2016). Sherman Sheffield (2018) notes many U.S. Quality Enhancement Plans written by or with Paul and Elder, typically on a five-year clock he judges too short. None of these is evidence that the *Miniature Guide* produces general critical thinking.

[122]Daniel T. Willingham, *How to Teach Critical Thinking*, Education: Future Frontiers Occasional Paper Series (Sydney: NSW Department of Education, 2019), © Willingham and the State of New South Wales. Full PDF fetched August 30, 2026, from education.nsw.gov.au. Also at danielwillingham.com. Adapted as Willingham, "Ask the Cognitive Scientist: How Can Educators Teach Critical Thinking?," *American Educator* (Fall 2020). Commonsensical three-part definition (novel, self-directed, effective) is the 2019 opening; what counts as effective varies by domain.

[123]Willingham, "Critical Thinking: Why Is It So Hard to Teach?," *American Educator* 31, no. 2 (Summer 2007). Full text verified August 30, 2026, against the Reading Rockets authorized reprint,

can design: domain-by-domain skill-and-knowledge maps; multi-year revisiting; infused practice; a modest role for metacognitive cues. What he cannot: a one-semester generic course that transfers; a single test of "the" skill; a content-free life manual. The 2019 punchline is the veto on Ennis-style general courses as a standalone bet: "We are not even sure the general skills exist, but we are quite sure there is no proven way to teach them directly. In contrast, we have a pretty good idea of how to teach students the more specific critical thinking skills. I suggest we do so."[124] Do not cite Willingham as "critical thinking cannot be taught." He says generic bicycle-critical-thinking cannot; domain-specific critical thinking can; maxims help *some*.

**McPeck** (1981). *Critical Thinking and Education* is the book the 1980s movement had to answer. Definition, widely quoted from page 8: "the propensity and skill to engage in an activity with reflective scepticism."[125] The conceptual keystone, quoted in Paul's *Informal Logic* critique:

> it is a matter of conceptual truth that thinking is always thinking about X, and that X can never be 'everything in general' but must always be something in particular. Thus the claim 'I teach my students to think' is at worst false and at best misleading. … To the extent that critical thinking is not about a specific subject X, it is both conceptually and practically empty.[126]

Consequences McPeck actually draws: no unitary general reasoning skill; informal-logic courses and generic critical-thinking courses are the wrong unit of instruction; criteria of good thinking are field-dependent, the epistemology of the parent discipline; transfer of training "cannot be assumed … but must be established in each case by empirical tests"; distinguished logicians who cannot say whom to vote for, or why, are Exhibit A. Dispositions, especially to suspend judgment under cognitive dissonance, *can* be more general than abilities; Hitchcock notes that McPeck did not extend strong subject-specificity to all dispositions.[127] What McPeck can design: subject teaching that brings out a field's cognitive structure; skepticism about generic courses and about Watson–Glaser and Cornell as *the* evidence. What he cannot: a constructive life-skill syllabus; an account of cross-field principles (the hole Ennis, Siegel, and Paul press); a full story about dispositions. He is not a crank. A life-skill book that promises a content-free general power is making the claim he called conceptually empty. This book can reject his *ban on general courses* without rejecting his *knowledge constraint*.

**Lipman / Philosophy for Children** (1988; *Thinking in Education*, 1991/2003). Lipman in *Educational Leadership*, September 1988: "Skillful, responsible thinking that facilitates good judgment

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https://www.readingrockets.org/topics/comprehension/articles/critical-thinking-why-it-so-hard-teach. Prefer the article body over the AFT issue-PDF deck "There's no such thing as critical thinking 'skills.' " The bicycle passage is quoted at length in Chapter 3, not paraphrased here.

[124] Willingham, *How to Teach Critical Thinking* (2019). Do not cite Willingham as "CT cannot be taught."

[125] John E. McPeck, *Critical Thinking and Education* (New York: St. Martin's Press, 1981), 8. Full book not independently fetched. Definition confirmed from *Informal Logic* reviews (Stephen P. Norris 1985; Paul 1985), McPeck's own later precis, and secondary extracts.

[126] McPeck 1981, as quoted in Richard W. Paul, "McPeck's Mistakes," *Informal Logic* 7, no. 1 (1985).

[127] Hitchcock, "Critical Thinking," on McPeck's consequences and on the exception for some dispositions. McPeck, "The Evaluation of Critical Thinking Programs: Dangers and Dogmas," *Informal Logic* 6, no. 2 (1984), restates field-dependent knowledge as a major ingredient and rejects general reasoning skills and Watson–Glaser/Cornell as *the* evidence standard.

because it (1) relies upon criteria, (2) is self-correcting, and (3) is sensitive to context."[128] What it *is*: Philosophy for Children — community of inquiry, judgment as outcome, K–12 dialogue; the IAPC and *Philosophy in the Classroom* (1977) lineage. What it can design: a pedagogy (community of inquiry); a reason to start before college; a link from critical thinking to *judgment*, not only to argument analysis. What it cannot: a cognitive-science transfer theory; a psychometric definition; a substitute for subject knowledge. Lipman is closer to Dewey than to Watson–Glaser. Delphi included Lipman; P4C is not a side-hobby. It keeps *children*, *dialogue*, *criteria*, and *responsibility* on the table. Chapter 5 will have to say what the trials actually moved. EEF 2021's **0 months** on Key Stage 2 attainment is not a reason to drop Lipman from the conception table; it is a reason not to use P4C as proof that generic critical thinking raises test scores.

Adjacent accounts this chapter knows exist and does not treat as a seventh of the six: Harvey Siegel, *Educating Reason* (1988) — critical thinking as educational ideal, reasons plus "critical spirit," four justifications (respect for persons, self-sufficiency, initiation into rational traditions, democratic citizenship); constrains the *value* chapter more than this one. Bailin and colleagues (1999a, 1999b) — against generic discrete skills; the three-feature core Hitchcock uses. Diane Halpern (1998) — the most optimistic cognitive psychologist in the family: teach for transfer via disposition, skills, structure training, metacognitive monitoring. Alec Fisher and Michael Scriven (1997): "skilled, active interpretation and evaluation of observations, communications, information, and argumentation." Keith Stanovich: critical thinking as override of the autonomous mind in the service of epistemic plus instrumental rationality. Deanna Kuhn (2019): critical thinking as dialogic practice of advancing and responding to arguments, more than an individual ability. Freire and bell hooks: "critical" as politically engaged pedagogy — Hitchcock's non-example for the mainstream concept, and a live competitor in folk use.[129]

The table is the apparatus. It is not a finding those six signed.

[128] Matthew Lipman, "Critical Thinking—What Can It Be?," *Educational Leadership* 46, no. 1 (September 1988): 38–43. Full PDF fetched from ASCD, https://files.ascd.org/staticfiles/ascd/pdf/journals/ed_lead/el_198809_lipman.pdf. ERIC EJ376244 / ED352326. *Thinking in Education* (Cambridge: Cambridge University Press, 1991; 2nd ed. 2003) and the IAPC / *Philosophy in the Classroom* (1977) lineage.

[129] Harvey Siegel, *Educating Reason: Rationality, Critical Thinking, and Education* (New York: Routledge, 1988). Bailin et al. 1999a and 1999b. Diane F. Halpern, "Teaching Critical Thinking for Transfer across Domains," *American Psychologist* 53, no. 4 (1998): 449–55. Alec Fisher and Michael Scriven, *Critical Thinking: Its Definition and Assessment* (Point Reyes, CA: Edgepress, 1997). Keith E. Stanovich on override of the autonomous mind in the service of epistemic plus instrumental rationality. Deanna Kuhn, on dialogic practice (2019). Freire and hooks: Hitchcock's non-example for the mainstream concept. Adjacent; not a seventh of the six.

| Account | Core formula | What it *is* | What it can design | What it cannot design | Constraint it still exerts |
|---|---|---|---|---|---|
| **Ennis** (1962, then 1985–) | Reasonable reflective thinking focused on deciding what to believe or do; plus a long outline of dispositions and abilities | Philosophical-educational conception; curriculum-and-assessment spec | CT course or CTAC goals; Cornell / Ennis–Weir style assessment; a public one-sentence definition | A theory of transfer or of knowledge; a reason to ignore McPeck/Willingham | The book needs a definition this clean. Ennis's "believe *or do*" is why this can be a *life* book, not only an argument-appraisal book. |
| **Facione / APA Delphi 1990** | Purposeful, self-regulatory judgment resulting in interpretation, analysis, evaluation, inference, plus explanation of the grounds; skills + dispositions of an *ideal* | Expert-consensus construct *for assessment and lower-division instruction* | Skill taxonomies; outcome language for accreditation; CCTST/CCTDI as *one* operationalization | A settled meaning of "CT" (panel split on dispositions-in-the-meaning; rejected a normative component); proof of generality; a K–12 life curriculum | Any book that lists "the six skills" is speaking Delphi. Must say so, and must not confuse the test with the construct. |

| Account | Core formula | What it *is* | What it can design | What it cannot design | Constraint it still exerts |
|---|---|---|---|---|---|
| **Paul and Elder / Foundation** | Intellectually disciplined analysis and evaluation of thinking with a view to improving it; elements × standards leading to intellectual virtues; strong vs. weak sense | Curriculum brand + Socratic/dialectical moral psychology | Classroom vocabulary, checklists, virtue language, "strong sense" as a warning against sophistry | Meta-analytic evidence; a substitute for domain knowledge; historiography (the 2,500-year claim) | Popular footprint is too large to ignore. Use as pedagogy-adjacent vocabulary; do not treat criticalthinking.org as the literature. |
| **Willingham** (2007, 2019) | Novel, self-directed, effective thinking; *not* a context-free skill; maxims help only with knowledge and practice; domain-specific skills are teachable | Cognitive-science constraint on *any* educational conception | Domain-by-domain skill-and-knowledge maps; multi-year revisiting; infused practice; modest role for metacognitive cues | A one-semester generic course that transfers; a single test of "the" skill; a content-free life manual | Load-bearing. If the book promises bicycle-transfer, it is done. If it refuses all general principles, it has over-read him. |
| **McPeck** (1981) | Propensity and skill to engage in an activity with reflective scepticism; thinking always about X | Conceptual critique of the general-skills movement | Subject teaching that brings out a field's cognitive structure; skepticism about generic courses and about WG/Cornell as *the* evidence | A constructive life-skill syllabus; an account of cross-field principles; a full story about dispositions | Blocks the empty sentence "I teach thinking." Forces the book to say *about what*. |

| Account | Core formula | What it *is* | What it can design | What it cannot design | Constraint it still exerts |
|---|---|---|---|---|---|
| **Lipman / P4C** (1988; *Thinking in Education* 1991/2003) | Skillful, responsible thinking that facilitates good judgment because it (1) relies upon criteria, (2) is self-correcting, and (3) is sensitive to context | Philosophy for Children: community of inquiry, judgment as outcome, K–12 dialogue | A pedagogy (community of inquiry); a reason to start before college; a link from CT to *judgment*, not only to argument analysis | A cognitive-science transfer theory; a psychometric definition; a substitute for subject knowledge | Keeps *children*, *dialogue*, *criteria*, and *responsibility* on the table. Delphi included Lipman; P4C is not a side-hobby. |

Six accounts. Not one operational definition. The next section chooses a coalition and names the vetoes.

## 2.7 The coalition, and where they veto each other

This is the architectural decision of the book. It does not wait for a later draft.

The coalition this manuscript can honestly run is **Ennis** (believe *or do*) plus **Dewey** (inquiry, not only appraisal) plus **Willingham** (knowledge and practice) plus a **Paul-ish** warning about weak-sense sophistry plus **Lipman** for childhood judgment. McPeck is a constraint, not a designer: reject his *ban on general courses*; keep the *knowledge constraint*. That is a coalition. Coalitions hide contradictions. Each contradiction below is a veto that later chapters are not allowed to forget.

The replies to McPeck that still matter are on the table before the vetoes, because this book does not treat him as a crank and does not treat the replies as a wipeout. Paul (1985, "McPeck's Mistakes") and Siegel (1985) press the writing-and-speaking analogy: speech is always about something particular, yet general composition and speech courses are not empty.[130] Ennis (1989) distinguishes *conceptual*, *epistemological*, and *empirical* subject-specificity and argues that McPeck overreaches on the first two.[131] Hitchcock's verdict: McPeck's central argument "needs elaboration" and has not explained how thinking differs from writing and speaking; *nevertheless* "his position that the dispositions and abilities of a critical thinker are best developed in the context of subject-matter instruction is shared by many theorists," including Dewey, Glaser, Passmore, Weinstein, Bailin and

[130] Paul, "McPeck's Mistakes" (1985); Harvey Siegel, 1985 reply to McPeck, as reported by Hitchcock. Writing/speaking analogy: speech is always about something particular, yet general composition and speech courses are not empty.

[131] Ennis, "Critical Thinking and Subject Specificity" (1989). See n. 98.

colleagues, and Willingham (2019).[132] The writing/speaking analogies weaken the claim that there are *no general abilities at all*. They do not weaken the *about what?* demand. McPeck's conceptual point survives even if his ban on general courses does not. "I teach thinking" simpliciter is empty.

The vetoes that must appear on this page:

**1. Willingham vetoes Ennis-style general courses.** The 2019 punchline is not a mood. "We are not even sure the general skills exist, but we are quite sure there is no proven way to teach them directly." Ennis's mixed undergraduate design (2013/2018) — a separate critical-thinking strand plus subject-matter practice — is a *proposal*, not a completed multi-institution randomized trial. Abrami and colleagues (2015) found Ennis-type differences (mixed, infusion, general, immersion) **not statistically significant**. Design mixed anyway, for transfer reasons; do not cite 2015 as proof that mixed won a horse race.[133] Where Ennis's outline still earns pages: the public sentence; the *do*; the correlative-care warning; a goals list for *named* domains of a life, not a content-free semester. The life-book analogue of Willingham's four-step plan is architectural rather than definitional, but the definitions make it unavoidable: money, health, news, work, friendship, citizenship as the "subjects," with knowledge and practice specified per domain.

**2. Dewey vetoes reducing critical thinking to appraisal and fallacy-spotting.** Inquiry, five phases, the affective cost of suspended judgment. A life of deciding what to believe or do is not a life of spotting other people's informal fallacies. Paul's 1981 warning is the ally here: training on "neutral" cases can make students "more sophistic rather than less so." Steal the warning. Do not outsource theory to the Foundation. The 1987 Scriven and Paul definition is a brand statement with a real distinction inside it (skills plus the habit of using them; weak sense versus strong sense). It is not a substitute for Dewey's process picture, and it is not a meta-analysis of classroom effects.

**3. McPeck vetoes a content-free general power —** the claim the folk conjunction makes. This book agrees. It does not agree that no general course can ever be useful as a *cue-and-practice* strand beside domain work. Willingham's second point is the remaining room: there *are* metacognitive strategies that make critical thinking more likely. Maxims without knowledge and practice will not implement. Maxims with knowledge and practice, repeated until recognition is cheap, are not the empty sentence "I teach thinking." The ban is rejected. The knowledge constraint is kept.

**4. Delphi/Facione cannot be the meaning.** Assessment construct; 61/30 split on whether dispositions are in the meaning; majority rejection of ethics-in-the-meaning; philosophy-heavy 1988–89 panel; McPeck absent. The six-skills list is named as Delphi's list, not as nature. CCTST and CCTDI are downstream instruments. Jones and colleagues' Goals 2000 survey is uptake. Insight Assessment is a vendor. A chapter that lists interpretation, analysis, evaluation, inference, explanation, and self-regulation as "what critical thinking is," without naming Facione 1990, has laundered a consensus statement into a discovery.

[132] Hitchcock, "Critical Thinking." The shared claim that CT dispositions and abilities are best developed in subject-matter instruction is, on Hitchcock's list, Dewey, Glaser, Passmore, Weinstein, Bailin et al., and Willingham (2019).

[133] Willingham, *How to Teach Critical Thinking* (2019), punchline. Ennis 2013/2018 mixed undergraduate design: a *proposal*, not a completed RCT (n. 97). Philip C. Abrami et al., "Strategies for Teaching Students to Think Critically: A Meta-Analysis," *Review of Educational Research* 85, no. 2 (2015): 275–314; Ennis-type differences (mixed / infusion / general / immersion) were **not statistically significant**, as reported via Hitchcock. Design mixed anyway, for transfer reasons; do not cite 2015 as proof that mixed won a horse race. Effect sizes belong to Chapters 4–5.

**5. Lipman designs childhood pedagogy, not an attainment intervention.** Community of inquiry and judgment as the childhood chapter's object. The EEF 2021 effectiveness trial — 198 schools, 5/5 padlock, **0 months** on Key Stage 2 reading and maths — vetoes using Philosophy for Children as proof that generic critical thinking raises KS2 scores. Gorard, Siddiqui, and See (2015) found small attainment effects in an efficacy trial; those are not the 2021 result, and they were not large.[134] Lipman stays in the coalition as a reason to start before college and as a link from critical thinking to *judgment*. He does not stay as a test-score warrant.

**6. Paul–Elder as franchise is vetoed.** Classroom vocabulary can be borrowed. The three-layer schema is not a meta-analysis. List-variation of standards is a fact about their own materials. The 2,500-year blurb is brand-narrative. Moseley 2005 is a framework review. Payette and Ross 2016 is an adoption story. Sheffield 2018 is a warning about five-year clocks. None of those is evidence that the *Miniature Guide* produces general critical thinking. The 1981 strong-sense distinction is the piece this book keeps.

The following table is book synthesis, not a finding those authors signed.

| Coalition member (or constraint) | What this book takes | What vetoes it, on this page |
|---|---|---|
| Ennis | Public sentence including *do*; correlative care; goals list for named domains | Willingham 2019: no proven way to teach general skills directly. Ennis 2013/2018 mixed design is a proposal, not an RCT. Abrami 2015 Ennis-type differences NS. |
| Dewey | Inquiry, not only appraisal; lantern/specificity; affective cost of suspense | A skills-list or fallacy-spotting reduction of the 1910 definition. Treating 1933 as identical to 1910 on the word "critical." |
| Willingham | Knowledge and practice; modest role for maxims; domain-specific CT can be taught | Over-reading him as "CT cannot be taught." Collapsing him into Delphi's six skills. |
| Paul-ish warning | Weak sense vs. strong sense; "more sophistic rather than less so" | Paul–Elder as franchise, meta-analysis, or 2,500-year historiography. |

[134]Education Endowment Foundation, Philosophy for Children effectiveness trial (2021): 198 schools, 5/5 padlock, **0 months** on KS2 reading and maths. Stephen Gorard, Nadia Siddiqui, and Beng Huat See, *Philosophy for Children: Evaluation Report and Executive Summary* (EEF, 2015): small attainment effects (reading +0.12, maths +0.10) in an efficacy trial. Those are not the 2021 result. Full discussion in Chapter 5. Lipman stays in the coalition as pedagogy of judgment, not as a test-score warrant.

| Coalition member (or constraint) | What this book takes | What vetoes it, on this page |
|---|---|---|
| Lipman | Community of inquiry; judgment; a reason to start before college | EEF 2021 **0 months** as a warrant for generic-CT attainment claims. Treating P4C as a psychometric definition. |
| McPeck (constraint, not designer) | *About what?*; knowledge constraint | His ban on all general courses. The empty sentence "I teach thinking" is still empty. |

Open question 1 — which conception this book actually adopts when they conflict — is answered here as coalition-plus-vetoes, not as a single operational definition. Later chapters that lean on one member will say, on the page, where another vetoes it. The title-phrase stays. Dewey preferred reflective thinking. Some cognitive scientists prefer scientific thinking, historical thinking, rationality, judgment. No better public sentence than Ennis's has appeared. Hold the title. Spend the pages making the vetoes visible.

## 2.8 Dispositions in or out of the meaning

The live debate this chapter owns, and that Chapter 10 will close on, is not a definitional trick. It is whether "critical thinking" names a process you can run in a bad cause, or a process-plus-character that already includes the will to use it well, or a process-plus-ethics that already includes a political morality. The documents do not agree. This book has to say so, and then choose.

Delphi split. About 61 percent of the panel, in Round 5B, held that the affective dispositions are part of the *meaning*: a skilled but lazy closer of inquiry is not a critical thinker. About 30 percent held the strict-proceduralist line: "CT" is a judgmental process; the sophist is a critical thinker, just not a good one. Separately, 52 percent rejected building ethics into the meaning; only 17 percent wanted to deny the title to a defense attorney who uses the skills to acquit a guilty client. The report's own finding is that "the mistaken notion that CT has a normative component is rejected," and that what the term means, why it is of value, and the ethics of its use are "best regarded as three distinct concerns."[135] Those are not three wordings of one majority. The panel agreed on a list of dispositions of the *ideal* thinker. It did not agree that the list is in the meaning. It majority-rejected ethics-in-the-meaning. A writer who prints Table 1's ideal-thinker paragraph and then treats Delphi as having settled that critical thinking *is* fair-mindedness has flattened the split.

Ennis's 1996 move is the one this book adopts, and it is labelled as a choice. Care for persons' dignity is a *correlative* disposition, not a constitutive one. Without it a critical-thinking program "would be deficient and perhaps dangerous." It is not, thereby, what the word means. Constitutive versus correlative is the difference between smuggling a morality into the definition so that political

[135] Facione, ERIC ED315423 (1990), Round 5B and the ethics finding. See nn. 110–111. Book position in the text is labelled as a choice: meaning = Ennis's sentence; correlative care as Ennis 1996 (n. 96); constitutive partisan ethics refused.

opponents can be declared, by vocabulary, not to be thinking, and naming a danger that a program can fail even if the definition is being met. Folk use does the smuggling. “We are the critical thinkers; they are the sheeple” is Paul’s weak sense wearing a compliment. Delphi’s majority is the constraint against building a moral-political program into the meaning. This book follows that constraint and keeps Ennis’s correlative on the page so the danger does not disappear.

Paul treats intellectual virtues as the point. The three-layer schema is elements, standards, traits; the traits are supposed to grow from repeated application of standards to elements. Strong-sense critical thinking is a moral-psychological ambition: self-deception, world-views, a dialectical mode, the refusal to become a more able sophist. That ambition is why the Paul-ish warning belongs in the coalition. It is not why the Foundation’s virtue list belongs in the meaning. Virtues as the *point of a pedagogy* and virtues as the *definition of the term* are different claims. The first can design classroom talk. The second would make every dispute about what counts as fairmindedness into a dispute about whether the other side is thinking at all.

Willingham 2019 distinguishes a second informal sense of “critical thinking” — thinking when others might not; need for cognition, in the Cacioppo and colleagues (1996) sense — and sets it aside as under-researched for education.[136] Need for cognition may be personality. Limited educational evidence. A life book that treats “being the sort of person who likes hard problems” as what critical thinking *is* has changed the subject from a goal-directed activity to a temperament. Temperament is real. It is not Ennis’s sentence, not Dewey’s inquiry, and not Willingham’s domain-specific skill.

Book position, labelled as a choice. The *meaning* of critical thinking for this life book is Ennis’s sentence: reasonable reflective thinking focused on deciding what to believe or do. Care for persons is correlative, as Ennis said — without it a program would be deficient and perhaps dangerous. It is not smuggled into the definition so that political morality becomes “what critical thinking is.” Delphi’s majority is the constraint against that smuggling; folk use does it anyway. Dispositions of the ideal thinker (inquisitiveness, open-mindedness, fair-mindedness, honesty about biases) remain goals a program can name and try to cultivate. They are not, in this book, the test of whether the word applies.

Who is left out of expert panels remains a constraint, not a solved subsection. Delphi is North American, college-experienced, philosophy-heavy, 1988–89. P21 is employer-facing. UNESCO’s transversal competencies are Asia-Pacific ministerial, and they sit critical thinking in a bin with entrepreneurship. A global life book that treats those documents as “what critical thinking is” has a standpoint problem. Naming the standpoint is the honest move. Filling it with a seventh account invented for balance would be a different kind of collapse.

A chosen definition still cannot think about what it does not know. Ennis’s sentence includes action. Dewey’s lantern says the action is specific. Willingham will spend Chapter 3 showing why. Glaser already knew in 1941: the attitude of wanting evidence is more subject to general trans-

[136]Willingham, *How to Teach Critical Thinking* (2019), on a second informal sense (thinking when others might not; need for cognition) set aside as under-researched for education. John T. Cacioppo, Richard E. Petty, Jeffrey A. Feinstein, and W. Blair G. Jarvis, “Dispositional Differences in Cognitive Motivation: The Life and Times of Individuals Varying in Need for Cognition,” *Psychological Bulletin* 119, no. 2 (1996): 197–253, is the need-for-cognition citation Willingham uses.

fer; skill in applying methods is limited by pertinent knowledge. McPeck's *about what?* is the conceptual version of the same veto. The folk collapse — one general skill, standalone course, short test, political badge, already promised by the school — is the thing no serious account in this chapter fully endorses, and the thing the rest of this book exists to refuse. The phrase has meant Dewey's reflective thought, Glaser's three-part construct, Bloom's vocabulary, Ennis's two sentences, California's GE requirement, Paul's conference, Delphi's Table 1, a Foundation brand, a cognitive-science constraint, a conceptual critique, and a childhood pedagogy of judgment. Show the family. Then choose. The choice is a coalition. The vetoes are on the page.

# Chapter 3

## You Cannot Think About What You Do Not Know

Figure 4: A chessboard mid-game, a wine glass, and two books on a dark wooden table.

Knowledge is not the thing critical thinking is higher than. It is the medium of critical thought. A maxim that tells you to consider both sides cannot supply the sides. A strategy that tells you to look for deep structure cannot, by itself, make the structure visible. Working memory holds what you already know long enough to compare it. Without that store, "think critically" is an empty instruction, as empty as "look both ways" said to someone who cannot see the road.

Daniel Willingham stated the constraint in the Summer 2007 issue of the American Federation of

Teachers' *American Educator*, in sentences this book will not paraphrase away:

> After more than 20 years of lamentation, exhortation, and little improvement, maybe it's time to ask a fundamental question: Can critical thinking actually be taught? Decades of cognitive research point to a disappointing answer: not really. People who have sought to teach critical thinking have assumed that it is a skill, like riding a bicycle, and that, like other skills, once you learn it, you can apply it in any situation. Research from cognitive science shows that thinking is not that sort of skill. The processes of thinking are intertwined with the content of thought (that is, domain knowledge). Thus, if you remind a student to "look at an issue from multiple perspectives" often enough, he will learn that he ought to do so, but if he doesn't know much about an issue, he can't think about it from multiple perspectives. You can teach students maxims about how they ought to think, but without background knowledge and practice, they probably will not be able to implement the advice they memorize. Just as it makes no sense to try to teach factual content without giving students opportunities to practice using it, it also makes no sense to try to teach critical thinking devoid of factual content.[137]

That is the bicycle passage. The body is more careful than the AFT deck line sometimes pulled from the same issue — "There's no such thing as critical thinking 'skills.' " Critical thinking, Willingham says, "is not a skill"; it "does not have certain characteristics normally associated with skills."[138] Riding a bicycle, once learned, can be deployed whenever a bicycle is present. Thinking is not that sort of skill, because the processes of thinking are intertwined with what is being thought about.

He is not saying that critical thinking cannot be taught. The "not really" answers a different question: whether a context-free skill, acquired once and applied anywhere, is what the 1980s programs were selling. Twelve years later, in a paper commissioned by the New South Wales Department of Education, he will say the complementary sentence this chapter also has to keep: domain-specific critical thinking *can* be taught; generic add-on programs have had limited success; "we are not even sure the general skills exist, but we are quite sure there is no proven way to teach them directly."[139] The 2007 essay is the veto on the bicycle. The 2019 paper is the plan that survives the veto. Neither is a factional banner in a curriculum war. Both are cognitive-science constraints on a life book.

Chapter 2 already chose a public sentence — Robert Ennis's "reasonable reflective thinking focused on deciding what to believe or do" — and a coalition around it. That sentence does not settle

[137] Daniel T. Willingham, "Critical Thinking: Why Is It So Hard to Teach?," *American Educator* 31, no. 2 (Summer 2007): 8–19. Wording verified 30 August 2026 against the Reading Rockets authorized reprint of the *American Educator* article (https://www.readingrockets.org/topics/comprehension/articles/critical-thinking-why-it-so-hard-teach). The AFT issue page is https://www.aft.org/ae/summer2007/willingham; a 2014 AFT file path also exists (`/sites/default/files/media/2014/Crit_Thinking.pdf`). The 2023 AFT compiled-issue PDF was located but timed out in the research fetch; the reprint matches search-snippets of the issue PDF.

[138] Willingham, "Critical Thinking," 2007, article body. Prefer the body over the AFT deck/pull-quote "There's no such thing as critical thinking 'skills.' " The body says critical thinking "is not a skill" and "does not have certain characteristics normally associated with skills."

[139] Daniel T. Willingham, *How to Teach Critical Thinking*, Education: Future Frontiers Occasional Paper (Sydney: NSW Department of Education, 2019), educator-facing punchline. PDF fetched 30 August 2026 from education.nsw.gov.au. © Willingham and the State of New South Wales. Adapted for *American Educator*, Fall 2020, as "Ask the Cognitive Scientist: How Can Educators Teach Critical Thinking?"

pedagogy. This chapter does the next job. It installs the knowledge constraint in Willingham's own words, shows why surface structure captures attention and why a hint is not enough, walks the expertise evidence that makes "thinking like an expert" domain-bound twice, recovers Edward Glaser's 1941 split so 2007 is not treated as a novelty, quotes the 2019 four-step plan without turning it into this book's teaching chapter, keeps John McPeck's *about what?* demand while rejecting a total ban on general courses, and names the architectural decision the later life chapters stand on: the domains of a life as the "subjects," with knowledge and practice specified per domain.

## 3.1 The bicycle, the three points, and the three-year-old

Willingham's lay gloss, which this book can use without treating it as a definition to rival Ennis, is that critical thinking consists of "seeing both sides of an issue, being open to new evidence that disconfirms your ideas, reasoning dispassionately, demanding that claims be backed by evidence, deducing and inferring conclusions from available facts, solving problems, and so forth." There are also "specific types of critical thinking that are characteristic of different subject matter: That's what we mean when we refer to 'thinking like a scientist' or 'thinking like a historian.' "[140] The gloss is commonsensical. The argument is that the commonsensical goal has been translated into the wrong instructional object: a set of skills that can be acquired and then deployed regardless of context.

Three conclusions, after he reviews the problem-solving and scientific-thinking literatures. Do not drop the second. A life-skill book that is only maxims is the thing he is diagnosing. A life-skill book that is only "just teach history and the critical thinking will come" ignores the second point.

> First, critical thinking (as well as scientific thinking and other domain-based thinking) is not a skill. There is not a set of critical thinking skills that can be acquired and deployed regardless of context. Second, there are metacognitive strategies that, once learned, make critical thinking more likely. Third, the ability to think critically (to actually do what the metacognitive strategies call for) depends on domain knowledge and practice. For teachers, the situation is not hopeless, but no one should underestimate the difficulty of teaching students to think critically.[141]

Metacognitive strategies are the maxims: "look for a problem's deep structure"; "consider both sides of an issue." They are little chunks of knowledge that can cue a useful move. They do not complete the move. They "suggest what you ought to do"; they "don't provide the knowledge necessary to implement the strategy."[142] You may know that you ought not to accept the first reasonable-sounding solution. That does not mean you know how to generate alternatives, or how to weigh them. That requires domain knowledge and practice in putting the knowledge to work.

The same architecture explains a fact that otherwise looks like a paradox, and that this book will not resolve by flattering either children or experts. Critical thinking "is not a set of skills that can be deployed at any time, in any context. It is a type of thought that even 3-year-olds can engage in — and even trained scientists can fail in. And it is very much dependent on domain knowledge

[140] Willingham, "Critical Thinking," 2007, opening lay gloss.
[141] Willingham, "Critical Thinking," 2007, three-point boil-down, closing section.
[142] Willingham, "Critical Thinking," 2007, on metacognitive strategies: they suggest what you ought to do; they do not provide the knowledge necessary to implement the strategy.

and practice."[143] Scientific thinking, on the view Willingham reports from the developmental and science-education literatures, is not a different *kind* of reasoning. It is knowing *when* to use certain reasoning, plus enough relevant knowledge, and enough practice, to do it.

Conditional probability is the demonstration. If two things go together, one may cause the other — or the relationship may depend on a third factor. Adults sometimes use that logic successfully and fail to use it on problems that call for it. Physicians are known to discount or misinterpret new patient data that conflict with a diagnosis they have in mind. Ph.D.-level scientists are prey to faulty reasoning when a problem is embedded in an unfamiliar context.[144] And yet three-year-olds, in one experimental procedure, can use the same logic on a "blicket detector": a box said to play music if a blicket is placed on top. The child sees sequences of blocks on the box and is asked which block is a blicket. The association of each individual block with the music can be held constant across sequences; what differs is the *conditional* probability, given the other block. Three-year-olds tracked that difference.[145] The body of studies, as Willingham summarizes it, has a punchline this chapter will not improve: "Children are not as dumb as you might think, and adults (even trained scientists) are not as smart as you might think."[146]

That is not a sentimental claim about toddlers, and it is not a smear of scientists. It is the same point as the bicycle passage, now measured at both ends of expertise. When the domain is a causal toy the child has just been shown, the three-year-old can think. When the domain is an unfamiliar frame, the trained scientist can fail. Critical thinking does not have the property we expect of a skill such as reading music — that, once learned, you can use it whenever you want. People can engage in some types of it without training, and even with extensive training they will sometimes fail to think critically. The implication, which Willingham draws and which this book accepts, is that teaching students to think critically "probably lies in small part in showing them new ways of thinking, and in large part in enabling them to deploy the right type of thinking at the right time."[147]

The 2007 essay is already, in other words, a three-part design space, not a counsel of despair. Knowledge-rich practice, in domains, plus maxims that cue the right move, repeated until recognition is cheap. Drop the knowledge and the maxims do not implement. Drop the maxims and you are hoping that content alone will produce the recognitions — a hope a National Research Council science-education committee, as Willingham quotes it, had already refused: "Teaching content alone is not likely to lead to proficiency in science, nor is engaging in inquiry experiences devoid of meaningful science content."[148] The bicycle is refused. So is the opposite error, content without practice in using it. The rest of this chapter is what that refusal requires.

[143]Willingham, "Critical Thinking," 2007, introductory frame after the bicycle passage.

[144]Willingham, "Critical Thinking," 2007, scientific-thinking section, on physicians and Ph.D.-level scientists in unfamiliar frames.

[145]Willingham, "Critical Thinking," 2007, "blicket detector" procedure with three-year-olds and conditional probability. The original developmental papers sit behind Willingham's summary and were not independently re-opened for this chapter (**UNVERIFIED** as primary reports). This chapter follows his report of the design and the children's pattern of choices.

[146]Willingham, "Critical Thinking," 2007: "Children are not as dumb as you might think, and adults (even trained scientists) are not as smart as you might think."

[147]Willingham, "Critical Thinking," 2007, on what the "not a skill" claim implies for teaching.

[148]National Research Council science-education committee, as quoted by Willingham, "Critical Thinking," 2007: "Teaching content alone is not likely to lead to proficiency in science, nor is engaging in inquiry experiences devoid of meaningful science content."

## 3.2 Surface, deep, and the analog you do not notice

Why do maxims fail? Because the mind, reading a problem, is captured by what the problem is *about* — its surface — and does not spontaneously retrieve a structurally similar problem whose surface is different. Deep structure is what would transfer. Novices do not see it. A poster that says "look for deep structure" names the right search. It does not perform the search, and it does not supply the stored structures the search would need to find.

Willingham's classroom-shaped demonstration is a pair of word problems that share a deep structure and almost no surface. Subjects first read four problems with detailed explanations, ostensibly to rate the writing. One of the four is a garden problem whose solution is the least common multiple. Later they get a problem about the West High School Band, marching in rows of twelve, eight, three, and finally five, with Andrew left over until the last arrangement. The mathematics is the same. Few subjects — **19 percent** — saw that the band problem was similar and that they could use the garden solution. When experimenters told subjects working on the band problem that it was similar to the garden problem, more solved it: **35 percent** compared with 19 percent without the hint. Most subjects, even when told what to do, were not able to do it.[149] The hint raised the odds of search. It did not complete the search. Domain knowledge and practice do the rest.

The same noticing-versus-application gap is the model Mary Gick and Keith Holyoak established with Duncker's radiation problem. A tumor is to be destroyed by rays that, at full strength, would also destroy healthy tissue. About **10 percent** of subjects solve it spontaneously. After reading an analogous fortress-and-army story — disperse the forces, converge at the target — about **30 percent** produce the convergence solution *without* a hint. With a hint to use the story, rates rise to about **75 percent**; some later conditions reach the low 90s.[150] Willingham's 2019 restatement of the same pair is slightly more generous to the hint: "Merely mentioning that the story might help solve the problem boosted solution rates to nearly 100 per cent." Using the analogy was not hard; "the problem was thinking to use it in the first place."[151] Without the hint, the military story read moments earlier does not transfer. Surface structure — tumors, rays, or rebels, a fortress — captures memory search. Deep structure — disperse force, converge at the target — is not explicit. You can tell people to look for deep structure. You can even tell them *which* prior problem is similar. Most still cannot implement the strategy without the relevant knowledge.

That is why a poster of maxims is empty as a curriculum. The maxim is a hint you carry around. The band/garden numbers show what a hint is worth when the analog was studied minutes ago and the hint names it: a move from 19 percent to 35 percent, with most people still failing. The radiation numbers show what a hint is worth in the laboratory's best case: a move from about 10

[149] Willingham, "Critical Thinking," 2007, band/garden isomorphic word problems: 19 percent saw the analog; a hint that the band problem was similar to the garden problem raised solution to 35 percent.

[150] Mary L. Gick and Keith J. Holyoak, "Analogical Problem Solving," *Cognitive Psychology* 12, no. 3 (1980): 306–355; Gick and Holyoak, "Schema Induction and Analogical Transfer," *Cognitive Psychology* 15, no. 1 (1983): 1–38. Spontaneous solution of Duncker's radiation problem about 10 percent; analog without hint about 30 percent; analog plus hint about 75 percent (some later conditions reach the low 90s), as reported in the cognitive-science memo this chapter stands on.

[151] Willingham, *How to Teach Critical Thinking*, 2019, restating Gick and Holyoak: without the hint, the military story read moments earlier does not transfer; with a mention that the story might help, "nearly 100 per cent." The 2019 wording is Willingham's restatement, slightly more generous to the hint condition than the 10/30/75 band taken from the 1980 paper.

percent, or about 30 percent with an unhinted analog, toward a majority or a near-ceiling — *in that hour, on that pair*. Neither result is far transfer. Neither is a life. Chapter 4 will spend the transfer literature as a transfer literature. This chapter only needs the noticing fact. The gap that looks like a failure of "critical thinking skill" is often a failure to see that this problem is a problem you already know.

Familiarity with many surface variants of one deep structure *does* produce recognition. Willingham's illustration is a cave-and-treasure puzzle whose solution is to leave a trail of sand. About 75 percent of American college students thought of it; about 25 percent of Chinese students did. The experimenters suggested that Americans solved it because most grew up with Hansel and Gretel. A puzzle based on a Chinese folk tale reversed the percentages.[152] American subjects did not think of the problem in terms of sand, caves, and treasure; they thought of it in terms of finding something with which to leave a trail. The deep structure was so well represented in memory that they saw it when they read the problem. That is not a general power. It is a stored type, acquired by long, repeated experience with many surface variants — or, more cheaply, by growing up inside a story that *is* the type. Schooling that never shows the same deep structure in different clothes is not training recognition. It is training the surface.

The design implication is already visible, and it is not "try harder to look." It is the implication a later chapter will treat as pedagogy: interleave cases that share deep structure and differ in surface; force comparison; then test on a new surface without the hint. A life book that never does that, and instead repeats "consider both sides," is betting on the mechanism the band problem measured. The hint helps some. Most still cannot implement.

What experts actually have, if the noticing story is right, is not a general power they can swing onto any object. It is categories — problem types, patterns, vocabularies — that make deep structure cheap to see *inside a domain*, and that do not travel when the domain changes.

## 3.3 Expertise is encapsulated

"Thinking like an expert" is domain-bound twice. It is bound by the knowledge the expert has, and it is bound by the envelope of that knowledge. Chess thinking is not wine thinking is not physics thinking. A book that uses chess as a metaphor for "thinking" without this caveat is teaching the wrong lesson.

Michelene Chi, Paul Feltovich, and Robert Glaser had experts and novices sort physics problems. Experts grouped the problems by the physics principle needed for solution — conservation of energy, Newton's second law. Novices grouped them by literal surface features — inclines, pulleys, springs.[153] Representation, not raw computational power, was the difference. Completing the representation depended on the knowledge associated with the category. This is the same surface/deep split Willingham uses, now in a domain where "critical thinking" is obviously not general. It is the possession of the right problem categories. A novice looking at an inclined plane sees an inclined

[152] Willingham, "Critical Thinking," 2007, cave-and-treasure / Hansel-and-Gretel illustration of deep-structure familiarity (about 75 percent of American college students versus about 25 percent of Chinese students; the percentages reverse on a Chinese folk-tale puzzle).

[153] Michelene T. H. Chi, Paul J. Feltovich, and Robert Glaser, "Categorization and Representation of Physics Problems by Experts and Novices," *Cognitive Science* 5 (1981): 121–152.

plane. An expert looking at the same drawing sees a conservation-of-energy problem that happens to be dressed as a ramp. The expert is not more open-minded. The expert has the type.

William Chase and Herbert Simon, building on Adriaan de Groot, showed the same architecture on a chessboard. Chess masters reconstruct game positions after a five-second glance far better than novices — **and lose the advantage on random, illegal boards**.[154] The skill is chunked, domain-specific pattern knowledge: familiar sub-configurations bound by defense, proximity, attack. It is not a general visual memory. That is why treating chess as a gym for thinking fails far transfer, a result Chapter 4 will spend with the meta-analyses. The chunks do not exist in algebra. A master who "just sees" the structure of a mid-game position is seeing *chess*. Randomize the pieces so the position could never arise in play, and the seeing disappears. The expertise was never a stronger eye. It was a memory for legal structure.

Wine expertise is the same architecture in a perceptual-conceptual hybrid. Gregg Solomon found expert–novice differences in matching descriptions to wines.[155] Andrew Hughson and Robert Boakes, in "The knowing nose," showed that experts recall more wine-related words than novices *only when the words form possible varietal descriptions*, not when the same words are shuffled into meaningless combinations.[156] That is the chess-random-board analog, now in a tasting room. Novices given short variety-relevant descriptor lists could match wines above chance; the same lists did not help when the wines were the same variety. Expertise here is vocabulary plus knowledge of types, not a better nose in the abstract. Take away the legal combinations — the varietal-possible bundles — and the expert's recall advantage collapses. The nose was never the skill. The types were.

Science thinking is not wine thinking is not chess thinking. Controlling variables in an experiment, sourcing a historical document, and evaluating a forced move in a closed Sicilian are different recognitions attached to different knowledge. Willingham's 2019 paper makes the point with the historian/scientist contrast that every "think like a…" slogan blurs. If students are to read as historians do, they need specific skills: interpreting documents in light of their sources, corroborating them, putting them in historical context. Scientists, by contrast, treat the source of a document as irrelevant so long as it is trustworthy, and scientific documents are written in a consistent format that historical documents are not.[157] "Think like a historian" is not a goal until those moves are named. Those moves are **not** the ones a scientist needs. A maxim that pretended they were the same maxim would be a third surface, not a deep structure.

Expertise is also encapsulated *inside* a field. Willingham 2019, citing the expertise literature, is blunt. Knowledge of medicine transfers poorly among subspecialties: neurologists do not diagnose

[154] William G. Chase and Herbert A. Simon, "Perception in Chess," *Cognitive Psychology* 4, no. 1 (1973): 55–81. Masters reconstruct game positions after a five-second glance far better than novices, and lose the advantage on random (illegal) boards.

[155] Gregg E. A. Solomon, "Psychology of Novice and Expert Wine Talk," *American Journal of Psychology* 103 (1990): 495–517.

[156] Andrew L. Hughson and Robert A. Boakes, "The Knowing Nose: The Role of Knowledge in Wine Expertise," *Food Quality and Preference* 13 (2002): 463–472. Expert recall advantage only for possible varietal descriptions, not shuffled combinations — the chess-random-board analog.

[157] Willingham, *How to Teach Critical Thinking*, 2019, first step of the four-step plan: sourcing, corroboration, and contextualization as named historical-thinking skills, contrasted with scientific reading.

cardiac cases well.[158] Expertise in writing is similarly boxed: a technical writer who specializes in instruction pamphlets for home electronics does not thereby write newspaper articles.[159] Perhaps most surprisingly, the analytic abilities of professional philosophers do not extend to everyday judgments. Philosophers are no less susceptible than average adults to being swayed by irrelevant features of problems such as question order or wording.[160] An expert does not think as well outside her area of expertise, even in a closely related domain. She is still better than a novice. Her skills do not transfer completely.

Those three citations — Rikers, Schmidt, and Boshuizen on neurologists and cardiac cases; Ronald Kellogg on technical writers; Eric Schwitzgebel and Fiery Cushman on philosophers — are Willingham's, and they are enough for this chapter's purpose. They are not a claim that expertise is fake. They are a claim that expertise is *about something*, and that the something has a border. The neurologist's cardiac miss is not a failure of "critical thinking." It is the envelope of encapsulated knowledge, measured where the envelope ends. The philosopher's susceptibility to order and wording is the same envelope, now in a domain whose professionals are often recruited as Exhibit A for general reasoning. If professional philosophers remain swayable by irrelevant problem features in everyday moral judgment, a one-semester generic course is not going to mint a general reasoner. The honest reading is not "philosophers cannot think." It is that thinking, even among people whose job is thinking, remains tied to the content and the cues they have practiced.

Willingham 2019 names three ways knowledge helps on open-ended problems, and a life book needs all three. First, recognition of subproblems: complex critical thinking may be simpler solutions from memory "snapped together." Calculating the best value among vacation packages may be novel; if the comparison calls for long division, you do not need to invent long division. Chess experts, in the Chase-and-Simon result already cited, use recognition to evaluate positions — thousands of mid-play boards in memory — and so know where to look next.[161] Second, chunking that frees working memory. The king, a rook, and three pawns in a defensive corner become a unit, not five pieces. The same clumping has been observed in dance, circuit design, and programming. Space in working memory is thereby left for the next move rather than crowded with "what I am to do next."[162] Third, enabling the use of a known strategy. You can tell students to evaluate the logic of an author's argument when they read an op-ed, and they may have no trouble labeling the task. They still cannot use the strategy without the right domain knowledge. "Look for hidden assumptions" will not help much in sizing up an op-ed about a war in Afghanistan if you know very little about the topic. Never mind evaluating the argument: if you lack background knowledge, you will not comprehend the author's claims in the first place. Writers omit what they assume the audience

[158] Remy M. J. P. Rikers, Henk G. Schmidt, and Henny P. A. Boshuizen, "On the Constraints of Encapsulated Knowledge: Clinical Case Representations by Medical Experts and Subexperts," *Cognition and Instruction* 20, no. 1 (2002): 27–45, as cited by Willingham, 2019.

[159] Ronald T. Kellogg, "Professional Writing Expertise," in *The Cambridge Handbook of Expertise and Expert Performance* (Cambridge: Cambridge University Press, 2018), as cited by Willingham, 2019.

[160] Eric Schwitzgebel and Fiery Cushman, "Philosophers' Biased Judgments Persist Despite Training, Expertise, and Reflection," *Cognition* 141 (2015): 127–137, as cited by Willingham, 2019.

[161] Willingham, *How to Teach Critical Thinking*, 2019, on recognition of subproblems and Chase and Simon's chess result as enabling attention, not as a general memory.

[162] Willingham, *How to Teach Critical Thinking*, 2019, on chunking and working memory, including the observation of clumping in dance, circuit design, and programming.

knows.[163]

Background knowledge is what lets working memory hold the pieces long enough to compare them. Without it, "consider both sides" is an empty maxim. That sentence is this book's pedagogical constraint, not a slogan against knowledge. The next section is only to stop the reader from treating the constraint as a twenty-first-century discovery.

## 3.4 Glaser 1941, sixty-six years early

Edward Glaser already had the split. Chapter 2 quoted him as the writer who rewrote Dewey, named a three-part construct — attitude, knowledge of methods, skill in applying them — and, in the same 1941 dissertation, limited what general training can do. The sentence that belongs in *this* chapter is the knowledge-constraint finding, not the test lineage. Hitchcock quotes it from page 175 of *An Experiment in the Development of Critical Thinking*:

> The aspect of critical thinking which appears most susceptible to general improvement is the attitude of being disposed to consider in a thoughtful way the problems and subjects that come within the range of one's experience. An attitude of wanting evidence for beliefs is more subject to general transfer. Development of skill in applying the methods of logical inquiry and reasoning, however, appears to be specifically related to, and in fact limited by, the acquisition of pertinent knowledge and facts concerning the problem or subject matter toward which the thinking is to be directed.[164]

Disposition of wanting evidence transfers better than skill in applying logical methods. Skill is limited by pertinent knowledge. That is the 1941 version of Willingham 2007. The bicycle passage is not a novelty. It is the cognitive-science restatement of a split an educational experimenter had already drawn, in Dewey's vocabulary, before the 1980s movement industrialized the phrase.

A book that cites Glaser only as the father of the Watson–Glaser Critical Thinking Appraisal, and not as an early knowledge-constraint theorist, is cherry-picking. The Appraisal is a commercial descendant. It is an instrument, not a definition. The assessment chapter will have to say what that instrument does and does not measure. This chapter only needs Glaser's conclusion. Attitude is the part that looks more movable in general. Method, when it has to be applied to a problem, is limited by what you know about the problem. Willingham's three points do not replace that split. They explain it. Metacognitive strategies are closer to Glaser's attitude — they make the right move more likely. Doing what the strategies call for is Glaser's skill, and it is limited by knowledge. The 2007 essay is what the 1941 dissertation looks like after a century of problem-solving experiments.

Dewey's lantern, already quoted in Chapter 2, is the same constraint in 1910 prose: thinking "is specific, not a machine-like, ready-made apparatus to be turned indifferently and at will upon all sub-

[163] Willingham, *How to Teach Critical Thinking*, 2019, Afghanistan op-ed example; background knowledge and comprehension, citing Willingham, *The Reading Mind* (San Francisco: Jossey-Bass, 2017).

[164] Edward M. Glaser, *An Experiment in the Development of Critical Thinking*, Teachers College, Columbia University, Contributions to Education no. 843 (1941), 175, as quoted by David Hitchcock, "Critical Thinking > History," *Stanford Encyclopedia of Philosophy*, Winter 2024 edition. Full 1941 volume not independently fetched — **UNVERIFIED** at page level against the monograph; definition at p. 6 and conclusion at p. 175 via Hitchcock. See also Chapter 2.

jects, as a lantern may throw its light as it happens upon horses, streets, gardens, trees, or river."[165] Glaser limited the skill. Willingham said the processes are intertwined with the content. McPeck, in section 3.6, will say that thinking is always about X. The family is not a coincidence. It is the same fact, named from a process picture, from an experiment, from cognitive science, and from a conceptual claim. A life book that promises a content-free general power is contradicting all four.

The 2019 paper turns the constraint into a program. It does not turn it into this book's pedagogy chapter. Chapters 5 and 10 will have to say what teaching actually moves, and what a life of deeds looks like. What belongs here is the plan as a design constraint: domain by domain, knowledge specified, sequenced, revisited for years — and a punchline this book will quote rather than improve.

## 3.5 Willingham 2019: four steps, not a generic course

*How to Teach Critical Thinking*, published in the NSW Department of Education's Education: Future Frontiers Occasional Paper series, restates the 2007 architecture for a school system that had named critical thinking a "general capability." Willingham's commonsensical definition, which he flags as his own 2007 view, is that you are thinking critically if (1) the thinking is *novel* — not merely recalling a prior conclusion; (2) *self-directed* — not merely executing instructions; and (3) *effective* — it respects conventions that make useful conclusions more likely, "consider both sides of an issue," "offer evidence for claims made," "don't let emotion interfere with reason." What counts as effective *varies by domain*.[166] A second informal sense — thinking when others might not; need for cognition — he sets aside as under-researched for education, and possibly in part a matter of personality.[167] This chapter follows him in setting it aside. The life book's ethics chapter will have to return to dispositions. It will not smuggle need-for-cognition into the meaning of critical thinking and then pretend the evidence is in.

Scientists, he reports, are "united in their belief that content knowledge is crucial to effective critical thinking." They are "somewhat divided as to whether critical thinking is best characterised as a large number of more specific skills or a smaller number of more generic skills." His own judgment: the generic-skills framing is "not a fruitful way to conceptualise skills in education … as there is little theory to guide how to teach generic skills."[168] That division is a live debate this book will not pretend to close with a slogan. The educator-facing fact, which he thinks ought to be salient whether or not the general skills exist, is the punchline:

> We are not even sure the general skills exist, but we are quite sure there is no proven way to teach them directly. In contrast, we have a pretty good idea of how to teach

[165] John Dewey, *How We Think* (Boston: D. C. Heath, 1910), lantern/specificity passage, Gutenberg text, fetched for this book's definitional research. Quoted at length in Chapter 2.

[166] Willingham, *How to Teach Critical Thinking*, 2019, opening commonsensical definition (novel, self-directed, effective), with effectiveness varying by domain.

[167] Willingham, *How to Teach Critical Thinking*, 2019, setting aside need for cognition (Cacioppo et al., 1996) as under-researched for education.

[168] Willingham, *How to Teach Critical Thinking*, 2019, authors' abstract and body: scientists united on content knowledge; somewhat divided on specific versus generic skills; generic framing "not a fruitful way to conceptualise skills in education." Australian spelling in the source is preserved in quotation.

students the more specific critical thinking skills. I suggest we do so.[169]

Domain-specific critical thinking *can* be taught. Willingham's examples are not a meta-analysis, and this chapter will not launder them into one. D. Alan Bensley and Rachel Spero infused principles for evaluating evidence — correlational research versus true experiments, anecdote versus formal research — into a psychology class; compared with a control that learned principles of memory, the students performed better on a test that required evaluation of *psychology* evidence.[170] That is near transfer, explicit practice, domain-matched. Officers in the Royal Netherlands Navy received training in the analysis of complex battlefield problems in a high-fidelity tactical simulator — a sequence of steps, eight hours on surface-warfare problems with expert feedback — and outperformed controls on a new surface-warfare problem and on air-warfare problems, as judged by the quality of action contingency plans.[171] Willingham also points to a broader instructional literature, including Abrami and colleagues' 2008 meta-analysis, as evidence that critical-thinking skills are open to instruction.[172] The coefficients and the transfer limits of that literature are Chapter 4 and Chapter 5. The fact that belongs here is the one Willingham is willing to stand on: you tell students that this is a good strategy for this type of problem, you have them practice that strategy, and later they use that strategy when they encounter the problem. That is not a bicycle. It is a type.

Generic add-on programs have had "limited success." Evaluations often test the same puzzle-type they trained. If the training features logical and spatial puzzles, the measure of success tends to feature the same sort of puzzle. If the regimen entails argument and debate, the outcome is usually the ability to evaluate arguments or take both perspectives in debate. When investigators have tested for transfer in such curricular programs, "positive results have been absent or modest and quick to fade."[173] Formal discipline — Latin, geometry, then Logo and programming, Mozart, chess, music — does not survive as advertised. A computational-thinking meta-analysis looks better until active controls are required; remaining effects look like conceptual overlap, not general transfer, and there is no benefit to literacy.[174] Chapter 4 will spend Sala and Gobet and the active-control shrink. This chapter only needs the moral Willingham draws. It is no surprise that school programs meant to teach *general* critical thinking have had limited success. The surprise would be if they had not.

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[169] Willingham, *How to Teach Critical Thinking*, 2019, punchline before the four-step plan.

[170] D. Alan Bensley and Rachel A. Spero, "Improving Critical Thinking Skills and Metacognitive Monitoring through Direct Infusion," *Thinking Skills and Creativity* 12 (2014): 55–68, as cited by Willingham, 2019. Near transfer, psychology evidence, domain-matched.

[171] Anne S. Helsdingen, Karel van den Bosch, Tamara van Gog, and Jeroen J. G. van Merriënboer, "The Effects of Critical Thinking Instruction on Training Complex Decision Making," *Human Factors* 52, no. 4 (2010): 537–545, as cited by Willingham, 2019.

[172] Philip C. Abrami et al., "Instructional Interventions Affecting Critical Thinking Skills and Dispositions: A Stage 1 Meta-Analysis," *Review of Educational Research* 78, no. 4 (2008): 1102–1134, cited by Willingham, 2019, as evidence that critical-thinking skills are open to instruction. Effect sizes are not dumped in this chapter; they belong to Chapters 4–5.

[173] Willingham, *How to Teach Critical Thinking*, 2019, on generic add-on programs: "limited success"; when transfer is tested, "positive results have been absent or modest and quick to fade," citing Ron Ritchhart and David N. Perkins, "Learning to Think: The Challenges of Teaching Thinking," in *The Cambridge Handbook of Thinking and Reasoning*, ed. Keith J. Holyoak and Robert G. Morrison (Cambridge: Cambridge University Press, 2005), 775–802.

[174] Willingham, *How to Teach Critical Thinking*, 2019, on Scherer, Siddiq, and Viveros (cited there as 2018; journal publication 2019) programming meta-analysis: overall $g = 0.47$, smaller with active controls, no benefit to literacy. The full transfer treatment of chess, music, working-memory training, and programming as "thinking gyms" is Chapter 4.

There are principles that *do* carry across domains. “A” and “not A” cannot be simultaneously true in mathematics or in history. Denial of the consequent is always wrong; straw-person arguments are always weak; a conflict of interest always makes an argument suspect. The law of large numbers is a law wherever samples are. The problem is deployment. People who learn these broadly applicable principles in one situation often fail to apply them in a new situation. They understand the law of large numbers when the context is loaded dice, and are ready to say that two poor maths tests mean the student is simply bad at maths.[175] Understanding offers no guarantee that the student will recognise new situations in which the idea will be useful. Previous critical-thinking successes seem encapsulated in memory. That is Gick and Holyoak again, now as a theory of the skill rather than as one experiment: the problem in transfer is not only that different domains have different norms. It is that the success does not come to mind when the surface changes.

Diane Halpern is the most optimistic cognitive psychologist in this family. Her 1998 *American Psychologist* paper proposes teaching for transfer via disposition, skills, structure training, and metacognitive monitoring.[176] Willingham cites her among the sources for explicit instruction plus practice. He does not adopt her optimism as a finding that general skills have been taught. The constraint remains his: we are not even sure the general skills exist; we are quite sure there is no proven way to teach them directly. This book can steal Halpern’s four ingredients as a design hypothesis for *named* domains. It cannot treat them as a demonstration that a content-free course transfers.

The four-step plan is the positive program. Quote it as a design constraint, not as this book’s teaching chapter.

**First**, identify what is meant by critical thinking *in each domain*. Be specific. It is not useful to set a goal that students “think like historians,” or “learn the controversies surrounding historical events.” If students are to read as historians do, they need to learn specific skills: interpreting documents in light of their sources, corroborating them, putting them in historical context. Learning to read like a scientist means, in part, learning the conventions of a consistent format, and treating source as irrelevant so long as it is trustworthy. The skills should be explicitly taught and practiced; simple exposure without explicit instruction is less effective. Educators will have to pick and choose. Even across thirteen years of schooling, time is limited.[177]

**Second**, identify the domain content students must know. Sourcing a letter written by an army sergeant to his wife just before a named battle is not a pure process. The student must know enough historical context to understand how that sourcing information ought to influence interpretation. What knowledge is essential depends on educational goals — employment, citizenship, a better understanding of individual abilities and passions. Willingham is explicit that the prospect

[175] Christopher Jepson, David H. Krantz, and Richard E. Nisbett, “Inductive Reasoning: Competence or Skill?,” *Behavioral and Brain Sciences* 6, no. 3 (1983): 494, as cited by Willingham, 2019, on failure to deploy the law of large numbers across surfaces.

[176] Diane F. Halpern, “Teaching Critical Thinking for Transfer across Domains: Disposition, Skills, Structure Training, and Metacognitive Monitoring,” *American Psychologist* 53, no. 4 (1998): 449–455. Willingham 2019 cites Halpern among sources for explicit instruction plus practice; this chapter treats her as the most optimistic cognitive psychologist in the family, not as a finding that general skills have been taught directly.

[177] Willingham, *How to Teach Critical Thinking*, 2019, four-step plan, step 1, including the historian/scientist contrast and the claim that simple exposure without explicit instruction is less effective (citing Abrami et al. 2008; Halpern 1998; Heijltjes, van Gog, and Paas 2014).

of someone deciding which knowledge students ought to learn, and what they will not learn, makes people uneasy *because the decision depends on values*. "Selection of content is a critical way that values are expressed." Making the choice will lead to uncomfortable trade-offs. "But not choosing is still making a choice. It is choosing not to plan, and to let random forces determine what students learn."[178] That sentence is a gift to a *life* book. A life-skill without a domain is a life-skill without a life. Not naming the domains is still a choice. It is a choice to let the feed, the workplace, the clinic, and the family argument determine what the reader learns to think *about*, which is to say, what the reader can actually think.

**Third**, select the best sequence in which to learn the skills. Knowledge and skills build on one another. It is easier to understand why a constitution was approved if one already knows the regional fears that preceded it. We interpret new information in light of what we already know. The right preparation makes new learning easier.[179]

**Fourth**, plan which knowledge and skills will be revisited across years. Even if content is learned quite well over half a school year, about half will be forgotten in three years.[180] That does not mean there is no value in exposing students to content once; most will forget much and remember something, and for some an interest may be kindled. But when considering skills meant to stick for the long term, "we should plan on at least three to five years of practice." Most of the time the practice will look different — embedded in new skills and content. The revisiting should be assured and planned.[181]

Four steps. Not a generic course. Not a poster. Not a one-semester infusion that is then declared to have produced a thinker. The plan assumes subjects, years, and a curriculum. A book for adults needs an analogue, which section 3.7 will name. The plan also assumes patience. "The way the mind works, shallow is what you get first. Deep, critical thinking is hard-won." Designers of any program to improve it "must take the long view, both in the time frame over which the program operates, and especially the speed with which one expects to see results."[182] A life book that promises a converted reader by Chapter 10 has not read the forgetting curve.

Do not pick a factional banner in the English and Australian knowledge-rich curriculum wars. Use Willingham as a cognitive-science constraint. E. D. Hirsch's equity argument for a knowledge-rich curriculum is an *argument*, not a fetched randomized trial that racial or socioeconomic gaps closed; it is planted here and spent, if it is spent, in the last chapter.[183] The 2019 paper sits inside that

[178] Willingham, *How to Teach Critical Thinking*, 2019, step 2: "Selection of content is a critical way that values are expressed" (citing Willingham, *When Can You Trust the Experts?*, 2012). "But not choosing is still making a choice. It is choosing not to plan, and to let random forces determine what students learn."

[179] Willingham, *How to Teach Critical Thinking*, 2019, step 3, sequence; the Australian constitution / South Pacific example is his.

[180] Andrew Pawl, Analia Barrantes, David E. Pritchard, and Rudolph Mitchell, "What Do Seniors Remember from Freshman Physics?," *Physical Review Special Topics—Physics Education Research* 8 (2012): 020118, as cited by Willingham, 2019: about half of well-learned content forgotten in three years.

[181] Harry P. Bahrick, "Semantic Memory Content in Permastore: Fifty Years of Memory for Spanish Learned in School," *Journal of Experimental Psychology: General* 113, no. 1 (1984): 1–29; Harry P. Bahrick and Lynda K. Hall, "Lifetime Maintenance of High School Mathematics Content," *Journal of Experimental Psychology: General* 120, no. 1 (1991): 20–33. Willingham, 2019: plan on at least three to five years of practice for skills meant to stick.

[182] Willingham, *How to Teach Critical Thinking*, 2019, closing: "The way the mind works, shallow is what you get first. Deep, critical thinking is hard-won."

[183] E. D. Hirsch Jr.'s knowledge-rich equity argument (2016) is planted, not tried. This chapter does not claim a

broader debate. This book's job is not to join a camp. It is to refuse a content-free general power, and to refuse the opposite error that content without named, practiced, revisited skills will produce the recognitions by itself.

McPeck is the conceptual partner of this constraint, not its crank. The book can reject his ban on general courses without rejecting his demand that thinking always be about something.

## 3.6 McPeck as partner, not crank

John McPeck's *Critical Thinking and Education* (1981) is the book the 1980s movement had to answer. His definition, widely quoted from page 8, is "the propensity and skill to engage in an activity with reflective scepticism."[184] The conceptual keystone, as quoted in Richard Paul's critique and in McPeck's own later restatements, is sharper:

> it is a matter of conceptual truth that thinking is always thinking about X, and that X can never be "everything in general" but must always be something in particular. Thus the claim "I teach my students to think" is at worst false and at best misleading. … To the extent that critical thinking is not about a specific subject X, it is both conceptually and practically empty.[185]

That is the *about what?* demand. A life-skill book that promises a content-free general power is making the claim McPeck called conceptually empty. This book agrees. Thinking is always about X. X is never everything in general. "I teach thinking" simpliciter is empty. The bicycle passage is the cognitive-science version of the same sentence. Glaser's "limited by" is the experimental version. Dewey's lantern is the 1910 version. McPeck is not an outlier. He is the philosopher who said the constraint out loud, as a conceptual truth, while the movement was scaling a general course.

He drew consequences this book does not all accept. No unitary general reasoning skill; informal-logic courses and generic critical-thinking courses are the wrong unit of instruction; criteria of good thinking are field-dependent, the epistemology of the parent discipline; transfer of training cannot be assumed but must be established in each case by empirical tests; distinguished logicians who cannot say whom to vote for, or why, are Exhibit A.[186] Dispositions, especially the disposition to suspend judgment under cognitive dissonance, can be more general than abilities. Hitchcock notes that McPeck did not extend strong subject-specificity to all dispositions.[187] Do not cite him as

named U.S. knowledge-rich curriculum has closed racial or SES gaps; no such RCT was fetched in the research this book stands on. Spend, if at all, in Chapter 10.

[184] John E. McPeck, *Critical Thinking and Education* (New York: St. Martin's Press, 1981), 8, as quoted in Informal Logic reviews and McPeck's later restatements. Full 1981 volume not independently re-opened for this chapter — **UNVERIFIED** against the monograph; wording as in the definitional research memo.

[185] McPeck, *Critical Thinking and Education*, 1981, conceptual keystone, as quoted in Richard Paul's Informal Logic critique of McPeck and in the definitional memo. Quotation marks around "everything in general" follow the memo's wording. Page-level wording **UNVERIFIED** against the 1981 monograph.

[186] McPeck's consequences as summarized in the definitional research (01-definitions §2.6): no unitary general reasoning skill; field-dependent criteria; transfer must be established empirically; distinguished logicians as Exhibit A.

[187] David Hitchcock, "Critical Thinking," *Stanford Encyclopedia of Philosophy* (substantive revision 12 October 2022), on McPeck's not extending strong subject-specificity to all dispositions.

holding that there are no general thinking abilities of any kind without noting that exception, and without noting the replies.

The replies that still matter are the writing and speaking analogies. Paul (1985, "McPeck's Mistakes") and Harvey Siegel (1985) press the point: speech is always about something particular, yet general composition and speech courses are not empty.[188] Ennis (1989), in "Critical Thinking and Subject Specificity," distinguishes *conceptual*, *epistemological*, and *empirical* subject-specificity, and argues that McPeck overreaches on the first two.[189] Hitchcock's verdict is the one this book can live with: McPeck's central argument "needs elaboration" and has not explained how thinking differs from writing and speaking; *nevertheless* "his position that the dispositions and abilities of a critical thinker are best developed in the context of subject-matter instruction is shared by many theorists," including Dewey, Glaser, Passmore, Weinstein, Bailin and colleagues, and Willingham (2019).[190] Shared, that is, as a development claim, not as a proof that no general course can ever be useful.

This book's choice, labelled as a choice, is the one Chapter 2 already named. Keep the knowledge constraint. Reject the *ban on general courses* as a design rule. A general strand can still be useful as a *cue-and-practice* strand beside domain work: maxims that make the right move more likely, comparison of structurally same and surface-different cases, a shared vocabulary for "what would falsify this" and "name a person who disagrees." Willingham's second point is exactly that permission. Metacognitive strategies, once learned, make critical thinking more likely. They do not implement themselves. McPeck's ban would treat the general strand as empty because it is not about a specific X. The writing/speaking replies, and Willingham's second point, are why this book will not treat it as empty. The knowledge constraint is why this book will not treat it as sufficient. A standalone semester of maxims, with no domain and no years of revisiting, is the thing both of them, in different vocabularies, have already refused.

The live debate this chapter leaves standing is how general is general enough for a *life* skill. Everyday life is not a school subject and not "everything in general." Ennis's everyday-life examples, Dewey's transit and ferryboat and bubbles, and Willingham's Afghanistan op-ed pull in different directions. Ennis wants a conception that covers deciding what to believe or do in ordinary affairs, and that is why his sentence is this book's public definition. Dewey's examples are specific perplexities, each with its own subject-matter. Willingham's op-ed is the comprehension constraint in one object: you cannot look for hidden assumptions in a piece you cannot read. Those three are not the same grain of "general." Dose–response — how much domain knowledge is enough for a given critical-thinking move, and how that mix should be set against maxims and against incentives — is an open question in the cognitive-science record this book stands on.[191] It will not be filled

[188] Richard Paul, "McPeck's Mistakes," *Informal Logic* 7, no. 1 (1985); Harvey Siegel's 1985 reply in the same Informal Logic exchange. Writing/speaking analogy: speech is always about something particular, yet general composition and speech courses are not empty. Siegel's 1985 full text not independently fetched for this chapter (**UNVERIFIED** as a primary PDF).

[189] Robert H. Ennis, "Critical Thinking and Subject Specificity: Clarification and Needed Research," *Educational Researcher* 18, no. 3 (1989). Conceptual, epistemological, and empirical subject-specificity.

[190] Hitchcock, "Critical Thinking," SEP, verdict on McPeck: central argument "needs elaboration"; the subject-matter-instruction claim is shared by Dewey, Glaser, Passmore, Weinstein, Bailin et al., and Willingham (2019).

[191] Open question in the cognitive-science stream (02-cogsci): dose–response of domain knowledge versus maxims versus incentives is not fitted. Not filled here with analogy.

here with an analogy. Schraw's claim that metacognitive knowledge is relatively domain-general and teachable is the author's theoretical preference, not a settled finding; implementation still depends on domain knowledge.[192] Flag that. A later chapter that treated journaling as a substitute for knowledge would be ignoring the flag.

What McPeck still exerts, after the ban is refused, is the question the folk collapse cannot answer. About what? If the answer is "everything," the course is empty. If the answer is "history, or science, or chess," the life book has not yet said how a person who is not in school is supposed to think. The next section is the architectural answer, not a curriculum.

## 3.7 Domains of a life as the "subjects"

Willingham's four steps assume subjects, years, and a curriculum. A book for adults needs an analogue. Everyday life is not a school subject. It is also not everything in general. The honest unit is the domains of a life, treated as the "subjects," with knowledge and practice specified per domain.

Name them so the later chapters can spend them rather than invent them: **money**, **health**, **news**, **work**, **friendship**, **citizenship**. Those six are a design decision, not a fitted syllabus and not a randomized trial. They are this book's answer to McPeck's *about what?* for a reader who is not enrolled in a history course. They are also this book's answer to Willingham's warning that not choosing is still making a choice. A life-skill book that never names the domains is choosing to let whoever shouts next determine the content of thought. The processes will then be intertwined with whatever content happens to arrive — which is to say, they will not be critical.

Each domain has its own recognitions, its own content, and its own practiced moves. News is not health. Lateral reading of a source — leaving the page; asking who is behind it, what the evidence is, what other sources say — is a civic habit with trial evidence, and it is Chapter 7's object. It will not automatically appear in a clinic when a relative-risk poster is on the wall. Health has its own knowledge constraint: base rates, natural frequencies, the difference between a diagnostic test's sensitivity and what a patient needs to hear. Work has case knowledge and the envelope of expertise the neurologist/cardiac result already named. Money has compounding, fees, and the difference between a story about a stock and a sample. Friendship has other people, which laboratory tasks typically omit. Citizenship has identity, which can recruit intelligence against the evidence — a result Chapter 4 will spend, and which this chapter only plants. Science thinking is not wine thinking is not chess thinking. Civic thinking is not clinical thinking is not managerial thinking. The names are different because the types are different.

This is not a claim that the six domains have been evaluated as a curriculum. They have not. It is not a claim that a person who gets good at one becomes good at the others. The encapsulation evidence says otherwise. It is a claim that a life book has to specify *something*, and that "critical thinking" as a mass noun specifies nothing. Preview, without writing them: Chapters 7 and 8 are the first two domains written out — civic and media judgment, then clinic, work, and court. Chapter 9 is

[192] Gregory Schraw, "Promoting General Metacognitive Awareness," *Instructional Science* 26 (1998): 113–125. Domain-generality of metacognitive knowledge is the author's theoretical preference, not a settled finding. Implementation still depends on domain knowledge (Willingham). Flagged as Open question 23 in the book outline.

the life condition that now sits across all of them: a machine will finish the sentence. Chapter 10 is deeds, not a badge, each deed empty without the domain knowledge this chapter installed.

Hirsch's knowledge-rich argument, already planted, is an equity claim about school content: if school is the chief venue through which some students meet advanced vocabulary, rich content, and demands for high-level thinking, reducing those opportunities is not mercy.[193] Willingham 2019 makes a related point about tracking and socioeconomic status, citing OECD indicators and the association between access to challenging content and family status.[194] Those sentences belong to a later chapter's fairness argument. They are not a fetched trial that a named United States knowledge-rich curriculum closed racial or SES gaps. Do not upgrade an argument to a finding. Do not skip the argument because the trial is missing. For *this* chapter the load-bearing sentence is still Willingham's on choice: not choosing is still making a choice. The six domains are this book's choice. A reader who wants a seventh is not refuting the architecture. The reader is doing the fourth step — deciding what to revisit — in a life rather than in a school.

How much knowledge is enough, in each domain, for a given move, remains open. The Afghanistan op-ed is a minimum: enough to comprehend the claims, or the evaluation never starts. The chess master's thousands of boards are a maximum this book will not demand for citizenship. Between those poles there is no fitted dose. What the record does support is directional. More relevant knowledge makes the maxim more implementable. Practice with varied surfaces makes the type cheaper to see. Years of revisiting, not a workshop, are what the forgetting numbers imply. A book that offers a weekend of maxims and calls it a life skill has not read Pawl, and has not read Bahrick.

The architectural decision can be stated in one paragraph so later chapters can point at it rather than re-litigate it. Critical thinking for life is not a faculty you acquire and then apply to money, health, news, work, friendship, and citizenship as a lantern swung onto new objects. It is knowledge plus evaluation habits, practiced on real stakes, *in* those domains. Ennis's sentence still names the aim: deciding what to believe or do. Willingham still names the constraint: you cannot do what the aim requires without the content of thought and the practice that makes the content usable. McPeck still names the question: about what? The domains are the answer this book is willing to put on the page. They are not everything in general. They are not nothing. They are a life, specified enough to think about.

That is the constraint every later chapter will cite rather than re-argue. If the bicycle passage is paraphrased instead of quoted, the rest of the book becomes maxims again. It has now been quoted. The three-point boil-down is on the page, second point included. The noticing numbers are on the page. Encapsulated expertise is on the page, with the Chase-and-Simon caveat attached to any chess metaphor. Glaser is on the page as a knowledge-constraint theorist, sixty-six years early. The 2019 punchline and the four steps are on the page. McPeck's *about what?* is kept; his ban on general courses is not adopted as a design rule. The domains of a life are named as a choice.

What this chapter did not claim is as important as what it did. It did not claim that critical thinking cannot be taught. Domain-specific critical thinking can be taught; the 2019 paper is explicit, and

[193] Hirsch 2016 as equity *argument* for knowledge-rich curriculum, not a fetched RCT. See note 182.

[194] Willingham, *How to Teach Critical Thinking*, 2019, on access to challenging content and socioeconomic status, citing OECD, *Education at a Glance 2018*, and Parker et al. 2016 on tracking. Planted for a later fairness chapter; not a U.S. gap-closing trial.

Bensley and Spero, Helsdingen, and the instructional literature Willingham cites are the existence proof at the grain of a type, not at the grain of a life. It did not dump the generic-course effect sizes; those are transfer and teaching results, and they belong in the next two chapters. It did not claim that McPeck showed there are no general thinking abilities of any kind. The writing and speaking replies, and his own exception for some dispositions, stay attached to his name. It did not use chess as a metaphor for thinking without the random-board result. It did not claim that a named United States knowledge-rich curriculum has closed racial or socioeconomic gaps. It did not treat the six domains as an evaluated syllabus. It did not treat Willingham as a faction. It did not drop the metacognitive strategies because knowledge is the medium. Knowledge without the cue, and the cue without knowledge, are both the NRC's refused half.

The open questions this chapter leaves standing are the ones it would otherwise fill with analogy. How general is general enough for a life skill — Ennis's ordinary affairs, Dewey's specific perplexities, Willingham's op-ed — is not settled by naming six domains; it is made tractable. How much domain knowledge is enough for a given move, versus maxims versus incentives, is a dose–response question the record has not fitted. Knowledge-rich curriculum as equity is planted, not tried. Whether metacognition is domain-general remains Schraw's preference, not a finding. Year-scale unaided far transfer of a specific principle into a workplace or a civic decision, scored independently of a critical-thinking test, is not this chapter's result, because it is not anyone's clean result yet.

If knowledge is the medium, why do maxims, dual-process posters, and "be smarter" still fail even among people who *have* knowledge? That is a different failure, and it is the next chapter's. Near is not far. Type 2 is not the hero. Smarter is not a myside cure. The bicycle has been refused. The remaining work is to refuse the other shortcuts without pretending that refusing them produces a thinker.

# Chapter 4

## Why Maxims Fail: Transfer, Dual Process, Myside

Figure 5: A chalkboard, a torn list on parchment, wooden puzzle cubes, and an old desk clock.

Near is not far. Type 2 is not the hero. Smarter is not a myside cure.

A poster of maxims is the public face of critical-thinking instruction: consider both sides; look for deep structure; slow down and use System 2; name the bias of the week. The poster is not nothing. It is also not a theory of mind, not a curriculum, and not a life. Three distinctions from cognitive science constrain everything that follows. A skill is not a bicycle. Near is not far. Performance is not later unaided judgment. A book that treats the first member of each pair as evidence for the second will overclaim. This chapter is the constraint chapter. It is not a tour of named fallacies and it is not a claim that critical thinking cannot be taught. It is the record of why maxims fail when they are asked to travel — across domains, across months, across rooms, and across the identity of the person who would have to use them. The popular story treats a named skill as a faculty that, once strengthened, applies everywhere. Thorndike already tested that story. The rest of the chapter is what the century since has added: noticing is the bottleneck, high-road transfer is rare unless cued, thinking gyms shrink under active controls, Type 2 is not a hero, myside does not yield to IQ, and life is not a laboratory.

## Three distinctions that run the chapter

The first distinction is already on the table from Chapter 3, and it can be restated in one sentence. Riding a bicycle, once learned, can be deployed whenever a bicycle is present. Critical thinking is not that sort of skill.[195] Thought processes are intertwined with the content of thought. Maxims can cue useful moves. They do not supply the knowledge those moves require. You can tell people to look for deep structure, and even tell them which prior problem is similar, and most still cannot implement the strategy without the relevant knowledge. That is Daniel Willingham's architecture, not a slogan: there are metacognitive strategies that, once learned, make critical thinking more likely; the ability to *do* what those strategies call for depends on domain knowledge and practice.[196]

The second distinction is the one this chapter exists to enforce. Near is not far. Susan Barnett and Stephen Ceci taxonomize transfer along content — what is transferred: the learned skill, the performance change, the memory demands — and context: knowledge domain, physical setting, temporal distance, functional demand, social context, and modality.[197] "Near" and "far" are not a binary. They are positions on those dimensions. A student who uses the law of large numbers on a second coin-flip problem in the same class is near. A student who uses it, unprompted, a year later, at work, in a hiring conversation, while identity is on the line, is far. A tidy essay written under a prompt that names the bias is near. Spotting the same structure in a family argument, a clinic, or a feed, with no examiner in the room, is far. A book that treats a moved score on a critical-thinking inventory, in the same semester, in the same academic frame, as evidence that a person will think when the prompt, the time, the other people, and the identity stake change, has already collapsed the taxonomy.

The third distinction is the one fluency conceals. Performance is not later unaided judgment. Nicholas Soderstrom and Robert Bjork's integrative review is the warning: conditions that worsen immediate performance can improve later retention and transfer; conditions that make performance look fluent — massed practice, complete answers, a smooth recap — can leave later unaided skill untouched.[198] A reader who can recite "confirmation bias" after a smooth chapter has demonstrated

[195]Daniel T. Willingham, "Critical Thinking: Why Is It So Hard to Teach?," *American Educator* 31, no. 2 (2007): 8–19. The bicycle passage is the constraint this book restates from Chapter 3.

[196]Willingham, "Critical Thinking," 2007; Daniel T. Willingham, "How to Teach Critical Thinking," NSW Department of Education, Education: Future Frontiers Occasional Paper, 2019. Metacognitive strategies make critical thinking more likely; implementing them depends on domain knowledge and practice.

[197]Susan M. Barnett and Stephen J. Ceci, "When and Where Do We Apply What We Learn? A Taxonomy for Far Transfer," *Psychological Bulletin* 128, no. 4 (2002): 612–637.

[198]Nicholas C. Soderstrom and Robert A. Bjork, "Learning versus Performance: An Integrative Review," *Perspec-*

performance, not later spotting of confirmation bias in their own political feed. Deslauriers and colleagues, in a physics-classroom experiment, found that the feeling of learning tracked the *opposite* of actual learning in active versus passive instruction.[199] Fluency of rereading is the enemy of later judgment. The implication for a life book is not a classroom method. It is a reading rule. Fluent output — a named bias, a confident oral recap, a tidy paragraph a model will finish — is not a check that the person can think when the conditions change.

Those three distinctions are the reading rules. They also explain the industry's love of maxims. A maxim is cheap, portable, and photographable. It can be printed on a syllabus, a poster, and a quality-enhancement plan without specifying the knowledge, the practice, or the later unaided test. "Consider both sides" does not tell you what the sides are, or what would count as evidence for either. "Look for deep structure" does not tell you which structures a domain contains. "Slow down and use System 2" does not tell you whether the problem is missing mindware, a weak logical intuition, identity load, or Type 2 writing a press release. "Name the bias" does not produce the recognition that Gick and Holyoak showed is the bottleneck. The industry is not dishonest for wanting a portable object. It is dishonest when it treats portability as evidence of transfer.

Barnett and Ceci's dimensions make the honesty test operational. Content can move — a skill, a performance change, a memory demand — while context stays put: same knowledge domain, same room, same week, same academic function, same classmates, same paper-and-pencil modality. That is near on every context axis. Far on the context axes is a different object: a different domain, a different place, months or years later, a decision that is not a test, an audience that is not a grader, a modality that is not the worksheet. A meta-analysis that averages near outcomes and reports them as "critical thinking improved" has not lied about the tests. It has invited the reader to treat near as far. This book will not.

Transfer first, because the whole generic-course industry is a transfer claim.

## Transfer: default failure, noticing, high road

In 1901, Edward Thorndike and Robert Woodworth tested the doctrine of formal discipline — the claim that hard subjects, Latin and geometry among them, train general faculties of attention and logic that then apply everywhere. Their estimation experiments, on areas, lengths, and weights, found that improvement in one mental function rarely brings equal improvement in any other, even when the two look similar. Spread of practice, they concluded, occurs only where **identical elements** are concerned. The working of every function-group is conditioned by the nature of the data in each particular case.[200] That finding is the century-old constraint on every generic critical-thinking course. Teaching "reasoning" as a faculty is the modern form of formal discipline. The elements that actually transfer are specific habits, ideas, and associations — a mental standard

*tives on Psychological Science* 10, no. 2 (2015): 176–199.

[199]Louis Deslauriers, Logan S. McCarty, Kelly Miller, Kristina Callaghan, and Greg Kestin, "Measuring Actual Learning versus Feeling of Learning in Response to Being Actively Engaged in the Classroom," *Proceedings of the National Academy of Sciences* 116, no. 39 (2019): 19251–19257.

[200]Edward L. Thorndike and Robert S. Woodworth, "The Influence of Improvement in One Mental Function upon the Efficiency of Other Functions. (I)," *Psychological Review* 8 (1901): 247–261.

for area, a discount for a known constant error, a named statistical principle — not a mysterious strengthening of the mind.

Douglas Detterman, opening *Transfer on Trial* in 1993, put the prosecution case in one paragraph that later reviewers still quote. Most studies fail to find transfer. Studies that claim it do so by generous criteria that would not meet the classical definition — behavior repeated in a new situation. Nothing in the literature contradicts Thorndike's general conclusions: **transfer is rare, and its likelihood is directly related to the similarity between two situations**. He treats general transfer as an *epiphenomenon* of lower-level processes, not a fundamental mechanism of intelligence.[201] Not every contributor to that volume was as pessimistic. The point for this book is not that transfer never happens. It is that **the default prediction is failure**, and success has to be earned by design — shared elements, hints, comparison, varied practice — not assumed because the skill was named "critical thinking."

Default failure is an engineering prior, not a counsel of despair. If you are designing a course, a poster, or a life-book chapter, you should predict that a named skill will *not* appear in a new surface, a new room, a later month, or a hotter identity, unless you have built the identical elements and cued the high road. That prior changes what counts as evidence. A post-test in the same week, on items that resemble training, with the bias named in the stem, is not a transfer result. It is a near-performance result. Detterman's classical definition — behavior repeated in a new situation — is stricter than the industry's, and it is the definition this book will keep. Generous criteria are how transfer becomes an epiphenomenon with a press release.

The canonical demonstration that the gap is noticing, not application, is Mary Gick and Keith Holyoak's analogical-problem-solving work. Duncker's radiation problem — a tumor that must be destroyed by rays that, at full strength, would also destroy healthy tissue — is solved spontaneously by about **10 percent** of subjects. After reading an analogous fortress-and-army story, about **30 percent** produce the convergence solution *without* a hint. With a hint to use the story, rates rise to about **75 percent**; some later conditions reach the low 90s.[202] The gap is not application skill. It is **noticing**. Surface structure — tumors, rays — captures memory search. Deep structure — disperse force, converge at the target — is not explicit. Willingham uses this pattern as the model of why maxims fail. You can tell people to look for deep structure. You can even tell them which prior problem is similar. Most still cannot implement the strategy without the relevant knowledge. His band-marching word problem, isomorphic to a garden least-common-multiple problem solved minutes earlier, was seen as an analog by **19 percent**; a hint that "this is like the garden problem" raised solution only to **35 percent**. Most still could not implement.[203] Few-shot analogical transfer is mostly a noticing problem. One analog plus a hint can raise solution from about 10 percent to about 75 percent. The analog alone yields about 30 percent. Far transfer without cues is the exception.

[201] Douglas K. Detterman, "The Case for the Prosecution: Transfer as an Epiphenomenon," in *Transfer on Trial*, ed. Douglas K. Detterman and Robert J. Sternberg (Ablex, 1993), 1–24.

[202] Mary L. Gick and Keith J. Holyoak, "Analogical Problem Solving," *Cognitive Psychology* 12 (1980): 306–355; Gick and Holyoak, "Schema Induction and Analogical Transfer," *Cognitive Psychology* 15 (1983): 1–38. Spontaneous solution of Duncker's radiation problem about 10 percent; analog without hint about 30 percent; analog plus hint about 75 percent (hint condition in 1980; some later conditions reach the low 90s).

[203] Willingham, "Critical Thinking," 2007. Band-marching word problem isomorphic to a garden/least-common-multiple problem: 19 percent saw the analog; hint raised solution to 35 percent.

David Perkins and Gavriel Salomon distinguish two mechanisms, and only one of them is a plausible route to far. **Low-road** transfer is the automatic triggering of well-practiced routines by similar stimulus conditions; it is mostly near. **High-road** transfer is mindful abstraction and deliberate search for connections; it is the only plausible route to far. High-road transfer is effortful, time-consuming, and rare unless the learner is cued to abstract.[204] A critical-thinking book that never forces comparison of structurally same, surface-different cases is betting on a mechanism the literature says is weak. Low-road transfer will give you the second coin-flip problem in the same class. It will not give you the hiring conversation a year later. High-road transfer is the only plausible route to that conversation, and it is effortful, time-consuming, and rare unless the learner is cued to abstract — which is another way of saying that the maxim "look for deep structure" is a cue that most people cannot execute without the knowledge Gick and Holyoak already showed is missing. The pedagogical moral is not "tell them to look for deep structure." It is: force the comparison, vary the surface, cue the abstraction, and then test later, unaided, in a new context. That is design. It is not a maxim.

Richard Nisbett, Geoffrey Fong, Darrin Lehman, and Patricia Cheng's "Teaching Reasoning," often cited as the optimistic case, is Thorndikean on a careful reading. Transfer applies only insofar as there are common identical elements. Gains on one trained skill were not associated with gains on other skills. The law of large numbers, trained in one domain, shows some transfer to another domain — sports versus ability testing — that **fades over two weeks**.[205] Nathan Kuncel, summarizing this line, splits what people mean by critical thinking into two senses: field-specific expertise, which does not readily transfer — veterinary diagnosis is not restaurant management — and a finite set of specific reasoning skills, the gambler's fallacy, the law of large numbers, correlation versus causation. The second set is useful *for problems that share those elements*. It is not a general solvent.[206]

The "thinking gym" story is the popular form of formal discipline, and it shrinks once the controls are honest. Giovanni Sala and Fernand Gobet's 2016 meta-analysis of chess instruction found an overall **g = 0.338** across 24 studies and 40 effect sizes — mathematics **g = 0.382**, reading **g = 0.248** — with duration predicting larger effects. No study used an ideal design with both a do-nothing and an active control.[207] Their 2017 paper, pooling chess, music, and working-memory training, found that effect sizes **shrink toward zero once active controls are required**: music versus active control about **0.03 SD**; working-memory training versus active control about **0.05 SD**; chess had only one active-control study, about **0.10 SD**. They conclude that far transfer of this kind rarely occurs, in line with Thorndike.[208] Willingham reads the computational-thinking meta-analysis of Scherer,

[204] David N. Perkins and Gavriel Salomon, "Are Cognitive Skills Context-Bound?," *Educational Researcher* 18, no. 1 (1989): 16–25; Salomon and Perkins, "Rocky Roads to Transfer: Rethinking Mechanisms of a Neglected Phenomenon," *Educational Psychologist* 24, no. 2 (1989): 113–142. Low-road versus high-road.

[205] Richard E. Nisbett, Geoffrey T. Fong, Darrin R. Lehman, and Patricia W. Cheng, "Teaching Reasoning," *Science* 238 (1987): 625–631. Transfer only insofar as identical elements; law-of-large-numbers transfer fades over two weeks.

[206] Nathan R. Kuncel, "Measurement and Meaning of Critical Thinking" (working paper discussed in Huber and Kuncel 2016). Two senses of critical thinking; little discriminant validity of common tests against general cognitive ability.

[207] Giovanni Sala and Fernand Gobet, "Do the Benefits of Chess Instruction Transfer to Academic and Cognitive Skills? A Meta-Analysis," *Educational Research Review* 18 (2016): 46–57. Overall g = 0.338 (24 studies, 40 effect sizes); mathematics g = 0.382; reading g = 0.248. No study used an ideal do-nothing *and* active-control design.

[208] Sala and Gobet, "Does Far Transfer Exist? Negative Evidence from Chess, Music, and Working Memory Train-

Siddiq, and Viveros the same way: overall **g = 0.47**, considerably smaller with active controls, and no benefit to literacy — consistent with conceptual overlap, not mental muscle.[209] Chess chunks do not exist in algebra. Chase and Simon showed that chess masters reconstruct game positions after a five-second glance far better than novices, **and lose the advantage on random, illegal boards**. The skill is chunked, domain-specific pattern knowledge, not a general visual memory.[210] That is why chess-as-critical-thinking-training fails far transfer. The chunks do not travel.

That is the same architecture in other expertises, and it is why "thinking" is a bad metaphor for what experts do. Chi, Feltovich, and Glaser had experts and novices sort physics problems. Experts grouped by the **physics principle** needed for solution — conservation of energy, Newton's second law. Novices grouped by **literal surface features** — inclines, pulleys, springs. Representation, not raw computational power, was the difference. Completion of the representation depended on the knowledge associated with the category.[211] This is the same surface/deep split Willingham uses, now in a domain where "critical thinking" is obviously not general: it is the possession of the right problem categories. Wine expertise is the same architecture in a perceptual-conceptual hybrid. Hughson and Boakes, "The knowing nose," showed that experts recall more wine-related words than novices **only when the words form possible varietal descriptions**, not when they are shuffled into meaningless combinations — the chess-random-board analog.[212] Science thinking is not wine thinking is not chess thinking. Controlling variables, sourcing a document, and evaluating a forced-move in a closed Sicilian are different recognitions attached to different knowledge. A critical-thinking book that uses chess as a metaphor for "thinking" without this caveat is teaching the wrong lesson.

Expertise is encapsulated. Neurologists do not diagnose cardiac cases well. Technical writers of appliance pamphlets do not thereby write newspaper articles. Professional philosophers remain susceptible to order and wording effects in everyday moral judgment.[213] The knowledge that makes Type 1 accurate in a benign, practiced environment does not travel to an unfamiliar one. That is Thorndike in modern dress: identical elements, not mental muscle.

Willingham's 2019 program for New South Wales is the positive design this constraint implies, and it is four steps, not a slogan: list the *specific* critical-thinking skills for each subject; identify the

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ing," *Current Directions in Psychological Science* 26, no. 6 (2017): 515–520. Active-control music about 0.03 SD; working-memory training about 0.05 SD; chess one active-control study about 0.10 SD.

[209] Ronny Scherer, Fazilat Siddiq, and Bárbara Sánchez Viveros, "The Cognitive Benefits of Learning Computer Programming: A Meta-Analysis of Transfer Effects," *Journal of Educational Psychology* 111, no. 5 (2019): 764–792, as read by Willingham, "How to Teach Critical Thinking," 2019: overall g = 0.47, smaller with active controls, no benefit to literacy.

[210] William G. Chase and Herbert A. Simon, "Perception in Chess," *Cognitive Psychology* 4, no. 1 (1973): 55–81.

[211] Michelene T. H. Chi, Paul J. Feltovich, and Robert Glaser, "Categorization and Representation of Physics Problems by Experts and Novices," *Cognitive Science* 5 (1981): 121–152.

[212] A. L. Hughson and R. A. Boakes, "The Knowing Nose: The Role of Knowledge in Wine Expertise," *Food Quality and Preference* 13 (2002): 463–472. See also Gregg E. A. Solomon, "Psychology of Novice and Expert Wine Talk," *American Journal of Psychology* 103 (1990): 495–517.

[213] Remy M. J. P. Rikers, Henk G. Schmidt, and Henny P. A. Boshuizen, "On the Constraints of Encapsulated Knowledge: Clinical Case Representations by Medical Experts and Subexperts," *Cognition and Instruction* 20, no. 1 (2002): 27–45. Eric Schwitzgebel and Fiery Cushman, "Philosophers' Biased Judgments Persist despite Training, Expertise, and Reflection," *Cognition* 141 (2015): 127–137. Encapsulated-expertise citations as used in Willingham, "How to Teach Critical Thinking," 2019.

content those skills need; sequence knowledge and skills; plan multi-year revisiting. "Think like a historian" is not a goal until sourcing, corroboration, and contextualization are named. Those moves are **not** the ones a scientist needs.[214] Generic add-on programs: limited success; when transfer is tested, positive results have been absent or modest and quick to fade. The punchline belongs in this chapter because it is a transfer claim: we are not even sure the general skills exist, but we are quite sure there is no proven way to teach them directly. Chapter 5 will spend the teaching trials. The constraint is already in.

The implication is not despair. It is design. Generic critical-thinking courses can move scores. The next section puts the numbers on the page as transfer constraints, not as "courses work."

## What the tests move: Abrami, Huber and Kuncel

The evidence is not zero. It is **modest, task-similar, and poorly incremental over a normal education**. Philip Abrami and colleagues' 2008 Stage-1 meta-analysis in *Review of Educational Research* pooled 117 studies, 20,698 participants, and 161 effects. The average effect on critical-thinking skills and dispositions was **g+ = 0.341**, with a standard deviation of **0.610**, highly heterogeneous. Type of intervention and pedagogical grounding accounted for 32 percent of variance. Improvement, they wrote, "cannot be a matter of implicit expectation." Objectives have to be explicit.[215] That sentence is the load-bearing correction for every campus that stamps "critical thinking" on a syllabus and hopes.

The 2015 successor is the better-designed paper, and it is the number this book will keep using. Abrami, Bernard, Borokhovski, Waddington, Wade, and Persson pooled **341 effect sizes** — not 341 studies; the distinction matters — from quasi- or true experiments using standardized critical-thinking measures. Pre-experiments were dropped. The weighted random-effects mean was **g+ = 0.30**, heterogeneous, with a range of about −1 to +2.[216] Cohen's conventional bands put 0.20 small, 0.50 medium, 0.80 large. A 0.30 is real and small. It is the average over a junk-drawer of treatments. The interesting variance is in the moderators.

Dialogue, authentic or situated problems, and mentoring each help. The combination of all three, in 19 studies, reached **g = 0.57**. Content-specific critical-thinking outcomes were larger — **g = 0.57**, 97 effects — than disposition, **g = 0.23**, 25 effects. Subject-content learning sat at **g = 0.33**.[217] Those

[214] Willingham, "How to Teach Critical Thinking," 2019. Four-step program; "think like a historian" requires named moves; generic add-on programs limited; no proven way to teach general skills directly.

[215] Philip C. Abrami, Robert M. Bernard, Evgueni Borokhovski, Anne Wade, Michael A. Surkes, Rana Tamim, and Dai Zhang, "Instructional Interventions Affecting Critical Thinking Skills and Dispositions: A Stage 1 Meta-Analysis," *Review of Educational Research* 78, no. 4 (2008): 1102–1134. 117 studies, 20,698 participants, 161 effects; g+ = 0.341 (SD 0.610); type of intervention plus pedagogical grounding 32 percent of variance; improvement "cannot be a matter of implicit expectation."

[216] Philip C. Abrami, Robert M. Bernard, Evgueni Borokhovski, David I. Waddington, C. Anne Wade, and Tonje Persson, "Strategies for Teaching Students to Think Critically: A Meta-Analysis," *Review of Educational Research* 85, no. 2 (2015): 275–314. 341 *effect sizes* (not 341 studies) from quasi- or true experiments using standardized measures; weighted random-effects g+ = 0.30; heterogeneous, range about −1 to +2.

[217] Abrami et al., "Strategies," 2015. Dialogue, authentic/situated problems, and mentoring combined g = 0.57 (k = 19, as summarized by TUM); content-specific CT g = 0.57 (97 effects); disposition g = 0.23 (25 effects); subject-content learning g = 0.33 (140 effects). Outcome splits as summarized in the TUM Clearing House Unterricht short review of

are not interchangeable numbers. 0.30 is not 0.57. Content-specific thinking is not a disposition inventory. A book that mashes them has already overclaimed.

Ennis's four course types — general, infusion, immersion, mixed — did **not** differ significantly from one another in the 2015 analysis. Hitchcock's methods supplement reports the numerical order: mixed +0.38, infusion +0.29, general +0.26, immersion +0.23. The difference was not statistically significant.[218] That finding undercuts both "a standalone critical-thinking course is enough" and "only infusion works." Mixed was numerically largest. Immersion — critical thinking as a hoped-for by-product of rigorous content, without naming the principles — was smallest, as it was in 2008. Design mixed anyway, for transfer reasons this chapter already gave: identical elements have to be practiced in the places a life actually contains, and high-road abstraction has to be cued. Do not cite 2015 as proof that mixed won a horse race. It did not.

A college-only meta sits lower still. Niu, Behar-Horenstein, and Garvan, in 2013, pooled 40 effect sizes of instructional interventions in higher education and found a mean of **0.195** (SE 0.055), significant and heterogeneous, with discipline and treatment length — more than twelve weeks better — explaining some variance.[219] Pair it with Abrami rather than averaging it in. It is the same direction, a smaller college-specific estimate, and another reminder that 0.30 is not a law of nature.

Christopher Huber and Nathan Kuncel's 2016 *Review of Educational Research* paper is a different object. It is not an intervention meta. It is 71 reports of *gains over college*, not of short treatments. The mixed-design four-year estimate for critical-thinking *skills* is **0.59 SD** — roughly 50th to 72nd percentile. The purely longitudinal four-year estimate is **0.46 SD**. Disposition, on the California Critical Thinking Disposition Inventory, is **0.55 SD** over four years.[220] Two constraints matter more than the headline gain.

First, **curriculum-wide efforts to improve critical thinking did not necessarily produce incremental long-term gains** above a normal college experience. Second, nursing programs — which the National League for Nursing requires to include formal critical-thinking training — did **not** outperform other majors on domain-general critical-thinking tests. Huber and Kuncel treat this as evidence against pouring more time into generic critical thinking at the expense of reading, mathematics, and field-specific expertise.[221] Ordinary college already moves the generic tests about half a standard deviation in four years. That is not nothing. It is also not evidence that adding a first-

Abrami 2015. TUM calls the 341 effect sizes "studies"; the k counts sit in the same units as those 341 effects.

[218] David Hitchcock, "Critical Thinking > Educational Methods," *Stanford Encyclopedia of Philosophy*, methods supplement, https://plato.stanford.edu/entries/critical-thinking/methods.html, re-fetched 30 August 2026 (the Winter 2023 `/supplement.html` path 404s; this is the live Educational Methods page). Abrami et al. 2015 mean effects by Ennis type: mixed +0.38, infusion +0.29, general +0.26, immersion +0.23; difference not statistically significant. Hitchcock also confirms the 341 *effect sizes* and the range about −1 to +2.

[219] Lian Niu, Linda S. Behar-Horenstein, and Cyndi W. Garvan, "Do Instructional Interventions Influence College Students' Critical Thinking Skills? A Meta-Analysis," *Educational Research Review* 9 (2013): 114–128. 40 effect sizes; mean 0.195 (SE 0.055).

[220] Christopher R. Huber and Nathan R. Kuncel, "Does College Teach Critical Thinking? A Meta-Analysis," *Review of Educational Research* 86, no. 2 (2016): 431–468. Mixed-design four-year CT-skill gain 0.59 SD; purely longitudinal 0.46 SD; disposition (CCTDI) 0.55 SD over four years. 71 reports; not an intervention meta.

[221] Huber and Kuncel, "Does College Teach Critical Thinking?," 2016. Curriculum-wide efforts not necessarily incremental; nursing mandate did not outperform other majors on domain-general CT tests.

year "critical thinking" seminar, or stamping the words on every syllabus, adds a second half-sigma. A campus that adds the words and does nothing else has implemented immersion-by-committee. Abrami 2008 predicts that will not move the needle.

Gains also appear to have **shrunk across publication year**. Huber and Kuncel's model-predicted four-year gain is about **1.22 SD** for a 1963 study versus about **0.33** for 2011. They flag the pattern as multiply interpretable — measurement, student mix, curriculum — not as a clean indictment of "kids today."[222] Do not launder it into one.

Kuncel's own split, as summarized in that paper, is the transfer constraint in two clauses. Common critical-thinking tests show little discriminant validity against general cognitive ability, and little evidence that those tests predict grades or job performance *better than IQ*.[223] Field-specific expertise does not readily transfer. A finite set of specific reasoning skills is useful for problems that share those elements. The tests mostly capture something close to the second set, overlapping g, under instructions that tell you Type 2 is required. Incremental validity after controlling g is an open question this book plants here and spends in Chapter 6. It is not a finding that the tests measure a transferable life faculty.

How to use 0.30 in a book is therefore narrower than the industry wants. Cohen's 0.20 / 0.50 / 0.80 bands are a convention, not a law, but they keep a 0.30 from being sold as a transformation. It is real and small. It is the average over a junk-drawer of treatments. The interesting variance is in the moderators: explicitness, dialogue, authenticity, mentoring, intensity of practice. It is also an average over **standardized critical-thinking tests**, which is exactly the general-skill assumption the rest of this chapter doubts. Abrami shows you can move those tests. Abrami does not show that moving them is the educational goal, and does not show far transfer, job performance, or civic wisdom. Huber and Kuncel show that college is associated with about half a sigma on those tests over four years, that curriculum-wide campaigns are not a reliable increment, and that a mandated professional curriculum built around the tests did not beat other majors on the tests.

The opportunity cost is the part campus slogans omit. If generic-test gains from extra instruction sit around 0.20 on top of a college slope of about 0.50, what is crowded out — writing, quantitative reasoning, disciplinary cases — and which of those better predicts later judgment? Huber and Kuncel's advice is to stop pouring time into generic critical thinking at the expense of reading, mathematics, and field-specific expertise. That is a transfer conclusion, not a culture-war one. The gains that survive are mostly **near**: same test family, same semester or same college years, same academic frame. Far transfer has to be built — varied surface, same deep structure, high-road abstraction, later unaided tests in new contexts. It cannot be declared.

Dual process is the public language for "why reflection fails." Do not print a poster.

[222] Huber and Kuncel, "Does College Teach Critical Thinking?," 2016. Model-predicted four-year gain about 1.22 SD for 1963 versus about 0.33 for 2011; multiply interpretable (measurement, student mix, curriculum), not a clean "kids today" claim.

[223] Kuncel, "Measurement and Meaning of Critical Thinking," via Huber and Kuncel 2016. Little discriminant validity against g; little evidence common CT tests predict grades or job performance better than IQ.

## Dual process: doorway, then the machinery

Daniel Kahneman's *Thinking, Fast and Slow* (2011) is the public doorway, and it is a masterful essay, not a methods text.[224] It popularized a **default-interventionist** picture already in the heuristics-and-biases program — Tversky and Kahneman 1974; Kahneman and Frederick 2002 — and in Jonathan Evans's and Keith Stanovich's dual-process work: fast, autonomous "System 1" generates a default; slower, capacity-limited "System 2" may or may not intervene.[225] Using the book as a syllabus — System 1 equals bias, System 2 equals truth, always — is the "received view" Evans and Stanovich spent the 2010s dismantling. This chapter will not print that poster.

Evans and Stanovich, in *Perspectives on Psychological Science* in 2013, respond to the critics — Keren and Schul, Kruglanski and Gigerenzer, Osman — without defending a generic two-system package. Defining features, not the long cluster of correlates: **Type 1** is autonomous; it does not require working memory; it is multiple systems, TASS, the autonomous *set* of systems, not one System 1. **Type 2** requires working memory and supports **cognitive decoupling** and hypothetical thinking — simulating possibilities without confusing them with the real world.[226] Fallacies they name in the received view: Type 1 is always the source of bias and Type 2 of correct answers; Type 1 is contextualized and Type 2 abstract; speed is diagnostic of type. None of those is a defining feature. Type 2 processing can be biased — motivated search of models; belief bias under instruction. Type 1 processing can be accurate in **benign** environments: valid cues, practiced, no adversary exploiting the cue. Normativity is an organism-level concept. Subprocesses are efficient or not, not "rational." Modes of thought — Need for Cognition, actively open-minded thinking — are *continuous styles within Type 2*, not a second type. Their preferred architecture is **default-interventionist**, not the parallel-competitive picture associated with Sloman.[227]

Stanovich's later tripartite model splits Type 2 into an **algorithmic mind** — fluid intelligence, decoupling capacity — and a **reflective mind** — thinking dispositions: information gathering, calibration, seeking other views.[228] IQ tests measure the algorithmic mind under instructions that *tell you* Type 2 is required. Heuristics-and-biases tasks do not. That is why smart people can be irrational: the test never asked them to detect that the intuitive answer was inadequate. "Be smarter" is not a thinking curriculum. It is a measure of the wrong mind.

[224] Daniel Kahneman, *Thinking, Fast and Slow* (Farrar, Straus and Giroux, 2011). Public doorway, not a methods text.

[225] Amos Tversky and Daniel Kahneman, "Judgment under Uncertainty: Heuristics and Biases," *Science* 185 (1974): 1124–1131; Kahneman and Shane Frederick, "Representativeness Revisited: Attribute Substitution in Intuitive Judgment," in *Heuristics and Biases*, ed. Thomas Gilovich, Dale Griffin, and Daniel Kahneman (Cambridge University Press, 2002), 49–81.

[226] Jonathan St. B. T. Evans and Keith E. Stanovich, "Dual-Process Theories of Higher Cognition: Advancing the Debate," *Perspectives on Psychological Science* 8, no. 3 (2013): 223–241. Type 1: autonomous, does not require working memory, TASS. Type 2: requires working memory; supports cognitive decoupling.

[227] Evans and Stanovich, "Dual-Process Theories," 2013. Received-view fallacies; default-interventionist architecture, not Sloman's parallel-competitive picture. Critics addressed include Gideon Keren and Yaacov Schul (2009), Arie W. Kruglanski and Gerd Gigerenzer (2011), and Magda Osman (2004).

[228] Keith E. Stanovich, *What Intelligence Tests Miss: The Psychology of Rational Thought* (Yale University Press, 2009); Stanovich, *Rationality and the Reflective Mind* (Oxford University Press, 2011); Stanovich, "The Comprehensive Assessment of Rational Thinking," *Educational Psychologist* 51, no. 1 (2016): 23–34. Algorithmic mind versus reflective mind.

The two rationalities are the split a life book needs, and they are not IQ. Stanovich, West, and Toplak, in the Comprehensive Assessment of Rational Thinking and in *The Rationality Quotient*, distinguish **epistemic rationality** — how well beliefs map onto the world, what is true — from **instrumental rationality** — choosing actions that are the best means to one's goals, what to do.[229] Epistemic failures include incoherent probabilities, overconfidence, myside bias, contaminated mindware: superstition, conspiracy, anti-science attitudes. Instrumental failures include expected-utility inconsistency, framing, temporal discounting. The two are related — actions need true beliefs — but they are not the same, and IQ tests measure neither comprehensively. CART is a prototype, not a school-ready instrument; several subtests have limited dimensionality. The *conceptual* split is what this book keeps. A person can be instrumentally sharp, getting what they want, and epistemically sloppy, believing what their group needs — or the reverse. Instrumental rationality is relative to *the agent's* goals. A book that pretends goals are given, as exam keys are given, is training a laboratory creature.

The 2010s did not kill dual-process theory. They **upgraded Type 1**. Wim De Neys, and Gordon Pennycook with Jonathan Fugelsang and Derek Koehler, accumulate evidence that people often **detect conflict intuitively**. Reasoners show longer response times, lower confidence, and anterior-cingulate activation on conflict items *even when they give the biased answer*.[230] The implication: bias is frequently a failure to *override* a dominant heuristic intuition, not a failure to notice that something is wrong. Hybrid models, Dual Process Theory 2.0, posit **multiple simultaneous intuitions**, including "logical intuitions" cued by elementary probabilistic and logical knowledge. Relative strength of those intuitions determines whether conflict is registered and whether deliberation is called. Two-response paradigms — a fast initial answer under load, then a final answer — show that some correct answers are already present at the intuitive stage.[231] De Neys and Pennycook's "Logic, Fast and Slow" is the accessible statement of this revision.[232]

For a critical-thinking book this revision matters, and it is an **active debate**, not a settled replacement of default-interventionism. "Slow down and use System 2" is incomplete advice. Sometimes the problem is missing mindware: the logical intuition was never compiled. Sometimes it is a weak logical intuition losing to a strong stereotypical one. Sometimes Type 2 is recruited to **rationalize** the intuition. Wason and Evans, in 1975, already showed subjects explaining matching-bias selection-task choices as if they had reasoned. Type 2 can write the press release for Type 1.[233] Teaching people to "be more reflective" can produce better-sounding justifications of the same er-

[229] Stanovich, "Comprehensive Assessment of Rational Thinking," 2016; Keith E. Stanovich, Richard F. West, and Maggie E. Toplak, *The Rationality Quotient: Toward a Test of Rational Thinking* (MIT Press, 2016). Epistemic versus instrumental rationality. CART is a research prototype, not a school-ready instrument.

[230] Wim De Neys, "Bias and Conflict: A Case for Logical Intuitions," *Perspectives on Psychological Science* 7 (2012): 28–38; De Neys, ed., *Dual Process Theory 2.0* (Routledge, 2018); Gordon Pennycook, Jonathan A. Fugelsang, and Derek J. Koehler, "What Makes Us Think? A Three-Stage Dual-Process Model of Analytic Engagement," *Cognitive Psychology* 80 (2015): 34–72.

[231] Bence Bago and Wim De Neys, "Fast Logic?: Examining the Time Course Assumption of Dual Process Theory," *Cognition* 158 (2017): 90–109; Bago and De Neys, "The Smart System 1: Evidence for the Intuitive Nature of Correct Responding on the Bat-and-Ball Problem," *Thinking & Reasoning* (2019).

[232] Wim De Neys and Gordon Pennycook, "Logic, Fast and Slow: Advances in Dual-Process Theorizing," *Current Directions in Psychological Science* 28, no. 5 (2019): 503–509. Accessible statement of the DPT 2.0 revision. An active debate, not a settled replacement of default-interventionism.

[233] P. C. Wason and Jonathan St. B. T. Evans, "Dual Processes in Reasoning?," *Cognition* 3 (1975): 141–154.

ror. Whether DPT 2.0's logical intuitions can be trained as mindware, so conflict detection becomes the default rather than a rare gift, is open. This chapter will not fill that opening with a slogan.

Gigerenzer's ecological-rationality critique is a live dissent, not a fight this book must pick a winner for. Many "biases" are tools matched to environments, and some disappear when information is presented in natural frequencies.[234] Both sides are constraints. Environments can make heuristics smart. Modern hostile environments can make them expensive. Tversky and Kahneman's 1974 core — representativeness, availability, anchoring-and-adjustment — and the 1983 conjunction fallacy are real, replicable effects on well-specified tasks. They are not a complete theory of everyday error, and they are not all "System 1 bugs."[235] A book should present both the effects and the boundary conditions, not a zoo of named fallacies.

The poster is pedagogically harmful for a reason that is not pedantry. If Type 1 is always the villain, then practice that compiles accurate Type 1 recognitions — Chi's problem categories, Chase and Simon's chunks, Klein's recognition-primed first move in a valid environment — looks like a failure of "critical thinking." If Type 2 is always the hero, then motivated search, belief bias under instruction, and Kahan's identity-protective numeracy look like success, because they are slow, effortful, and articulate. If speed is diagnostic, then every expert who is fast is suspect and every novice who is slow is virtuous. Evans and Stanovich named those fallacies because they are the received view the public doorway invites. A life book that uses Kahneman as a doorway and then prints the poster has used the doorway to leave the building.

Use Kahneman as the public doorway. Teach Evans and Stanovich's types, Stanovich's two rationalities, and DPT 2.0's logical intuitions as the actual machinery. System 1 is not the villain. Type 2 is not the hero. Myside is the bias "be smarter" does not cure.

## Myside, identity, and the U.S. pattern that is not a human universal

Myside bias is the tendency to evaluate evidence, generate arguments, and test hypotheses in a manner biased toward one's own prior opinions and attitudes. Stanovich and Richard West, in 2007 and 2008, showed that this bias is often **independent of cognitive ability**.[236] That is a book-level constraint, not a personality footnote. "Be smarter" is not a myside cure. The algorithmic mind, the thing IQ tests measure, does not automatically recruit the reflective mind to seek the other side. Heuristics-and-biases tasks that *tell* you Type 2 is required are the wrong probe for a bias that lives in the decision to look.

[234] Gerd Gigerenzer, "On Narrow Norms and Vague Heuristics: A Reply to Kahneman and Tversky," *Psychological Review* 103 (1996): 592–596; Gigerenzer and Peter M. Todd, *Simple Heuristics That Make Us Smart* (Oxford University Press, 1999). Ecological-rationality critique; do not pick a winner against Kahneman/Tversky. Both are constraints.

[235] Tversky and Kahneman, "Judgment under Uncertainty," 1974; Tversky and Kahneman, "Extensional versus Intuitive Reasoning: The Conjunction Fallacy in Probability Judgment," *Psychological Review* 90 (1983): 293–315; Kahneman and Tversky, "Prospect Theory," *Econometrica* 47 (1979): 263–291.

[236] Keith E. Stanovich and Richard F. West, "Natural Myside Bias Is Independent of Cognitive Ability," *Thinking & Reasoning* 13 (2007): 225–247; Stanovich and West, "On the Failure of Intelligence to Predict Myside Bias and One-Sided Bias," *Thinking & Reasoning* 14 (2008): 129–167.

Ziva Kunda's 1990 *Psychological Bulletin* review is the mechanism: people can be motivated either to be accurate or to reach a particular conclusion. Directional goals bias the *selection* of beliefs and inferential rules, constrained by the need to construct a justification that would look reasonable to others.[237] Lord, Ross, and Lepper, in 1979, is the demonstration that mixed evidence does not moderate. Mixed scientific evidence on the deterrent effect of capital punishment made proponents and opponents **more polarized**. The same methods were judged adequate when they supported one's view and inadequate when they did not.[238] Raymond Nickerson's 1998 review treats confirmation bias as a family of phenomena, not a single bug.[239] Emily Pronin, Thomas Gilovich, and Lee Ross's bias blind spot is the metacognitive cousin: people see bias in others more readily than in themselves.[240] These are classics. They are not a zoo, and they are not all "System 1." Motivated reasoning, in the Kahan results that follow, is skilled Type 2 processing in service of identity.

Dan Kahan, Ellen Peters, Maggie Wittlin, Paul Slovic, Lisa Larrimore Ouellette, Donald Braman, and Gregory Mandel, in *Nature Climate Change* in 2012, is the finding a critical-thinking book cannot dodge. N = **1,540** U.S. adults. Science literacy and numeracy were **not** associated with greater concern about climate change. Literacy correlated $\mathbf{r = -0.05}$ ($p = 0.05$); numeracy $\mathbf{r = -0.09}$ ($p < 0.01$). The people highest in literacy and numeracy were the **most culturally polarized**: hierarchical individualists more dismissive, egalitarian communitarians more concerned.[241] The science-comprehension, or deficit, model failed. The cultural-cognition, identity-protective model fit: people use their reasoning capacity to form risk perceptions that express loyalty to affinity groups. "Learn more science" is not a depolarization program, at least not on this issue, in this sample, in this country.

Kahan 2013, in *Judgment and Decision Making*, sharpens the dual-process point. Conservatives and liberals did not differ on the Cognitive Reflection Test. **Higher CRT predicted more, not less, ideologically motivated cognition** on politicized facts.[242] Motivated reasoning here is not a System 1 glitch. It is skilled Type 2 processing in service of **identity-protective, or expressive, utility**. Conservatives did no worse than liberals on CRT. The reflective mind can be recruited to defend a badge. Westen, Blagov, Harenski, Kilts, and Hamann showed neural signatures of motivated reasoning in a 2004-election study; treat it as converging, not as the primary design constraint.[243]

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[237] Ziva Kunda, "The Case for Motivated Reasoning," *Psychological Bulletin* 108 (1990): 480–498.

[238] Charles G. Lord, Lee Ross, and Mark R. Lepper, "Biased Assimilation and Attitude Polarization," *Journal of Personality and Social Psychology* 37 (1979): 2098–2109.

[239] Raymond S. Nickerson, "Confirmation Bias: A Ubiquitous Phenomenon in Many Guises," *Review of General Psychology* 2 (1998): 175–220.

[240] Emily Pronin, Thomas Gilovich, and Lee Ross, "Objectivity in the Eye of the Beholder: Divergent Perceptions of Bias in Self versus Others," *Psychological Review* 111 (2004): 781–799.

[241] Dan M. Kahan, Ellen Peters, Maggie Wittlin, Paul Slovic, Lisa Larrimore Ouellette, Donald Braman, and Gregory Mandel, "The Polarizing Impact of Science Literacy and Numeracy on Perceived Climate Change Risks," *Nature Climate Change* 2 (2012): 732–735. N = 1,540 U.S. adults; literacy r = −0.05, numeracy r = −0.09 with climate concern; highest literacy = greatest cultural polarization.

[242] Dan M. Kahan, "Ideology, Motivated Reasoning, and Cognitive Reflection," *Judgment and Decision Making* 8 (2013): 407–424. Higher CRT predicted more, not less, ideologically motivated cognition; conservatives and liberals did not differ on CRT.

[243] Drew Westen, Pavel S. Blagov, Keith Harenski, Clint Kilts, and Stephan Hamann, "Neural Bases of Motivated Reasoning: An fMRI Study of Emotional Constraints on Partisan Political Judgment in the 2004 U.S. Presidential

**Do not universalize the 2012 U.S. pattern.** Henrik Y. L. Pröpper, Sandra Geiger, Tessa F. Blanken, and Cameron Brick, in the *Journal of Environmental Psychology* in 2022, tested cultural cognition across 23 countries — 21 European countries, Russia, and Israel — with N = **44,378**. Climate concern correlated only weakly with left identification (*r*s = 0.04–0.13), egalitarianism (*r*s = 0.04–0.13), and communitarianism (*r*s = 0.01–0.07). Cultural cognition was a weak predictor of climate beliefs (**$R^2$ = 3.82 percent**), policy preferences (**$R^2$ = 2.09 percent**), and actions (**$R^2$ = 0.62 percent**). Polarization was **not** greatest among the highly educated. Education was positively associated with climate beliefs (*r*s = 0.07–0.17) irrespective of political affiliation. The authors' conclusion: cultural cognition may not be central to climate attitudes in Europe.[244] The U.S. finding remains a constraint for U.S. identity-laden issues. It is not a species-level law. Abrami, Huber and Kuncel, Kahan 2012, and Morewedge — which this chapter is about to reach — are heavily WEIRD, U.S.-college, or U.S.-adult. Transfer of the *constraints* is plausible. Transfer of the effect sizes is not established.

The implication for a life of critical thinking is a design parameter, not a sermon. On identity-charged issues, "think harder" and "learn more science" can **widen** disagreement — at least in the U.S. samples Kahan used. Delay the tribal cue. Change who is seen as the messenger. Ask for private rather than public judgment. Pick tasks where status is not at stake, before asking people to update. A chapter that demonstrates base-rate neglect with cancer screening will transfer better than the same chapter with climate or guns — not because the logic differs, but because the identity load differs. Debiasing under identity load is an open question. Almost all successful training uses non-tribal content. Kahan's paradigm is the transfer test that has not been passed.

That is also why this chapter will not universalize a U.S. correlational pattern into a theory of the species. Pröpper, Geiger, Blanken, and Brick did not "disprove Kahan." They showed that cultural cognition, measured as they measured it, was a weak predictor of climate beliefs, policy preferences, and actions in 23 mostly European countries, and that polarization was not greatest among the highly educated. Education there tracked climate belief irrespective of political affiliation. A book that needs Kahan 2012 to be a human universal, in order to say that literacy can serve identity, does not need a human universal. It needs the U.S. finding as a constraint on U.S. identity-laden issues, and the weak replication as a constraint on overclaim. Both stay on the page.

Everyday argument skill is weaker than critical-thinking slogans imply. That is the next constraint, and it is not a claim that people cannot think.

## Kuhn: evidence, alternatives, explanation as argument

Deanna Kuhn interviewed people aged 14 to 69 on everyday causal questions: school failure, unemployment, criminal recidivism. Jonathan Baron, reviewing *The Skills of Argument* in *Informal*

Election," *Journal of Cognitive Neuroscience* 18 (2006): 1947–1958. Converging, not the primary design constraint.

[244] Henrik Y. L. Pröpper, Sandra Geiger, Tessa F. Blanken, and Cameron Brick, "Truth over Identity? Cultural Cognition Weakly Replicates across 23 Countries," *Journal of Environmental Psychology* 83 (2022): 101865. N = 44,378; 21 European countries, Russia, Israel. Cultural cognition $R^2$ = 3.82 percent for climate beliefs, 2.09 percent for policy preferences, 0.62 percent for actions. Do not universalize Kahan 2012. (Ondřej Kácha, Jáchym Vintr, and Cameron Brick 2022, *JEP* 101815, is a different paper — a latent-class segmentation, "Four Europes," N = 44,387 — not this CCT replication.)

*Logic*, reports the design from the text: **160 main subjects**, balanced by sex and by college versus noncollege — college-bound for adolescents — plus five teachers, five parole officers, and five philosophy graduate students. Topics included "What causes prisoners to return to crime after they're released?", school failure, and unemployment.[245]

The findings that should constrain a book, with the source of each percentage named, are not a portrait of stupidity. They are a generation deficit on unsupported everyday social issues.

Baron, quoting Kuhn: successful and unsuccessful answers appeared in all ages and both education levels; **about half** of answers in each category classified as successful; **age and sex had essentially no effect**; college responses were consistently more successful than noncollege, **on the order of 25 percent**. Domain familiarity — teachers, parole officers — had no effect. Philosophy students succeeded on all questions. Baron's educational moral: something about early education or family background of college-bound students moves these measures; little improvement in informal reasoning occurs in high school or college for most U.S. students, a conclusion he also ties to Perkins 1985.[246] Evaluativist epistemology — Kuhn's term, Baron quoting page 188 — is the stance that viewpoints can be compared and evaluated for relative adequacy even if certainty is impossible. **14 percent of the college group, 5 percent of the noncollege group.** The majority were absolutist, even in the college group.[247]

Petra Barchfeld and Beate Sodian, summarizing Kuhn 1991 as the generation-task baseline, report that only **16 percent** of Kuhn's subjects consistently generated **genuine evidence** for their own theories; about **30 percent** generated **pseudoevidence** — a descriptive instance or script that elaborates the theory without testing it; only about **one third** consistently generated alternative theories or a counterargument to their own or an alternative theory; a mature epistemological understanding appeared consistently in only **9 to 22 percent** of adult subjects. Performance covaried with **education**, not sex. Adolescents and older adults performed worse than young adults.[248]

Do not collapse that deficit with "people cannot think." Sarah Brem and Lance Rips showed that Kuhn may underestimate ability when no data are available. Prompting subjects to imagine the strongest supporting evidence one could provide yielded genuine-evidence production in **68 percent** — still far from ceiling.[249] Walter Sá, Caitlin Kelley, Caroline Ho, and Stanovich found that **74 percent** of adult participants were unable to create a covariation comparison in an argument.[250]

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[245] Deanna Kuhn, *The Skills of Argument* (Cambridge University Press, 1991). Design as reported by Jonathan Baron, review of *The Skills of Argument*, *Informal Logic* 14, no. 1 (1992): 59–67: 160 main subjects, balanced by sex and college versus noncollege, plus five teachers, five parole officers, and five philosophy graduate students.

[246] Baron, review of Kuhn, 1992, quoting the text: about half of answers classified as successful; age and sex essentially no effect; college more successful than noncollege on the order of 25 percent; domain familiarity no effect; philosophy students succeeded on all questions. Educational moral also citing David N. Perkins, "Postprimary Education Has Little Impact on Informal Reasoning," *Journal of Educational Psychology* 77 (1985): 562–571.

[247] Baron, review of Kuhn, 1992, quoting p. 188: evaluativist epistemology 14 percent of the college group, 5 percent of the noncollege group.

[248] Petra Barchfeld and Beate Sodian, "Differentiating Theories from Evidence: The Development of Argument Evaluation Abilities in Adolescence and Early Adulthood," *Informal Logic* 29, no. 4 (2009): 396–416. Summarizing Kuhn 1991: about 16 percent genuine evidence consistently; about 30 percent pseudoevidence; about one third alternatives/counterarguments; mature epistemology 9 to 22 percent of adult subjects.

[249] Sarah K. Brem and Lance J. Rips, "Explanation and Evidence in Informal Argument," *Cognitive Science* 24 (2000): 573–604. Prompting imagined strongest evidence raised genuine-evidence production to 68 percent.

[250] Walter Sá, Caitlin N. Kelley, Caroline Ho, and Keith E. Stanovich, "Thinking about Personal Theories: Individual

The deficit is real. The ceiling depends on task support. Barbara Koslowski criticized Kuhn's emphasis on covariation and noted that theory knowledge constrains what counts as evidence — another domain-knowledge point, not a refutation of the skill deficit.[251] Jonathan Osborne, Sibel Erduran, and Shirley Simon drew the pedagogical moral: **valid argument does not come naturally**; it has to be taught with task structure and modelling. "Just discuss the controversy" is not instruction.[252]

Explanation is not argument, and the conflation damages both research and classroom design. Osborne and Amanda Patterson, in *Science Education* in 2011, insist on the distinction. An **explanation** accounts for a phenomenon — why this happens; causal or mechanistic adequacy. An **argument** justifies a claim under uncertainty or disagreement — why this claim rather than that; evidence, warrants, rebuttals.[253] You can have a causal story with no evidence, and a well-evidenced claim that does not explain. For a life book the distinction is practical. "Explain why you believe X" elicits Kuhnian pseudoevidence, a story. "What would you need to see to abandon X, and what does the other side say?" elicits argument. Kuhn's 2010 "Teaching and learning science as argument" treats argumentation as a core epistemic practice of science, not a speech-and-debate add-on, which is compatible with Osborne and Patterson if the labels stay straight.[254] Glassner, Weinstock, and Neuman found that eighth-graders correctly generated explanations when asked to explain, but the majority also generated explanations, rather than evidence, when asked to *prove* a claim.[255] Ask for both, named separately: a mechanism, then the evidence that this mechanism rather than another.

Kruger and Dunning close the metacognitive loop. Bottom-quartile performers on humor, grammar, and logic tests — actual about the **12th percentile** — estimated themselves at about the **62nd**. The same incompetence that caused the errors blocked recognition of the errors. Improving the skill improved the monitoring.[256] Metacognitive theater — "rate your confidence" without teaching the skill — would have left the miscalibration in place. Reflection helps when it is tied to a standard, to feedback, and to knowledge the person can actually use. It is theater when it is **uninformed** — a journal prompt to "consider both sides" of a topic the student does not know produces fluent vacuity; **fluency-fooled** — immediate performance, complete answers, and polished prose raise judgments

Differences in the Coordination of Theory and Evidence," *Personality and Individual Differences* 38 (2005): 1149–1161. 74 percent of adult participants unable to create a covariation comparison. (Sometimes cited 2004/2005.)

[251] Barbara Koslowski, *Theory and Evidence: The Development of Scientific Reasoning* (MIT Press, 1996). Theory knowledge constrains what counts as evidence — a domain-knowledge point, not a refutation of Kuhn's skill deficit.

[252] Jonathan Osborne, Sibel Erduran, and Shirley Simon, "Enhancing the Quality of Argumentation in School Science," *Journal of Research in Science Teaching* 41, no. 10 (2004): 994–1020. Valid argument does not come naturally; "just discuss the controversy" is not instruction.

[253] Jonathan F. Osborne and Amanda Patterson, "Scientific Argument and Explanation: A Necessary Distinction?," *Science Education* 95 (2011): 627–638. See also Leema K. Berland and Brian J. Reiser, "Making Sense of Argumentation and Explanation," *Science Education* 93, no. 1 (2009): 26–55.

[254] Deanna Kuhn, "Teaching and Learning Science as Argument," *Science Education* 94 (2010): 810–824. Compatible with Osborne and Patterson if the labels stay straight. See also Kuhn, "Thinking as Argument," *Harvard Educational Review* 62, no. 2 (1992): 155–178.

[255] Amnon Glassner, Michael Weinstock, and Yair Neuman, "Pupils' Evaluation and Generation of Evidence and Explanation in Argumentation," *British Journal of Educational Psychology* 75 (2005): 105–118.

[256] Justin Kruger and David Dunning, "Unskilled and Unaware of It," *Journal of Personality and Social Psychology* 77 (1999): 1121–1134. Bottom-quartile performers actual about 12th percentile, estimated about 62nd; improving the skill improved the monitoring.

of learning and lower later unaided skill; **identity-serving** — more reflection can increase polarized use of evidence; **post-hoc** — Type 2 writes the press release; or **unscored** — reflection without a later unaided test, a counterargument requirement, or an external standard is cheap talk.

The book's reflection apparatus, when it has one, should be conditional and checked: What would falsify this? Name a person who disagrees and steel-man them. What knowledge would you need that you do not have? Then a later test without notes. Bare "what did you think about this chapter?" is theater. Whether later online media environments have changed Kuhn's 1991 base rates is not well measured on her tasks. That life-span question stays open.

## Reflection is not a substitute for knowledge

John Flavell's classical apparatus is still the right skeleton: monitoring of cognitive enterprises proceeds through interplay among metacognitive knowledge — persons, tasks, strategies — metacognitive experiences, goals, and actions. Young children often say they are ready to recall and are not; they miss blatant gaps in instructions. The developmental fact is the adult fact in weaker form: people mis-monitor.[257] Gregory Schraw and Rayne Dennison built the Metacognitive Awareness Inventory around two factors, knowledge of cognition and regulation of cognition. Schraw argued that metacognitive knowledge is multidimensional, relatively domain-general, and teachable, while still being used *with* domain knowledge.[258] Flag the domain-generality claim as **somewhat speculative**. It is the author's own theoretical preference, not a settled finding. Implementation still depends on domain knowledge. That last clause is the one critical-thinking programs drop.

Nelson and Narens split monitoring — judgments of learning, feeling of knowing, confidence — from control: what to study, when to stop, whether to answer. Asher Koriat showed that judgments of learning are often **experience-based**, fluency of encoding or retrieval, rather than **theory-based**. Fluency is a treacherous cue. Massed practice and complete answers both feel like knowing.[259] A model that finishes the sentence will raise the same cue. Offloading — the use of physical action to alter the information-processing requirements of a task so as to reduce cognitive demand — is old and often adaptive. The critical-thinking problem begins when the thing offloaded is the *evaluation* itself, and the metacognitive cue that evaluation has occurred is the fluency of the external product.[260] That constraint is Chapter 9's object. The monitoring error belongs here.

[257] John H. Flavell, "Metacognition and Cognitive Monitoring," *American Psychologist* 34, no. 10 (1979): 906–911. See also Flavell, Friedrichs, and Hoyt (1970) and Markman (1977) as cited there on children's mis-monitoring.

[258] Gregory Schraw and Rayne Sperling Dennison, "Assessing Metacognitive Awareness," *Contemporary Educational Psychology* 19 (1994): 460–475; Schraw, "Promoting General Metacognitive Awareness," *Instructional Science* 26 (1998): 113–125. Domain-generality of metacognitive knowledge is Schraw's theoretical preference, not a settled finding. Flag as such (Open question 23).

[259] Thomas O. Nelson and Louis Narens, "Metamemory: A Theoretical Framework and New Findings," *The Psychology of Learning and Motivation* 26 (1990): 125–173; Asher Koriat, "Monitoring One's Own Knowledge during Study: A Cue-Utilization Approach to Judgments of Learning," *Journal of Experimental Psychology: General* 126 (1997): 349–370. Experience-based versus theory-based JOLs; fluency as a treacherous cue. See also Alan S. Benjamin, Robert A. Bjork, and Bennett L. Schwartz, "The Mismeasure of Memory," *Journal of Experimental Psychology: General* 127 (1998): 55–68.

[260] Evan F. Risko and Sam J. Gilbert, "Cognitive Offloading," *Trends in Cognitive Sciences* 20, no. 9 (2016): 676–688. Offloading is metacognitively governed and often adaptive; the CT problem begins when evaluation itself is offloaded. Full life-condition in Chapter 9. Do not retell Bastani / Khanmigo / Kestin here.

Reflection helps when it is **tied to a standard, to feedback, and to knowledge the person can actually use**. Metacognitive strategies in Willingham's sense — "this might share a deep structure with a problem I know" — raise the odds of search; they do not complete the search. The band-problem hint raised solve rates from 19 percent to 35 percent. Still most failed. Bensley and Spero, infusing evaluation-of-evidence principles into a psychology course, improved evaluation of psychology evidence relative to a memory-principles control — near transfer, explicit practice, domain-matched.[261] Self-explanation and error-driven revision work when the learner can generate a relevant explanation; they stall when the knowledge is absent. Morewedge's after-action reviews, which this chapter is about to reach, are metacognition with teeth: you just committed the bias, here is the name, here is your score, try again. Thompson's "feeling of rightness," as used in Dual Process Theory 2.0, is the monitoring cue that can trigger Type 2 override — when the cue is calibrated.

Cognitive load is the design constraint when the "content" is a judgment. Intrinsic load is element interactivity among claims, evidence, sources, and values. Extraneous load is split attention, jargon dumps, and bias-of-the-week without a case. Germane load is schema construction — the Chi-style problem category. The testing effect shrinks or reverses for high element-interactivity material among novices.[262] A book that opens with a twelve-variable policy case and asks for unguided generation will overload, not train. Worked examples, then fading, then generation, is the right sequence for judgment too: show a fully worked evaluation, then a partially worked one, then an isomorphic case without the solution. Interleave cases that share deep structure and differ in surface — the Gick–Holyoak medicine/military pair, the garden/band pair. For this book, every chapter's exercises should be generation-first on a *new surface*, with feedback, not restudy of the chapter's own examples. Fluency of rereading is the enemy of later judgment. Desirable difficulties — spacing, interleaving, generation, retrieval — worsen immediate performance and can improve later retention and transfer.[263] The dissociation is the warning. Reciting a definition is weak retrieval. Generating an alternative hypothesis on a new case is the right grain.

Can you train the biases away? The reviews are honest.

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[261] D. Alan Bensley and Rachel A. Spero, "Improving Critical Thinking Skills and Metacognitive Monitoring through Direct Infusion," *Thinking Skills and Creativity* 12 (2014): 55–68, as cited in Willingham, "How to Teach Critical Thinking," 2019. Near transfer, explicit practice, domain-matched.

[262] Tamara van Gog and John Sweller, "Not New, but Nearly Forgotten: The Testing Effect Decreases or Even Disappears as the Complexity of Learning Materials Increases," *Educational Psychology Review* 27 (2015): 247–264; José Hanham, Wayne Leahy, and John Sweller, "Cognitive Load Theory, Element Interactivity, and the Testing and Reverse Testing Effects," *Applied Cognitive Psychology* 31, no. 3 (2017): 265–280. See also John Sweller, "Cognitive Load during Problem Solving," *Cognitive Science* 12, no. 2 (1988): 257–285; Sweller, Paul Ayres, and Slava Kalyuga, *Cognitive Load Theory* (Springer, 2011). Worked-example to fading sequence: Sweller and G. A. Cooper, "The Use of Worked Examples as a Substitute for Problem Solving in Learning Algebra," *Cognition and Instruction* 2 (1985): 59–89; Kalyuga, Ayres, Chandler, and Sweller, "The Expertise Reversal Effect," *Educational Psychologist* 38, no. 1 (2003): 23–31.

[263] Robert A. Bjork, "Memory and Metamemory Considerations in the Training of Human Beings," in *Metacognition*, ed. Janet Metcalfe and Arthur P. Shimamura (MIT Press, 1994), 185–205; Elizabeth L. Bjork and Robert A. Bjork, "Making Things Hard on Yourself, but in a Good Way," in *Psychology and the Real World*, ed. Morton Ann Gernsbacher et al. (Worth, 2011), 56–64; Robert A. Bjork and Elizabeth L. Bjork, "Desirable Difficulties in Theory and Practice," *Journal of Applied Research in Memory and Cognition* 9, no. 4 (2020): 475–479; Soderstrom and Bjork, "Learning versus Performance," 2015.

## Debiasing: consider-the-opposite, not a civic vaccine

Psychology has catalogued biases far more thoroughly than it has learned to correct them. Scott Lilienfeld, Rachel Ammirati, and Kristin Landfield's "Giving Debiasing Away," in *Perspectives on Psychological Science* in 2009, is the honest review. Their PsycINFO counts at the time: **1,211** hits for "cognitive bias(es)" versus **158** for "debias/debiasing."[264] The ratio is the finding. Techniques with some support against confirmation-related errors are few, specific, and mostly laboratory.

**Consider-the-opposite**, or consider-an-alternative, has the best classical support — Lord, Lepper, and Preston 1984; Hirt and Markman 1995; Anderson 1982 on inoculation and counter-explanation against theory perseverance.[265] **Perspective-taking** helps some stereotype measures (Galinsky and Moskowitz 2000).[266] **Accountability** and delayed decision making help in some professional-judgment settings (Tetlock and Kim 1987).[267] **Instruction** about a named bias is mixed. Hal Arkes, in 1981, called psychoeducation alone "absolutely worthless." Others find modest effects on belief-bias syllogisms.[268] One-shot psychoeducation is the weakest cell in Lilienfeld's table. A curriculum that names a new bias every Monday is a poor bet for reasons this chapter already gave. Naming a bias does not produce the knowledge needed to see it in a new surface. Willingham and Gick and Holyoak already said that. The bias-of-the-week lecture is the weak cell.

Barriers Lilienfeld and colleagues list are all still live. Bias blind spot: "I don't need this." Low personal stake. **Domain-specificity of critical thinking**, citing Willingham 2007. Possible **backfire**: Sanna, Schwarz, and Stocker, on hindsight, found that generating many alternatives *increased* certainty.[269] Hot affect and motivated reasoning, Kunda 1990, that purely cognitive methods will not reach. Individual differences: need for closure, dogmatism, integrative complexity. Their goal is modest: plant a seed of doubt, not convert fanatics. Critical-thinking transfer, they note, is weak.[270] That last clause is this chapter's transfer section, restated as a debiasing constraint.

[264] Scott O. Lilienfeld, Rachel Ammirati, and Kristin Landfield, "Giving Debiasing Away: Can Psychological Research on Correcting Cognitive Errors Promote Human Welfare?," *Perspectives on Psychological Science* 4, no. 4 (2009): 390–398. PsycINFO counts: 1,211 "cognitive bias(es)" versus 158 "debias/debiasing."

[265] Charles G. Lord, Mark R. Lepper, and Elizabeth Preston, "Considering the Opposite: A Corrective Strategy for Social Judgment," *Journal of Personality and Social Psychology* 47 (1984): 1231–1243; Edward R. Hirt and Keith D. Markman, "Multiple Explanation: A Consider-an-Alternative Strategy for Debiasing Judgments," *Journal of Personality and Social Psychology* 69 (1995): 1069–1086; Craig A. Anderson, "Inoculation and Counter-Explanation: Debiasing Techniques in the Perseverance of Social Theories," *Social Cognition* 1 (1982): 126–139. Best classical support against confirmation-related errors.

[266] Adam D. Galinsky and Gordon B. Moskowitz, "Perspective-Taking: Decreasing Stereotype Expression, Stereotype Accessibility, and In-Group Favoritism," *Journal of Personality and Social Psychology* 78 (2000): 708–724.

[267] Philip E. Tetlock and Jae Il Kim, "Accountability and Judgment Processes in a Personality Prediction Task," *Journal of Personality and Social Psychology* 52 (1987): 700–709.

[268] Hal R. Arkes, "Impediments to Accurate Clinical Judgment and Possible Ways to Minimize Their Impact," *Journal of Consulting and Clinical Psychology* 49 (1981): 323–330. Psychoeducation alone "absolutely worthless"; later work finds modest effects on some belief-bias syllogisms. Mixed cell.

[269] Lawrence J. Sanna, Norbert Schwarz, and Sarah L. Stocker, "When Debiasing Backfires," *Journal of Experimental Psychology: Learning, Memory, and Cognition* 28 (2002): 497–502. Generating many alternatives increased hindsight certainty.

[270] Lilienfeld, Ammirati, and Landfield, "Giving Debiasing Away," 2009. Barriers: bias blind spot, low personal stake, domain-specificity (citing Willingham 2007), possible backfire, hot affect and motivated reasoning, individual differences. Goal: plant a seed of doubt, not convert fanatics. CT transfer weak.

The experimental counterweight is real and must not be overread. Carey Morewedge, Haewon Yoon, Irene Scopelliti, Carl Symborski, James Korris, and Karim Kassam, in *Policy Insights from the Behavioral and Brain Sciences* in 2015, ran two longitudinal experiments, IARPA-funded, targeting biases relevant to intelligence analysis. A single session — a serious game of about 60 minutes, or an instructional video of about 30 minutes — produced **medium-to-large** reductions immediately that persisted **at least two to three months**. Games **≥ −31.94 percent** immediately and **≥ −23.57 percent** at two to three months; videos **≥ −18.60 percent** immediately and **≥ −19.20 percent** at follow-up. Experiment 1 overall *d*: game **1.68** immediate **/ 1.11** at eight weeks; video **0.69 / 0.66**. Games beat videos, consistent with feedback and practice. Effects showed some domain-generality *within the test family*: reduction on both trained and untrained facets of confirmation bias, and on problem formats not taught.[271]

**Do not overread this.** Outcomes were researcher-built bias scales, not job performance or civic behavior. The population was a paid convenience sample in Pittsburgh, not students in a critical-thinking course. The authors themselves recommend training *alongside* incentives, information presentation, and nudges — not instead of knowledge. Do not claim Morewedge-style games have been shown to reduce ideological extremism or far-life bias. Items resemble training. Lilienfeld asked the extremism question and left it open. Successful training is mostly non-tribal content. Kahan's paradigm remains the transfer test that has not been passed.

Katherine Milkman, Dolly Chugh, and Max Bazerman, in 2009, issued a sourced call to shift from cataloguing bias to testing improvement strategies: moving people from intuitive to deliberative processing *when it matters*, outside view, analogical reasoning, consider-the-opposite, and choice architecture so that automatic processes work in the decision maker's favor.[272] It is a research agenda, not a finding that "bias of the week" curricula work. Incentives and representations often beat training. Natural frequencies versus conditional probabilities, Gigerenzer. Outside view versus inside view, Kahneman and Lovallo 1993. Accountability, Tetlock. Huber and Kuncel's nursing result is the curricular version of the same point: a required critical-thinking program did not add long-run generic-test gains.

The implication is a small set, not a bestiary. Teach a small set of high-leverage, element-sharing principles — base rates, sample size, confound, alternative hypothesis, consider the opposite — **inside domains the reader will actually use**, with feedback. Morewedge's games worked because they **elicited the bias, gave personalized feedback, and practiced mitigating strategies**, closer to worked-example-plus-retrieval than to a lecture on a named fallacy. That is the same architecture Willingham and Perkins and Salomon already required: identical elements, high-road abstraction, later unaided tests. A "bias of the week" lecture is the weak cell because it supplies names without

[271] Carey K. Morewedge, Haewon Yoon, Irene Scopelliti, Carl W. Symborski, James H. Korris, and Karim S. Kassam, "Debiasing Decisions: Improved Decision Making with a Single Training Intervention," *Policy Insights from the Behavioral and Brain Sciences* 2, no. 1 (2015): 129–140. Games ≥ −31.94 percent immediate / ≥ −23.57 percent at 2–3 months; videos ≥ −18.60 percent / ≥ −19.20 percent. Experiment 1 d: game 1.68 / 1.11; video 0.69 / 0.66. Researcher-built bias scales; paid Pittsburgh convenience sample; not job performance, not civic behavior, not ideological extremism.

[272] Katherine L. Milkman, Dolly Chugh, and Max H. Bazerman, "How Can Decision Making Be Improved?," *Perspectives on Psychological Science* 4 (2009): 379–383. Research agenda, not a finding that bias-of-the-week curricula work. Outside view: Kahneman and Dan Lovallo, "Timid Choices and Bold Forecasts," *Management Science* 39 (1993): 17–31.

the knowledge, the comparison, or the later unaided check. Active-control trials of such curricula against knowledge-rich comparison arms are popular in schools and under-tested in the literature this book fetched. Until those trials exist, the honest prior is Lilienfeld's: psychoeducation alone is the weak cell, consider-the-opposite has the best classical support, and transfer is weak.

Do not confuse a moved lab-bias score with a civic vaccine. Morewedge moved researcher-built scales, in a paid sample, on items that resemble training, for months. That is not nothing. It is also not depolarization, not job performance, and not unaided spotting of myside in an identity-laden argument a year later. The Kahan transfer test has not been passed. A book that sells one-shot games as the cure for tribal epistemology has overread the paper the same way a book that sells chess as a thinking gym overreads Sala and Gobet before the active controls.

Teaching is the next chapter's object. The constraint is already in: explicit, dialogic, authentic, mentored, domain-embedded — and still near until you test far. Before that chapter, one more constraint this book cannot park until the clinic, the workplace, and the court. Life is not a lab.

## Life is not a laboratory

Laboratory critical-thinking and heuristics tasks typically provide a closed problem, sufficient information, a defined correct answer, no audience, no ticking clock that matters, and no cost to the self if one updates. Life typically provides the opposite: incomplete information, time pressure, other people, and values.

Gary Klein's naturalistic decision making, and the Lipshitz, Klein, Orasanu, and Salas stock-taking, study fireground commanders, military officers, and similar experts under time pressure, high stakes, ambiguous data, unstable goals, and organizational constraints. The recognition-primed decision model: experts often do **not** generate and compare option sets. They recognize a situation as a type, simulate the first plausible action, and modify or reject it.[273] That is not a failure of critical thinking. It is how expertise works when the environment will not wait for a matrix.

Kahneman and Klein, in 2009, wrote a joint paper whose subtitle is the finding: "Conditions for intuitive expertise: a failure to disagree." They agreed on the boundary. Intuition is trustworthy when the environment is **valid** — stable cues, feedback — and the person has **learned** those cues.[274] Many life domains fail validity. Geopolitics, some medicine, personnel. Tetlock's expert-political-judgment work, and the later superforecasting tournaments, are the demonstration that prestige is not calibration.[275] The open question this chapter plants, and Chapter 8 will spend,

[273] Gary Klein, "Recognition-Primed Decisions," in *Advances in Man-Machine Systems Research*, vol. 5, ed. William B. Rouse (JAI Press, 1989), 47–92; Klein, "Naturalistic Decision Making," *Human Factors* 50, no. 3 (2008): 456–460; Raanan Lipshitz, Gary Klein, Judith Orasanu, and Eduardo Salas, "Taking Stock of Naturalistic Decision Making," *Journal of Behavioral Decision Making* 14 (2001): 331–352.

[274] Daniel Kahneman and Gary Klein, "Conditions for Intuitive Expertise: A Failure to Disagree," *American Psychologist* 64 (2009): 515–526. Intuition trustworthy when the environment is valid (stable cues, feedback) and the person has learned those cues. Joint statement, not a victory lap. Open question 20 — can recognition-primed experts be trained to notice invalid environments without wrecking speed — is planted here and spent in Chapter 8.

[275] Philip E. Tetlock, *Expert Political Judgment* (Princeton University Press, 2005); Barbara Mellers, Lyle Ungar, Jonathan Baron, et al., "Psychological Strategies for Winning a Geopolitical Forecasting Tournament," *Psychological Science* 25 (2014): 1106–1115. Prestige is not calibration; superforecasting hygiene (base rates, update, outside view,

is whether recognition-primed experts can be trained to notice when their environment is invalid without wrecking the speed that makes them expert. This chapter will not fill that opening with a maxim.

The joint paper is a failure to disagree, not a victory lap for either program. Kahneman's heuristics-and-biases tradition documented systematic departures from normative models on well-specified tasks. Klein's naturalistic tradition documented how experts actually decide when the world will not wait. The agreed boundary is validity plus learning. A fireground with recurring cue-outcome pairs, and commanders who have seen those pairs with feedback, is a high-validity environment. A geopolitical forecast, a personnel decision with delayed and noisy outcomes, a diagnostic puzzle the physician has never seen, is often not. Intuition in the second set is not "System 1 being itself." It is Type 1 running in an environment that does not support it. The maxim "trust your gut" is as empty as "slow down and use System 2." Both depend on whether the cues are valid and whether they have been learned.

Time pressure is the default, not the exception, and it interacts with the dual-process machinery already on the table. Evans and Curtis-Holmes, and De Neys, show that speeding and load increase belief bias and heuristic answers.[276] A life of untimed reflection is not available in a fire, a clinic, a news cycle, or a family argument. High-road transfer needs time. Low-road transfer needs varied practice until the recognition is fast. A curriculum that only trains the first, then sends people into the second, has trained a laboratory creature. Superforecasting hygiene — base rates, update, outside view, scorekeeping — is closer to thinking for life than Wason-card performance, though the Wason card remains a useful Type-2 gym, because the tournament is scored in an open world against a base rate and a Brier score, not against a grader who already named the bias.

Other people are the default. Kuhn's argument skills are social: alternative theories and counter-arguments are generated more readily in dialogue than in solitary essays.[277] Abrami 2015 found dialogue among the few robust instructional ingredients. Life judgments are made in families, workplaces, parties, and feeds, where the audience is not a grader and the identity stake is not optional. Myside bias is largely independent of cognitive ability — another reason IQ-plus-reflection is not a life curriculum. Values are the default. Kahan's result is the extreme: when a fact is a badge, reasoning capacity is recruited to defend the badge. Instrumental rationality is relative to *the agent's* goals. A book that pretends goals are given, as exam keys are given, is training the same laboratory creature.

The implication is not that laboratory tasks are worthless. Exercises that want to be for life have to include missing data, a clock, a real audience, and at least some items where identity is touched — then, separately, items where it is not — so the book can *show* the Kahan effect rather than only describe it. Clinic, work, and court are Chapter 8's domains. The constraint belongs here: lab-normative answers trained in closed worlds will not automatically appear in open ones.

scorekeeping) is closer to thinking for life than Wason-card performance.

[276] Jonathan St. B. T. Evans and Jodie Curtis-Holmes, "Rapid Responding Increases Belief Bias," *Thinking & Reasoning* 11 (2005): 382–389; Wim De Neys, "Dual Processing in Reasoning: Two Systems but One Reasoner," *Psychological Science* 17 (2006): 428–433.

[277] Deanna Kuhn and Amanda Crowell, "Dialogic Argumentation as a Vehicle for Developing Young Adolescents' Thinking," *Psychological Science* 22, no. 4 (2011): 545–552.

## What this chapter has not shown

This chapter has not shown that critical thinking cannot be taught. Abrami's 0.30 and Huber and Kuncel's college gains say it can, *on the tests used*. A 0.30 is real and small. A four-year 0.46 to 0.59 is real and modest. Dialogue plus authentic problems plus mentoring, in 19 studies, reached 0.57. Content-specific thinking, 0.57, outran disposition, 0.23. Those are not interchangeable, and they are not far transfer, job performance, or civic wisdom. Kuncel's discriminant-validity warning still stands. Nursing's mandated curriculum did not beat other majors on generic tests. Sala and Gobet kill the chess-gym story once active controls are required. Abrami's 2008 standard deviation, 0.61, is the heterogeneity: g = 0.30 is not "critical-thinking courses work."

This chapter has not shown that a small set of generic skills exists, or that it does not. Willingham's 2019 judgment is the one this book will keep using: scientists are "united" that content knowledge is crucial and "somewhat divided" on whether critical thinking is a large set of specific skills or a small set of generic ones; the generic-skills framing is "not a fruitful way to conceptualise skills in education," because there is little theory to guide how to teach them; generic add-on programs have "limited success"; when transfer is tested, "positive results have been absent or modest and quick to fade." The punchline a book can quote: "We are not even sure the general skills exist, but we are quite sure there is no proven way to teach them directly."[278] That is a live debate, not a filled hole. Year-scale unaided far transfer of specific reasoning principles — the law of large numbers, confound, consider-the-opposite — into workplace and civic decisions, scored independently of critical-thinking tests, is not in the fetched literature. How general is general enough for a life skill remains Open question 2.

This chapter has not shown that Dual Process Theory 2.0 has replaced default-interventionism. It is an active debate. Logical intuitions as trainable mindware is Open question 19. This chapter has not picked a winner between Gigerenzer and Kahneman. Both are constraints.

This chapter has not shown that Morewedge-style games reduce ideological extremism. Items resemble training. Paid samples. Researcher-built scales. Debiasing under identity load is Open question 5. This chapter has not treated Kahan 2012 as a human universal. Pröpper, Geiger, Blanken, and Brick 2022 is a weak cross-national replication, and it stays on the page as a weak replication. This chapter has not collapsed Kuhn's generation deficit with "people cannot think." Brem and Rips raised genuine-evidence rates with prompts and imagined data. The deficit is real; the ceiling depends on task support. Life-span argument skill after 1991 is Open question 22.

This chapter has not printed a System 1 / System 2 poster and called it a theory chapter. Type 1 is autonomous. Type 2 is working-memory-demanding decoupling. Type 1 is not always wrong. Type 2 is not always right. Epistemic rationality is not instrumental rationality. Myside is ability-independent. "Be smarter" is not a myside cure.

The three distinctions this chapter opened on are the ones it can close on. A skill is not a bicycle: maxims cue; knowledge and practice implement. Near is not far: Abrami's 0.30, Huber and Kuncel's college slope, Morewedge's months-later bias scales, Gick and Holyoak's 75 percent with a

[278] Willingham, "How to Teach Critical Thinking," 2019. Scientists "united" that content knowledge is crucial and "somewhat divided" on generic versus specific skills; "We are not even sure the general skills exist, but we are quite sure there is no proven way to teach them directly."

hint, and Sala and Gobet's near-zero under active controls are not the same object, and a book that mashes them has already failed the honesty test Barnett and Ceci supplied. Performance is not later unaided judgment: a fluent recap, a named bias, a tidy paragraph, a CRT score used to defend a badge — none of those is the check. The check is whether the person can think when the prompt, the time, the other people, and the identity stake change. Life is not a lab. Klein's commanders do not generate option sets. Kahneman and Klein agreed on validity plus learning. That is the boundary, not a maxim.

Replication geography stays standing. Abrami, Huber and Kuncel, Kahan 2012, and Morewedge are heavily WEIRD, U.S.-college, or U.S.-adult. Transfer of the constraints is plausible. Transfer of the effect sizes is not established. Incremental validity of common tests after g is planted for Chapter 6. Kahneman and Klein's joint boundary waits for Chapter 8's domains, where incomplete information, time, other people, and values are the point rather than the constraint.

What remains, and what Chapter 5 has to spend, is the narrower empirical question: which teaching moves, on which tasks, with which controls, move which scores — and which of those scores are near. Explicit beats implicit. Dialogue plus authentic problems plus mentoring is a combination, not a course title. Domain-embedded practice is the beam. Naming is not teaching. The constraint chapter's job was to make those later claims expensive to overread. That job is done if the reader can no longer treat a poster, a 0.30, a System 2 sermon, and a life of judgment as the same object. What teaching actually moves is an empirical question with named trials. That is Chapter 5.

# Chapter 5

## What Teaching Actually Moves

Figure 6: A teacher's desk with books, a tied manuscript, pencils, and a chalkboard diagram in window light.

Improvement in students' critical thinking "cannot be a matter of implicit expectation." That is Philip Abrami and colleagues, writing in 2008 in the *Review of Educational Research*, after pooling 117 studies, 20,698 participants, and 161 effects.[279] The mean was g+ = 0.341, with a standard

[279] Philip C. Abrami, Robert M. Bernard, Evgueni Borokhovski, Anne Wade, Michael A. Surkes, Rana Tamim, and Dai Zhang, "Instructional Interventions Affecting Critical Thinking Skills and Dispositions: A Stage 1 Meta-Analysis,"

deviation of 0.610 and a 95 percent confidence interval from 0.31 to 0.37. Heterogeneity was not a footnote. It was the result: Q_T = 1,767.86, p < .001. Type of intervention plus pedagogical grounding together accounted for 32 percent of the variance in effects. Immersion — critical thinking as a hoped-for by-product of rigorous content — was the weakest cell. The sentence about implicit expectation is worth keeping almost verbatim, because it is the one finding in this literature that can actually design a school. Hoping that a general skill will leak from "good teaching" is a design. It is the weakest one they measured.

The popular story about teaching critical thinking is a different sentence: put the words on the syllabus, run a discussion, buy a stand-alone course, wait. This chapter is the empirical record of that wait. It is not a catalogue of programs that "work." It is a record of what moved, on which outcome, under which design, and what did not replicate when someone finally ran a trial large enough to matter. A moved test score is not better judgment. A named hour is not a treatment until someone specifies the practice, the dose, the feedback, and the outcome. Domain-embedded practice is a different claim from generic-skill transfer. Those three distinctions, already in the teaching record, will do more work here than any slogan.

Most of the experimental literature measures one of three things: a commercial multiple-choice inventory; a school attainment test that was never designed as a critical-thinking instrument; or a researcher-built task aligned to the intervention, which inflates effects. None of those, by itself, is evidence that a person will make a better decision in an unseen case in a new domain. Mission language is not a treatment either. First-year seminars, "critical thinking across the curriculum," Quality Enhancement Plans, and K–12 standards that *name* the phrase are not instructional interventions until someone specifies the practice. Huber and Kuncel found that curriculum-wide campaigns do not reliably add long-term increment over ordinary college. That finding is more important than any catalogue of campus slogans, and it arrives later in this chapter as a measured slope, not as a complaint.

The numbers in this chapter do not average. Abrami's 2015 mean on standardized critical-thinking tests is g+ = 0.30. The pedagogical combination of dialogue, authentic or anchored problems, and mentoring is g = 0.57 from 19 studies. Gorard's 2015 Philosophy for Children efficacy trial moved reading by +0.12. The 2021 effectiveness trial, larger and more secure, moved reading and maths by 0 months. Van Gelder's argument-mapping pool is a pre–post figure around 0.7, and it is not a randomized trial. The control-of-variables meta-analysis is g = 0.61. Reisman's document-based history curriculum is a quasi-experiment with a significant MANCOVA and no effect size this book will invent. Those are different objects. Heterogeneity is the story. Averaging them is how a book becomes a vendor slide.

Robert H. Ennis, in 1989 in *Educational Researcher*, gave the taxonomy the later metas used. Four approaches, not a slogan and a hope.[280] **General:** a stand-alone critical-thinking or informal-logic

*Review of Educational Research* 78, no. 4 (2008): 1102–1134. 117 studies, 20,698 participants, 161 effects; mean g+ = 0.341, SD 0.610, 95% CI [.31, .37]; Q_T = 1,767.86, p < .001. Type of CT intervention plus pedagogical grounding accounted for 32% of variance. Headline: improvement "cannot be a matter of implicit expectation." Educators must make CT objectives explicit and put them in pre-service and in-service training.

[280] Robert H. Ennis, "Critical Thinking and Subject Specificity: Clarification and Needed Research," *Educational Researcher* 18, no. 3 (1989): 4–10. Four approaches: general, infusion, immersion, mixed. Ennis, "Critical Thinking Across the Curriculum: The Wisdom CTAC Program," *Inquiry* 28, no. 2 (2013); Ennis, "Critical Thinking Across the

course, separate from subject matter; principles made explicit; content incidental. **Infusion:** deep subject-matter instruction in which students are encouraged to think critically *and* general principles are named. **Immersion:** the same depth and encouragement, without naming the principles. **Mixed:** a general thread or course plus infusion or immersion in the subjects. Ennis later argued, in 2013 and 2018, for a mixed undergraduate design — a separate strand plus subject-matter practice. That is a proposal. It is not a completed multi-institution randomized trial, and this chapter will not promote it to one.

Daniel Willingham's 2007 *American Educator* essay is the cognitive-science counterweight, and it is not optional here. Critical thinking is not a bicycle skill that transfers once learned. "You can teach students maxims about how they ought to think, but without background knowledge and practice, they probably will not be able to implement the advice they memorize."[281] Metacognitive prompts help *if* the student already has the domain knowledge the prompt requires. Ennis never denied that knowledge is necessary. John McPeck denied that a general skill is sufficient. The experimental literature has not resolved that debate. It has shown that implicit immersion is the worst of Ennis's four, and that generic-test gains from stand-alone courses are typically small. Willingham still vetoes a stand-alone general course as the engine of a life skill. Ennis's mixed proposal still earns a design paragraph, because transfer reasons — high-road abstraction, cued practice, the same moves used in history and lab and clinic — are not the same thing as a horse race among course titles. The 2015 meta did not run that horse race to a winner.

A campus that adds "critical thinking" to every syllabus and does nothing else has implemented immersion-by-committee. Abrami 2008 predicts that will not move the needle. A campus that buys a stand-alone course and never requires students to use the same moves in history, lab, clinic, or studio has implemented general-only, and should expect modest inventory-test gains with unknown transfer. The mixed design is the honest pedagogical target. It is also the expensive one, which is why mission language substitutes for it.

The coalition this book runs has to be visible on this page, because Chapter 5 is where the coalition's vetoes become expensive. Ennis gives the public sentence — reasonable reflective thinking focused on deciding what to believe or do — and a taxonomy of course architectures. Willingham gives the constraint: thinking processes are intertwined with the content of thought; domain-specific critical thinking can be taught; there is no proven way to teach a content-free general skill directly. Lipman gives childhood a pedagogy of judgment in a community of inquiry. Paul, used here only for the warning, reminds a writer that atomistic fallacy-spotting can make students more sophistic rather than less so. McPeck's *about what?* demand is a constraint, not a designer: this book rejects his ban on any general course and keeps the knowledge constraint. Where they veto each other in this chapter is specific. Willingham vetoes treating a stand-alone general course as the engine. Lipman does not veto dialogue; the 2021 trial vetoes using Philosophy for Children as proof that generic thinking raises Key Stage 2 scores. Ennis's mixed proposal is not licensed by Abrami 2015's non-

Curriculum: A Vision," *Topoi* 37 (2018): 165–184. The mixed undergraduate design is a proposal, not a completed multi-institution RCT.

[281] Daniel T. Willingham, "Critical Thinking: Why Is It So Hard to Teach?," *American Educator* 31, no. 2 (2007): 8–19. Authorized reprint at Reading Rockets, fetched 30 August 2026. 2019 restatement: domain-specific CT can be taught; there is no proven way to teach general skills directly. John E. McPeck, *Critical Thinking and Education* (Oxford: Martin Robertson, 1981): thinking is always about X.

significant type differences. The rest of the chapter is what survives those vetoes.

## Four approaches, and a combination that is not a horse race

The 2008 paper is Stage 1. Abrami, Bernard, Borokhovski, Wade, Surkes, Tamim, and Zhang did not separate or adjust by research-design quality, because little variation by design showed up at that stage. What did show up was the type of intervention and whether teachers had been grounded in it. Mixed was the largest cell in 2008. General and infusion were moderate. Immersion was weakest. The headline was not a ranking of course catalogues. It was a veto on hope.

The 2015 paper is the better-designed successor, and a book that treats it as 2008 with a bigger N will overclaim.[282] Abrami, Bernard, Borokhovski, Waddington, Wade, and Persson pooled **341 effect sizes** — not 341 studies — from quasi-experiments and true experiments that used standardized critical-thinking measures. Pre-experiments were dropped. The weighted random-effects mean was g+ = 0.30, $p < .001$, heterogeneous, with a range of about −1 to +2. That 0.30 is the average over a junk-drawer of treatments, scored on standardized tests that already assume a general skill. Cohen's conventional 0.2 / 0.5 / 0.8 makes 0.30 real and small. Short interventions can register. That is a psychometric smallness, not a reason to run a one-hour workshop and declare victory.

Stanford Encyclopedia of Philosophy, in David Hitchcock's methods supplement, reports the 2015 mean effects by Ennis type: mixed +0.38, infusion +0.29, general +0.26, immersion +0.23.[283] **The difference was not statistically significant.** A book that treats "mixed beats everything" as settled is citing 2008's pattern and ignoring 2015's test. Design mixed anyway, for the transfer reasons Ennis cares about and Willingham will not let a writer forget. Do not cite 2015 as proof that a stand-alone course plus infusion is reliably superior. Numerically largest is not a horse race won.

What *is* settled across both Abrami papers is more useful than a ranking. Making critical thinking an explicit objective beats hoping it emerges. Training the teachers matters. And the pedagogical combination with the largest reported effects is not a course title. It is dialogue plus authentic or anchored problems plus mentoring. Combined, in the 2015 analysis as summarized by the TUM Clearing House Unterricht short review, that combination is g = 0.57 from 19 studies.[284] Each element helps; the combination is larger. "We use discussion" is not that combination. A discussion without an authentic problem and without mentoring is talk. Talk is a later section of this chapter,

[282] Philip C. Abrami, Robert M. Bernard, Evgueni Borokhovski, David I. Waddington, C. Anne Wade, and Tonje Persson, "Strategies for Teaching Students to Think Critically: A Meta-Analysis," *Review of Educational Research* 85, no. 2 (2015): 275–314. 341 *effect sizes* from quasi- or true-experiments using standardized CT measures; pre-experiments dropped. Weighted random-effects g+ = 0.30, $p < .001$, heterogeneous, range about −1 to +2. Do not collapse 2008 and 2015.

[283] David Hitchcock, "Critical Thinking > Educational Methods," *Stanford Encyclopedia of Philosophy*, methods supplement, https://plato.stanford.edu/entries/critical-thinking/methods.html, re-fetched 30 August 2026 (the Winter 2023 `/supplement.html` path 404s; this is the live Educational Methods page). Abrami et al. (2015) mean effects by Ennis type: mixed +0.38, infusion +0.29, general +0.26, immersion +0.23. Difference not statistically significant. Hitchcock also confirms 341 effect sizes and range about −1 to +2.

[284] TUM Clearing House Unterricht, Short Review 18 of Abrami et al. 2015, fetched 30 August 2026. Pedagogical combination of dialogue, anchored/authentic problems, and mentoring: combined g = 0.57, 19 studies. Outcome splits: content-specific CT g = 0.57 (97 effects); disposition g = 0.23 (25); subject-content learning g = 0.33 (140). Generic CT is the 0.30. The TUM short review slips on "341 studies"; SEP and the RER abstract say effect sizes.

and it is not automatically thinking.

Outcome splits from the same short review of the 2015 paper should sit next to the mean, not inside it. Content-specific critical thinking: g = 0.57 from 97 effects. Disposition: g = 0.23 from 25. Subject-content learning: g = 0.33 from 140. Generic critical thinking is the 0.30. Those are not the same construct. A treatment that moves a content-specific measure of scientific reasoning is not, by that fact, a treatment that moves a Watson–Glaser. A treatment that moves a disposition inventory a little is not a life of judgment. Abrami shows you can move standardized critical-thinking tests. Abrami does not show that moving them is the educational goal. Chapter 6 will spend that sentence. This chapter has to keep it on the table, or the rest of the programs will be misread as engines of a faculty they were not built to prove.

Niu, Behar-Horenstein, and Garvan (2013), in *Educational Research Review*, is the college-only companion, not a third Abrami.[285] Forty effect sizes of instructional interventions in higher education; mean 0.195, standard error 0.055; significant, heterogeneous. Discipline and treatment length — more than twelve weeks better — explained some variance. Smaller than Abrami, as a college-restricted set should be allowed to be. Pair them. Do not average them with Gorard, and do not use 0.20 as if it were 0.57.

Ortiz (2007), an unpublished master's meta sometimes cited in this neighborhood, reported about 0.12 SD per semester for non-philosophy samples, with philosophy samples small (k = 6) and overlapping confidence intervals.[286] It is not a third Abrami. It is a reminder that the college slope, which Huber and Kuncel will measure in a later section, is already doing some of the work that stand-alone courses claim.

The brutal reading of Ennis's taxonomy, after both metas, is therefore not "pick mixed and you are done." It is this. Implicit expectation fails. Explicitness is the one slogan with a meta-analytic spine. The pedagogical combination that actually shows up is specified practice: dialogue, a problem that is real to the student, and mentoring. Course architecture — general, infusion, mixed — did not win a significance test in 2015. Willingham still vetoes the fantasy that a general course, by itself, produces a transferable life skill. Ennis's mixed design remains the honest target for anyone who wants both named principles and somewhere to use them. Honesty is not evidence. The next sections are the programs that have been tried.

One more caution before the named trials, because the 0.30 will be misread the moment it leaves this page. Abrami's mean is an average over standardized critical-thinking tests. That is exactly the general-skill assumption Chapter 4 installed and Chapter 6 will take apart. A small positive on those tests is consistent with "you can move the inventories a little if you make the objective explicit and if you use dialogue, authentic problems, and mentoring." It is not consistent with "critical thinking, the bicycle skill, has now been taught." Heterogeneity is large because the treatments are a junk drawer and because the tests are a family, not a single organ. Content-specific 0.57 and

[285] Lian Niu, Linda S. Behar-Horenstein, and Cyndi W. Garvan, "Do Instructional Interventions Influence College Students' Critical Thinking Skills? A Meta-Analysis," *Educational Research Review* 9 (2013): 114–128. 40 effect sizes; mean 0.195, SE 0.055; significant, heterogeneous. Discipline and treatment length (>12 weeks better) explained some variance.

[286] A. M. Ortiz, unpublished master's meta-analysis (2007), cited by Huber and Kuncel (2016) and by van Gelder (2015): about 0.12 SD per semester, non-philosophy; philosophy samples k = 6, overlapping CIs. Do not treat as a third Abrami.

generic 0.30 are adjacent rows, not a pooled truth. A writer who wants a bigger number can quote the combination cell or the content-specific split and forget to say k = 19, or forget to say which construct moved. This chapter will keep saying it.

## Philosophy for Children: two independent RCTs, different stories

Matthew Lipman founded the Institute for the Advancement of Philosophy for Children at Montclair in 1970. The original model used purpose-written philosophical novels and a classroom community of enquiry. UK SAPERE practice — the version actually trialled at scale — dropped the novels, kept the enquiry sequence, and used stories, images, and film as stimuli. Lipman himself taught a nine-week, n = 40 matched study in Montclair district schools, reported in Lipman, Sharp, and Oscanyan (1980) and in later reviews: gains on the California Test of Mental Maturity in logical reasoning and reading.[287] That is a developer-run, tiny, non-random study. It is origin, not evidence. This book keeps Lipman for childhood pedagogy — judgment, criteria, self-correction, context-sensitivity, a community of inquiry — and will not launder that pedagogy as an attainment intervention. The trials decide the attainment claim. Lipman's definition does not.

Trickey and Topping (2004), in *Research Papers in Education*, is a systematic review, not a randomized trial.[288] Ten controlled outcome studies met inclusion. Mean effect size 0.43, described as "consistent moderate" with low variance. Outcomes mixed: reading, reasoning, cognitive ability, self-esteem, child behaviour, questionnaires. Gorard, Siddiqui, and See, in the 2015 Education Endowment Foundation report, flag two problems a book must not bury. The studies were not commensurate — different tests, different durations, different ages. And the New Jersey Test of Reasoning Skills was built for Lipman and IAPC to measure reasoning taught in the P4C curriculum. Citing Moriyón and Tudela (2004), Gorard and colleagues note that NJTRS studies showed larger effects than literacy and numeracy tests.[289] Alignment bias is not a scandal. It is also not a generic-critical-thinking result. A test built for the curriculum will move when the curriculum is taught. That is what it is for.

Topping and Trickey (2007), in the *British Journal of Educational Psychology*, is the strongest small UK study, and it is still small.[290] Eight schools, eight classes, Dundee. 177 pupils (105 ex-

[287] Matthew Lipman, Ann Margaret Sharp, and Frederick S. Oscanyan, *Philosophy in the Classroom*, 2nd ed. (Philadelphia: Temple University Press, 1980). Nine-week, n = 40 matched study in Montclair district schools; gains on the California Test of Mental Maturity in logical reasoning and reading, as reported in later reviews. Developer-run, tiny, non-random. Origin, not evidence. Lipman, "Critical Thinking—What Can It Be?," *Educational Leadership* 46, no. 1 (1988): 38–43, for the judgment/criteria/self-correction/context definition.

[288] Steve Trickey and Keith J. Topping, "Philosophy for Children: A Systematic Review," *Research Papers in Education* 19, no. 3 (2004): 365–380. Ten controlled outcome studies; mean ES 0.43. Outcomes mixed. Not an RCT.

[289] Stephen Gorard, Nadia Siddiqui, and Beng Huat See, *Philosophy for Children: Evaluation Report and Executive Summary* (London: Education Endowment Foundation, 2015), ERIC ED581147, fetched 30 August 2026. On Trickey and Topping: studies not commensurate; NJTRS built for Lipman/IAPC; Moriyón and Tudela (2004) noted larger NJTRS effects than literacy/numeracy tests.

[290] Keith J. Topping and Steve Trickey, "Collaborative Philosophical Inquiry for Schoolchildren: Cognitive Gains at 2-Year Follow-up," *British Journal of Educational Psychology* 77 (2007): 787–796. Eight Dundee schools/classes; 177 pupils (105 experimental, 72 control); complete CAT data for 115; one hour per week for 16 months; ES = 0.7.

perimental, 72 control), matched and randomised at class level; complete Cognitive Abilities Test data for 115. One hour per week of collaborative enquiry for 16 months. Treatment pupils improved on the CAT; controls declined. Reported ES = 0.7. Two-year follow-up: the advantage was maintained for more able pupils, not for the lowest-achieving. Attrition was material. The outcome is a cognitive-abilities battery, not a critical-thinking inventory and not Key Stage 2 attainment. This book will not promote 0.7 on a complete n of 115 into a national claim.

Gorard, Siddiqui, and See (2015) is the real efficacy trial.[291] Independent evaluation for the Education Endowment Foundation. School-level randomized controlled trial. 48 English primary schools, volunteer, above-average free-school-meals eligibility. 22 treatment, 26 waitlist control. Teachers trained by SAPERE — a charity training package, which is to say a vendor in the only sense that matters here: the training is a product, not a finding. About one P4C period per week, January through December 2013. Impact sample: original Year 5 Key Stage 2, n = 1,529. CAT4 pre/post for Years 4–5, n = 2,821 of 3,159, about 11 percent pupil attrition on the CAT, no school dropout. The primary outcome was KS2 attainment, *not* a critical-thinking test.

Headline effects, Hedges' g on KS1-to-KS2 gain:

| Outcome | All pupils | FSM-eligible |
|---|---|---|
| Reading | +0.12 (~2 months) | +0.29 (~4 months) |
| Maths | +0.10 (~2 months) | +0.20 (~3 months) |
| Writing | +0.03 (~0 months) | +0.17 (~2 months) |
| CAT4 overall | +0.07 | −0.02 |

Almost all CAT gain was on the verbal subscale. Year 5 starters showed CAT +0.14; Year 4 starters about 0. Teacher and pupil reports of confidence, listening, and self-esteem were positive and were *not* the confirmatory outcome. Fidelity scores from SAPERE trainers did not correlate with attainment gains (Pearson r approximately 0 for CAT and reading). That negative is load-bearing. "Better implementation" as judged by the trainer did not predict bigger KS2 movement. Teacher glow is not an outcome. Trainer fidelity is not an outcome. KS2 is the outcome they registered.

Gorard, Siddiqui, and See also published the trial in the *Journal of Philosophy of Education* in 2017, same numbers.[292] Their own interpretation was modest, and a book that makes it immodest is not following them. For people who want short-term attainment, P4C is "promising, especially

Two-year follow-up: advantage maintained for more able pupils, not the lowest-achieving (their Table 3, p. 794, as quoted in Gorard et al. 2015). Attrition material. Outcome is CAT, not a CT inventory and not KS2.

[291] Gorard, Siddiqui, and See, *Philosophy for Children* (EEF, 2015). School-level RCT; 48 English primary schools, volunteer, above-average FSM; 22 treatment, 26 waitlist; SAPERE training (**vendor/charity package**); about one P4C period per week, January–December 2013. KS2 impact n = 1,529; CAT4 n = 2,821 of 3,159 (~11% pupil attrition; no school dropout). Primary outcome: KS2 attainment, not a CT test. Hedges' g on KS1 → KS2 gain as in the table in the text. Almost all CAT gain on the verbal subscale. Year 5 CAT +0.14; Year 4 ~0. Trainer fidelity Pearson r ≈ 0 with CAT and reading.

[292] Stephen Gorard, Nadia Siddiqui, and Beng Huat See, "Can 'Philosophy for Children' Improve Primary School Attainment?," *Journal of Philosophy of Education* 51, no. 1 (2017). Same numbers as the 2015 EEF report. Authors' modest interpretation quoted in the text: promising for the poorest if the goal is short-term attainment, "but perhaps not the most effective way forward"; does not damage attainment for those who value reasoning for its own sake.

for the poorest students, but perhaps not the most effective way forward.” For people who value reasoning for its own sake, the trial shows that using curriculum time this way **does not damage** attainment. That is an honest pair of sentences. It is not “P4C works.”

The 2021 Education Endowment Foundation effectiveness trial is real, and it did not replicate the attainment claim.[293] Independent evaluator: National Foundation for Educational Research. 198 schools (75 intervention, 123 control), 3,601 pupils. Whole-school SAPERE “Going for Gold” (Bronze to Gold) over two years. Primary outcome: KS2 reading for FSM pupils. Security rating: **5/5 padlock**. Result: **0 months** additional progress on FSM reading; **0 months** on reading and maths for the whole cohort. Pupil-survey single items on explaining ideas and working with different opinions: no significant improvement. 35 of 75 intervention schools were not implementing at the expected level after two years; six did not implement at all. Teacher surveys remained enthusiastic — 96 percent thought pupils improved respect for others’ opinions. EEF’s published conclusion: no plans for a further P4C trial on attainment; schools that want P4C for non-attainment reasons should not be discouraged, because the trial also showed **no harm** to reading and maths.

That is the rare “thinking” program with two large independent randomized trials. The 2015 efficacy trial moved reading and maths a little, more for FSM-eligible pupils, and barely moved CAT. The 2021 effectiveness trial, larger and more secure, moved neither. Implementation at scale was incomplete, which is itself a finding about what a whole-school enquiry program is in the wild, not a reason to throw out the 0. Teacher enthusiasm survived. The attainment effect did not. A CT-specific outcome was never the primary endpoint. Better measures of dialogic skill than two survey items still do not exist in the EEF reports. What P4C actually does when attainment is zero at scale is an open question this chapter will leave standing. Dialogic practice may still be worth the hour if the goal is the hour of dialogue. It is not a proven engine of generic critical thinking, and it was never measured that way at scale.

Colom and colleagues’ 2014 Madrid longitudinal study is sometimes cited as if it closed the question. Gorard and colleagues already treated the preliminary analysis as high-caution: two private schools, not randomised, relatively prosperous families.[294] It is not added here as a confirming RCT.

Lipman remains. Community of inquiry, judgment, criteria, self-correction, sensitivity to context: those are a childhood pedagogy, and they do not require KS2 months to be worth an hour. What they do not earn is a sentence that says the trials proved a general thinking skill, or that 2015’s two months on reading survived contact with 198 schools. Gorard showed a small efficacy effect. The effectiveness trial showed 0 months. Both paths are the record. A book that reports only the first has chosen a vendor.

[293] Education Endowment Foundation, Philosophy for Children effectiveness trial, completed November 2021; independent evaluator NFER. Project page fetched 30 August 2026. 198 schools (75 intervention, 123 control), 3,601 pupils; whole-school SAPERE “Going for Gold” over two years; primary outcome KS2 reading for FSM pupils; security **5/5 padlock**. **0 months** additional progress on FSM reading; **0 months** on reading and maths for the whole cohort. Pupil-survey items on explaining ideas and working with different opinions: no significant improvement. 35 of 75 intervention schools not implementing at expected level after two years; six did not implement at all. Teacher surveys: 96% thought pupils improved respect for others’ opinions. EEF: no plans for a further P4C attainment trial; no harm to reading and maths.

[294] Roberto Colom et al., Madrid longitudinal P4C study (2014), treated as high-caution by Gorard et al. (2015): two private schools, not randomised, relatively prosperous families. Not a confirming RCT.

The two trials also teach a measurement lesson this book will need again in Chapter 6. Efficacy asked about attainment, not about a critical-thinking inventory, and found small reading and maths movement, larger for FSM-eligible pupils, almost none on CAT, none that tracked trainer fidelity. Effectiveness asked the same attainment question at a scale and security rating that can actually inform a national conversation, and found 0 months, with incomplete implementation and surviving teacher glow. A writer who wanted to say "P4C teaches critical thinking" would have needed a critical-thinking outcome at scale. That outcome was never the primary endpoint. A writer who wanted to say "therefore dialogue is a waste" would have needed a trial that harmed reading or maths. It did not. The hour can still be justified as an hour of enquiry. It cannot be justified as a proven engine of a generic faculty, and it cannot be justified as the most effective available use of the hour if the goal is KS2 scores. Gorard already said the modest version in 2017. 2021 made the modest version cheaper to ignore and more important not to.

Childhood still needs a page. Willingham's three-year-olds can think critically in a domain they know. Lipman's community of inquiry is a way of practicing criteria, self-correction, and context-sensitivity with children, on stories and claims they can actually handle. That page does not get written by laundering EEF months. It gets written by refusing to treat a pedagogy as an attainment intervention, and by refusing to treat 0 months as a reason to stop talking with children about reasons. The veto is directional. Do not use P4C as proof that generic critical thinking raises test scores. Do not use 0 months as a reason to empty childhood of judgment.

## Argument mapping: a candidate method without a large RCT

Argument mapping is box-and-arrow depiction of claims, reasons, and objections. Textbook diagramming is old. Software-supported intensive practice is the modern claim. Reason!Able, in the late 1990s, was the University of Melbourne Reason! Project's tool. **Rationale** and **bCisive** were developed by Tim van Gelder and sold through Austhink, later purchased by Kritisch Denken BV. Both names are vendors. Classroom enthusiasm and product copy are not trials.

Van Gelder, in a 2015 chapter in Davies and Barnett's *Palgrave Handbook of Critical Thinking in Higher Education*, is explicit that **no large randomized trial** of argument mapping versus matched traditional critical-thinking instruction has been conducted.[295] The empirical base is pre–post testing, mostly in one-semester university critical-thinking subjects, often using the California Critical Thinking Skills Test. Van Gelder, Bissett, and Cumming (2004), in the *Canadian Journal of Experimental Psychology*, reported about 20 percent CCTST gains, converted to d ≈ 0.87, in Reason!-based subjects where 5 percent of course grade was assigned to best-of-two test sittings.[296] That 5

[295]Tim van Gelder, "Using Argument Mapping to Improve Critical Thinking Skills," in *The Palgrave Handbook of Critical Thinking in Higher Education*, ed. Martin Davies and Ronald Barnett (New York: Palgrave Macmillan, 2015). PDF fetched from reasoninglab.com, 30 August 2026. Explicit: **no large RCT** of AM versus matched traditional CT instruction. Pool of 26 pre–post studies, weighted ES around 0.7; 15 high-intensity studies ~0.85. Includes unpublished studies; not an independent *RER* meta-analysis. Reason!Able / **Rationale** / **bCisive** / Austhink / Kritisch Denken BV: **vendor**.

[296]Tim van Gelder, Melanie Bissett, and Geoff Cumming, "Cultivating Expertise in Informal Reasoning," *Canadian Journal of Experimental Psychology* 58, no. 2 (2004): 142–152. ~20% CCTST gains, d ≈ 0.87, in Reason!-based subjects; 5% of course grade assigned to best-of-two test sittings — a motivation control, not a comparison group. Project Report 2002, studies 3 and 4, is a project document, not a refereed multi-lab replication, on the written-test

percent is a motivation control, not a comparison group. Students who know the posttest is graded will try harder than students who do not. Chapter 6 will spend motivation as a design constraint on the whole gain literature. It already belongs here as a warning: a pre–post d of 0.87 with a grade incentive is not a treatment effect against a well-taught control.

Van Gelder's own in-progress pooling, as of that 2015 chapter: 26 pre–post studies of semester argument-mapping-based critical thinking, Australia, Europe, United States, a mix of published and unpublished. Weighted effect size around 0.7 overall; 0.85 for 15 high-intensity studies — mapping as the central activity, substantial homework, instructors proficient in the method. Published ingredients of that pool include Twardy (2004), Harrell (2011), Dwyer, Hogan, and Stewart (2011, 2012), and Butchart and colleagues (2009). He then subtracts Alvarez (2007) semester-at-college baseline of about 0.11 and Abrami (2008) college critical-thinking instruction of about 0.34, and infers mapping's "value add."

What that is not: an independent meta-analysis in the *Review of Educational Research*. It is a developer-adjacent, incomplete, pre–post pool. Pre–post without a concurrent control confounds mapping with taking any critical-thinking course, with instructor skill, with test motivation, and with teaching to the CCTST. Van Gelder argues that studies 3 and 4 of the Reason! Project Report 2002 cut against teaching-to-the-test, because gains appeared on a written test as well. That report is a project document, not a refereed multi-lab replication. A 2025 systematic review in *Higher Education Quarterly* found mixed controlled studies, most often using Rationale, and called for a proper meta-analysis.[297] That is the right next sentence, not "argument mapping produces 0.8σ."

The mechanism, stated modestly, is still worth having. Mapping externalizes argument structure. That is a plausible way to get deliberate practice on the part of critical thinking that *is* argument analysis. Diane Halpern's Critical Thinking Assessment has five subscales; van Gelder himself notes that mapping should load more on verbal reasoning and argument analysis than on probability or decision making. If a book claims mapping trains "critical thinking," it should say **which slice**. The slice is real. The solvent is not.

A semester randomized trial of high-intensity argument mapping versus a well-taught non-mapping informal-logic course, with both a CCTST *and* a far-transfer task, still has not been done. Van Gelder said so in 2015. The teaching record this book draws on did not find one since. That open question is not a smear on the method. It is the difference between a candidate and a proof. Treat mapping as a method for the argument-analysis slice, practiced with feedback, in a course that also has somewhere to use the slice. Do not treat 0.85 as an independent, published, RCT-based meta-analysis. It is not.

Intensity is the other confound, and van Gelder is frank about it. The 0.85 cell is high-intensity mapping: central activity, substantial homework, instructors who know the method. Compare that with a one-hour-a-week P4C period, or with a QEP that stamps a word, and the number is no

argument against teaching-to-the-test.

[297] Systematic review in *Higher Education Quarterly* (2025), abstract fetched 30 August 2026: mixed controlled studies of argument mapping, most often using Rationale; calls for a proper meta-analysis. Published ingredients of van Gelder's 2015 pool include Charles Twardy (2004); Maralee Harrell (2011); Christopher Dwyer, Michael Hogan, and Ian Stewart (2011, 2012); Sam Butchart et al. (2009). Alvarez (2007) semester-at-college baseline ~0.11, used by van Gelder as a subtraction, is not a concurrent AM control.

longer mysterious. Deliberate practice on a named slice, at a dose that is actually a dose, on a test that samples that slice, will move the test. The CCTST is not a far-transfer task. Subtracting Abrami's 0.34 and Alvarez's 0.11 from a pre–post pool is an inference, not a control group. The 2025 systematic review's call for a proper meta-analysis is the adult sentence. Until that meta-analysis exists, mapping is a candidate method with promising pre–post numbers, a vendor attached, and no gold-standard RCT. This book can recommend the slice. It cannot print 0.8 as a settled teaching effect.

## Domain-embedded: named moves, measured as themselves

This is the section that should discipline the rest of the book. Where researchers taught a *named disciplinary practice* and measured *that practice*, effects are often larger and cleaner than generic critical-thinking courses. Where they then claimed transfer to a Watson–Glaser, the claim usually thins out. The successful treatments share a shape: a named move, deliberate practice with feedback, and an outcome that is the move itself. Relabeling those moves "critical thinking" and then testing them with a decontextualized inventory is how the literature became noisy.

Start with science, because the logic of unconfounded tests can be taught, and the record says so without needing a general faculty.

Chen and Klahr (1999), in *Child Development*, n = 87, ages 7–10: explicit training plus probe questions beat probes without direct instruction on designing unconfounded experiments; probes-only did not.[298] Direct instruction also aided conceptual change in the domain, because unconfounded tests of the materials were now possible. Klahr and Nigam (2004) found direct-instruction control of variables outperformed discovery on near transfer. Kuhn (2005) and Hmelo-Silver, Duncan, and Chinn (2007) criticised the comparison as a straw "discovery." The productive reading is not "direct instruction wins forever." It is: **the logic of unconfounded tests can be taught explicitly, in a simplified domain, to elementary students, and it does not routinely appear from unguided inquiry.** Remote transfer in Chen and Klahr was limited and age-dependent; it improved with age. Near is not far. The named move is still the thing that moved.

Schwichow, Croker, Zimmerman, Höffler, and Hartig (2016) meta-analysed 72 intervention studies, 226 pairwise comparisons, and reported mean g = 0.61 (95 percent CI 0.53–0.69) after excluding outliers.[299] Moderators: cognitive conflict and demonstrations were related to larger effects. **Assessment format was a major source of between-study variance.** That last clause is the

[298] Zhe Chen and David Klahr, "All Other Things Being Equal: Acquisition and Transfer of the Control of Variables Strategy," *Child Development* 70, no. 5 (1999): 1098–1120. n = 87, ages 7–10. Explicit training plus probes beat probes-only; remote transfer limited and age-dependent. David Klahr and Milena Nigam, "The Equivalence of Learning Paths in Early Science Instruction," *Psychological Science* 15, no. 10 (2004): 661–667. Deanna Kuhn (2005) and Cindy E. Hmelo-Silver, Ravit Golan Duncan, and Clark A. Chinn (2007) criticised the "discovery" comparison; the productive reading is that unconfounded tests can be taught explicitly and do not routinely appear from unguided inquiry.

[299] Martin Schwichow, Christoph Croker, Corinne Zimmerman, Tim Höffler, and Hendrik Hartig, "Teaching the Control-of-Variables Strategy: A Meta-Analysis," *Developmental Review* 39 (2016): 37–85. 72 intervention studies, 226 pairwise comparisons, mean g = 0.61 (95% CI 0.53–0.69) after excluding outliers. Cognitive conflict and demonstrations related to larger effects; assessment format a major source of between-study variance. Scientific reasoning, not a general CT course.

critical-thinking-assessment problem in miniature: what you count as "having the skill" depends on how you ask. This is scientific reasoning, not a general critical-thinking course. Transfer to remote domains is limited. The 0.61 is a control-of-variables effect. It is not Abrami's 0.30, and it is not Gorard's +0.12. Do not mash them.

Argument-Driven Inquiry, as Sampson, Grooms, and Walker (2011) reported it in *Science Education*, is an exploratory 18-week, 15-lab, pre–post study of disciplinary engagement and written scientific arguments on a discrepant-event task.[300] Students improved; the authors report remaining learning problems. Subsequent chemistry-lab work (Sampson and Walker, 2012) is also exploratory, one section, community-college chemistry. Later small quasi-experiments, many from thesis literature, and some very large meta-analytic *d*s from regional journals, should not be promoted to clearinghouse-grade evidence. ADI is a lab-structure claim — investigation, argument, peer review, revision — not a generic-critical-thinking claim. It belongs in this section because the object of practice is named. It does not belong in a table of proven general-skill engines.

The IES What Works Clearinghouse products that sit near this science work evaluate science achievement, not generic critical-thinking tests. LASER — Smithsonian Science and Technology Concepts curriculum plus leadership professional development — has a WWC intervention report on science achievement. STeLLA has a May 2021 WWC primary-science report on teacher lesson analysis and student science achievement.[301] Useful for "inquiry can be a real treatment." Useless as a proxy for a Watson–Glaser. They will be named again only to keep the clearinghouse from being over-invoked. The negative search for named critical-thinking programs is a later paragraph.

History is the other existence proof, and it is a quasi-experiment, not a randomized trial.

Sam Wineburg distilled three historian heuristics: sourcing, contextualization, corroboration.[302] Those are named moves. They are not "think like a historian" as a poster. Avishag Reisman (2012), in *Cognition and Instruction*, reported "Reading Like a Historian: A Document-based History Curriculum Intervention in Urban High Schools." Quasi-experiment. 236 eleventh-graders. Five San Francisco high schools. Six-month document-based curriculum versus textbook instruction. Outcomes: historical thinking, transfer of those strategies to contemporary issues, factual knowledge, general reading comprehension. MANCOVA: significant main effects for treatment on **all four**.[303]

[300]Victor Sampson, Jonathon Grooms, and Joi Phelps Walker, "Argument-Driven Inquiry as a Way to Help Students Learn How to Participate in Scientific Argumentation and Craft Written Arguments: An Exploratory Study," *Science Education* 95, no. 2 (2011): 217–257. Exploratory 18-week, 15-lab, pre–post; remaining learning problems reported. Sampson and Walker, *International Journal of Science Education* (2012): also exploratory, one community-college chemistry section. Not WWC-grade evidence; not a generic-CT claim.

[301]IES What Works Clearinghouse intervention reports: LASER (Smithsonian STC curriculum + leadership PD) on **science achievement**; STeLLA, May 2021, primary science, teacher lesson analysis and student science achievement. Not generic CT tests. Named so WWC is not over-invoked as a CT clearinghouse.

[302]Samuel S. Wineburg, "Historical Problem Solving: A Study of the Cognitive Processes Used in the Evaluation of Documentary and Pictorial Evidence," *Journal of Educational Psychology* 83, no. 1 (1991): 73–87, and subsequent Stanford History Education Group / Digital Inquiry Group materials. Named moves: sourcing, contextualization, corroboration. Dissemination of SHEG/DIG materials is not a second trial.

[303]Avishag Reisman, "Reading Like a Historian: A Document-Based History Curriculum Intervention in Urban High Schools," *Cognition and Instruction* 30, no. 1 (2012): 86–112. Quasi-experiment, 236 eleventh-graders, five San Francisco high schools, six-month document-based curriculum versus textbook. MANCOVA significant on historical thinking, transfer to contemporary issues, factual knowledge, and general reading comprehension. **Not an RCT; not a generic-CT result.** Full PDF, Stanford stacks, HTTP 500 in the 30 August 2026 research pass; no effect

This is one of the better existence proofs that *disciplinary* reading can be taught in ordinary urban high schools, including to students below grade-level reading.

What it is not: an RCT; a generic-critical-thinking result; a demonstration that sourcing transfers to medical diagnosis or legal discovery without further training. The full PDF, from Stanford stacks, returned HTTP 500 in the research pass this book stands on. Claims are therefore limited to the abstract and to secondary descriptions. **No effect size is invented.** Significance on four outcomes is the finding. Magnitude is not a number this chapter will fabricate to make a table row look complete. SHEG and Digital Inquiry Group materials are widely downloaded; that is dissemination, not a second trial. American Institutes for Research is presently evaluating newer Reading-Like-a-Historian-plus-digital-literacy lessons under a U.S. Department of Education American History and Civics grant (S422B2300049); results were not a completed RCT as of 30 August 2026.

Mathematics is the honest negative in the same shape. Andreas Stylianides's program of design experiments treats proof as a school-mathematics practice — investigating whether and why "things work" — not as a module in a critical-thinking course.[304] A 2017 *Educational Studies in Mathematics* special issue includes **failed** experimental and quasi-experimental interventions: Fan and colleagues on auxiliary lines; Roy, Inglis, and Alcock on multimedia proof presentation. Well-motivated treatments that did not beat controls. That is the honest state of the art. There is no large randomized trial showing that "teaching proof" raises Watson–Glaser or CLA+ scores. Common Core Mathematical Practice 3 — "construct viable arguments and critique the reasoning of others" — names the practice. Naming is not teaching, and teaching is not yet a robust experimental literature on generic transfer. Failed experiments belong in the beam. They keep the beam from becoming a slogan.

Medical diagnosis is the most important negative result for generic critical thinking in a profession. Dual-process models — System 1 pattern recognition, System 2 analytic checking — are widely taught. Geoff Norman, Sandra Monteiro, Jonathan Sherbino, Jonathan Ilgen, Henk Schmidt, and Silvia Mamede (2017), in *Academic Medicine*, argue that diagnostic error is not primarily a story of uncorrected System-1 bias.[305] Errors arise in both systems. The binding constraint is **access to the relevant knowledge**, not a missing general "critical thinking" trait. Teaching "avoid bias" as a decontextualized skill is therefore the wrong intervention. Teaching is: many cases, including simulated and written cases, with feedback on whether the right diagnosis was reached. Assessment is: many cases, because a general inventory will not capture case-specific knowledge. This is domain-embedded judgment. It is also a direct critique of the general-skill assumption in CCTST

sizes invented; claims from abstract. AIR evaluation of newer RLH-plus-digital-literacy lessons, U.S. Department of Education American History and Civics grant S422B2300049: not a completed RCT as of the access date.

[304] Andreas J. Stylianides, "Proof and Proving in School Mathematics," *Journal for Research in Mathematics Education* 38, no. 3 (2007): 289–321; Stylianides and Stylianides, eds., special issue, *Educational Studies in Mathematics* (2017), including failed experimental/quasi-experimental interventions (Fan et al. on auxiliary lines; Roy, Inglis, and Alcock on multimedia proof presentation). No large RCT of "teaching proof" → Watson–Glaser or CLA+. Common Core Mathematical Practice 3 names the practice; naming is not teaching.

[305] Geoffrey R. Norman, Sandra D. Monteiro, Jonathan Sherbino, Jonathan S. Ilgen, Henk G. Schmidt, and Silvia Mamede, "The Causes of Errors in Clinical Reasoning: Cognitive Biases, Knowledge Deficits, and Dual Process Thinking," *Academic Medicine* 92, no. 1 (2017): 23–30. Diagnostic error is not primarily uncorrected System-1 bias; binding constraint is access to relevant knowledge, not a missing general CT trait. Teaching: many cases with feedback. Assessment: many cases, because of case specificity.

and Watson–Glaser use for clinical programs. Huber and Kuncel's nursing result, later in this chapter, is the outcome-level cousin: a profession that mandated generic critical thinking and measured it on generic tests did not beat ordinary college on those tests. Norman says why, from inside the clinic. Diagnosis is knowledge plus cases. It is not a trait you install in a first-year seminar.

Legal reasoning makes the same bet without a clean trial this book can cite. The case method and analogical reasoning are the profession's pedagogy. Comparative work stresses that legal reasoners construct a virtual fact pattern to fit or not fit a precedent — *accommodatio factorum* — which is not the same as medical explanation.[306] There is a large literature on whether law school improves generic critical-thinking tests. Watson–Glaser is used in some UK and US legal recruitment; that is a hiring screen, a vendor product, not a learning outcome. This chapter did not locate a clean, large randomized trial of "legal reasoning course → generic critical-thinking gain → better later briefs." What can be said without invention: legal education's bet is apprenticeship on cases, not a standalone critical-thinking module, and that bet is at least consistent with Norman and Willingham.

The payload of this section is a design rule, not a mean. Teach a named move. Practice it with feedback. Measure the move. Control of variables, sourcing a document, constructing a proof, generating a differential, analogizing to a precedent: those are objects of practice. "Critical thinking," as a course title or a test name, is not. When the move transfers, the transfer will be similarity-bound and often age-dependent, and most studies never look six months later in another domain. That open question is Chapter 4's transfer constraint applied to teaching. It is not a reason to stop teaching the move.

Notice what the successful rows share, and what they refuse. Chen and Klahr named the move — control of variables — taught it explicitly, and measured unconfounded tests. Schwichow's 0.61 is a mean over that family of interventions, and even there assessment format was a major source of variance. Reisman named sourcing, contextualization, and corroboration, taught them in a document-based curriculum, and measured historical thinking, a transfer task to contemporary issues, facts, and reading. The full paper was not recovered in this research pass; the abstract is enough to say the MANCOVA was significant on all four and not enough to print a d. Norman named the binding constraint as knowledge access and refused the missing-trait story. Stylianides named proof and then published failures. That pattern is the load-bearing beam: a named move, practice with feedback, an outcome that is the move, and a willingness to report when the move does not move. Generic inventories sit off to the side. They can be used. They cannot be allowed to rename the object.

The temptation, after a paragraph like that, is to declare that "critical thinking is just subject teaching" and sit down. McPeck would be pleased; Ennis would not, and neither would this book. Cross-field principles exist — correlation is not causation; sources have interests; unconfounded tests beat confounded ones; a fluent answer can be wrong. The question is whether those principles become usable without the knowledge that makes them bite, and whether they transfer unprompted a year later into a domain that was not the training domain. Chapter 4 already answered: transfer is rare and similarity-bound; noticing is the gap. This chapter's contribution is narrower. When you

[306] Geoffrey Samuel, comparative legal-reasoning work (e.g. *Legal Studies*, 2015): *accommodatio factorum* — constructing a virtual fact pattern to fit or not fit a precedent — is not medical explanation. Watson–Glaser in some UK/US legal recruitment: hiring screen, **vendor**, not a learning outcome. No clean large RCT of legal-reasoning course → generic CT gain → better later briefs was located.

teach and measure the principle *in a domain*, you often get a cleaner effect than when you teach "critical thinking" and measure a commercial inventory. That is not a proof that general principles are empty. It is a proof that implicit expectation and decontextualized testing are the wrong way to find them.

## Classroom talk is not automatically thinking

"We do discussion" is the cheapest version of Abrami's dialogue cell, and it is the version a 2009 meta-analysis already chastened.

Murphy, Wilkinson, Soter, Hennessey, and Alexander, in the *Journal of Educational Psychology*, meta-analysed nine discussion approaches on teacher and student talk, text comprehension, and critical-thinking/reasoning outcomes.[307] Several approaches increased student talk and reduced teacher talk and improved text comprehension. **Few approaches increased literal or inferential comprehension and critical thinking/reasoning.** Effects were moderated by design, outcome measure, and student ability; most studies were grades 4–6. The author PDF returned HTTP 500 in the research pass. Claims here are limited to ERIC, the abstract, and later Quality Talk papers that cite, from this meta-analysis, Philosophy for Children ES ≈ 0.214 and Collaborative Reasoning ES ≈ 0.260 on argumentation outcomes.[308] Those are not large generic-critical-thinking effects. They are the right order of magnitude: talk can move *talk* and sometimes *written argument about the discussed text*. It does not automatically move a Cornell or a Watson–Glaser.

Accountable Talk is a trademarked design language, not a clearinghouse-endorsed critical-thinking intervention. Michaels, O'Connor, and Resnick (2008), in *Studies in Philosophy and Education*, give three accountabilities: to the community, to reasoning, to knowledge.[309] UNESCO IBE Educational Practices Series 29 (Resnick, Asterhan, and Clarke, 2018) restates eight principles. This is a long research-and-professional-development program at the University of Pittsburgh, not a single multi-state randomized trial against a critical-thinking inventory. The authors themselves describe challenges when discourse norms are not shared. Treat it as specified talk, with supporting small studies and a theoretical lineage. Do not treat "we do Accountable Talk" as Abrami's combination.

Quality Talk is teacher-facilitated small-group discussion plus mini-lessons on questioning and argumentation. Murphy, Greene, and colleagues have run year-long quasi-experiments in grade 4–5 language arts. A later quasi-experiment — treatment n = 133, comparison n = 155, two schools,

[307] P. Karen Murphy, Ian A. G. Wilkinson, Anna O. Soter, Maeghan N. Hennessey, and John F. Alexander, "Examining the Effects of Classroom Discussion on Students' Comprehension of Text: A Meta-Analysis," *Journal of Educational Psychology* 101, no. 3 (2009): 740–764. Nine discussion approaches. Several increased student talk and text comprehension; **few** increased literal or inferential comprehension *and* critical thinking/reasoning. Most studies grades 4–6. Author PDF HTTP 500 in the 30 August 2026 research pass; claims limited to ERIC/abstract and later Quality Talk citations.

[308] Later Quality Talk papers (Murphy and colleagues) cite, from the 2009 meta-analysis, P4C ES ≈ 0.214 and Collaborative Reasoning ES ≈ 0.260 on argumentation outcomes. Not large generic-CT effects.

[309] Sarah Michaels, Catherine O'Connor, and Lauren B. Resnick, "Deliberative Discourse Idealized and Realized: Accountable Talk in the Classroom and in Civic Life," *Studies in Philosophy and Education* 27 (2008): 283–297. Accountable Talk[SM]: accountability to community, reasoning, and knowledge. Lauren B. Resnick, Christa S. C. Asterhan, and Sherice N. Clarke, *Accountable Talk: Instructional Dialogue That Builds the Mind*, UNESCO IBE Educational Practices Series 29 (2018). Design language for talk, not a WWC-endorsed CT intervention.

district literacy program in both arms — found no difference on basic comprehension relative to an already-enriched comparison, and a significant advantage for Quality Talk on **written argumentation** from Time 2 to Time 3.[310] Davies and colleagues, in secondary classrooms in the *British Educational Research Journal*, reported d = 0.92 on critical-analytic writing in seven intervention classes versus one comparison — a large number from a small, non-randomized design. Promising for written argument about discussed text. Not a national RCT. Not a critical-thinking-inventory result.

Collaborative Reasoning, in the Anderson / Institute of Education Sciences line, has been funded as development and a pilot: pair-randomised classrooms, outcomes including arguments, counter-arguments, rebuttals, text evidence in essays, a standardized reading test, and a critical-thinking test.[311] That is the right outcome mix. This chapter treats it as a funded research program, not as a completed WWC intervention report.

The brutal reading is the one Murphy already wrote. More student talk is not more critical thinking. What moves written argument is a specified talk structure, taught discourse moves, text or data on the table, and an outcome that is the argument, not a vibe. Abrami's dialogue cell, when it is real, looks like that specification plus an authentic problem plus mentoring. A fishbowl with no text and no criterion is not the cell.

The talk literature is also a warning about slogans that sound like Abrami's combination. "Accountable Talk," "Quality Talk," "community of enquiry," "Socratic seminar": each of those names *can* be a specified structure with taught moves, text on the table, and an outcome that is the argument. Each of them can also be a poster. Murphy's 2009 result is what you get when the research looks at the outcome instead of the poster. Comprehension sometimes moves. Critical-thinking and reasoning outcomes, on the measures those studies used, mostly do not. Later Quality Talk work is more encouraging for written argumentation about the discussed text, in quasi-experiments, not for a Cornell score. If this book later asks schools to spend time on dialogue, it will mean Abrami's combination as a design — dialogue plus authentic problems plus mentoring — and Murphy as the veto on volume. More talk is the easy variable. It is not the construct.

## Higher-ed slogans, first-year seminars, and ordinary college

Faculty already believe the work is being done. DeAngelo and colleagues (2009), in the Higher Education Research Institute's survey, found that more than 99 percent of faculty say developing critical thinking is very important or essential.[312] Tsui (1998): 92 percent of students in a multi-

[310] Quality Talk quasi-experiment (Murphy, Greene, and colleagues): treatment n = 133, comparison n = 155, two schools, district literacy program in both arms; no difference on basic comprehension relative to an already-enriched comparison; significant advantage on written argumentation from Time 2 to Time 3. Not an RCT. Davies and colleagues, *British Educational Research Journal*: d = 0.92 on critical-analytic writing, seven intervention classes vs one comparison — large number, small non-randomized design.

[311] Institute of Education Sciences award, Collaborative Reasoning units ("Improving Comprehension and Writing Through Reasoned Argumentation"): pair-randomised classrooms; outcomes including arguments, counterarguments, rebuttals, text evidence, standardized reading, and a critical-thinking test. Funded research program, not a completed WWC intervention report for CR.

[312] Linda DeAngelo et al., Higher Education Research Institute faculty survey (2009): >99% of faculty say developing CT is very important or essential. Lisa Tsui, "A Review of Research on Critical Thinking," paper and related multi-

institution study believed they had made critical-thinking gains. Self-report is not a test. It is the slogan, internalized.

Ordinary college already moves generic critical-thinking tests about half a standard deviation in four years. Huber and Kuncel (2016), in the *Review of Educational Research*, is not an intervention meta. It is a synthesis of longitudinal and cross-sectional *gains during college* on generic tests.[313] Seventy-one studies; about 16,185 longitudinal and 9,392 cross-sectional participants for skills; much smaller for dispositions, CCTDI only. Model-predicted four-year skill gain: 0.59 SD in mixed designs, 0.46 longitudinal, 0.73 cross-sectional. Disposition: 0.55 over four years. Pascarella and Terenzini (2005) had about 0.50 in 1990s samples. Arum and Roksa's four-year CLA figure is 0.47 — a different instrument, a later chapter's problem, and not a third copy of the same 0.5. That slope is not nothing. It is also not evidence that adding a first-year "critical thinking" seminar, or stamping the words on every syllabus, adds a second half-sigma.

Curriculum-wide critical-thinking campaigns did not necessarily produce incremental long-term gains relative to ordinary college.[314] Nursing versus non-nursing: no significant difference, despite the National League for Nursing's historical critical-thinking requirement and heavy CCTST use. If the flagship professional curriculum built around a generic instrument cannot beat "ordinary college" on the instrument's own terms, the instrument-plus-curriculum package is not a miracle. Why nursing did not beat other majors is an open question this chapter plants and Chapter 6 will spend. Domain-specific measures might have moved. They were mostly not in Huber and Kuncel's inclusion set. The publication-year moderator is multiply interpretable and is not a clean "college got worse" causal claim: predicted four-year gain 1.22 for a 1963 study versus 0.33 for 2011, with large residual heterogeneity.

First-year seminars are the wrong citation for a critical-thinking claim. Permzadian and Credé (2016) quantitative review: first-year seminars average $\delta = 0.02$ on first-year GPA ($k = 89$, $N = 52{,}406$) and $\delta = 0.11$ on one-year retention ($k = 195$, $N = 169{,}666$).[315] Academic-type seminars — the ones that *claim* critical thinking, writing, oral communication — are a moderator of those tiny grade and retention effects. This literature does not establish critical-thinking-test gains. A book that cites the first-year seminar as a critical-thinking intervention is citing the wrong outcome.

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institution study (1998): 92% of students believed they had made CT gains. Self-report is not a test.

[313] Christopher R. Huber and Nathan R. Kuncel, "Does College Teach Critical Thinking? A Meta-Analysis," *Review of Educational Research* 86, no. 2 (2016): 431–468. 71 studies; ~16,185 longitudinal and 9,392 cross-sectional participants for skills. Model-predicted 4-year skill gain 0.59 SD (mixed designs), 0.46 longitudinal, 0.73 cross-sectional; disposition 0.55 over 4 years. Ernest T. Pascarella and Patrick T. Terenzini, *How College Affects Students*, vol. 2 (San Francisco: Jossey-Bass, 2005), ~0.50 in 1990s samples. Richard Arum and Josipa Roksa, *Academically Adrift* (Chicago: University of Chicago Press, 2011), 0.47 SD on CLA over four years — criterion contamination; spent in Chapter 6.

[314] Huber and Kuncel (2016): curriculum-wide CT efforts do not necessarily produce incremental long-term gains relative to ordinary college. Nursing vs non-nursing: no significant difference, despite NLN's CT requirement and heavy CCTST use. Publication-year moderator: predicted 4-year gain 1.22 for a 1963 study vs 0.33 for 2011; large residual heterogeneity; not a clean "college got worse" causal claim. Why nursing did not beat other majors: open; domain-specific measures might have moved; mostly not in the inclusion set.

[315] Vahe Permzadian and Marcus Credé, "Do First-Year Seminars Improve College Students' Academic Performance and Retention? A Meta-Analytic Review of the Longitudinal Evidence," *Review of Educational Research* / related quantitative review (2016): $\delta = 0.02$ on first-year GPA ($k = 89$, $N = 52{,}406$); $\delta = 0.11$ on one-year retention ($k = 195$, $N = 169{,}666$). Academic-type seminars moderate those tiny grade/retention effects. Wrong outcome for a CT-test claim.

What "critical thinking across the curriculum" looks like, as implemented, is often the word. Mark Weinstein (2013) on Montclair State, 1987 through the 1990s: a requirement to include critical thinking in all general-education courses produced the word on syllabi without a unifying theoretical basis. The approval committee looked for term papers, projects, group work, and dialogue — surface features — not curricular outcomes tied to critical thinking.[316] Eric Sheffield (2018) on Rochester Institute of Technology's Fram Chair in Applied Critical Thinking, 2012–2015: a cross-disciplinary advisory group would not accept the institution's approved definition and never reached consensus on another. He notes many U.S. Quality Enhancement Plans devoted to critical thinking, many written by or with Richard Paul and Linda Elder and the Foundation for Critical Thinking — a vendor — typically on a five-year clock he judges too short for institutional change.[317] Payette and Ross (2016) report widespread *acceptance* of the Paul–Elder framework. Acceptance is not a randomized trial.[318]

Stamping "critical thinking" on every syllabus is immersion-by-committee. Abrami 2008 already said implicit expectation fails. Huber and Kuncel measured the long-term version: the campaign does not reliably add increment over the college slope. The opportunity cost is not abstract. If generic-test gains from extra instruction sit around 0.20 — Niu's college mean; Abrami's 2015 0.30 as the broader average — on top of a college slope of about 0.5, the hour spent on a generic add-on is an hour not spent on writing, quantitative reasoning, and domain cases. Those are the things Willingham and Norman say make judgment possible. Which of them better predicts later judgment is an open question. It is a better question than whether the QEP used the words.

Nursing is the existence proof of a full-dress professional attempt. A mandated critical-thinking curriculum. Heavy use of the CCTST, because accreditation historically demanded a critical-thinking outcome. Huber and Kuncel: no significant difference from other majors on long-term generic-test gains. Two readings are available, and this chapter will not pick the cheap one. The cheap reading is that critical thinking cannot be taught, which is false on Abrami's 0.30 and false on the college slope. The better reading is that a generic inventory, used as the outcome of a professional curriculum, may be the wrong instrument for the judgment the profession actually needs — which is Norman's point from inside diagnosis, and Chapter 6's point about what the tests measure. Domain-specific measures might have moved. They were mostly not in the inclusion set. Until someone shows they did, nursing is not a miracle and not a proof of impossibility. It is a null on the generic tests, in the profession that tried hardest to move them.

Montclair and RIT are not random samples of American higher education. They are what implementation looks like when the word is required and the conception is not shared. Weinstein's approval committee looked for surface features. Sheffield's advisory group could not agree on

[316] Mark Weinstein on Montclair State (1987 through the 1990s), as reported in Hitchcock, SEP methods supplement, fetched 30 August 2026: CT required in all general-education courses produced the word on syllabi without a unifying theoretical basis; approval committee looked for term papers, projects, group work, and dialogue — surface features, not curricular outcomes tied to CT. Weinstein (2013).

[317] Eric C. Sheffield on the Fram Chair in Applied Critical Thinking at RIT (2018): 2012–2015 advisory group would not accept the institution's approved definition and never reached consensus on another. Many U.S. QEPs devoted to CT, many written by or with Paul and Elder (Foundation for Critical Thinking — **vendor**), typically on a five-year clock he judges too short.

[318] Paul Payette and Ed Ross, on widespread acceptance of the Paul–Elder framework (2016). Acceptance is not an RCT. Do not outsource this book's theory to criticalthinking.org.

the definition the institution had already approved. Paul–Elder acceptance, in Payette and Ross, is a vocabulary spreading. Vocabularies are not treatments. Five-year QEP clocks, in Sheffield's judgment, are too short for institutional change. Ordinary college, meanwhile, is already doing half a sigma in four years, by the boring means of reading, writing, quantitative work, and disciplinary practice. The increment the campaigns wanted is the thing Huber and Kuncel did not find. That is the higher-education version of implicit expectation, measured over years instead of over a semester.

## Standards distributed the practices; they did not create a course

K–12 standards writers learned, correctly, not to add a fifty-first-state "Critical Thinking" course. They distributed practices into mathematics, science, and social studies. That is the right architecture. It does not implement itself. Naming is not teaching.

Common Core State Standards for Mathematics, Standards for Mathematical Practice (2010), eight practices, of which the critical-thinking-relevant are especially: make sense of problems and persevere in solving them; reason abstractly and quantitatively; **construct viable arguments and critique the reasoning of others**; model with mathematics; attend to precision; look for and make use of structure.[319] Mathematical Practice 3 is the named argument practice. It is not a Watson–Glaser syllabus.

Common Core English Language Arts, including literacy in history, science, and technical subjects, speaks in capacities rather than "practices": students who comprehend as well as critique, who value evidence, who build strong content knowledge. The C3 Framework and the Next Generation Science Standards were written to align with this literacy layer.

NGSS science and engineering practices, from the National Research Council Framework (2012) and NGSS Lead States (2013), eight, of which the relevant are especially: asking questions and defining problems; planning and carrying out investigations; analyzing and interpreting data; constructing explanations and designing solutions; **engaging in argument from evidence**; obtaining, evaluating, and communicating information.[320] Practice 7 is not "generic debate." It is argument from **empirical** evidence about phenomena or designs. Equating it with Mathematical Practice 3 erases the difference between empirical and deductive argument. The NGSS documents warn against that conflation. A school that cites Practice 7 and then runs cookbook labs has the slogan, not the practice.

The C3 Framework (National Council for the Social Studies, 2013) is a four-dimension Inquiry Arc: developing questions and planning inquiries; applying disciplinary concepts and tools in civics, economics, geography, and history; **evaluating sources and using evidence**; communicating con-

[319] National Governors Association Center for Best Practices and Council of Chief State School Officers, *Common Core State Standards for Mathematics*, Standards for Mathematical Practice (2010). Practice 3: construct viable arguments and critique the reasoning of others. Naming is not teaching.

[320] National Research Council, *A Framework for K–12 Science Education* (Washington, DC: National Academies Press, 2012); NGSS Lead States, *Next Generation Science Standards* (2013). Practice 7: engaging in argument from evidence — empirical evidence about phenomena or designs, not generic debate. NGSS documents warn against conflating this with Mathematical Practice 3.

clusions and taking informed action.[321] C3's own overview states an objective to "build critical thinking, problem solving, and participatory skills to become engaged citizens." That is mission language. The operational content is the inquiry arc, Wineburg-adjacent in Dimension 3, not a Watson–Glaser syllabus. State uptake varies; Swan, Grant, and Lee (2023) surveyed state standards. Uptake is not outcome evidence.[322] Saye (2017) and others have argued that disciplined inquiry at this grain requires expertise and scaffolding novices do not have — a cognitive-load warning, not a randomized trial.

The evidence that *practices can be taught* sits in the domain-embedded section: control-of-variables lessons, document-based history, proof instruction, argument-driven labs, specified discussion. It does not sit in the standards PDF. Citing the standards as if they were an intervention confuses a framework with a treatment.

The IES What Works Clearinghouse, named once so it is not over-invoked: as of 30 August 2026, WWC has **no intervention report** for Philosophy for Children, argument mapping, Watson–Glaser coaching, CCTST-prep courses, Paul–Elder, Thinking Maps, Building Thinking Classrooms, or "critical thinking" as a named program.[323] Related WWC products that do exist — LASER, STeLLA — evaluate inquiry science and teacher professional development against *science achievement*, not generic critical-thinking tests. They are not evidence that a critical-thinking course works. The clearinghouse's empty shelf is a finding. It is not a license to treat the absence as a secret proof that nothing works. It is a license to stop citing WWC as if it had endorsed a named-CT curriculum.

## What to spend the hour on

A table, not an average. Different rows, different objects. Heterogeneity is the story.

| What was measured | Design | Result | What it is not |
|---|---|---|---|
| Standardized CT tests, mixed treatments | Abrami 2015, 341 effect sizes, experiments and QES | g+ = 0.30 | Far transfer, job performance, civic wisdom |
| Dialogue + authentic problems + mentoring | Abrami 2015 moderator | g = 0.57, 19 studies | "We use discussion" |

[321] National Council for the Social Studies, *The College, Career, and Civic Life (C3) Framework for Social Studies State Standards* (2013). Four-dimension Inquiry Arc. Dimension 3: evaluating sources and using evidence. Overview language includes "critical thinking"; operational content is the inquiry arc, Wineburg-adjacent in Dimension 3, not a Watson–Glaser syllabus.

[322] Kathy Swan, S. G. Grant, John Lee, et al., survey of state standards uptake (2023): uptake, not outcome evidence. John Saye (2017) and others: disciplined inquiry at this grain requires expertise and scaffolding novices do not have — cognitive-load warning, not an RCT.

[323] IES What Works Clearinghouse intervention-reports index, searched 30 August 2026, for Philosophy for Children, argument mapping, Watson–Glaser coaching, CCTST-prep, Paul–Elder, Thinking Maps, Building Thinking Classrooms, and "critical thinking" as a named program: **no matching intervention report**. LASER and STeLLA exist and evaluate science achievement, not generic CT tests. Negative search is the finding. Do not say WWC endorses a named CT curriculum.

| What was measured | Design | Result | What it is not |
|---|---|---|---|
| P4C, KS2 reading | Gorard et al. 2015, 48-school RCT | +0.12 (~2 months) | The 2021 result |
| P4C, KS2 reading and maths | EEF 2021, 198 schools, 5/5 padlock | **0 months** | Proof that dialogue is worthless |
| Argument mapping, mostly CCTST | Van Gelder 2015 pre–post pool | ~0.7 overall; ~0.85 high-intensity | An RCT; an independent *RER* meta |
| Control of variables | Schwichow et al. 2016, 72 studies | g = 0.61 (CI 0.53–0.69) | A general CT course |
| Reading Like a Historian | Reisman 2012, QES, n = 236 | MANCOVA significant on four outcomes; ES not invented | An RCT; a generic-CT result |
| College, four-year generic-CT slope | Huber and Kuncel 2016 | ~0.5–0.6 SD | Increment from CT campaigns |
| First-year seminars | Permzadian and Credé 2016 | GPA δ = 0.02; retention δ = 0.11 | A CT-test gain |

Do not average the rows. A writer who averages 0.30 with 0.57 with +0.12 with 0 months with 0.7 with 0.61 has produced a number that does not name an object.

The design rule this chapter can defend, without a fitted syllabus, is narrower than the popular story and more expensive. Domain-embedded named moves are the load-bearing beam: control of variables, sourcing and corroboration, diagnosis as knowledge plus cases, proof as a mathematics practice. Dialogue plus authentic problems plus mentoring is the pedagogical combination with the largest Abrami cell, k = 19, and it is a teaching design, not a course title. Generic add-ons, first-year seminars cited as critical-thinking interventions, and syllabus-stamping are the weak cells. Lipman remains for childhood judgment and community of inquiry, not for KS2 months. Ennis's mixed undergraduate design remains a proposal. Willingham's three-to-five-year revisiting is the time scale, not a workshop.[324] Explicit beats implicit. That is the one slogan with a spine.

The open questions this chapter will not fill with analogy: what Philosophy for Children actually does when attainment is zero at scale, with better dialogic measures than two survey items; argument mapping with a real control, CCTST *and* far transfer, still not done; why nursing did not beat other majors on generic tests, planted for Chapter 6; the opportunity cost of extra generic instruction versus writing, quantitative reasoning, and disciplinary cases; Ennis mixed as an untested multi-institution design. Those are not holes in a vendor pitch. They are the edge of the record.

Spend the hour, then, as if opportunity cost were real, because it is. An hour of generic add-on that moves an inventory 0.20 on top of a college slope of 0.5 is an hour not spent on the named move. In science that hour is unconfounded tests, with feedback, on materials the student actually has to

[324] Willingham, "How to Teach Critical Thinking," NSW Department of Education (2019): identify what CT means in each domain; identify the content those tasks need; sequence; plan 3–5 years of revisiting. Time scale, not a workshop. Abrami 2015 combination g = 0.57, k = 19, remains a teaching design, not a course title.

understand. In history it is sourcing the document, with a contemporary transfer task that is not a trick. In clinic it is cases, many of them, with the right diagnosis as the outcome. In mathematics it is proof as a practice, knowing that some well-motivated interventions fail. In childhood it is a community of inquiry that is not being asked to raise KS2. In a university it is Ennis mixed as a *design* — named principles plus somewhere to use them — without pretending Abrami 2015 proved the horse race, and without pretending a QEP that stamps the word has implemented the design. Willingham's three-to-five-year revisiting is the calendar. A workshop is not.

None of that is a fitted syllabus. A fitted syllabus would pretend the open questions were closed. This chapter's job was to say what teaching actually moved, on which measure, under which design, and what did not replicate. Explicit beats implicit. Domain-embedded is the beam. Naming is not teaching. The test is not yet the thing. Turn the page.

If teaching moves tests, the next question is whether the test is the thing. Abrami's 0.30 is a movement on standardized critical-thinking inventories. Huber and Kuncel's college slope is a movement on the same family. Neither is a life of judgment until someone says what the score was a score *of*, and what it was not. That is Chapter 6.

# Chapter 6

## The Test Is Not the Thing

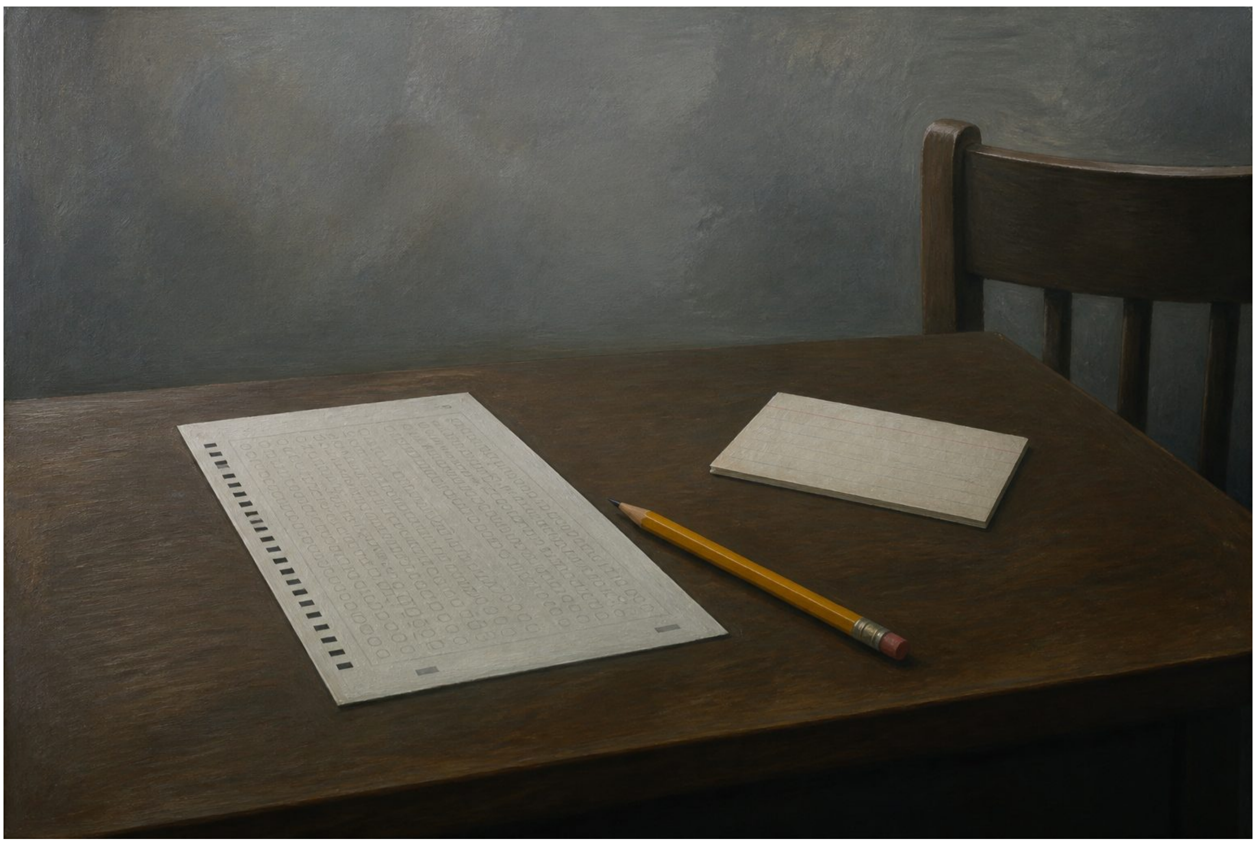

Figure 7: A scantron sheet, a yellow pencil, and a stack of index cards on a dark desk.

If this book refuses the folk collapse — a single, general, content-free skill, teachable in a standalone course, measurable by a short test, possessed as a personal trait — it owes an account of what would count as improvement in a life, without defining critical thinking as the test. Chapter 5 showed that teaching can move standardized inventories a little, that domain-embedded named moves move more cleanly when you measure the move, and that curriculum-wide campaigns do not reliably add increment over ordinary college. This chapter asks what those inventories were inventories *of*. The short answer is: not the thing. CCTST is not the meaning. A Watson–Glaser is often one factor. Independent CCTST subscale alphas have been reported as low as 0.21. CLA+ constructed-response reliability is 0.43. No instrument named HALO was found. Nathan Kuncel found little discriminant validity of common critical-thinking tests from general cognitive ability. Richard Arum and Josipa Roksa's 45 percent "no gain" is a CLA finding, partly motivation, and not a warrant for "critical thinking cannot be taught."

Liu, Frankel, and Roohr (2014), ETS Research Report RR-14-10, is the single best map of the commercial instruments, and a next-generation conversation that does not start there is improvising.[325] This chapter walks the map, then the cross-cutting threats that make "the campus went up four points" uninterpretable, then what judgment evidence looks like when the construct is deciding what to believe or do in an unseen case. Redesign — unseen cases, process evidence, oral defense — is language, not a fitted standard operating procedure and not a randomized trial of "oral defense raises CCTST." The Artificial Intelligence Assessment Scale is language in the same sense, and it is Mike Perkins, not David Perkins. A take-home essay is no longer, by itself, evidence of the student's judgment. That sentence is older than generative AI. AI made it unskippable.

Facione's 1990 Delphi report is expert consensus *for purposes of educational assessment and instruction*: 46 panelists, philosophy-heavy, McPeck absent, Siegel not listed, split 61/30 on whether dispositions are in the meaning, majority rejecting an ethical component in the meaning.[326] From that construct Facione built the California Critical Thinking Skills Test and the California Critical Thinking Disposition Inventory. Insight Assessment sells them. A book that treats a CCTST score as "what critical thinking is" has confused an instrument with a concept. Delphi did not discover a mental organ. It operationalized a consensus statement for testing. The rest of this chapter is what happens when schools, accreditors, and employers treat the operationalization as the thing.

The confusion is not a secret of psychometrics. It is the public meaning of the phrase, as Chapter 1 named it: a short test, a personal trait, a license. Accreditors asked nursing programs for a critical-thinking outcome and got CCTST sittings. Law firms asked for a reasoning screen and got Watson–Glaser. Campuses asked for "value added" and got CLA. Each of those choices is intelligible as an operationalization. None of them is Ennis's sentence. None of them is Willingham's constraint. None of them is Lipman's judgment in a community of inquiry. A book that wants to talk about a life has to say, on the page, that the score is a score of the items, under the conditions of the sitting, in the presence of *g*, motivation, coaching, and rater effects, and that "critical thinking" as a life is a different object. Chapter 5 moved the tests a little. This chapter refuses to let that movement rename the object.

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[325] Ou Lydia Liu, Lois Frankel, and Katrina Crotts Roohr, "Assessing Critical Thinking in Higher Education: Current State and Directions for Next-Generation Assessment," ETS Research Report RR-14-10 (Princeton, NJ: Educational Testing Service, 2014). ERIC PDF fetched 30 August 2026. The single best map of the commercial instruments. Independent psychometric summaries in the text (CCTST alphas, Watson–Glaser alphas, CLA+ CR reliability) are from this report unless a later note names another source.

[326] Peter A. Facione, *Critical Thinking: A Statement of Expert Consensus for Purposes of Educational Assessment and Instruction* (The Delphi Report), ERIC ED315423 (1990), fetched 30 August 2026. 46 panelists, six rounds (11 Feb 1988–25 Sept 1989); philosophy-heavy; McPeck absent; Siegel not listed. Quote Table 1; do not paraphrase. ~61% dispositions-in-the-meaning vs ~30% strict proceduralism; 52% reject ethics in the meaning. CCTST/CCTDI are one operationalization of one consensus statement; not "the APA's official definition."

# The map, and the general-skill assumption

Kuncel's 2011 review, as Huber and Kuncel restated it in 2016, is the constraint the map has to live under.[327] Common critical-thinking tests show little discriminant validity from general cognitive ability. There is little evidence that they predict grades or job performance *better than IQ*. Liu, Frankel, and Roohr asked for incremental predictive validity and noted that it is largely missing. Two senses of "critical thinking" are in play, and they are not the same claim. One is field-specific expertise that does not readily transfer. The other is a finite set of specific reasoning skills useful for problems that share those elements — large-numbers, correlation versus causation, unstated assumptions. If a test samples that thin set, gains mean "better at those items." McPeck's *about what?*, Willingham's knowledge constraint, and Norman's diagnosis-as-knowledge result are conceptual partners here, not decorations. The general-skill assumption is load-bearing for every vendor table of contents. It is also the assumption the rest of this book has been refusing.

That does not make the tests useless. It makes them tests. A reasoning inventory that correlates with *g* and with SAT and with job performance in the range typical of a *g*-saturated cognitive test is doing what cognitive tests do. Ejiogu and colleagues (2006), cited by Liu et al., reported corrected correlations of .32 to .52 between Watson–Glaser and job performance.[328] Moderate. Typical. Not incremental. Teaching to such a test is a real commercial industry. Score gains under coaching are not automatically skill gains. Daniel Koretz's CRESST work, later in this chapter, is the name for that distinction.

The map, compressed, so the walk that follows has a legend. Watson–Glaser: selected-response, five advertised buckets, employment screen, often one factor, coachable, Pearson / TalentLens, **vendor**. Cornell Critical Thinking Test, Ennis, Millman, and Tomko: multiple-choice closer to Ennis's conception; open-ended cousin the Ennis–Weir essay; thinner independent psychometrics. CCTST and CCTDI: Delphi operationalization, Insight Assessment **vendor**, subscale alphas independently reported as low as 0.21, CCTDI factor structure unstable. Halpern Critical Thinking Assessment: everyday scenarios, open-ended plus forced-choice, the most interesting design, **retired** by 2024. CLA / CLA+: performance task plus selected-response, writing-plus-reasoning blend, constructed-response reliability 0.43. Holistic Critical Thinking Scoring Rubric and AAC&U VALUE: rating tools, rater halo, weak high-stakes measures. HALO: not found as a named instrument. CART, Keith Stanovich's Comprehensive Assessment of Rational Thinking (2016), is a research prototype, not a school-ready instrument, and will not be promoted to one.[329]

None of these is the meaning. Each operationalizes *a* conception under psychometric constraint.

[327] Nathan R. Kuncel, review of CT tests and general cognitive ability (2011), as restated in Christopher R. Huber and Nathan R. Kuncel, "Does College Teach Critical Thinking? A Meta-Analysis," *Review of Educational Research* 86, no. 2 (2016): 431–468. Little discriminant validity of common CT tests from *g*; little evidence they predict grades or job performance *better than IQ*. Two senses of CT: field-specific expertise that does not readily transfer vs a finite set of specific reasoning skills useful for problems that share those elements. Incremental predictive validity largely missing (Liu et al. 2014 asked for it).

[328] Kizzy M. Ejiogu, Z. Yang, J. Trent, and M. Rose, corrected r = .32–.52 between Watson–Glaser and job performance (2006), cited by Liu, Frankel, and Roohr (2014). Typical of a *g*-saturated reasoning test, not incremental validity over general cognitive ability.

[329] Keith E. Stanovich, *The Comprehensive Assessment of Rational Thinking* (CART) (2016). Research prototype, not a school-ready instrument. Named so it is not mistaken for a fielded CT test.

The product category teaches the public that critical thinking is whatever the timed items are. That is the third folk use Chapter 1 named. This chapter is the autopsy.

Kuncel's two senses deserve a slower walk, because the map is unreadable if they are collapsed. Field-specific expertise that does not readily transfer: the historian's sourcing, the physician's differential, the physicist's conservation-law sort. Chapter 3 and Chapter 5 already spent that sense. Tests of it look like Reisman's historical thinking measures, like unconfounded experiments, like many diagnostic cases. They do not look like a 40-minute selected-response inventory of unstated assumptions. The second sense is a finite set of specific reasoning skills useful for problems that share those elements. Some of those skills are real: sample size, base rates, correlation versus causation, the difference between a reason and a restatement. A test that samples them will correlate with *g*, because *g* is, among other things, the ability to handle novel items of that kind. Gains on the test can be real and still be "better at those items," especially if the course taught those items. Incremental validity over *g* — does the test tell you anything about grades, job performance, or civic outcomes *beyond* IQ? — is the question Liu et al. asked and found largely unanswered. This book will not answer it with a vendor table. It will leave it standing, and it will not let a *g*-saturated inventory pose as a life skill.

McPeck, Willingham, and Norman are the conceptual partners because they predict the missing increment. If thinking is always about X, if processes are intertwined with content, if diagnosis is knowledge access, then a thin sample of logical moves should not predict clinical or civic or workplace judgment after you already know how generally able the person is and how much they know of the domain. The prediction could be wrong. The studies that would show it is wrong, with *g* controlled, on a life outcome, are the studies Liu and Kuncel both say are scarce. Butler's $r = -.38$ for HCTA and negative life events is the nearest neighbor, and it did not control *g*, and the instrument is retired. Absence of incremental validity is not proof that the tests measure nothing. It is proof that they have not earned the claim that they measure a faculty *beyond* general cognitive ability. A hiring screen can still be a screen. A life book needs the distinction.

## Watson–Glaser, Cornell, CCTST: lineages, not a mental organ

Goodwin Watson and Edward Glaser; Glaser (1941) used early forms to test his own teaching materials. Current forms — WGCTA, Watson–Glaser II, short form — use selected-response items in five buckets: inference, recognition of assumptions, deduction, interpretation, evaluation of arguments.[330] Widely used in employment, especially law-firm screening, which is a different validity argument from "college taught critical thinking." A hiring screen can be useful without being a learning outcome. It can also be a *g* test with a critical-thinking label, which is the discriminant-validity problem again.

Critiques that are sourced, not folkloric. Subscale internal consistencies in independent studies have been reported as low and wide: 0.17–0.74, Loo and Thorpe (1999), cited by Liu et al.[331] Bernard,

[330] Goodwin Watson and Edward M. Glaser, Watson–Glaser Critical Thinking Appraisal; current forms WGCTA, Watson–Glaser II, short form. Pearson / TalentLens — **vendor**. Five advertised buckets: inference, recognition of assumptions, deduction, interpretation, evaluation of arguments. Employment screen, especially law-firm hiring — a different validity argument from college learning. Glaser (1941) used early forms to test his own teaching materials.

[331] Robert Loo and K. Thorpe, independent Watson–Glaser subscale internal consistencies 0.17–0.74 (1999), cited by

Zhang, Abrami, Sicoly, Borokhovski, and Surkes (2008) meta-analysed 60 published studies and found **a single-component structure**, undermining the five-subscale story.[332] If the test is one factor, selling five scores as instructional steering is a vendor convenience. Report a total, or build a longer test. Moderate correlations with SAT, GPA, and job performance are typical of a *g*-saturated reasoning test (Kuncel 2011; Ejiogu et al. 2006). Incremental validity over general cognitive ability is the missing piece Liu and Kuncel both flag. Teaching to the Watson–Glaser is a real commercial industry — practice-test vendors. Score gains under coaching are the Koretz problem, not a skill gain, until someone shows otherwise with a far task the coach did not teach.

The Cornell Critical Thinking Test, Ennis, Millman, and Tomko, commercially distributed, is closer to Ennis's conception than Facione's Delphi list.[333] Level X, grades roughly 4–14: induction, deduction, credibility, assumption identification. Level Z, college and adult: those plus fallacies, definition, prediction, experimental planning. Multiple-choice. Open-ended cousin: the Ennis–Weir Critical Thinking Essay Test (1985), a letter-to-the-editor essay scored on argument moves. Independent psychometric work is thinner than for Watson–Glaser and CCTST; Liu et al. include it in the inventory table without a thick validity dossier. Closer to Ennis is not the same as validated for high-stakes decisions. Thinner independent work is a fact, not a smear. A life book that uses Ennis's public sentence as its definition does not therefore inherit Cornell as its outcome. The sentence includes *do*. A multiple-choice inventory of induction and assumption identification does not.

Ennis–Weir is the closer cousin for a life book, and it is still not the thing. A letter-to-the-editor essay scored on argument moves is at least about argument, open-ended, and nearer Ennis's conception than a selected-response inventory. It is also a writing sample, a rater-dependent score, and a single letter. Possin's warning about rhetoric versus argument quality, written for CLA graders, applies to any essay scored as "critical thinking" by readers who may be scoring fluency. Halo, restriction of range, 36 percent perfect agreement on VALUE: those are the rater facts waiting for Ennis–Weir at scale as well. Closer to the conception is a reason to prefer it as a *teaching* task. It is not a reason to treat a scored letter as the meaning of deciding what to believe or do, in a domain, on a stake.

The California Critical Thinking Skills Test and the CCTDI are the Delphi operationalization, Insight Assessment, **vendor**. CCTST is multiple-choice, about 34 items in about 45 minutes, widely used in nursing because NLN accreditation historically demanded a critical-thinking outcome. CCTDI measures dispositions: truth-seeking, open-mindedness, analyticity, systematicity, confidence, inquisitiveness, maturity of judgment. Those seven empirically factored constructs are *not* identical to Delphi Table 5. Collapsing them is how a skills list becomes a personality inventory without anyone noticing. The Delphi panel split 61/30 on whether dispositions are in the meaning of critical thinking at all. CCTDI then factored seven dispositions that are not the panel's Table 5 list, and independent samples would not restabilize the factors. A program that

Liu, Frankel, and Roohr (2014). Low and wide; not instructional steering.

[332] Robert M. Bernard, Dai Zhang, Philip C. Abrami, F. Sicoly, Evgueni Borokhovski, and Michael A. Surkes, meta-analysis of 60 published Watson–Glaser studies (2008): **single-component structure**, undermining the five-subscale story.

[333] Robert H. Ennis, Jason Millman, and Thomas N. Tomko, Cornell Critical Thinking Test, Levels X and Z, commercially distributed. Ennis and Eric Weir, *The Ennis–Weir Critical Thinking Essay Test* (1985): letter-to-the-editor scored on argument moves. Closer to Ennis's conception than Facione's Delphi list; independent psychometric work thinner than for Watson–Glaser and CCTST (Liu et al. 2014 inventory table without a thick validity dossier).

reports "open-mindedness went up" from a CCTDI sitting is reporting a subscale of an unstable structure, on a construct the panel did not even agree was inside the meaning. Dispositions matter to this book — Ennis's correlative care, Paul's intellectual traits as the point of strong-sense work, Willingham's need-for-cognition caution that the trait may be personality with limited educational evidence. They do not matter as seven CCTDI scores. Chapter 2 already chose: dispositions are in the coalition as aims and warnings, not as a factored inventory that Insight Assessment sells.

Independent psychometrics are weaker than author-reported figures, and Ku (2009) noted the pattern: independent researchers report weaker psychometrics than author-affiliated studies.[334] Leppa (1997): CCTST subscale alphas **0.21–0.51**, against author-reported about 0.68–0.70.[335] Form B reported more difficult than Form A (Jacobs 1999); low form comparability (Bondy et al. 2001).[336] CCTDI factor structure unstable across samples (Kakai 2003; Walsh and Hardy 1997; Walsh, Seldomridge, and Badros 2007).[337] An alpha of 0.21 on a subscale is not a subscale. It is noise with a name. Vendor copy that steers instruction from those names is not a finding.

Huber and Kuncel's nursing analysis is the outcome-level critique, and Chapter 5 already planted it. Despite NLN's requirement and heavy CCTST use, nursing students did not show larger long-term generic-critical-thinking gains than other majors.[338] If the flagship professional curriculum built around this instrument cannot beat ordinary college on the instrument's own terms, the instrument-plus-curriculum package is not a miracle. Two readings remain open, and this chapter will not close them with analogy. Domain-specific measures of clinical judgment might have moved; they were mostly not in Huber and Kuncel's inclusion set. Or the generic inventory does not capture the judgment nursing actually needs, which is Norman's point from inside diagnosis. Incremental validity after *g* is still under-shown for CCTST as for Watson–Glaser. Nursing's null is not "critical thinking cannot be taught." It is a null on these tests, in the profession that tried hardest to move them. Norman, from inside diagnosis, already said why a general inventory is the wrong outcome for clinical judgment: the binding constraint is knowledge access, and the right assessment is many cases. If nursing moved case-based judgment and did not move CCTST more than other majors, Huber and Kuncel's inclusion set would not have seen it. That is the open question, not a hidden RCT. Until domain-specific measures are in the synthesis, the honest sentence is the null on the generic tests, plus Norman's prediction, plus the refusal to treat the null as impossibility.

The publisher of the Holistic Critical Thinking Scoring Rubric — Facione and Facione, Insight Assessment, **vendor** — itself says rating tools are generally weaker than the validated standardized instruments, and that high-stakes use requires trained raters; Kappa is the relevant reliability

[334] Kelly Y. L. Ku, "Assessing Students' Critical Thinking Performance: Urging for Measurements Using Multi-Response Format," *Thinking Skills and Creativity* 4 (2009): 70–76, cited by Liu et al. (2014): independent researchers report weaker CCTST/CCTDI psychometrics than author-affiliated studies.

[335] C. J. Leppa, CCTST subscale alphas **0.21–0.51** (1997), against author-reported ~0.68–0.70 (Ku 2009, cited by Liu et al. 2014). An alpha of 0.21 is not a subscale.

[336] S. S. Jacobs, Form B reported more difficult than Form A (1999); K. N. Bondy et al., low CCTST form comparability (2001). Both cited by Liu, Frankel, and Roohr (2014).

[337] CCTDI factor structure unstable across samples: H. Kakai (2003); C. M. Walsh and R. C. Hardy (1997); C. M. Walsh, L. A. Seldomridge, and K. K. Badros (2007). Cited by Liu et al. (2014). CCTDI's seven empirically factored dispositions are not identical to Delphi Table 5.

[338] Huber and Kuncel (2016): nursing vs non-nursing, no significant difference on long-term generic-CT-test gains, despite NLN's historical CT requirement and heavy CCTST use. Domain-specific measures might have moved; mostly not in the inclusion set. Not "CT cannot be taught."

statistic.[339] Hold that sentence for the HALO confusion later. Subscores, across Watson–Glaser, CCTST, and CLA+ performance tasks, are mostly unusable for steering instruction. Independent work repeatedly finds they are too unreliable. Report a total, or build a longer test. Vendor copy does not enter this chapter as a finding.

A note on what "independent" means here, because Ku's 2009 observation is easy to read as an accusation. Author-affiliated studies of a commercial instrument often report better psychometrics than independent studies. That pattern is not unique to CCTST. It is a reason to prefer the independent numbers when they exist, and to say so. Leppa's 0.21–0.51 against author-reported 0.68–0.70 is the load-bearing contrast for CCTST subscales. Loo and Thorpe's 0.17–0.74 is the corresponding contrast for Watson–Glaser subscales. Bernard et al.'s one-component structure, from 60 published studies, is the corresponding contrast for the five-bucket story. None of these independent results says the total score is meaningless. They say the *subscores* are not instructional instruments, and that the advertised structure is not the measured structure. A program that changes its syllabus because "evaluation of arguments" was low on a Watson–Glaser has been steered by a name the factor analysis does not support.

Employment use deserves a sentence that assessment-for-learning does not. A law firm that uses Watson–Glaser as a screen is making a selection decision, typically with other information, typically in a population already truncated on cognitive ability. The validity argument is "does this test add anything to the hiring decision," not "did three years of college produce critical thinking." Those arguments have different criteria, different stakes, and different cheating-and-coaching ecologies. Practice-test vendors exist because the employment use is real. Coaching to a hiring screen is not a learning gain. Koretz again. This chapter will not tell firms to stop screening. It will tell a college not to treat the same instrument as proof that a general-education outcome has been met, and it will tell a reader not to treat a coached gain as the life skill the title named.

## CLA+, HCTA retired, and the HALO that is not a test

The Collegiate Learning Assessment, and CLA+, promised authenticity: a constructed-response performance task, a document-based memo, plus selected-response items. Council for Aid to Education, **vendor**. The construct is a blend of critical thinking, problem solving, and **writing**.[340] That blend is not a defect in a writing-and-reasoning exam. It is criterion contamination if someone treats the score as a pure critical-thinking test and then compares it to Watson–Glaser or CCTST syntheses. Huber and Kuncel correctly call it that.

Arum and Roksa (2011), *Academically Adrift*: 24 institutions, more than 2,300 students; **45 percent** showed no statistically significant CLA gain in the first two years; 0.18 SD over three semesters;

[339] Peter A. Facione and Noreen C. Facione, Holistic Critical Thinking Scoring Rubric (HCTSR), Insight Assessment — **vendor**. Publisher itself: rating tools generally weaker than validated standardized instruments; high-stakes use requires trained raters; Kappa is the relevant reliability statistic. Useful teaching tool; weak high-stakes measure.

[340] Collegiate Learning Assessment / CLA+, Council for Aid to Education — **vendor**. Constructed-response performance task (document-based memo) plus selected-response. Blend of critical thinking, problem solving, and **writing**. Huber and Kuncel (2016): criterion contamination — CLA is not a pure CT test and cannot be compared directly to Watson–Glaser/CCTST syntheses.

0.47 SD over four years.[341] That book is a serious book about effort and curriculum. It is a bad citation for “critical thinking cannot be taught.” The 45 percent is a CLA finding, not a Watson–Glaser finding. It is a two-year window, not a four-year life. It is partly a motivation-and-effort number, as the next section will show, because CLA administrations are typically low-stakes for the student. And CLA is not a pure critical-thinking test. Four reasons, any one of which would be enough to stop the slogan. Together they make the slogan a category error.

*Academically Adrift* earned its audience by saying something American higher education did not want said: a large share of students, on this performance task, did not show statistically significant gain in the first two years, and the four-year gain was 0.47 SD — in the neighborhood of Huber and Kuncel’s college slope on other instruments, and not a second half-sigma. Effort, curriculum, and time-on-task are the book’s subjects. They are good subjects. They are not the same subject as “the CCTST cannot be moved,” which Abrami 2015 already contradicts at g+ = 0.30, or “critical thinking cannot be taught,” which Willingham already refused for domain-specific thinking, or “college does nothing,” which Huber and Kuncel’s 0.59 mixed-design four-year gain already refuses. Criterion contamination, motivation, a two-year window, and a writing-plus-reasoning blend are not nitpicks. They are why a serious book about drifting is a bad warrant for a slogan about impossibility. This chapter’s job is to keep Arum and Roksa as a book about effort and curriculum, and to stop other writers from stealing the 45 percent as a theory of mind.

Psychometric critiques of CLA+ are not a vendetta. They are the reliability and validity the authenticity promise bought. Zahner (2013, CAE): 60-minute constructed-response reliability **0.43**; test-level reliability 0.87, “largely driven” by the 30-minute multiple-choice section, as Liu et al. report.[342] A 0.43 on the performance task is the authenticity-reliability trade in a single number. The part that looks like judgment is the unreliable part. The part that looks like a test is the multiple-choice. Possin (2013), in *Informal Logic*: graders without critical-thinking expertise can score **rhetoric** rather than argument quality — a validity attack, not a reliability attack.[343] Aloisi and Callaghan (2018), in UK Learning Gain work: student-level reliability still too low for individual decisions; the performance task and the selected-response should be interpreted separately.[344] School-level decisions and student-level decisions are different validity arguments. Klein et al. (2009) found school-level correlations around .93 between multiple-choice and constructed-response critical-thinking tests (EPP, CLA, CAAP).[345] That is not a reason to treat the individual performance-task score as a diagnosis.

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[341] Richard Arum and Josipa Roksa, *Academically Adrift: Limited Learning on College Campuses* (Chicago: University of Chicago Press, 2011). 24 institutions, >2,300 students; **45%** no statistically significant CLA gain in the first two years; 0.18 SD over three semesters; 0.47 SD over four years. A serious book about effort and curriculum. A bad citation for “critical thinking cannot be taught.” CLA finding, two-year window, writing-plus-reasoning blend, partly motivation (see note 352).

[342] Dirk Zahner (CAE, 2013), as reported by Liu, Frankel, and Roohr (2014): 60-minute constructed-response reliability **0.43**; test-level reliability 0.87, “largely driven” by the 30-minute multiple-choice section.

[343] Kevin Possin, “A Serious Flaw in the Collegiate Learning Assessment [CLA] Test,” *Informal Logic* 33, no. 3 (2013). Graders without CT expertise can score **rhetoric** rather than argument quality. Validity, not reliability.

[344] C. Aloisi and A. Callaghan, UK Learning Gain work (2018): student-level CLA+ reliability still too low for individual decisions; performance task and selected-response should be interpreted separately.

[345] Stephen Klein et al. (2009), school-level r ≈ .93 between multiple-choice and constructed-response critical-thinking tests (EPP, CLA, CAAP), cited by Liu, Frankel, and Roohr (2014). Authenticity and reliability trade; they are not synonyms. School-level agreement is not student-level diagnosis.

The Halpern Critical Thinking Assessment — HCTA, sometimes "Halpern HOTS," Schuhfried, **vendor**, later retired — was the most interesting design in the drawer, and it is gone.[346] Halpern (2003, 2010): critical thinking as purposeful, reasoned, goal-directed thinking for problems, inferences, likelihoods, decisions. HCTA used everyday scenarios with both open-ended and forced-choice items on five subscales: verbal reasoning, argument analysis, hypothesis testing, likelihood and uncertainty, decision-making and problem-solving. Butler (2012) reported that HCTA scores predicted fewer negative real-world outcomes, $r = -.38$ — one of the rare *life-event* correlations.[347] Liu et al. note the study did **not** control general cognitive ability, so the incremental claim is open. Do not treat −.38 as incremental validity after $g$. By 2024, a review in the *Journal of Intelligence* described HCTA as **retired**.[348] The interesting design is a historical object. A book that needs a life-event correlation will not find a currently fielded instrument that has shown one after $g$ is controlled.

HALO, as a named critical-thinking test, battery, or curriculum, was searched and not found.[349] Do not invent one. What exists, and is often confused with it, is three different objects.

First, the Holistic Critical Thinking Scoring Rubric (HCTSR), Facione and Facione, Insight Assessment, **vendor**: a four-level holistic rating of reasoning in writing or talk. The publisher itself, already quoted, says rating tools are generally weaker than validated standardized instruments. Useful as a teaching tool. Weak as a high-stakes measure.

Second, the **halo effect** in rubric scoring: one strong trait coloring the rest. VALUE Institute / AAC&U VALUE rubrics for Critical Thinking and Written Communication have been studied with Many-Facets Rasch models specifically for rater severity, halo, and restriction of range (Shapovalov, JMU thesis).[350] Finley (2012) reported only **36 percent** perfect agreement across raters on the VALUE critical-thinking rubric.[351] Holistic scoring of "critical thinking" in student work is exactly where halo is a known rater effect. 36 percent perfect agreement is not a scoring system you want making a graduation decision.

Third, HCTA, sometimes misheard or miscited as HALO. Retired, as above.

A book should not invent a HALO test. It should treat holistic rubrics as useful teaching tools and weak high-stakes measures, and name halo as a scoring threat, not as an instrument. Oral defense

[346] Diane F. Halpern, Halpern Critical Thinking Assessment (HCTA / "Halpern HOTS"), Schuhfried — **vendor**. Everyday scenarios; open-ended plus forced-choice; five subscales: verbal reasoning, argument analysis, hypothesis testing, likelihood/uncertainty, decision-making/problem-solving. Halpern (2003, 2010) conception: purposeful, reasoned, goal-directed thinking.

[347] Heather A. Butler, "Halpern Critical Thinking Assessment Predicts Real-World Outcomes of Critical Thinking," *Applied Cognitive Psychology* 26, no. 5 (2012): 721–729. $r = -.38$ with negative real-world outcomes. Liu et al. (2014): study did **not** control general cognitive ability. Do not treat as incremental validity after $g$.

[348] Review in *Journal of Intelligence* (2024), PMC10890380: HCTA described as **retired**. Access date 30 August 2026.

[349] Searches for a critical-thinking test, battery, or curriculum named **HALO**, comparable to Watson–Glaser, Cornell, CCTST, HCTA, or CLA+: not found, 30 August 2026 research pass. Do not invent a HALO test. What exists: HCTSR (note 338); halo *effect* in VALUE scoring (notes 349–350); HCTA, sometimes misheard (notes 345–347).

[350] Y. Shapovalov, James Madison University thesis (2021): Many-Facets Rasch on AAC&U VALUE rubrics (Critical Thinking and Written Communication) — rater severity, halo, restriction of range.

[351] Ashley Finley (2012): **36%** perfect agreement across raters on the AAC&U VALUE critical-thinking rubric. Holistic scoring of "critical thinking" in student work is where halo is a known rater effect.

and process rubrics, later in this chapter, will have the same rater-severity and halo problem at scale. That is an open question, not a reason to skip redesign. It is a reason not to pretend redesign is a solved technology.

The HALO confusion is worth a paragraph of hygiene because acronyms travel faster than instruments. A reader who heard "HALO" in a workshop may have heard HCTSR, or HCTA, or halo-as-bias, or a local rubric with a local name. Searches for a standard critical-thinking test, battery, or curriculum named HALO, comparable to Watson–Glaser, Cornell, CCTST, HCTA, or CLA+, did not return one. What returned were the rubric, the rater effect, and the retired Halpern. A manuscript that invented a HALO test to complete a table would be doing the thing this book exists not to do. The table has a blank. The blank is the finding.

## Teaching to the test, motivation, and the authenticity–reliability trade

Koretz, McCaffrey, and Hamilton (2001), and Koretz (2005/2008), distinguish meaningful gain — more teaching, harder work, better methods — from reallocation, alignment, coaching, and cheating.[352] The distinction is not unique to critical thinking. Critical-thinking inventories are highly coachable, which makes the distinction load-bearing. A semester of CCTST practice items is coaching. Score inflation is a rival explanation for every campus press release that begins "we went up four points." Van Gelder's argument-mapping studies, as Chapter 5 noted, tried to block teaching-to-the-test by using a second written test; most campus reports do not. A gain on the trained item type, without a far task, is the Koretz problem in a blue book.

Motivation on low-stakes tests can flip the conclusion. Liu, Bridgeman, and Adler (2012), in *Educational Researcher*, showed that conclusions about college learning gains **flip** depending on whether students are induced to try.[353] CLA, CLA+, and end-of-semester CCTST administrations are typically low-stakes for the student. Arum and Roksa's 45 percent is partly a motivation-and-effort number, not a pure skill number. A student who does not try on a posttest that does not affect a grade has not demonstrated that two years of college taught nothing. The student has demonstrated that the testing occasion did not require trying. Huber and Kuncel's college slope of about 0.5–0.6 SD over four years was measured on the same family of tests, with the same motivation problem in unknown and varying degrees. Liu, Bridgeman, and Adler should be a design constraint on the whole gain literature, not a footnote. How much of the college-gain and intervention-gain record is effort on a low-stakes posttest is an open question this chapter will leave standing.

Constructed-response authenticity and reliability trade; they are not synonyms. Liu et al.: given

[352] Daniel Koretz, D. F. McCaffrey, and L. S. Hamilton, CRESST work on score inflation (Koretz, McCaffrey, and Hamilton 2001; Koretz 2005/2008). Meaningful gain (more teaching, harder work, better methods) vs reallocation, alignment, coaching, and cheating. A semester of CCTST practice items is coaching. Van Gelder's AM studies tried to block this with a second written test; most campus "we went up 4 points" reports do not.

[353] Ou Lydia Liu, Brent Bridgeman, and Rachel M. Adler, "Measuring Learning Outcomes in Higher Education: Motivation Matters," *Educational Researcher* 41, no. 9 (2012): 352–362. Conclusions about college learning gains **flip** depending on whether students are induced to try. CLA/CLA+ and end-of-semester CCTST are typically low-stakes for the student. Arum and Roksa's 45% is partly a motivation-and-effort number, not a pure skill number — and not "CT cannot be taught." Design constraint on the whole gain literature, not a footnote.

the same testing time, constructed response is less reliable than multiple choice. Klein et al. (2009) nonetheless found school-level r ≈ .93 between multiple-choice and constructed-response critical-thinking tests on EPP, CLA, and CAAP. At the institution level, the two formats largely agree. At the student level, CLA+'s performance-task reliability of 0.43 is why Aloisi and Callaghan say the score is too low for individual decisions. Authenticity is a validity argument about the task. Reliability is a property of the scores. A memo that looks like work is not, by looking like work, a precise measure of a person. The performance task bought a writing-plus-reasoning blend, a motivation problem, and a reliability trade-off. It did not buy a pure window onto judgment.

Subscores, already flagged, belong in this design-constraint list. Almost every vendor sells them. Independent work repeatedly finds they are too unreliable to steer instruction — CCTST, Watson–Glaser, CLA+ performance task. An alpha of 0.21 is not a diagnostic. A one-factor Watson–Glaser is not five skills. Report a total, or build a longer test. Do not let a dashboard of noisy subscales become a curriculum.

Put the threats together and "the campus went up four points on the CCTST" is uninterpretable without answers this chapter has already named. Which form? Form B is harder. Was the posttest motivated? Liu, Bridgeman, and Adler can flip the sign. Was the gain coaching? Koretz. Was the subscale used as a steering number? Too unreliable. Did the gain beat ordinary college? Huber and Kuncel. Did it beat *g*? Kuncel, mostly no. Did it transfer to an unseen case in a new domain? Usually unmeasured. Chapter 5's 0.30 is real and small and is a movement on these instruments. This section is why that movement is not yet a life.

Motivation and coaching interact, which is why they are one section and not two slogans. A student who is graded on the posttest will try; van Gelder's 5 percent of course grade on best-of-two sittings is a motivation control, not a comparison group, and it sits inside a pre–post d of 0.87 that Chapter 5 already refused to promote to an RCT. A student who is not graded, at the end of a semester, on a CCTST the program needs for accreditation, is Liu, Bridgeman, and Adler's situation. Induce effort and the gain appears; do not induce it and the gain can vanish. A campus that reports a gain without saying which of those sittings it ran has reported a number that cannot be interpreted. A campus that reports a gain after a semester of practice items has reported coaching until a far task says otherwise. Koretz's meaningful-gain cell — more teaching, harder work, better methods — is the cell this book wants. Alignment, reallocation, coaching, and cheating are the rival cells. Critical-thinking inventories, being short, selected-response, and commercially practiced, live closer to the rival cells than a six-month document-based history curriculum with four outcomes does. That is another reason domain-embedded measurement, in Chapter 5, was cleaner. The test was the move.

The authenticity–reliability trade is the other reason a performance task did not rescue the construct. A memo looks like work. Looking like work is a face-validity argument. Reliability 0.43 on the constructed response is a score-property argument. Possin's rhetoric-versus-argument warning is a validity argument about what graders actually reward. Klein's school-level r ≈ .93 says that at the institution level, the cheap format and the expensive format largely agree, which is either reassuring (you can use multiple choice for school-level monitoring) or damning (the expensive format is not buying a different construct at the level you can trust). Both readings can be true for different decisions. Individual diagnosis needs reliability the performance task does not have. School-level monitoring may not need the performance task at all. Neither reading makes CLA+ "the" critical-thinking test. Huber and Kuncel's criterion-contamination label remains the right one when the blend is used as if it were CCTST.

## Redesign when the construct is judgment

If the target is **judgment** — deciding what to believe or do in an unseen, under-specified case — then a 40-minute multiple-choice inventory is the wrong genre. The redesign literature is mostly design language plus professional-education custom, not randomized trials of "oral defense raises CCTST." Cite it as language. Do not print it as a fitted SOP, and do not cite it as evidence that a new assessment *improves* critical thinking. Validity is the claim. Acceleration of a generic trait is not.

What judgment evidence looks like, without inventing trials:

**Unseen cases.** Medicine already does this, and Norman's assessment implication is "lots of cases," not one heroic capstone, because of case specificity.[354] A general inventory will not capture case-specific knowledge; a single case will not capture the next case. Law does a version in issue-spotting exams, which are also gameable. History document-based questions do a version: the student cannot have pre-drafted the answer if the documents are new. The point is not that any of these is a critical-thinking test. The point is that the student cannot have pre-drafted the answer, and that sampling many cases is the design, not a single performance that a rater can halo.

**Process evidence.** Drafts, argument maps, lab notebooks, sourcing annotations, search logs, peer-review memos. These are closer to the *move* than a final essay a model can generate. They are also forgeable, which is why they are evidence *with* a live check, not instead of one. Chapter 5's argument-mapping slice belongs here as process, not as a CCTST substitute: the map is the work of analysis, visible. A folder of maps without a live check is theatre.

**Oral defense.** Joughin (2010) is the assessment-theory source commonly cited for viva as verification.[355] A 2026 four-pillar conceptual paper on AI-resilient assessment — not a randomized trial — proposes an 8–12 minute protocol: summary, probing from process documents, extension to a novel scenario.[356] Health-sciences OSCEs are the existence proof that live performance can be a standard, not a stunt. Oral defense without sampling many cases reintroduces case specificity and rater halo. It is a check on the artefact, not a new mental organ made visible.

Abrami 2015's "anchored instruction" moderator is the experimental cousin: problems that make sense to the student, including simulations, role-play, ethical or medical dilemmas. Role-play g = 0.61 from 5 studies in that meta — small k, large number, treat as suggestive of anchored instruction,

[354] Geoffrey R. Norman, Sandra D. Monteiro, Jonathan Sherbino, Jonathan S. Ilgen, Henk G. Schmidt, and Silvia Mamede, "The Causes of Errors in Clinical Reasoning," *Academic Medicine* 92, no. 1 (2017): 23–30. Assessment implication: many cases, because of case specificity. A general CT inventory will not capture case-specific knowledge. Unseen cases without domain knowledge become IQ-with-a-cover-story (Willingham 2007/2019).

[355] Gordon Joughin, assessment-theory source commonly cited for viva as verification (2010). Oral defense as a check that the artefact had a student attached, not as a new trait. OSCEs in the health sciences are the existence proof that live performance can be a standard, with many stations because of case specificity.

[356] 2026 four-pillar conceptual paper on AI-resilient assessment (not an RCT): proposes an 8–12 minute protocol — summary, probing from process documents, extension to a novel scenario. Cite as language, not as a district SOP and not as evidence that oral defense raises CCTST.

not as a proven oral-defense accelerator of a generic critical-thinking trait.[357] Do not promote k = 5 into a policy.

CART, Stanovich's Comprehensive Assessment of Rational Thinking (2016), is a research prototype that tries to measure a broader rationality construct than the commercial inventories sample. It is not a school-ready instrument. This chapter names it so a reader who has heard the acronym does not think the problem has been solved in a lab and ignored by schools. Prototypes are not fielded tests.

Do not overclaim the redesign. Oral defense without many cases reintroduces case specificity and rater halo. Process logs without a live check become theatre. Unseen cases without domain knowledge become IQ-with-a-cover-story. Redesign is necessary for validity. It is not a proven accelerator of a generic critical-thinking trait. The open question — does requiring process plus oral defense recover construct validity for judgment, or shift the theatre? — has no randomized trial this book found. AIAS, next, is a framework awaiting that research.

Language, not SOP, means something specific in a book that will be stolen by workshops. A workshop wants a protocol: 8–12 minutes, three prompts, a rubric, a recording. The 2026 conceptual paper offers that protocol as a sketch. This chapter will not print it as a district policy. OSCEs exist because health sciences already needed live performance as a standard, with many stations because of case specificity, with trained raters, with a validity argument that is about clinical competence, not about a generic faculty. Importing the OSCE *shape* — many cases, live, sampled — is the transferable design idea. Importing a single 10-minute viva as "we now assess critical thinking" is how halo and case specificity re-enter through a more authentic door. Joughin is theory for viva as verification of the artefact. Verification is not a new score of a new trait. It is a check that the artefact had a student attached.

Process evidence has the same shape of temptation. Argument maps, search logs, sourcing annotations, lab notebooks: closer to the move. Also forgeable, and more forgeable now that a model can generate a plausible log. The live check is not a moralism about honesty. It is the only thing that keeps process from becoming a second artefact, equally unmoored. Chapter 5's named moves are the content of the process: control of variables in the notebook, sourcing in the annotation, the differential in the case write-up. A process folder that does not contain a named move is a vibe, as Murphy already said about talk.

Unseen cases without knowledge are the third temptation, and Willingham vetoes them. An unseen case in a domain the student does not know is an intelligence test with a cover story. Norman's many cases are many cases *in medicine*, with feedback on whether the right diagnosis was reached. Reisman's documents are history documents, with historical thinking as the outcome. The unseenness is what blocks pre-drafting. The domain is what makes judgment possible. Strip the domain and you are back to the bicycle.

[357] Abrami, Bernard, Borokhovski, Waddington, Wade, and Persson, *Review of Educational Research* 85, no. 2 (2015): role-play g = 0.61, k = 5. Small k, large number; treat as suggestive of anchored instruction, not as a proven oral-defense accelerator of a generic CT trait.

## AIAS as language: the artefact cannot be the whole evidence

Perkins, Furze, Roe, and MacVaugh (2024), *Journal of University Teaching and Learning Practice* 21(6) — the AI Assessment Scale — is **Mike Perkins et al.**, not David Perkins.[358] Cite as **language, not a randomized trial**. Five levels from "No AI" (controlled environment) to "AI Exploration." The 2024 paper and the 2025/2.1 revision argue that the scale's job is **assessment redesign**: align outcomes, task, evidence, rubric, and allowed AI so the grade still means what it claims. Level 4 in the original wording emphasised human evaluation of AI output — that is, **student judgment** as the assessed object. The official site, fetched 30 August 2026, is explicit: choosing a level is only the start; the work is redesigning evidence (process, artefacts, reflection, demonstration, live elements) and making rubrics score judgment and disciplinary standards. TEQSA, Australia's Tertiary Education Quality and Standards Agency, noted AIAS in 2024 as an option.[359] "Used in more than 350 institutions" is a dissemination claim, not an effect size.

AIAS is relevant to this book because generative AI collapsed the validity of take-home prose as evidence of unaided judgment. It does not tell you Philosophy for Children works. It does not tell you oral defense raises a CCTST. It tells you that if judgment is the construct, **the artefact cannot be the whole evidence**. That sentence is older than AI. Norman Frederiksen (1984), "The Real Test Bias," already argued that the tests we use shape the skills we get, and that the convenient artefact is not the construct.[360] AI made the sentence unskippable. A take-home essay a model can finish is not, by itself, evidence that the student decided what to believe or do.

Two-lane design — formation versus performance — waits for Chapter 9. Here the payload is validity, not tutor architecture, not Khanmigo, not classroom-agent design. This book is not a sequel to *Autonomous AI and Education*. Willingham, Bastani, and Fan already live there. The life condition is simpler and harder: a machine will finish the sentence. Evaluation is the remaining human work. The assessment implication, in this chapter's language, is that the finished sentence is not the evidence. Process, unseen cases, and a live check are how you keep the grade meaning what it claims. Whether that recovers construct validity or shifts the theatre is the open question already named. No randomized trial was found. Label the redesign as language. Do not print a Monday SOP. Do not cite AIAS as evidence that oral defense *improves* critical thinking.

The division of labor with the education book is therefore a constraint on the prose, not a slight. This chapter takes the validity implication. It does not retell the field experiments. A reader who wants the tutor designs knows where they are. A reader who wants to know whether a CLA+ memo,

[358] Mike Perkins, Leon Furze, Jasper Roe, and Jason MacVaugh, "The Artificial Intelligence Assessment Scale (AIAS): A Framework for Ethical Integration of Generative AI in Educational Assessment," *Journal of University Teaching and Learning Practice* 21, no. 6 (2024). **Mike Perkins, not David Perkins.** Language, not RCT. Five levels, "No AI" to "AI Exploration." Job is assessment redesign: align outcomes, task, evidence, rubric, and allowed AI. Level 4 original wording: human evaluation of AI output — student judgment as the assessed object. 2025/2.1 revision: choosing a level is only the start. Official site fetched 30 August 2026. "Used in more than 350 institutions" is a dissemination claim, not an effect size. Do not cite as evidence that oral defense *improves* CT. Two-lane formation/performance waits for Chapter 9; no Khanmigo, Kestin, or tutor-architecture tour.

[359] Tertiary Education Quality and Standards Agency (TEQSA, Australia) noted AIAS in 2024 as an option. An option is not a trial.

[360] Norman Frederiksen, "The Real Test Bias: Influences of Testing on Teaching and Learning," *American Psychologist* 39, no. 3 (1984): 193–202. The artefact cannot be the whole evidence; tests shape the skills instruction aims at. Older than generative AI; AI made the sentence unskippable for take-home prose.

a CCTST sitting, or a take-home essay is evidence of judgment now has the answer this book can defend: not by themselves. Not because AI arrived. Because they were never the thing, and AI made the mismatch impossible to ignore.

Frederiksen's "real test bias" is the older name for a problem the inventories already had. Tests sample a convenient slice. Instruction then aims at the slice. The slice becomes the construct in public. Watson–Glaser's five buckets, CCTST's Delphi skills, CLA+'s memo: each is a convenient slice. Coaching industries grow up around convenient slices. Koretz is the score-inflation name for the same loop. Generative AI did not invent the loop. It closed the take-home-prose hole so completely that a writer who still treats the essay as unaided judgment is not being traditional. The writer is being uninterpretable. AIAS's contribution is to make the allowed-AI decision visible, so the grade can still mean something, and to insist that choosing a level is not the work. The work is evidence: process, artefacts, reflection, demonstration, live elements. That list is this chapter's unseen cases, process, and oral defense, with an honesty rule about the machine. It is not a fifth AI chapter. It is assessment validity under the life condition the title named.

## What would count as improvement in a life

Improvement in a life is better judgment on named stakes: a source left, a base rate used, a fluent wrong answer refused, a mind changed in public including one's own. Those are Chapter 10's deeds. They are not CCTST points. They are not a Watson–Glaser percentile. They are not a CLA+ performance-task score with reliability 0.43. They are not a VALUE rubric with 36 percent perfect agreement. A book that ends its assessment chapter by recommending a better inventory has not refused the folk collapse. It has upgraded the third folk use.

The instruments remain usable as what they are. A Watson–Glaser can screen for *g*-saturated reasoning in a hiring pool if the employer knows that is what is being screened, and does not pretend five subscales are five skills. A CCTST can be a program's accreditation number if the program knows the number is an operationalization of Delphi, not the meaning of the work, and does not steer instruction from an alpha of 0.21. A CLA+ can be a writing-plus-reasoning performance at school level if no one uses the constructed-response score as an individual diagnosis. HCTA's life-event correlation was a hint, uncontrolled for *g*, and the instrument is retired. Holistic rubrics can teach. They cannot, on present rater-agreement evidence, carry high stakes alone.

What would count, if the construct is Ennis's sentence — reasonable reflective thinking focused on deciding what to believe or do — is evidence that the person decided, in a domain they actually know, on a stake that can hurt, including when the sentence arrives already finished. Unseen cases, sampled many times because of case specificity. Process that makes the move visible, with a live check because process is forgeable. Oral defense as verification, not as a new trait. Domain knowledge in the case, because Willingham and Norman will not let a writer forget. Motivation to try, because Liu, Bridgeman, and Adler can flip the conclusion if the occasion is low-stakes. Incremental validity after *g*, still under-shown, still the question Liu and Kuncel asked. Nursing's null on generic tests, still open as to domain-specific movement. None of that is a new commercial battery. None of it is a fitted life-span schedule.

The live debates this chapter will not close: incremental validity of CCTST, Watson–Glaser, and

CLA+ after *g*; why nursing did not beat other majors; how much of the gain literature is effort on a low-stakes posttest; whether process plus oral defense recovers construct validity or shifts theatre — planted for Chapter 9. CART is a prototype. AIAS is a framework. HCTA is retired. HALO is not a test.

A life is not a test. Leave the page. Chapter 7 is the first life domain written out: civic and media, where the named move is leaving the page, and the contrast case is a checklist that trains fakeable signals. The deeds will wait until the domains have been named. The inventories will not be invited along as if they were the meaning.

One more refusal, because it will be the pull-quote if this chapter is careless. Abrami moved the tests. Huber and Kuncel measured a college slope on the tests. Niu measured college interventions on the tests. Those movements are real and small and heterogeneous. They are not far transfer, job performance, or civic wisdom. They are not proof of impossibility when they are small, and they are not proof of a bicycle skill when they are larger in a combination cell of 19 studies. The test is not the thing. A life of critical thinking, in this book's coalition, is Ennis's believe-or-do, under Willingham's knowledge constraint, with Lipman for childhood judgment, and with Paul's warning that skilled defense of one's side is not the goal. Measuring that life with a 40-minute selected-response inventory, or with a performance task whose constructed-response reliability is 0.43, or with a rubric on which raters perfectly agree 36 percent of the time, is a category error that happens to be common. Common is not correct.

What a program can do on Monday, without a fitted SOP: stop treating CCTST as the meaning; stop steering from subscores; stop citing Arum and Roksa as impossibility; stop inventing a HALO test; stop calling a coached gain a skill; induce effort if you are going to interpret a low-stakes sitting; sample many unseen cases in the domain you actually teach; keep process with a live check; use oral defense as verification, knowing halo and case specificity; name the AI level so the grade still means what it claims. That list is this chapter's design language. It is not a trial. It is not a promise that the list will raise a Watson–Glaser. It is the condition under which a grade, or a life, could be evidence of judgment rather than evidence of a sitting.

# Chapter 7

## Leave the Page: Civic and Media Life

Figure 8: Open tabbed books, a stack of newspapers, and eyeglasses under a banker's lamp.

The civic habit that has trial evidence is not a checklist. It is leaving the page.

In 2019 Sam Wineburg and Sarah McGrew published, in *Teachers College Record*, a comparison that should have retired a generation of library handouts. They watched three groups evaluate live websites: ten professional fact checkers, ten Ph.D. historians, and twenty-five Stanford undergraduates. The fact checkers left. Before they invested attention in an unfamiliar page, they opened new tabs and searched the organization and its principals. Wineburg and McGrew named the move *lateral reading*. The historians and the undergraduates stayed. They read *vertically*: About pages, `.org` domains, prose quality, internal consistency. They were slower *and* less accurate.[361]

That is not a finding about "kids and fake news." Historians were in the room. Stanford undergraduates were in the room. The failure mode was a practiced habit of staying put, and the habit had been taught. Civic Online Reasoning, the Stanford History Education Group curriculum that grew out of this work, distills three questions in the project's own words:

1. Who's behind the information?
2. What's the evidence?
3. What do other sources say?[362]

Not a thirty-item rubric. A short heuristic, practiced on real pages. This chapter is the first *life* chapter of the book, and it has a narrower job than a media-literacy sermon. It names the civic habit with trial evidence. It replaces the slogan "kids believe fake news" with the actual failure: look, top-level domain, About pages, quantity of information — exactly the signals feature-based checklists train students to trust. It reports what six roughly fifty-minute Civic Online Reasoning lessons actually moved in a required high-school government course, and it refuses to upgrade a credibility-task gain into civic-outcome transformation. It treats inoculation and accuracy prompts as what they are: technique recognition in the lab, a small field effect on YouTube, a sharing nudge under inattention — not a belief vaccine and not a turnout trial. It spends pages on what to do with a correction, because the reason people skip correction is a finding that did not survive. And it keeps frameworks and laws in their lane. The National Association for Media Literacy Education's Core Principles are a professional association's position. Twenty-five state media-literacy laws are laws. Media Literacy Now has not evaluated the classrooms. A statute that says "teach critical thinking" is not lateral reading, not inoculation, and not an accuracy prompt.

The thesis of the book does not change when the stake is a feed instead of a test. Critical thinking is not a transferable bicycle-skill. It is domain knowledge plus evaluation habits practiced on real stakes. On the live web, the domain is civic: who is speaking, what would count as evidence, what other sources say. The habit is leaving. Everything else in this chapter is a constraint on that claim.

Ennis's public sentence — reasonable reflective thinking focused on deciding what to believe *or do* — is why civic life belongs here rather than in an assessment chapter. Sharing is a doing. Voting is a doing. Declining to amplify a video is a doing. A book that treated critical thinking as the correct assessing of statements could stop at accuracy judgments. This one cannot. Pennycook and Rand will appear later as the demonstration that accuracy judgments and sharing intentions can come apart; Wineburg's program is the demonstration that the assessing itself, on the live web, is a practiced departure from the page rather than a careful stay. Both are life moves. Neither is a score on a Watson–Glaser.

[361] Sam Wineburg and Sarah McGrew, "Lateral Reading and the Nature of Expertise: Reading Less and Learning More When Evaluating Digital Information," *Teachers College Record* 121 (2019). ERIC EJ1262001. Sample: 10 professional fact checkers, 10 Ph.D. historians, 25 Stanford undergraduates. Fact checkers arrived at more warranted conclusions in a fraction of the time. The 2017 precursor is Wineburg and McGrew, "Lateral Reading: Reading Less and Learning More When Evaluating Digital Information," SHEG working paper 2017-A1, SSRN 3048994.

[362] Stanford History Education Group, Civic Online Reasoning curriculum site, https://cor.stanford.edu/, fetched 30 August 2026. Three questions in the project's own words. Free lessons; not an outcome trial.

The chapter also inherits two vetoes already written. Transfer is rare and similarity-bound: six COR lessons that move SHEG credibility tasks are near transfer to similar pages, not far transfer to turnout. Implicit expectation fails: a government class, a library guide, or a state statute that hopes evaluation will appear as a by-product has chosen the weakest design Abrami documented. Those vetoes are not decorations. They are why this chapter will refuse to narrate civic-outcome RCTs the sources did not contain, and why it will keep NAMLE and the twenty-five laws in the framework column.

## Lateral reading is the signature move

Wineburg and McGrew's 2019 paper is the signature comparison; a 2017 working paper already named the move.[363] The design is simple enough to steal and strict enough to trust. Participants faced unfamiliar sites on the open internet, the same internet a voter, a parent, or a patient actually uses. Fact checkers did not begin by reading the page. They left it. They searched the organization's name. They searched the people named on the masthead. They used Wikipedia as a first hop, not as a banned last hop — harvesting references, reading Talk pages, treating the encyclopedia as a lead generator rather than as an authority to be cited or forbidden.[364] Historians, who spend professional lives sourcing documents, stayed on the page and read it the way one reads a monograph: carefully, vertically, looking for internal signs of quality. Undergraduates did the same, with less patience and worse accuracy. Reading carefully was not the skill. Knowing when *not* to read was.

The historian result is the one a critical-thinking book is most tempted to bury. These were not novices. They were professionals of sourcing. Their expertise was real, and it was bound to a medium: the document in hand, the archive, the footnote trail *inside* the work. The live web is a different medium. An About page is not a colophon. A `.org` is not a university press. Internal consistency is what a competent cloaked site is built to display. Expertise that is encapsulated in one domain does not automatically deploy in another. Chapter 3 already spent that finding on neurologists and cardiac cases, technical writers and newspaper articles, professional philosophers and irrelevant problem features. Wineburg and McGrew is the civic instance. The maxim "source your documents" did not fail because historians cannot think. It failed because the object of practice had changed and the practiced move had not.

Wikipedia as a first hop still needs a sentence of hygiene, because a generation of teachers banned it for reasons that were not crazy and are now the wrong reasons. The encyclopedia is editable. It is also, for a first look at *who an organization is*, faster than reading the organization's self-description. Fact checkers harvested references and read Talk pages — they used the map, including its disputes, and then left again. That is not "Wikipedia is a reliable source." It is "Wikipedia is a lead." A classroom that bans the first hop and assigns the About page has chosen the adversary's preferred document.

[363] Wineburg and McGrew, "Lateral Reading," SHEG working paper 2017-A1; Wineburg and McGrew, *Teachers College Record* 121 (2019).

[364] Sarah McGrew, Teresa Ortega, Joel Breakstone, and Sam Wineburg, "The Challenge That's Bigger Than Fake News: Civic Reasoning in a Social Media Environment," *American Educator*, Fall 2017, https://www.aft.org/ae/fall2017/mcgrew_ortega_breakstone_wineburg. Fetched in full 30 August 2026. Classroom translation of the 2016 report; click restraint; wise Wikipedia use; checklist critique. Not an RCT.

Sarah McGrew, Teresa Ortega, Joel Breakstone, and Wineburg, writing for *American Educator* in Fall 2017, named two companion moves. *Click restraint*: Google's ranking is not a trust ranking; the first result is a commercial and algorithmic fact, not an epistemic one. *Wise Wikipedia use*: the page is a map, not a destination. The same essay already named the failure of the alternative. Ten-to-thirty-item checklists — contact person, typos, banner ads, `.org` — train attention onto features an adversary can fake. "Providing an author, throwing up a reference list, and ensuring a site is free of typos hardly establishes it as a credible source."[365]

Civic Online Reasoning is what you get when you take those three fact-checker moves and ask what a high-school government class can actually practice. The curriculum site, fetched 30 August 2026, does not sell a thirty-item list. It sells three questions, in that order, on live pages.[366] Who's behind the information? — not "is there an About page," but *leave and find out*. What's the evidence? — not "does the page have citations," but *what kind of evidence is this, and does it show what the headline claims*. What do other sources say? — not "is the site internally consistent," but *what do independent accounts, including ones that disagree, report*. That is a different object of practice from Currency, Relevance, Authority, Accuracy, Purpose. The contrast is not a sibling intervention. It is the chapter's load-bearing distinction, and Section 7.4 will spend it. First the baseline, because the slogan about children and fake news is the wrong diagnosis of a real problem.

A note on what this section is not claiming. Lateral reading is a civic *habit* with classroom trial evidence. It is not a general critical-thinking skill that, once learned on websites, transfers unprompted to a clinic, a hiring story, or a fluent paragraph a model will finish. Willingham's constraint from Chapter 3 still holds: the processes of thinking are intertwined with the content of thought; maxims without background knowledge and practice will not be executed.[367] You cannot laterally read a climate-advocacy site if you have no idea what a fossil-fuel trade association is. You cannot evaluate a claim about voter fraud if you do not know what would count as evidence of it. The three questions are metacognitive cues of the kind Willingham allows. Doing what they call for depends on civic knowledge and on practice with real pages. That is why the curriculum lives in a government course and not in a standalone "critical thinking" period, and why six lessons, as the next sections will show, move a test of the same skill without producing informed citizens.

## The baseline: look, TLD, About pages, quantity

The popular story is that young people believe fake news. The record is worse, and differently worse. Students are not primarily failing a truth-or-lie quiz about cartoon hoaxes. They are trusting the signals that school-taught checklists told them to trust, on cloaked real sites with real agendas.

Wineburg, McGrew, Breakstone, and Ortega's 2016 report, *Evaluating Information: The Cornerstone of Civic Online Reasoning*, field-tested fifty-six tasks between January 2015 and June 2016.

[365] McGrew et al., "The Challenge That's Bigger Than Fake News." The quoted sentence is from that essay: "Providing an author, throwing up a reference list, and ensuring a site is free of typos hardly establishes it as a credible source."

[366] Stanford History Education Group, Civic Online Reasoning curriculum site, https://cor.stanford.edu/, fetched 30 August 2026. Three questions in the project's own words. Free lessons; not an outcome trial.

[367] Daniel T. Willingham, "Critical Thinking: Why Is It So Hard to Teach?," *American Educator* 31, no. 2 (Summer 2007). Load-bearing for this book in Chapter 3; cited here as the constraint, not re-argued.

Twelve states. 7,804 responses. Inner-city Los Angeles through well-resourced Minneapolis suburbs through Stanford and large state universities, middle school through selective college. The authors' own summary: at every level, students fell below a reasonable bar.[368] That is the baseline portrait, not a children's-internet panic. Selective-college students were in the sample. The bar was not "spot the tree octopus."

Breakstone, Smith, Wineburg, Amie Rapaport, Jill Carle, Marshall Garland, and Anna Saavedra published the national portrait in *Educational Researcher* in 2021. It is the largest investigation of its kind in the sources this book fetched: 3,446 high-school students, a live internet connection, six constructed-response tasks, a sample built to reflect U.S. high-school demographics. Three numbers from the abstract are load-bearing, and they should be kept together rather than cherry-picked.

Asked to investigate a site claiming to "disseminate factual reports" on climate science, **96 percent** never learned the organization's fossil-fuel ties.

On a popular homepage, **two thirds** could not distinguish news stories from advertisements.

**More than half** treated an anonymously posted Facebook video, shot in Russia, as "strong evidence" of U.S. voter fraud.[369]

Those are not "kids believe fake news." They are three different failures, and they share a mechanism. Students were duped by look, top-level domain, About pages, and quantity of information. A site that *looks* finished — professional design, a `.org`, an About page naming an author, a dense page of prose and references — collects trust. Quantity of information is taken as a proxy for quality. An anonymous video is treated as strong evidence because it is a video, which feels like seeing. The signals are exactly the ones CRAAP-style checklists train students to inspect *on the page*. The adversary who wants to be believed will provide them. The student who was taught to look for them will find them, stay, and score the site as credible.

Keep the three tasks from collapsing into one moral. Missing a fossil-fuel tie is a *who's behind this* failure: the page performed "factual reports," and 96 percent never left far enough to find the sponsor. Missing news versus ads on a homepage is a *what is this genre* failure: native advertising is designed to share the look of reporting, and two thirds did not separate the costumes. Treating an anonymous Facebook video as strong evidence of voter fraud is a *what would count as evidence* failure: a video is not a chain of custody, an anonymous post is not a named witness, and "shot in Russia" is a sourcing fact that more than half of the sample did not make decisive. Constructed-response tasks, not multiple choice, produced those numbers. Students had to *write* what they would do and why. The failure is not a trick of distractors. It is the practiced stay.

[368] Sam Wineburg, Sarah McGrew, Joel Breakstone, and Teresa Ortega, *Evaluating Information: The Cornerstone of Civic Online Reasoning* (Stanford History Education Group, 2016), https://purl.stanford.edu/fv751yt5934. Fifty-six tasks, January 2015–June 2016, twelve states, 7,804 responses. Stanford Digital Repository. Metadata fetched; full PDF not re-read on this pass.

[369] Joel Breakstone, Mark Smith, Sam Wineburg, Amie Rapaport, Jill Carle, Marshall Garland, and Anna Saavedra, "Students' Civic Online Reasoning: A National Portrait," *Educational Researcher* 50, no. 8 (2021), https://doi.org/10.3102/0013189X211017495. 3,446 high-school students; live internet; six constructed-response tasks. Load-bearing numbers from the abstract: 96 percent, two thirds, more than half. SES, race, maternal education, and free-or-reduced-price-lunch gradients in the abstract.

"Digital native" is the folk explanation that these numbers kill. The 3,446 students had a live internet connection. They were not locked out of the medium. They were fluent in the surface of the medium — look, TLD, the feeling of a finished page — and they treated that fluency as evaluation. Chapter 4's fluency problem is already visible here, years before a model that finishes sentences. Ease of processing is misread as evidence. A professional layout is easy to process. So is a video. So is a dense paragraph with citations. Lateral reading is, among other things, a refusal to let ease finish the job.

Breakstone and colleagues also reported an equity gradient that this book will not save for a closing sermon. Multilevel regression: scores varied by grade, self-reported grades, locality, socioeconomic status, race, maternal education, and free-or-reduced-price lunch.[370] The students who most need a public civic skill are the ones the assessment says lack it. That gradient is *pre-ChatGPT*. Chapter 10 will return to it as an equity finding, not as a reason to issue a badge. Here it is a constraint on the "digital native" story. Growing up with a phone is not growing up with lateral reading. Access to the live web is not practice in leaving it. And a knowledge-poor classroom that substitutes a media-literacy poster for civic content will not close the gap the 2021 paper already measured.

McGrew and colleagues, in the 2017 *American Educator* essay, had already named a related failure that still gets assigned: the hoax-site lesson. The Pacific Northwest tree octopus is a useful joke and a useless civic education. The live web's problem is not cartoon fakes. It is cloaked real sites with real funders, real staff, and real agendas, wearing the costume of a public-interest `.org`.[371] A student trained on the tree octopus learns to spot the absurd. A student facing a climate site that "disseminates factual reports" needs a different move: leave, search the organization, find the trade-association tie. Ninety-six percent never made that move. The checklist had trained them to stay.

This is the failure mode Chapter 3 would have predicted. Surface structure captures attention. Deep structure — who funds this, what would count as evidence, what do independent sources say — is what would transfer, and novices do not see it. A maxim ("consider the source") without the knowledge of what a source *is* in this domain, and without practice leaving the page, will not be executed. The 2016 and 2021 portraits are not evidence that students cannot think. They are evidence that they were practicing the wrong object, in a domain whose knowledge they mostly lacked, on tasks that rewarded looking finished.

## What six COR lessons actually moved

Then there is a trial. It is a real trial. It is modest. It is incomplete. It is not a civic-outcome RCT, and this book will not write one into the record.

[370] Joel Breakstone, Mark Smith, Sam Wineburg, Amie Rapaport, Jill Carle, Marshall Garland, and Anna Saavedra, "Students' Civic Online Reasoning: A National Portrait," *Educational Researcher* 50, no. 8 (2021), https://doi.org/10.3102/0013189X211017495. 3,446 high-school students; live internet; six constructed-response tasks. Load-bearing numbers from the abstract: 96 percent, two thirds, more than half. SES, race, maternal education, and free-or-reduced-price-lunch gradients in the abstract.

[371] McGrew et al., "The Challenge That's Bigger Than Fake News." The quoted sentence is from that essay: "Providing an author, throwing up a reference list, and ensuring a site is free of typos hardly establishes it as a credible source."

Wineburg, Breakstone, McGrew, Mark D. Smith, and Ortega published in the *Journal of Educational Psychology* in 2022. Urban midwestern district. Six high schools. A semester-long required government course, spring 2019. Teachers received professional development. Six lessons of about fifty minutes each, Civic Online Reasoning, over three months. The published abstract reports a matched control: treatment $n$ = 271, control $n$ = 228. The COR project page and SSRN version describe a cluster-randomized, repeated-measures analysis; the paper reports a multilevel linear mixed model. Experimental classrooms grew significantly in judging digital credibility. Stanford GSE news called it the first randomized experiment in an urban district — journalism, not a substitute for the paper.[372] That is the paper.

Stanford Graduate School of Education news, 19 April 2022 — **journalism / university communications**, summarizing that paper, not a substitute for it — reported that students in the experimental classrooms roughly doubled their pre-test scores, or were "twice as likely to spot questionable websites," and still "earned only half the total number of possible points."[373] Funded by Google.org. The doubling is the sentence that will be quoted. The remaining miss rate is the story. Six hours of practice, in a course that already had a civic job, moved scores on SHEG credibility tasks. Even then, about half the points were still on the table. A book that upgrades "grew significantly" into informed citizens, higher turnout, or less polarization has left the paper.

The design details matter because they are the conditions under which the modest gain appeared. A required government course, not an elective "media literacy" club. Teachers who received professional development, not a PDF emailed in August. Six lessons spread over three months, not a Friday assembly. Live internet, constructed-response tasks, matched control. Clustered in classrooms, analyzed with a multilevel linear mixed model — the right family of model for students nested in teachers nested in schools. Spring 2019, which is to say before a pandemic rearranged school and before a generative model rearranged the default document. Those are features, not asterisks. They are also limits. A district that copies the three questions onto a slide and skips the PD, the government-course home, and the live pages has not copied the trial. A commentator who cites the doubling and forgets the half has not read the journalism, let alone the paper.

Government class is a natural home for a reason Willingham would recognize. Civic online reasoning needs civic knowledge: what a trade association is, what a primary is, what would count as evidence of fraud, what an ad is doing on a news homepage. A standalone critical-thinking course that practices the three questions on empty content will produce the maxim without the knowledge. Mixed design, in Ennis's taxonomy, is the honest architectural guess — a short explicit strand plus

[372] Sam Wineburg, Joel Breakstone, Sarah McGrew, Mark D. Smith, and Teresa Ortega, "Lateral Reading on the Open Internet: A District-Wide Field Study in High School Government Classes," *Journal of Educational Psychology* 114, no. 5 (2022): 893–909, https://doi.org/10.1037/edu0000740. Urban midwestern district; six high schools; required government course, spring 2019; six ~50-minute COR lessons over three months; treatment $n$ = 271, matched control $n$ = 228 (published abstract). COR project page and SSRN version: cluster-randomized, repeated-measures analysis. Abstract and COR project page fetched 30 August 2026; ERIC abstract re-checked 30 August 2026. Do not upgrade "grew significantly" into a civic-behavior RCT.

[373] Stanford Graduate School of Education news, "It doesn't take long to learn how to spot misinformation online, Stanford study finds," 19 April 2022, https://ed.stanford.edu/news/it-doesn-t-take-long-learn-how-spot-misinformation-online-stanford-study-finds. **Journalism / university communications** summarizing Wineburg et al. 2022. Source of "twice as well" / "twice as likely to spot questionable websites" and "earned only half the total number of possible points." Funded by Google.org. Not the paper.

subject-matter practice — and even then Chapter 5 already reported that Ennis-type differences in Abrami 2015 were not statistically significant. The 2022 paper is not a horse-race win for mixed over infusion. It is a demonstration that *this* object of practice, in *this* course, with *this* dose, moved *this* test. Steal the object. Do not steal a slogan about citizens.

McGrew's 2020 study in *Computers & Education* is the smaller layer underneath. An eight-lesson COR intervention; students improved on three of four online tasks — investigate backers, critique social-media evidence, locate reliable information on a contentious question.[374] Small-*N* relative to the district study; still a trial, not a slogan. McGrew, Breakstone, Ortega, Smith, and Wineburg (2018), in *Theory & Research in Social Education*, is the assessment-validity layer: the tasks measure something coherent enough to teach toward.[375] None of these papers measured whether students voted, changed a partisan identity, avoided a scam, took a vaccine, or shared less low-quality news in the wild over a year. Those civic *outcomes* are an open question this chapter leaves standing. They were not shown in the trials fetched for this book. They must not be invented.

Dose and durability are open in the same honest way. Wineburg 2022 is six lessons and an immediate-ish post-test. The inoculation literature in Section 7.5 will say that effects decay without boosters. What a K–12 *sequence* of lateral reading looks like, whether it lasts past the unit test, and whether it requires the civic knowledge a government course is supposed to supply, is not fitted. Willingham's 2019 advice for domain-specific critical thinking — identify what the skill means in the domain, identify the content those tasks need, sequence, plan three to five years of revisiting — is the pedagogical constraint, not a completed COR scope-and-sequence.[376] Six lessons in one semester of government is a start. It is not a life.

Two further constraints, so the trial cannot be misread later. First, the outcome is judging digital credibility on constructed-response tasks of the kind SHEG built. Near transfer to similar pages is what one should expect if the object of practice was those pages. Far transfer to turnout, polarization, or a clinic waiting room is what one should not claim. Chapter 4 already spent Thorndike, Detterman, and Gick and Holyoak: spread of practice tracks identical elements; the default prediction is failure; noticing is the gap. Second, Google.org funding is a fact to print, not a reason to discard the paper and not a reason to inflate it. Industry-adjacent funding is common in this literature — Jigsaw will appear in the inoculation section — and the honest move is to name it and then read the design.

What the 2022 trial licenses is therefore narrow, and useful because it is narrow. A short set of evaluation habits, practiced on real pages, in a civic domain, with teachers who received professional development, moves a test of those habits. The gain is real. It is modest. After six hours students still miss about half the points, if one trusts the university news summary for that remaining-miss

[374] Sarah McGrew, "Learning to Evaluate: An Intervention in Civic Online Reasoning," *Computers & Education* 145 (2020): 103711, https://doi.org/10.1016/j.compedu.2019.103711. Eight-lesson intervention; gains on three of four tasks. Abstract and secondary sources fetched; full PDF paywalled on this pass.

[375] Sarah McGrew, Joel Breakstone, Teresa Ortega, Mark Smith, and Sam Wineburg, "Can Students Evaluate Online Sources? Learning from Assessments of Civic Online Reasoning," *Theory & Research in Social Education* 46, no. 2 (2018). Assessment-validity layer under the 2022 district study.

[376] Daniel T. Willingham, "How to Teach Critical Thinking," NSW Department of Education, 2019. Four-step plan: identify what CT means in each domain; identify needed content; sequence; plan three to five years of revisiting. Pedagogical constraint, not a completed COR sequence.

figure.[377] That is the opposite of a bicycle. It is also the opposite of despair. The civic skill is teachable *as civic practice*. It is not produced by a poster of maxims, and it is not produced by a law that says the words.

## CRAAP trains fakeable signals

Checklist literacy is the contrast case, not a sibling intervention. The acronym is CRAAP: Currency, Relevance, Authority, Accuracy, Purpose. College library pages recommended it. Students used it. Wineburg, Breakstone, Nadav Ziv, and Smith, in a 2020 Stanford History Education Group working paper titled *Educating for Misunderstanding*, watched college students evaluate the live web with the strategies those library pages had taught.[378]

The paper's assumptions, in the authors' register, are the ones the checklist encodes. Websites are like print. Evaluate by reading the page carefully. `.org` and nonprofit status are trust proxies. Those "best practices" can make students *more* susceptible, because they train attention onto features adversaries can fake. The PDF fetch for this book timed out; extra percentages will not be invented from a working paper that was not re-read in full.[379] The claim that survives the fetch is the one the brief already carried: vertical, feature-based checklists are the wrong object of practice. Lateral reading is a different object. A habit of leaving.

Print is the wrong analogy, and it is the analogy a library handout is built on. A bound book has a publisher, a copyright page, a physical cost of production that once screened some of the worst actors. The live web has none of that screening at the point of encounter. Anyone can register a `.org`. Anyone can write an About page. Anyone can list an author, throw up a reference list, and hire a designer. Currency, in CRAAP, asks whether the page is recent — and a cloaked site will be recent. Authority asks who wrote it — and a cloaked site will name someone. Accuracy asks whether the information checks out *on the page* — and a cloaked site will be internally consistent. Purpose asks why the page exists — and a cloaked site will say education, or health, or "factual reports." The checklist is a scoring sheet for a medium that no longer has the gates the sheet assumes. Training students to fill it is training them to grade the costume.

College library pages recommended CRAAP because it looks like information literacy, and in a print world some of its cells were not crazy. The 2020 working paper's force is not that librarians

[377] Stanford Graduate School of Education news, "It doesn't take long to learn how to spot misinformation online, Stanford study finds," 19 April 2022, https://ed.stanford.edu/news/it-doesn-t-take-long-learn-how-spot-misinformation-online-stanford-study-finds. **Journalism / university communications** summarizing Wineburg et al. 2022. Source of "twice as well" / "twice as likely to spot questionable websites" and "earned only half the total number of possible points." Funded by Google.org. Not the paper.

[378] Sam Wineburg, Joel Breakstone, Nadav Ziv, and Mark Smith, *Educating for Misunderstanding: How Approaches to Teaching Digital Literacy Make Students Susceptible to Scammers, Rogues, Bad Actors, and Hate Mongers*, SHEG working paper A-21322 (2020), https://purl.stanford.edu/mf412bt5333. CRAAP / vertical-checklist critique. PDF fetch timed out; Stanford Digital Repository abstract and stacks snippet used. Do not invent extra percentages. Named as a working paper on first citation.

[379] Sam Wineburg, Joel Breakstone, Nadav Ziv, and Mark Smith, *Educating for Misunderstanding: How Approaches to Teaching Digital Literacy Make Students Susceptible to Scammers, Rogues, Bad Actors, and Hate Mongers*, SHEG working paper A-21322 (2020), https://purl.stanford.edu/mf412bt5333. CRAAP / vertical-checklist critique. PDF fetch timed out; Stanford Digital Repository abstract and stacks snippet used. Do not invent extra percentages. Named as a working paper on first citation.

are the adversary. It is that a best practice can become a vulnerability when the information ecology changes and the practice does not. McGrew's 2017 sentence is the classroom version of the same force. Author, reference list, no typos: cheap signals, high production value, low epistemic value. If the object of practice is those signals, students will get better at finding them. Getting better at finding fakeable signals is not civic reasoning. It is the misunderstanding the working paper is titled for.

McGrew and colleagues had already written the classroom sentence in 2017. Providing an author, a reference list, and a page free of typos does not establish credibility.[380] Currency can be faked with a recent date. Relevance can be faked with on-topic prose. Authority can be faked with a titled expert and a `.org`. Accuracy can be faked with internal consistency and a quantity of citations, some of them real. Purpose can be faked with an About page that says "education" or "factual reports." Every cell of CRAAP is a signal. Every one of those signals is cheap to produce on a cloaked site. The 2021 national portrait is what you get when you train 3,446 high-school students to inspect cheap signals and then hand them the live web.

The contrast is worth a table, labelled as this book's synthesis of the sources, not as a finding Wineburg signed in this form.

**Object of practice.** CRAAP: features of the page in front of you. Lateral reading: the organization's identity and the page's claims, checked off the page.

**Typical move.** CRAAP: read vertically; score currency, relevance, authority, accuracy, purpose. Lateral reading: leave; search the organization and its principals; ask COR's three questions.

**What adversaries can fake.** CRAAP: look, TLD, About page, author, reference list, recency, quantity of information, nonprofit costume. Lateral reading: they can still fake a first-order search result; they cannot as cheaply fake the *second* tab that finds a trade-association tie, a partisan sponsor, or a fact-checker's file.

**Trial evidence.** CRAAP: the failure mode in Breakstone 2021 and the working-paper claim in *Educating for Misunderstanding* (2020). Lateral reading: Wineburg and McGrew 2019 as the expert contrast; Wineburg et al. 2022 as the classroom trial; McGrew 2020 as the smaller intervention layer.

This is not a claim that all media-literacy education fails. It is a claim that *vertical, feature-based checklists* are the wrong object. A teacher who has students practice COR's three questions on live pages is not doing CRAAP under a new acronym. A teacher who has students score a page for typos and `.org` status is doing the 2017 failure mode, whatever the unit is named. Naming is not teaching. Chapter 5 already spent that sentence on standards documents. It returns here as a civic rule.

Hoax-site lessons fail for the same structural reason, and they should be retired from the "media literacy" drawer. The tree octopus is absurd on its face. Cloaked real sites are not. The student who laughs at a fake octopus and then trusts a fossil-fuel `.org` because it looks finished has not

[380] McGrew et al., "The Challenge That's Bigger Than Fake News." The quoted sentence is from that essay: "Providing an author, throwing up a reference list, and ensuring a site is free of typos hardly establishes it as a credible source."

failed a sense of humor. She has succeeded at the checklist. The live web rewards the adversary who invests in looking finished. The civic habit is to stop paying the page that attention until you know who is behind it.

Willingham again, because the maxim is tempting. "Consider the source" is a maxim. Without the knowledge of what a source is, and without the practiced move of leaving, it will not be executed. "Look for .org" is worse than a maxim. It is a fakeable signal taught as a skill. Lateral reading is closer to what this book can honestly recommend: a short heuristic, a domain (civic information on the live web), and practice on real pages. Even then, six hours doubles a still-failing score. The honest book prints that remainder.

## Inoculation works on techniques, decays without boosters

Technique inoculation is a different object again. It is not evaluation of a page. It is resistance to a *move* — emotionally manipulative language, incoherence, false dichotomies, scapegoating, ad hominem, fake experts — delivered as a weakened dose before the live encounter. William McGuire's 1960s medical analogy is the ancestor: forewarning plus a weakened dose produces resistance.[381] John Cook, Stephan Lewandowsky, and Ullrich Ecker, in *PLOS ONE* in 2017, showed that exposing misleading *argumentation techniques* reduced their later influence. Technique-based rather than claim-based, which is why the method can scale past a single rumor.[382] Sander van der Linden's climate-inoculation line (2017, with Leiserowitz, Rosenthal, and Maibach in *Global Challenges*, and a *Science* letter the same year) is the domain-specific cousin.[383] Lewandowsky and van der Linden's 2021 review in the *European Review of Social Psychology* is the honest summary at that date: inoculation is promising; it is not a completed public-health program.[384]

Roozenbeek and van der Linden's *Bad News* game (2019; Basol, Roozenbeek, and van der Linden 2020) is the active, gamified version: players learn the techniques by using them in a simulated feed.[385] Maertens, Roozenbeek, Basol, and van der Linden, in the *Journal of Experimental Psychology: Applied* in 2021, ran three longitudinal experiments on that game. Effects remained with regular testing for months. They decayed to nonsignificance over two months without it. The long-term effect was no longer significant in the no-testing condition.[386] Boosters matter. A one-shot

[381] William J. McGuire's 1960s inoculation theory is the medical-analogy ancestor (forewarning + weakened dose). Cited here as background named in the later empirical papers, not as a new fetch of the 1960s experiments.

[382] John Cook, Stephan Lewandowsky, and Ullrich K. H. Ecker, "Neutralizing Misinformation through Inoculation: Exposing Misleading Argumentation Techniques Reduces Their Influence," *PLOS ONE* 12, no. 5 (2017): e0175799, https://doi.org/10.1371/journal.pone.0175799. Technique-based inoculation. Open access.

[383] Sander van der Linden, Anthony Leiserowitz, Seth Rosenthal, and Edward Maibach, climate-inoculation work in *Global Challenges* (2017); van der Linden, Maibach, Cook, Leiserowitz, and Lewandowsky, "Inoculating against Misinformation," *Science* 358, no. 6367 (2017): 1141–42, https://doi.org/10.1126/science.aar4533. Letter/comment, not a full RCT report.

[384] Stephan Lewandowsky and Sander van der Linden, "Countering Misinformation and Fake News Through Inoculation and Prebunking," *European Review of Social Psychology* 32, no. 2 (2021): 348–84, https://doi.org/10.1080/10463283.2021.1876983. Review: promising, not a completed public-health program.

[385] Jon Roozenbeek and Sander van der Linden, "The Fake News Game: Actively Inoculating against the Risk of Misinformation," *Journal of Risk Research* (2019), https://doi.org/10.1080/13669877.2018.1443491. Melisa Basol, Roozenbeek, and van der Linden, 2020, is the accompanying gamified-inoculation paper in this line.

[386] Rakoen Maertens, Jon Roozenbeek, Melisa Basol, and Sander van der Linden, "Long-Term Effectiveness of Inoculation against Misinformation: Three Longitudinal Experiments," *Journal of Experimental Psychology: Applied* 27,

"media literacy day" is not this literature. Later work on "psychological booster shots" (*Nature Communications*, 2025) argues that memory, not only threat-motivation, drives longevity; this book treats that paper's coefficients as **UNVERIFIED** pending a full fetch. The decay finding to print is Maertens 2021.[387]

The load-bearing field paper is Roozenbeek, van der Linden, Beth Goldberg, Steve Rathje, and Lewandowsky, *Science Advances*, 2022. Five short videos, about ninety seconds each, against emotionally manipulative language, incoherence, false dichotomies, scapegoating, and ad hominem. Humorous, nonpolitical microdoses. A Google Jigsaw partnership. Six laboratory studies (five videos plus a replication of the emotional-language video), $n$ = 6,464. A YouTube field study, $n$ = 22,632 people who answered a single embedded item; roughly 967,000 watched at least thirty seconds. Laboratory Cohen's $d$ for technique recognition ranged from 0.28 (scapegoating) to 0.68 (false dichotomies). In the five-video series, sixteen of twenty preregistered laboratory outcomes were significant (nineteen of twenty-three when the replication is counted). The field effect, combined across six items, was $h$ = 0.09, narrowly missing the authors' preregistered smallest effect size of interest, which was 0.10. Cost was about five cents per thirty-second view. Robust across ideology in their moderation tests.[388]

Read those numbers without laundering them. Lab effect: medium, on *technique recognition*, in controlled experiments. Field effect: small, on a single embedded item, in a YouTube sample, narrowly missing the authors' own SESOI. Neither is a voting RCT. Neither is durable "critical thinking skill." Neither immunizes the public. The medical metaphor is a metaphor. The honest description is: a cheap, scalable, technique-focused prebunk, with a lab effect you would take and a field effect you would not overclaim, perishable without boosters.

Cohen's $d$ of 0.28 to 0.68 is not Abrami's $g$+ of 0.30, is not the dialogue-plus-authentic-plus-mentoring $g$ of 0.57, is not Roozenbeek's own field $h$ of 0.09, and this book will not average them. Chapter 5 already forbade mashing effect sizes. The inoculation numbers measure technique recognition after short videos. Abrami's numbers measure standardized critical-thinking tests after instructional interventions. Wineburg 2022 measures SHEG credibility tasks after six government lessons. Different objects, different outcomes, different claims. A superintendent who reports "we do media literacy, effect size medium" has not read any of those papers. The lab videos are worth running *as technique inoculation*. They are not a substitute for leaving the page, and they are not a K–12 thinking curriculum.

McGuire's analogy remains useful if it is kept in its place. A weakened dose of a technique, with a

no. 1 (2021), https://doi.org/10.1037/xap0000315. Bad News versus Tetris; stable with regular testing for months; significant decay over two months without it; long-term effect no longer significant in the no-testing condition. Abstract and search fetched; full PDF not re-read on this pass.

[387]"Psychological booster shots" work in *Nature Communications* (2025): coefficients **UNVERIFIED** pending a full fetch. Print Maertens 2021 for decay. Do not launder 2025 coefficients into body prose.

[388]Jon Roozenbeek, Sander van der Linden, Beth Goldberg, Steve Rathje, and Stephan Lewandowsky, "Psychological Inoculation Improves Resilience against Misinformation on Social Media," *Science Advances* 8, no. 34 (2022): eabo6254, https://pmc.ncbi.nlm.nih.gov/articles/PMC9401631/. Fetched in full via PMC 30 August 2026; re-read this pass. Six laboratory studies (five videos plus a replication), $n$ = 6,464; YouTube field $n$ = 22,632 (~967,000 watched ≥30 seconds). Lab technique-recognition $d$ = 0.28 (scapegoating) to 0.68 (false dichotomies). Studies 1–5: 16 of 20 preregistered outcomes significant; studies 1–6: 19 of 23. Field $h$ = 0.09, narrowly missing preregistered SESOI of 0.10. Cost ~$0.05 per 30-second view. Google Jigsaw funded. Open access.

warning that the technique exists, can raise resistance to that technique later. Cook, Lewandowsky, and Ecker's contribution was to inoculate against *moves* (fake experts, among others) rather than against a single claim, which is why a ninety-second video about false dichotomies can, in principle, travel past one rumor. The limit is the same limit every other chapter has met. Resistance to a named move, measured soon after, in a lab or on an embedded YouTube item, is not a disposition to evaluate the next cloaked site, and it is not a civic identity. Boosters are what keep the resistance from decaying. A school that cannot schedule boosters should not advertise immunization.

Two further constraints. First, technique recognition is not page evaluation. A student who can name "false dichotomy" in a ninety-second video has not yet left a cloaked `.org` and found a funder. COR and inoculation can sit in the same civic toolkit. They are not the same intervention and they do not have the same outcome. Second, Jigsaw funding, like Google.org funding of the COR district study, is a fact to print. The paper is open access; the design is preregistered; the field effect missed the SESOI and the authors said so. That is how you read an industry-adjacent RCT. You do not discard it, and you do not promote $h = 0.09$ into a public-health victory.

The decay finding is the one a school system should actually hear. Maertens 2021: with regular testing, months; without it, two months to nonsignificance.[389] Wineburg 2022: six lessons, immediate-ish post-test, remaining miss rate high.[390] A district that buys a one-hour assembly, a poster, and a video series, and then declares the civic job done, has not implemented this literature. It has implemented a slogan with a guest speaker. Willingham's revisiting rule — three to five years, because even well-learned content is about half forgotten in three years — applies to technique recognition as much as it applies to historical thinking.[391] Boosters are not a marketing upsell. They are what the longitudinal experiments found.

## Accuracy nudges, and a published dispute

Gordon Pennycook and David Rand's limited-attention account is narrower still, and it is the one most often inflated into a belief vaccine. The claim, in their papers, is not that people are ignorant of accuracy or that they prefer falsehood. Most people say they want to share only accurate news. Veracity of headlines has a large effect on accuracy *judgments* and little effect on sharing *intentions*. The feed distracts them. A prompt that puts accuracy in working memory raises the quality of

[389] Rakoen Maertens, Jon Roozenbeek, Melisa Basol, and Sander van der Linden, "Long-Term Effectiveness of Inoculation against Misinformation: Three Longitudinal Experiments," *Journal of Experimental Psychology: Applied* 27, no. 1 (2021), https://doi.org/10.1037/xap0000315. Bad News versus Tetris; stable with regular testing for months; significant decay over two months without it; long-term effect no longer significant in the no-testing condition. Abstract and search fetched; full PDF not re-read on this pass.

[390] Sam Wineburg, Joel Breakstone, Sarah McGrew, Mark D. Smith, and Teresa Ortega, "Lateral Reading on the Open Internet: A District-Wide Field Study in High School Government Classes," *Journal of Educational Psychology* 114, no. 5 (2022): 893–909, https://doi.org/10.1037/edu0000740. Urban midwestern district; six high schools; required government course, spring 2019; six ~50-minute COR lessons over three months; treatment $n = 271$, matched control $n = 228$ (published abstract). COR project page and SSRN version: cluster-randomized, repeated-measures analysis. Abstract and COR project page fetched 30 August 2026; ERIC abstract re-checked 30 August 2026. Do not upgrade "grew significantly" into a civic-behavior RCT.

[391] Daniel T. Willingham, "How to Teach Critical Thinking," NSW Department of Education, 2019. Four-step plan: identify what CT means in each domain; identify needed content; sequence; plan three to five years of revisiting. Pedagogical constraint, not a completed COR sequence.

subsequent shares, mostly by cutting false shares. Sharing discernment is the outcome. Belief is not.[392]

Pennycook, Jonathon McPhetres, Yunhao Zhang, Jackson Lu, and Rand, in *Psychological Science* in 2020, is the COVID-era accuracy-nudge paper.[393] Pennycook, Ziv Epstein, Mohsen Mosleh, Antonio Arechar, Dean Eckles, and Rand, in *Nature* in 2021, is the load-bearing one: four survey experiments plus a Twitter field experiment. "People often share misinformation because their attention is focused on factors other than accuracy." A subtle accuracy prompt raises the quality of subsequent shares.[394] Pennycook and Rand, in *Nature Communications* in 2022, argue the prompt is replicable and generalizable.[395] Their 2022 *ANNALS of the American Academy of Political and Social Science* review is the authors' own caveat list: persistence and habituation are open; cross-cultural work is thin.[396]

What this literature does not license is the sentence "accuracy nudges work for everyone" or the sentence "accuracy nudges change what people believe." They change, at the margin, what inattentive people *share*. Partisan moderation is a live dispute, not a settled moderator. Steve Rathje, Roozenbeek, Cecilie Traberg, Jay Van Bavel, and van der Linden sent a letter regarding the 2020 paper: a meta-analysis, they argued, reveals little or no effect for U.S. conservatives.[397] That is a published dispute, not a consensus finding. An adversarial collaboration, published in *Psychological Science* in 2024, treats partisanship as the unresolved moderator. The abstract and search results were fetched for this book; coefficients will not be invented.[398] Do not write that the nudge works across the aisle. Do not write that it fails for half the country. Write that the authors who disagree ran a collaboration and that the question is not closed.

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[392] Gordon Pennycook, Ziv Epstein, Mohsen Mosleh, Antonio A. Arechar, Dean Eckles, and David G. Rand, "Shifting Attention to Accuracy Can Reduce Misinformation Online," *Nature* 592 (2021): 590–95, https://www.nature.com/articles/s41586-021-03344-2. Four survey experiments plus a Twitter field experiment. Sharing intentions are not accuracy judgments. Competing-interest note on the paper: Rand gift from Google; Eckles Facebook funding/interest. Fetched 30 August 2026 (abstract and open figures/notes).

[393] Gordon Pennycook, Jonathon McPhetres, Yunhao Zhang, Jackson G. Lu, and David G. Rand, "Fighting COVID-19 Misinformation on Social Media: Experimental Evidence for a Scalable Accuracy-Nudge Intervention," *Psychological Science* 31, no. 7 (2020): 770–80, https://doi.org/10.1177/0956797620939054.

[394] Gordon Pennycook, Ziv Epstein, Mohsen Mosleh, Antonio A. Arechar, Dean Eckles, and David G. Rand, "Shifting Attention to Accuracy Can Reduce Misinformation Online," *Nature* 592 (2021): 590–95, https://www.nature.com/articles/s41586-021-03344-2. Four survey experiments plus a Twitter field experiment. Sharing intentions are not accuracy judgments. Competing-interest note on the paper: Rand gift from Google; Eckles Facebook funding/interest. Fetched 30 August 2026 (abstract and open figures/notes).

[395] Gordon Pennycook and David G. Rand, "Accuracy Prompts Are a Replicable and Generalizable Approach for Reducing the Spread of Misinformation," *Nature Communications* 13 (2022): 2333, https://doi.org/10.1038/s41467-022-30073-5. Authors' replication and generalization claim. Not a belief-change paper.

[396] Gordon Pennycook and David G. Rand, "Nudging Social Media toward Accuracy," *ANNALS of the American Academy of Political and Social Science* 700, no. 1 (2022), https://doi.org/10.1177/00027162221092342. Review; flags persistence, habituation, and cross-cultural gaps.

[397] Steve Rathje, Jon Roozenbeek, Cecilie S. Traberg, Jay J. Van Bavel, and Sander van der Linden, "Letter to the Editors of *Psychological Science*: Meta-Analysis Reveals that Accuracy Nudges Have Little to No Effect for U.S. Conservatives," 2022, https://doi.org/10.25384/SAGE.12594110.v2. Published dispute regarding Pennycook et al. 2020, not a consensus finding.

[398] Adversarial collaboration on partisan moderation of accuracy prompts, *Psychological Science* (2024), https://doi.org/10.1177/09567976241232905. Abstract and search fetched; **do not invent coefficients**. Treats partisanship as unresolved.

A neighboring intervention must not be mashed with this one. Mosleh, Cameron Martel, Eckles, and Rand, at CHI 2021, found *perverse* effects of being *publicly* corrected on Twitter: increased later sharing of low-quality partisan content.[399] That is social correction, in public, with identity on the line. It is not a private accuracy prompt that asks, before you share, whether the headline is accurate. Kahan's work in Chapter 4 already warned that Type 2 reasoning can do identity work. A public correction is a status event. A private prompt is an attention event. Collapsing them into "corrections backfire on social media" is how a CHI paper and a *Nature* paper get laundered into a reason to do nothing. They are different interventions. This chapter will not mash them.

Andrew Guess, Michael Lerner, Benjamin Lyons, Jacob Montgomery, Brendan Nyhan, Jason Reifler, and Neelanjan Sircar, in *PNAS* in 2020, is the neighboring "tips" paper, not a Pennycook nudge. Facebook-style digital media literacy tips — the platform's own "tips to spot false news" — improved discernment 26.5 percent in a U.S. national sample and 17.5 percent in a highly educated Indian online sample. The U.S. gain remained measurable several weeks later; the Indian online gain did not. There was **no effect** in a largely rural northern Indian sample with low social-media use.[400] Authors' own caveats: modest; did not eliminate belief in false headlines; needs reinforcement. Checklist tips help a little, fade, and fail where the information ecology is different. That is the closest thing this chapter has to a cross-cultural constraint, and it is the opposite of a universal media-literacy law. A tip sheet that moves discernment in a U.S. Facebook sample is not a civic education for a population that does not live on that feed.

Put the three objects on the table so they cannot collapse. Lateral reading: leave the page; who, evidence, other sources; classroom trial; remaining miss rate high. Inoculation: technique recognition; lab $d$ 0.28–0.68; YouTube $h$ = 0.09; decays without boosters. Accuracy nudge: sharing discernment under inattention; not belief; partisan moderation disputed. A law, a NAMLE principle, or a school-mission poster that says "media literacy" has not chosen among them. Choosing is the civic-education job. Implicit expectation — the hope that a feed, a government class, or a library guide will produce evaluation as a by-product — is the intervention Abrami 2008 already ruled out.[401]

Limited attention is a more respectful theory of the share button than ignorance, and it is also a more limited intervention. If people mostly prefer accuracy, and the feed distracts them, then a prompt is a working-memory event, not a lesson. It does not teach who is behind a site. It does not teach what would count as evidence. It does not travel with the person into a clinic or a jury room. It is a nudge at the point of sharing, for people who were going to share without asking whether the

[399] Mohsen Mosleh, Cameron Martel, Dean Eckles, and David G. Rand, CHI 2021. Public social correction on Twitter; perverse effects (increased later sharing of low-quality partisan content). A different intervention from a private accuracy prompt. Do not mash.

[400] Andrew M. Guess, Michael Lerner, Benjamin Lyons, Jacob M. Montgomery, Brendan Nyhan, Jason Reifler, and Neelanjan Sircar, "A Digital Media Literacy Intervention Increases Discernment between Mainstream and False News in the United States and India," *Proceedings of the National Academy of Sciences* 117, no. 27 (2020): 15536–45, https://doi.org/10.1073/pnas.1920498117. Facebook-style tips; +26.5 percent U.S. discernment; +17.5 percent educated India online; U.S. gain still measurable several weeks later, Indian online gain not; **null** in rural northern India. PNAS full PDF timed out; abstract via PubMed/DOI used this pass.

[401] Philip C. Abrami et al., "Instructional Interventions Affecting Critical Thinking Skills and Dispositions: A Stage 1 Meta-Analysis," *Review of Educational Research* 78, no. 4 (2008). Headline already spent in Chapters 4–5: improvement "cannot be a matter of implicit expectation." Cited here as the civic application of that constraint.

headline was true. That is worth doing at the margin of a platform. It is not a reason to skip COR in a government class, and it is not a reason to declare the public inoculated. The Pennycook and Rand program, read honestly, is an argument *against* collapsing sharing discernment with belief change, and *against* collapsing a prompt with an education. This chapter will take them at their word.

## Backfire is rare; continued influence is real; knowledge does not protect against fluency

The surviving reason people skip correction is a finding that did not survive. Brendan Nyhan and Jason Reifler, in *Political Behavior* in 2010, found, in limited conditions, that corrections could strengthen misperceptions among ideological subgroups. The Iraq WMD item is the famous one. The finding was never uniform across their own experiments.[402] It became, in public conversation, a reason not to correct: you will only make it worse. That is the sentence this section exists to retire.

Thomas Wood and Ethan Porter, in *Political Behavior* in 2019, ran five experiments, more than 10,000 participants, fifty-two issues. **No item triggered a worldview backfire.**[403] Nyhan's own later writing, in *PNAS* in 2020, said that interpretations of the 2010 paper outstripped the findings, and that backfire effects are "extremely rare in practice." The quoted phrasing is from search snippets; the PNAS full PDF was Cloudflare-blocked on this book's fetch pass and is **UNVERIFIED at full-PDF**. Durability of misperceptions, Nyhan argued in that paper as summarized in secondary sources and as consistent with the handbook treated next, is better explained by not reaching people, elite cues, and partisan heuristics than by ironic strengthening.[404] *The Debunking Handbook 2020* — twenty-two authors, including Nyhan's co-author Reifler, Pennycook, Rand, van der Linden, Lewandowsky, Briony Swire-Thompson, and Porter, fetched in full — is the consensus sentence to print: "Do not refrain from attempting to debunk or correct misinformation out of fear that doing so will backfire."[405] Swire-Thompson, Joseph DeGutis, and David Lazer, in 2020, concluded that backfire is not a robust empirical phenomenon.[406] Familiarity backfire and overkill backfire also

[402] Brendan Nyhan and Jason Reifler, "When Corrections Fail: The Persistence of Political Misperceptions," *Political Behavior* 32, no. 2 (2010): 303–30, https://doi.org/10.1007/s11109-010-9112-2. Origin of the popular backfire claim. Limited items and subgroups; authors later revised the interpretation.

[403] Thomas Wood and Ethan Porter, "The Elusive Backfire Effect: Mass Attitudes' Steadfast Factual Adherence," *Political Behavior* 41 (2019): 135–63, https://doi.org/10.1007/s11109-018-9443-y. Five experiments, more than 10,000 participants, 52 issues; no worldview backfire.

[404] Brendan Nyhan, "Why the Backfire Effect Does Not Explain the Durability of Political Misperceptions," *Proceedings of the National Academy of Sciences* 117, no. 21 (2020), https://www.pnas.org/doi/10.1073/pnas.1912440117. Author's revision: backfire extremely rare in practice; durability better explained by targeting failures and elite cues. PNAS fetch Cloudflare-blocked on this pass; quoted phrasing ("extremely rare in practice") from search snippets plus overlap with *The Debunking Handbook 2020*. **UNVERIFIED at full-PDF.**

[405] Stephan Lewandowsky, John Cook, Ullrich K. H. Ecker, et al., *The Debunking Handbook 2020*, DOI 10.17910/b7.1182, https://www.climatechangecommunication.org/wp-content/uploads/2023/09/DebunkingHandbook2020.pdf. Twenty-two-author consensus; fetched in full 30 August 2026. Sentence to print: "Do not refrain from attempting to debunk or correct misinformation out of fear that doing so will backfire." Continued-influence and illusory-truth sections used. Familiarity backfire and overkill backfire also failed later tests, per the handbook.

[406] Briony Swire-Thompson, Joseph DeGutis, and David Lazer, "Searching for the Backfire Effect: Measurement

failed later tests; the handbook says so.[407]

What survived is the continued-influence effect. People can accept a correction and still use the myth in adjacent reasoning. The Johnson and Seifert 1994 line is the laboratory ancestor; the 2020 handbook is the consensus restatement.[408] A reader can say "I know that is false" and still, three paragraphs later, reason as if the retracted cause were in place. That is not ironic strengthening of the original belief. It is a sticky causal slot. The recommended repair, in the handbook, is a detailed refutation — fact, myth, fallacy, fact — plus a *causal alternative*, not a simple "that's false." If you only negate, the slot stays empty and the myth returns to fill it. If you offer a true cause that does the explanatory work, the slot has somewhere else to go.[409]

Spend the pages on what *to* do, because the veto is already in Chapter 1 and this chapter is a life chapter. Correct. Do not skip for fear of ironic strengthening. Use a causal alternative, not a bare negation. Slow down; consider plausibility against alternatives — the handbook's own slowing rule, which is cousin to Pennycook and Rand's accuracy prompt.[410] Do not confuse a private correction with a public status fight. Do not expect a factual correction to move a vote in a polarized setting; Swire-Thompson and colleagues have already warned that corrections often fail to move votes even when they move facts a little.[411] The deed, which Chapter 10 will list, is still to say "I was wrong about the number." A critical-thinking identity that refuses to correct, on the strength of a 2010 finding the literature has walked back, is weak-sense sophistry of the kind Richard Paul named in 1981, and it does not become civic wisdom by being fashionable.[412]

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and Design Considerations," *Journal of Applied Research in Memory and Cognition* 9, no. 3 (2020): 286–99, https://doi.org/10.1016/j.jarmac.2020.06.006. Open access. Backfire is not a robust empirical phenomenon.

[407] Stephan Lewandowsky, John Cook, Ullrich K. H. Ecker, et al., *The Debunking Handbook 2020*, DOI 10.17910/b7.1182, https://www.climatechangecommunication.org/wp-content/uploads/2023/09/DebunkingHandbook2020.pdf. Twenty-two-author consensus; fetched in full 30 August 2026. Sentence to print: "Do not refrain from attempting to debunk or correct misinformation out of fear that doing so will backfire." Continued-influence and illusory-truth sections used. Familiarity backfire and overkill backfire also failed later tests, per the handbook.

[408] Hollyn M. Johnson and Colleen M. Seifert, continued-influence line (1994); restated in *The Debunking Handbook 2020*. People can accept a correction and still use the myth in adjacent reasoning. Detailed refutation (fact–myth–fallacy–fact) plus a causal alternative outperforms a simple "that's false."

[409] Stephan Lewandowsky, John Cook, Ullrich K. H. Ecker, et al., *The Debunking Handbook 2020*, DOI 10.17910/b7.1182, https://www.climatechangecommunication.org/wp-content/uploads/2023/09/DebunkingHandbook2020.pdf. Twenty-two-author consensus; fetched in full 30 August 2026. Sentence to print: "Do not refrain from attempting to debunk or correct misinformation out of fear that doing so will backfire." Continued-influence and illusory-truth sections used. Familiarity backfire and overkill backfire also failed later tests, per the handbook.

[410] Stephan Lewandowsky, John Cook, Ullrich K. H. Ecker, et al., *The Debunking Handbook 2020*, DOI 10.17910/b7.1182, https://www.climatechangecommunication.org/wp-content/uploads/2023/09/DebunkingHandbook2020.pdf. Twenty-two-author consensus; fetched in full 30 August 2026. Sentence to print: "Do not refrain from attempting to debunk or correct misinformation out of fear that doing so will backfire." Continued-influence and illusory-truth sections used. Familiarity backfire and overkill backfire also failed later tests, per the handbook.

[411] Swire-Thompson and colleagues on corrections often failing to move votes in polarized settings even when factual belief moves a little: the 04-life memo's reading of this line, used here as a constraint on what a correction is for. Do not upgrade a factual correction into a turnout or vote-choice result.

[412] Richard Paul, 1981, strong-sense versus weak-sense: students trained on " 'egocentrically' neutral" cases "become more sophistic rather than less so." Hitchcock's "more able sophists" is a paraphrase. Stolen as a warning, not as a Foundation franchise. Already in Chapter 2.

The fact–myth–fallacy–fact sequence is a representation, not a vibe. Lead with the fact, because the first slot is the one fluency will keep. Name the myth, explicitly, so the reader knows which sentence is being retired. Name the fallacy or the move that made the myth travel — a fake expert, a false dichotomy, a relative risk standing in for an absolute one — because technique recognition is what inoculation actually moves. Return to the fact, and supply a true cause that does the explanatory work the myth was doing. A bare "that's false" leaves a hole. Holes recruit the old filler. The handbook's consensus is not that correction always works. It is that fear of backfire is the wrong reason to skip it, and that a causal alternative is the repair continued influence requires. Nyhan 2020, with the full-PDF flag still on, is the author's own walk-back of the popular 2010 reading. Wood and Porter is the large-N demonstration that worldview backfire was elusive across fifty-two issues. The 2020 handbook is the sentence a teacher, a journalist, and a public-health office can steal without waiting for a civic-turnout RCT.

Illusory truth is the other surviving mechanism, and it is why fluency — including *machine* fluency — is a civic problem. Repetition increases judged truth even when you know better. Hasher, Goldstein, and Toppino (1977) named the effect; Fazio, Brashier, Payne, and Marsh (2015) showed that *knowledge does not protect*.[413] You can know the capital, know the fact, and still rate the repeated falsehood as more true. That is not a deficit of information. It is a fluency heuristic: the repeated sentence comes to mind easily, and ease is misread as evidence. A cloaked `.org` that looks finished is a fluency trick in design. A model that finishes a paragraph is a fluency trick in syntax. Chapter 9 will spend the machine. Here the civic claim is smaller and prior: a page that reads fluently, a headline seen twice, a video that "just shows you," will collect truth-ratings the author's knowledge cannot veto. Lateral reading is, among other things, a way of interrupting fluency before it hardens. You leave before the prose has time to feel true.

The continued-influence effect and illusory truth together explain why six COR lessons can double a score and still leave half the points on the table, and why an accuracy nudge can cut false shares without changing what people believe. Evaluation is effortful. Fluency is cheap. A myth that has filled a causal slot will be used in the next sentence even after a correction has been nodded at. The civic habit is not a one-time immunization. It is practiced interruption: leave the page, put accuracy in working memory, offer a true cause when you correct, and do not expect knowledge alone to protect you from a sentence you have seen before.

## Twenty-five state laws are laws; NAMLE is a framework; classrooms have not been evaluated

Policy said the words. Policy is not Civic Online Reasoning.

The National Association for Media Literacy Education's Core Principles, 2023 revision, fetched from namle.org on 30 August 2026, define media literacy education as the habits of inquiry and skills of expression necessary for people to be critical thinkers, thoughtful and effective communi-

[413]Lynn Hasher, David Goldstein, and Thomas Toppino, 1977, illusory-truth origin; Lisa K. Fazio, Nadia M. Brashier, B. Keith Payne, and Elizabeth J. Marsh, "Knowledge Does Not Protect against Illusory Truth," *Journal of Experimental Psychology: General* 144, no. 5 (2015): 993, https://doi.org/10.1037/xge0000098. Cited via *The Debunking Handbook 2020* (fetched). Knowledge does not protect. Why fluency, including machine fluency, is a civic problem (Chapter 9).

cators, and informed and responsible members of society. Principle 3: curious, open-minded, and self-reflective inquiry while emphasizing reason, logic, and evidence. Principle 4 asks learners to "practice active inquiry, reflection, and critical thinking." Principle 10.4 "incorporates specific approaches for helping individuals to identify quality, reliable, and accurate information." The 2023 revision committee lists SHEG among sources drawn on.[414] That is a professional association's position. It is not a trial. NAMLE does not report an RCT on that page. A framework can name the right family of habits. It cannot stand in for Wineburg's three questions practiced on live pages, or for Roozenbeek's videos, or for Pennycook's prompt. Principle 4's use of "critical thinking" is slogan language in a standards-adjacent document unless a teacher has chosen an object of practice. This book has already spent, in Chapters 1 and 5, what happens when the phrase is stamped on a syllabus without that choice. Immersion-by-committee does not become civic education by being adopted at a membership meeting.

Media Literacy Now's policy map, last updated January 2026, and its *U.S. Media Literacy Policy & Impact Report*, looking through 30 October 2025, are the legislative count this book will print. **Twenty-five states** have media-literacy laws on the books. Eleven took new steps since January 2024. Many pair phone and social-media restrictions with instructional requirements.[415] MLN states explicitly that it has **not evaluated** how state departments integrated the laws into standards or classrooms. The map is legislative process. It is not a classroom outcome. PR Newswire, 15 January 2026 — **journalism / press release** — calls this a "policy inflection point," with "more than half of U.S. states" having taken legislative action "even as implementation struggles to keep up."[416] Print the twenty-five-state count from MLN. Print implementation-unknown from MLN. Do not print "students in twenty-five states can now evaluate news."

State examples commonly named in secondary roundups — Illinois's high-school unit (2021), New Jersey's K–12 requirement (2022), Delaware's Digital Citizenship Education Act (2022) — are named here as legislative examples in those roundups. Statutory details are **UNVERIFIED** pending the bill text.[417] This book will not narrate what an Illinois government teacher did on a Tuesday. It will not invent a New Jersey scope-and-sequence. It will not treat a Delaware act's title as evidence that students there leave the page. Common Core and state ELA and social-studies standards routinely use "critical thinking" as a goal word. That is slogan language in a standards document. It is not Wineburg's three questions, not an inoculation RCT, and not an accuracy prompt.

[414] National Association for Media Literacy Education, "Core Principles of Media Literacy Education," 2023 revision, https://namle.org/resources/core-principles/, fetched in full 30 August 2026. Professional-association framework, not an RCT. SHEG listed among sources drawn on. Principle 4 and principle 10.4 quoted from that page.

[415] Media Literacy Now, "Current Policy" map, https://medialiteracynow.org/impact/current-policy/, fetched 30 August 2026. Last updated January 2026. Twenty-five states with laws; eleven new steps since January 2024. MLN states it has not evaluated implementation. Companion: Media Literacy Now, *U.S. Media Literacy Policy & Impact Report* (through 30 October 2025; released January 2026), https://medialiteracynow.org/wp-content/uploads/2026/01/MLN_Policy_Impact_Report-1.2026.pdf. Legislative tracking, not classroom outcomes.

[416] PR Newswire / Media Literacy Now, "New U.S. Media Literacy Report Finds State Policy Accelerating as Lawmakers Respond to Phones, Social Media, and AI," 15 January 2026, https://www.prnewswire.com/news-releases/new-us-media-literacy-report-finds-state-policy-accelerating-as-lawmakers-respond-to-phones-social-media-and-ai-302662137.html. **Journalism / press release.** Use only as a pointer to the MLN report.

[417] Illinois high-school unit (2021), New Jersey K–12 (2022), and Delaware Digital Citizenship Education Act (2022) are named in secondary roundups (Ad Fontes; Population Education — journalism/advocacy, not statutes fetched on this pass). Statutory details **UNVERIFIED** pending bill text.

A law can do useful work without being a trial. It can fund professional development, name a course as the home, require live-web practice rather than a worksheet about a tree octopus, and forbid the substitution of a filter for a lesson. It can also do nothing but add a phrase to a standards document and a box to an audit. Media Literacy Now's own disclaimer is the constraint: the map tracks legislative process; it does not evaluate classrooms. Eleven states took new steps since January 2024, some pairing phone and social-media restrictions with instructional requirements. A phone ban is a time-and-attention policy. An instructional requirement is a curriculum policy. They are not the same instrument, and neither is COR unless the requirement specifies the object of practice. Pairing them in a statute is a political fact. It is not evidence that students in those states can now distinguish a cloaked `.org` from a reporting organization.

NAMLE's 2023 revision is closer to the right family than a "21st-century skills" poster, and the fact that SHEG is listed among sources drawn on is worth printing. A professional association that names inquiry, evidence, and identifying reliable information is not the enemy of this chapter. The enemy is the collapse of that naming into an outcome. Principle 4's "critical thinking" does not become Wineburg's three questions by sharing a vocabulary. Principle 10.4's "quality, reliable, and accurate information" does not become lateral reading by sharing a goal. Frameworks are allowed to be frameworks. This book's job is to refuse the upgrade.

Guess et al. 2020 is the empirical neighbor that policy should have to sit with. Even a well-designed tips campaign is modest and perishable, and it is null where the information ecology is different.[418] Wineburg 2022 is what six hours of *lateral reading practice in a government class* does to a test of the same skill.[419] Those are different objects. A law that says "teach critical thinking / media literacy" has named a family. It has not chosen among CRAAP, COR, a one-shot assembly, a Jigsaw video, a Facebook tips card, and a poster. An audit of what classrooms actually do — COR versus CRAAP versus "be a critical thinker" on the wall — would change what this book can claim about policy. That audit has not been done. Media Literacy Now has not done it. This book will not fill the gap with a hopeful inference from the existence of statutes.

The contrast the chapter can print, then, is architectural rather than sarcastic. Frameworks explain demand. NAMLE is demand, in professional-association prose, for inquiry and for identifying reliable information. Twenty-five laws are demand, in statutory prose, for something called media literacy. Trials measure objects. COR lessons measure leaving the page. Inoculation videos measure technique recognition. Accuracy prompts measure sharing discernment. Tips cards measure a little discernment that fades. A policymaker who cannot tell those objects apart will fund the

[418] Andrew M. Guess, Michael Lerner, Benjamin Lyons, Jacob M. Montgomery, Brendan Nyhan, Jason Reifler, and Neelanjan Sircar, "A Digital Media Literacy Intervention Increases Discernment between Mainstream and False News in the United States and India," *Proceedings of the National Academy of Sciences* 117, no. 27 (2020): 15536–45, https://doi.org/10.1073/pnas.1920498117. Facebook-style tips; +26.5 percent U.S. discernment; +17.5 percent educated India online; U.S. gain still measurable several weeks later, Indian online gain not; **null** in rural northern India. PNAS full PDF timed out; abstract via PubMed/DOI used this pass.

[419] Sam Wineburg, Joel Breakstone, Sarah McGrew, Mark D. Smith, and Teresa Ortega, "Lateral Reading on the Open Internet: A District-Wide Field Study in High School Government Classes," *Journal of Educational Psychology* 114, no. 5 (2022): 893–909, https://doi.org/10.1037/edu0000740. Urban midwestern district; six high schools; required government course, spring 2019; six ~50-minute COR lessons over three months; treatment $n = 271$, matched control $n = 228$ (published abstract). COR project page and SSRN version: cluster-randomized, repeated-measures analysis. Abstract and COR project page fetched 30 August 2026; ERIC abstract re-checked 30 August 2026. Do not upgrade "grew significantly" into a civic-behavior RCT.

acronym and get the checklist. A teacher who can tell them apart still needs civic knowledge, live pages, and years of revisiting. Willingham's constraint does not relax because a legislature voted.

Equity belongs in this section as a constraint, not as a closing hymn. Breakstone 2021 already found that civic-online-reasoning scores varied by socioeconomic status, race, maternal education, and free-or-reduced-price lunch.[420] If the public skill is lateral reading, and if affluent students are more likely to have been given the knowledge that makes "who's behind this" answerable, then a law that mandates "media literacy" without specifying the object of practice will not close that gradient. It may widen it. Checklist literacy that trains `.org`-trust is cheaper to implement than COR in a government course with professional development. Cheap implementation is what under-resourced schools are handed. *Educating for Misunderstanding* is the paper that says those best practices can make students more susceptible.[421] Chapter 10 will spend Hirsch's knowledge-rich equity *argument* as an argument, not as a fetched RCT. Here the civic version is sufficient: a public skill that requires knowledge and practice will become a luxury good if the knowledge and the practice are private and the poster is public.

Older adults wait for Chapter 10 as well, with a label this chapter can plant. Guess, Jonathan Nagler, and Joshua Tucker, in *Science Advances* in 2019, linked a survey to Facebook profile data from the 2016 election. Sharing fake-news-domain articles was rare overall. Users over sixty-five shared nearly seven times as many as the youngest age group, holding after partisanship and ideology.[422] That is a *sharing* study from 2016, not a 2026 detection RCT, and not a finding that older adults "believe more fake news because of cognitive decline." Nadia Brashier and Daniel Schacter's 2020 review is explicit: do not reduce this to cognitive decline; fluency is intact; accumulated knowledge helps evaluation; social goals, trust, lie-detection, and digital-newcomer status are candidate mechanisms.[423] Grinberg, Kenneth Joseph, Lisa Friedland, Swire-Thompson, and Lazer, in *Science* in 2019, found users over fifty overrepresented among Twitter "supersharers."[424] Detection is not sharing. 2016 platforms are not 2026 short video. No older-adult COR RCT was fetched. The

[420]Joel Breakstone, Mark Smith, Sam Wineburg, Amie Rapaport, Jill Carle, Marshall Garland, and Anna Saavedra, "Students' Civic Online Reasoning: A National Portrait," *Educational Researcher* 50, no. 8 (2021), https://doi.org/10.3102/0013189X211017495. 3,446 high-school students; live internet; six constructed-response tasks. Load-bearing numbers from the abstract: 96 percent, two thirds, more than half. SES, race, maternal education, and free-or-reduced-price-lunch gradients in the abstract.

[421]Sam Wineburg, Joel Breakstone, Nadav Ziv, and Mark Smith, *Educating for Misunderstanding: How Approaches to Teaching Digital Literacy Make Students Susceptible to Scammers, Rogues, Bad Actors, and Hate Mongers*, SHEG working paper A-21322 (2020), https://purl.stanford.edu/mf412bt5333. CRAAP / vertical-checklist critique. PDF fetch timed out; Stanford Digital Repository abstract and stacks snippet used. Do not invent extra percentages. Named as a working paper on first citation.

[422]Andrew Guess, Jonathan Nagler, and Joshua Tucker, "Less than You Think: Prevalence and Predictors of Fake News Dissemination on Facebook," *Science Advances* 5 (2019): eaau4586, https://doi.org/10.1126/sciadv.aau4586. Survey linked to Facebook profile data, 2016 election. Sharing rare overall; over-65s nearly seven times the youngest group after partisanship controls. Science.org fetch Cloudflare-blocked; abstract via DOI/ScienceOpen used. Sharing study, not a 2026 detection RCT.

[423]Nadia M. Brashier and Daniel L. Schacter, "Aging in an Era of Fake News," *Current Directions in Psychological Science* (2020), https://doi.org/10.1177/0963721420915872. PMC7505057. Do not reduce older-adult sharing to cognitive decline.

[424]Nir Grinberg, Kenneth Joseph, Lisa Friedland, Briony Swire-Thompson, and David Lazer, "Fake News on Twitter during the 2016 U.S. Presidential Election," *Science* 363, no. 6425 (2019): 374–78, https://doi.org/10.1126/science.aau2706. Users over fifty overrepresented among supersharers. Sharing and exposure study, not a detection RCT.

life claim is only this: the civic habit is not a K–12 unit. Adults share. The evaluation habits this chapter named — leave, check, notice when you do not know — do not retire at commencement.

## Leave, then check what the trial actually bought

Lateral reading is the civic habit. Say it again so the chapter cannot be remembered as a tour of interventions. Fact checkers leave the unfamiliar page. Students and historians, in Wineburg and McGrew 2019, stayed, and were slower and less accurate.[425] Civic Online Reasoning's three questions are the short heuristic that makes leaving teachable in a government class.[426] The baseline is not "kids believe fake news." It is 96 percent missing a fossil-fuel tie, two thirds missing news versus ads, more than half treating an anonymous video as strong evidence of voter fraud — duped by look, TLD, About pages, and quantity, the signals CRAAP trains.[427] Six COR lessons moved credibility-task scores in a matched-control district study (cluster-randomized analysis), treatment $n = 271$, control $n = 228$; university communications reports they roughly doubled and still earned about half the points.[428] [429] That is a real, modest, incomplete gain after six hours. It is not informed citizens.

Inoculation works on techniques in the lab ($d$ 0.28–0.68) and more weakly on YouTube ($h = 0.09$),

[425] Sam Wineburg and Sarah McGrew, "Lateral Reading and the Nature of Expertise: Reading Less and Learning More When Evaluating Digital Information," *Teachers College Record* 121 (2019). ERIC EJ1262001. Sample: 10 professional fact checkers, 10 Ph.D. historians, 25 Stanford undergraduates. Fact checkers arrived at more warranted conclusions in a fraction of the time. The 2017 precursor is Wineburg and McGrew, "Lateral Reading: Reading Less and Learning More When Evaluating Digital Information," SHEG working paper 2017-A1, SSRN 3048994.

[426] Stanford History Education Group, Civic Online Reasoning curriculum site, https://cor.stanford.edu/, fetched 30 August 2026. Three questions in the project's own words. Free lessons; not an outcome trial.

[427] Joel Breakstone, Mark Smith, Sam Wineburg, Amie Rapaport, Jill Carle, Marshall Garland, and Anna Saavedra, "Students' Civic Online Reasoning: A National Portrait," *Educational Researcher* 50, no. 8 (2021), https://doi.org/10.3102/0013189X211017495. 3,446 high-school students; live internet; six constructed-response tasks. Load-bearing numbers from the abstract: 96 percent, two thirds, more than half. SES, race, maternal education, and free-or-reduced-price-lunch gradients in the abstract.

[428] Sam Wineburg, Joel Breakstone, Sarah McGrew, Mark D. Smith, and Teresa Ortega, "Lateral Reading on the Open Internet: A District-Wide Field Study in High School Government Classes," *Journal of Educational Psychology* 114, no. 5 (2022): 893–909, https://doi.org/10.1037/edu0000740. Urban midwestern district; six high schools; required government course, spring 2019; six ~50-minute COR lessons over three months; treatment $n = 271$, matched control $n = 228$ (published abstract). COR project page and SSRN version: cluster-randomized, repeated-measures analysis. Abstract and COR project page fetched 30 August 2026; ERIC abstract re-checked 30 August 2026. Do not upgrade "grew significantly" into a civic-behavior RCT.

[429] Stanford Graduate School of Education news, "It doesn't take long to learn how to spot misinformation online, Stanford study finds," 19 April 2022, https://ed.stanford.edu/news/it-doesn-t-take-long-learn-how-spot-misinformation-online-stanford-study-finds. **Journalism / university communications** summarizing Wineburg et al. 2022. Source of "twice as well" / "twice as likely to spot questionable websites" and "earned only half the total number of possible points." Funded by Google.org. Not the paper.

and it decays without boosters.[430] [431] Accuracy nudges change sharing under inattention, not belief; partisan moderation is a published dispute.[432] [433] Backfire is rare; continued influence is real; knowledge does not protect against fluency.[434] [435] [436] Twenty-five state laws are laws; NAMLE is a framework; MLN has not evaluated classrooms.[437] [438] A law that says the words is not the habit.

What this chapter did not show, because the sources did not show it: turnout, vote quality, scam victimization, vaccine uptake, or wild sharing over a year, as outcomes of COR or of inoculation. Dose and durability past the unit test. Implementation of the twenty-five laws. Settled partisan

[430] Jon Roozenbeek, Sander van der Linden, Beth Goldberg, Steve Rathje, and Stephan Lewandowsky, "Psychological Inoculation Improves Resilience against Misinformation on Social Media," *Science Advances* 8, no. 34 (2022): eabo6254, https://pmc.ncbi.nlm.nih.gov/articles/PMC9401631/. Fetched in full via PMC 30 August 2026; re-read this pass. Six laboratory studies (five videos plus a replication), $n$ = 6,464; YouTube field $n$ = 22,632 (~967,000 watched ≥30 seconds). Lab technique-recognition $d$ = 0.28 (scapegoating) to 0.68 (false dichotomies). Studies 1–5: 16 of 20 preregistered outcomes significant; studies 1–6: 19 of 23. Field $h$ = 0.09, narrowly missing preregistered SESOI of 0.10. Cost ~$0.05 per 30-second view. Google Jigsaw funded. Open access.

[431] Rakoen Maertens, Jon Roozenbeek, Melisa Basol, and Sander van der Linden, "Long-Term Effectiveness of Inoculation against Misinformation: Three Longitudinal Experiments," *Journal of Experimental Psychology: Applied* 27, no. 1 (2021), https://doi.org/10.1037/xap0000315. Bad News versus Tetris; stable with regular testing for months; significant decay over two months without it; long-term effect no longer significant in the no-testing condition. Abstract and search fetched; full PDF not re-read on this pass.

[432] Gordon Pennycook, Ziv Epstein, Mohsen Mosleh, Antonio A. Arechar, Dean Eckles, and David G. Rand, "Shifting Attention to Accuracy Can Reduce Misinformation Online," *Nature* 592 (2021): 590–95, https://www.nature.com/articles/s41586-021-03344-2. Four survey experiments plus a Twitter field experiment. Sharing intentions are not accuracy judgments. Competing-interest note on the paper: Rand gift from Google; Eckles Facebook funding/interest. Fetched 30 August 2026 (abstract and open figures/notes).

[433] Steve Rathje, Jon Roozenbeek, Cecilie S. Traberg, Jay J. Van Bavel, and Sander van der Linden, "Letter to the Editors of *Psychological Science*: Meta-Analysis Reveals that Accuracy Nudges Have Little to No Effect for U.S. Conservatives," 2022, https://doi.org/10.25384/SAGE.12594110.v2. Published dispute regarding Pennycook et al. 2020, not a consensus finding.

[434] Thomas Wood and Ethan Porter, "The Elusive Backfire Effect: Mass Attitudes' Steadfast Factual Adherence," *Political Behavior* 41 (2019): 135–63, https://doi.org/10.1007/s11109-018-9443-y. Five experiments, more than 10,000 participants, 52 issues; no worldview backfire.

[435] Stephan Lewandowsky, John Cook, Ullrich K. H. Ecker, et al., *The Debunking Handbook 2020*, DOI 10.17910/b7.1182, https://www.climatechangecommunication.org/wp-content/uploads/2023/09/DebunkingHandbook2020.pdf. Twenty-two-author consensus; fetched in full 30 August 2026. Sentence to print: "Do not refrain from attempting to debunk or correct misinformation out of fear that doing so will backfire." Continued-influence and illusory-truth sections used. Familiarity backfire and overkill backfire also failed later tests, per the handbook.

[436] Lynn Hasher, David Goldstein, and Thomas Toppino, 1977, illusory-truth origin; Lisa K. Fazio, Nadia M. Brashier, B. Keith Payne, and Elizabeth J. Marsh, "Knowledge Does Not Protect against Illusory Truth," *Journal of Experimental Psychology: General* 144, no. 5 (2015): 993, https://doi.org/10.1037/xge0000098. Cited via *The Debunking Handbook 2020* (fetched). Knowledge does not protect. Why fluency, including machine fluency, is a civic problem (Chapter 9).

[437] National Association for Media Literacy Education, "Core Principles of Media Literacy Education," 2023 revision, https://namle.org/resources/core-principles/, fetched in full 30 August 2026. Professional-association framework, not an RCT. SHEG listed among sources drawn on. Principle 4 and principle 10.4 quoted from that page.

[438] Media Literacy Now, "Current Policy" map, https://medialiteracynow.org/impact/current-policy/, fetched 30 August 2026. Last updated January 2026. Twenty-five states with laws; eleven new steps since January 2024. MLN states it has not evaluated implementation. Companion: Media Literacy Now, *U.S. Media Literacy Policy & Impact Report* (through 30 October 2025; released January 2026), https://medialiteracynow.org/wp-content/uploads/2026/01/MLN_Policy_Impact_Report-1.2026.pdf. Legislative tracking, not classroom outcomes.

moderation of accuracy prompts. *Nature Communications* 2025 booster coefficients. Those remain open. Filling them with analogy would be the folk collapse in civic dress.

Civic life is one domain. The next chapter is three more, with bodies, jobs, and liberty on the line. The structure will not change. Named knowledge, plus a representation or habit that makes a move executable, plus real error-costs. On the feed, the habit was leaving. In a clinic, it will be a natural-frequency tree and a shared decision. At work, it will be a situated story, not a recruiter's "number-one skill." In court, it will be a designed burden of proof and a named fallacy, taught as a teaching object, not as a courtroom RCT. The bicycle still does not exist. The page is still there to leave.

# Chapter 8

## Stakes: Clinic, Workplace, Court

Figure 9: A stethoscope, a gavel, a stack of folders, and bound books on a dark desk.

In 1978 Ward Casscells, Arno Schoenberger, and Thomas Graboys asked Harvard Medical School faculty, staff, and students a question that still belongs on the first page of a life chapter. If a disease has a prevalence of one in a thousand, and a test for it has a false-positive rate of five percent, and you have no other information, what is the chance that a person with a positive result has the disease? Twenty-seven of sixty said 95 percent. The correct answer is about 2 percent. Eleven of sixty gave it.[439]

Most people will experience at least one diagnostic error in their lifetime. That sentence is not a blog. It is the National Academies of Sciences, Engineering, and Medicine, in *Improving Diagnosis in Health Care*, 2015: diagnostic error as failure to establish an accurate and timely explanation *or* to communicate it to the patient; about 5 percent of U.S. adults seeking outpatient care each year; postmortem work implying about 10 percent of deaths; medical-record reviews putting diagnosis in 6 to 17 percent of hospital adverse events.[440] Two documents, a year and a generation apart: a base-rate item that experts fail, and a consensus synthesis that makes the failure a life event. This chapter is about the conditions under which those documents are the same problem.

Judgment in a life does not look like a laboratory critical-thinking task. Incomplete information, time, other people, and values are the default, not the noise. Gary Klein's recognition-primed decision model, and the naturalistic-decision-making program around it, describes experts who often do *not* generate and compare option sets. They recognize a situation as a type, simulate the first plausible action, and modify or reject it.[441] Daniel Kahneman and Klein, in a 2009 joint statement, agreed on the boundary rather than on a victory: intuition is trustworthy when the environment is valid — stable cues, usable feedback — and the person has learned those cues. Many life domains fail validity. Philip Tetlock's work on expert political judgment is the demonstration that prestige is not calibration.[442] A book that only trains untimed, paper-ready reflection trains a mode people will not have when a body, a job, or a liberty is on the line.

Three domains, one structure. Named knowledge, plus a representation or habit that makes a move executable, plus real error-costs. **Health:** statistical illiteracy in patients, journalists, *and physicians*; a mammography "25 percent reduction" that is 1 in 1,000 in absolute risk; natural frequencies that restore insight; shared decision-making as a practice model that is empty without numeracy; diagnostic error as a National Academies synthesis and as a 2023 modeling estimate that must not be printed as a census. **Work:** recruiters rate critical thinking important and then ask a situated interview story. NACE *Job Outlook 2025* puts communication at 4.57 and critical thinking at 4.49 on a five-point importance scale. The World Economic Forum's *Future of Jobs 2025* puts *analyt-*

[439]Ward Casscells, Arno Schoenberger, and Thomas B. Graboys, "Interpretation by Physicians of Clinical Laboratory Results," *New England Journal of Medicine* 299 (1978): 999–1001, https://doi.org/10.1056/NEJM197811022991808. Harvard Medical School faculty, staff, and students. Prevalence 1/1,000, false-positive 5 percent; 27 of 60 said 95 percent; 11 of 60 said about 2 percent. Cited via the AMA *Journal of Ethics* 2009 restatement (fetched in full; re-read this pass) and the 04-life memo.

[440]National Academies of Sciences, Engineering, and Medicine, Committee on Diagnostic Error in Health Care, *Improving Diagnosis in Health Care* (Washington, DC: National Academies Press, 2015). Report brief fetched in full 30 August 2026: https://nap.nationalacademies.org/resource/21794/DiagnosticError_ReportBrief.pdf. NCBI Bookshelf edition: https://www.ncbi.nlm.nih.gov/books/NBK338596/. Definition from the patient's side; ~5 percent outpatient; ~10 percent of deaths; 6–17 percent of hospital adverse events; most people at least one diagnostic error. Synthesis report, not an RCT. Committee orbit includes Mark Graber and Patrick Croskerry; this chapter does not dump a second Croskerry review.

[441]Gary Klein, recognition-primed decision model (1989 paper); *Sources of Power* (1998; 2nd ed. 2008). Experts often do not generate and compare option sets; they recognize a type, simulate the first plausible action, and modify or reject it.

[442]Daniel Kahneman and Gary Klein, "Conditions for Intuitive Expertise: A Failure to Disagree," *American Psychologist* 64, no. 6 (2009): 515–26. Intuition trustworthy in high-validity environments with feedback, for people who have learned the cues. Philip E. Tetlock, *Expert Political Judgment* (Princeton: Princeton University Press, 2005): prestige is not calibration.

*ical thinking* at seven of ten companies — a survey, not a skill test, and not the phrase "critical thinking." Unsourced "number-one skill" is folklore. Sourced, it is still a recruiter survey. **Law:** teaching examples, not courtroom RCTs. Burden of proof as a designed error-cost asymmetry. The prosecutor's fallacy, named by William Thompson and Edward Schumann in 1987, as the inversion of P(evidence | innocence) with P(innocence | evidence). The cab problem as a teaching object whose "correct" 41 percent is a live academic dispute. Natural-frequency trees as the representation that bridges clinic, court, and civic life — domain content plus a format, which is this book's thesis in one object, not a transferable faculty.

The chapter will not claim that these examples prove a general critical-thinking skill. It will not dump a second literature review of dual-process diagnosis. It will not pretend goals are given, as exam keys are given. It will not print 795,000 diagnostic harms as a headcount. It will not say employers test critical thinking. It will not promote a legal-education course into a generic-CT gain. Willingham's constraint travels: you cannot implement "consider the base rate" without the knowledge and the format. Ennis's sentence travels: the decision is what to believe *or do*, and in a clinic the doing is consent, refusal, a second question asked before the fluent label sticks.

Chapter 7's civic habit was leaving the page. This chapter's habits are recoding a number, asking a situated question, and refusing an inversion that feels like arithmetic. The objects look different. The architecture does not. A cloaked `.org` collected trust because it looked finished; a relative-risk poster collects consent because it looks like a magnitude; a match statistic collects guilt because it looks like a probability of innocence inverted. Fluency is the through-line. Domain knowledge plus a representation is the interruption. The bicycle still does not exist. The error-costs are now a body, a paycheck, and a verdict.

## Life versus lab

Laboratory critical-thinking tasks strip out the conditions that make a life a life. Time is plentiful. Information is on the page. Other people are not in the room, or they are confederates. Values are the experimenter's values, coded as a key. The student who does well is the student who can decouple from a tempting answer when there is no cost to decoupling and no identity on the line. Chapter 4 already spent myside bias as largely independent of cognitive ability, and time pressure as a condition that increases belief bias. Evans and Curtis-Holmes (2005) and De Neys (2006) are the papers; they are cited here as the constraint, not re-analyzed.[443] A CT curriculum that only trains the untimed mode has trained a performance that will not be available in a triage bay, a hiring meeting, or a deliberation room.

Naturalistic decision making was built to study the other mode. Raanan Lipshitz, Klein, Judith Orasanu, and Eduardo Salas, taking stock in 2001, described the field: fireground commanders, military officers, and similar experts under time pressure, high stakes, ambiguous data, unstable goals, and organizational constraints.[444] Klein's recognition-primed model is the cognitive claim. Experts often do not generate three options and score them. They recognize a *type*. They run a

[443] Jonathan St. B. T. Evans and Jodie Curtis-Holmes, 2005; Wim De Neys, 2006: time pressure increases belief bias. Cited from the 02-cogsci memo as constraint; not re-analyzed here.

[444] Raanan Lipshitz, Gary Klein, Judith Orasanu, and Eduardo Salas, "Taking Stock of Naturalistic Decision Making," *Journal of Behavioral Decision Making* 14 (2001): 331–52.

mental simulation of the first plausible action. If the simulation fails, they modify or reject and simulate again. That is not laziness. It is what high-validity practice plus feedback produces. It is also what a generic "consider all sides" maxim would wreck if it were inserted, untested, into a fireground.

Kahneman and Klein (2009), "Conditions for Intuitive Expertise: A Failure to Disagree," is the joint statement a critical-thinking book actually needs. They did not split the difference between heuristics-and-biases and naturalistic expertise. They named the boundary. Intuition is trustworthy in environments with stable relationships between cues and outcomes, and for people who have had the opportunity to learn those cues with feedback. Chess, under Chase and Simon's caveat about random boards, is such an environment. Some medicine is. Some firegrounds are. Geopolitics, much personnel selection, and some diagnostic frontiers are not. Where validity is low, confidence is not a cue of accuracy. Tetlock (2005) is the political-judgment demonstration: prestige, media presence, and theoretical commitment did not buy calibration. Mellers, Ungar, Baron, and colleagues (2014), on superforecasting, is the hygiene cousin: base rates, updating, the outside view, scorekeeping — practices closer to thinking-for-life than Wason-card performance, and still not a transferable bicycle.[445]

The open question this section plants and does not fill is the naturalistic one Chapter 4 already named. Can recognition-primed experts be trained to notice when their environment is invalid — Kahneman and Klein's joint condition — without wrecking the speed that makes them expert? A CT book that answers "always slow down" has not read the fireground. A CT book that answers "trust the expert" has not read Tetlock. The honest intermediate is domain-specific: learn where the cues are valid, keep score, and have a representation ready when the environment does not support intuition. Natural frequencies, in the next sections, are that representation for a class of medical and legal problems. They are not a general slowdown. They are a format that makes a Bayesian move executable when the fluent answer is 95 percent and the right count is 2 in 100.

Time, other people, and values complete the stripping. Time pressure increases belief bias; the papers are in Chapter 4. Other people introduce accountability, status, and myside in public — Mosleh and colleagues' perverse effects of public Twitter correction, already distinguished in Chapter 7 from a private accuracy prompt, are the social-media version of a courtroom's audience. Values are not given. Instrumental rationality, in Stanovich's sense, is relative to *the agent's* goals.[446] A shared decision in clinic is not a matter of hitting a key. A hiring story is not a Watson–Glaser item. A jury's burden of proof is a designed value about error-costs, not a computational default. A book that pretends the goals are given has smuggled an exam into a life.

What follows from the stripping is architectural. Train the mode people will have, on the knowledge they will need, with the representation that makes the move executable, under the error-costs that make the move matter. That is not a slogan. It is why this chapter is three domains rather than a list of "real-world applications of critical thinking." Applications language assumes a skill that then gets applied. The record assumes the opposite. Diagnosis is knowledge plus cases (Norman and

[445] Barbara Mellers, Lyle Ungar, Jonathan Baron, et al., "Psychological Strategies for Winning a Geopolitical Forecasting Tournament," *Psychological Science* 25 (2014): 1106–15, https://doi.org/10.1177/0956797614524255. Base rates, updating, outside view, scorekeeping. Superforecasting hygiene, not a workplace CT RCT.

[446] Keith E. Stanovich on epistemic versus instrumental rationality, and instrumental rationality as relative to the agent's goals: constraint from Chapter 4. Do not pretend goals are given as exam keys are given.

colleagues, 2017, waiting in Section 8.3). Hiring, when it is not folklore, is a situated story. Legal reasoning constructs a virtual fact pattern to fit or not fit a precedent (Geoffrey Samuel, 2015) — not the same as medical explanation, and not a generic-CT course.[447] Three domains. One structure. Named knowledge, a usable representation, real error-costs.

A further constraint, so this section cannot be remembered as anti-expertise. Recognition-primed decision making is *expertise*. It is what you want in a valid environment. The fireground commander who generates a decision tree of five options while the building burns has not become a critical thinker. She has become a delay. Kahneman and Klein's joint statement is the adult version of Willingham's bicycle passage: the processes of thinking are intertwined with the environment of thought, not only with its content. High-validity, learned cues, feedback — then intuition is a form of evaluation, fast because it has been paid for in cases. Low-validity, no feedback, prestige standing in for a scoring rule — then intuition is fluency with a title. Tetlock measured the second. Klein described the first. A life chapter that can only praise slowing down, or only praise experts, has not yet chosen a domain.

Myside does not clock out at the clinic door or the conference table. Chapter 4 spent Stanovich and West: myside bias largely independent of cognitive ability. "Be smarter" is not a myside cure, and it is not a diagnostic-error cure. A physician who is good at probability items can still protect a first label. A recruiter who rates critical thinking 4.49 can still hire for resemblance. A lawyer who can state Thompson and Schumann's inversion can still say the match statistic as if it were a posterior. The structure this chapter is arguing for — knowledge, representation, error-costs — is a way of making the better move *available*. It is not a personality transplant. Open question 20 stays planted: noticing invalid environments without wrecking expert speed is not a fitted training program in the sources fetched here.

## Statistical literacy is the health-CT medium

Gerd Gigerenzer, Wolfgang Gaissmaier, Elke Kurz-Milcke, Lisa Schwartz, and Steven Woloshin, in *Psychological Science in the Public Interest* in 2008, is the load-bearing numeracy paper. Statistical illiteracy is common in patients, journalists, *and physicians*. It is created by nontransparent framing as much as by "cognitive limitations." Relative risk is the headline trick: a "25 percent reduction" for mammography screening is 1 in 1,000 in absolute risk. Survival rates are not longer life. Conditional probabilities — the language of sensitivity, false-positive rate, and posterior — hide the base rate that Casscells already showed experts drop.[448] Shared decision-making, the authors and their restaters argue, *requires* statistical literacy; without it the "informed" in informed consent is empty. The AMA *Journal of Ethics* in 2009 restated the same claim as a teaching article,

[447] Geoffrey Samuel, 2015: legal reasoners construct a virtual fact pattern to fit or not fit a precedent — not the same as medical explanation. No clean large RCT of "legal reasoning course → generic CT gain → better later briefs" was located in the 03-teaching memo.

[448] Gerd Gigerenzer, Wolfgang Gaissmaier, Elke Kurz-Milcke, Lisa M. Schwartz, and Steven Woloshin, "Helping Doctors and Patients Make Sense of Health Statistics," *Psychological Science in the Public Interest* 8, no. 2 (2008): 53–96, https://doi.org/10.1111/j.1539-6053.2008.00033.x. Statistical illiteracy in patients, journalists, and physicians; relative versus absolute risk (mammography 25 percent versus 1 in 1,000); survival rates are not longer life; conditional probabilities versus natural frequencies. Abstract and search fetched; AMA 2009 restatement fetched in full.

not as new data.[449]

Casscells, Schoenberger, and Graboys (1978, *New England Journal of Medicine*) is the item to keep on the page, not as a gotcha against Harvard Medical School but as the demonstration that prestige and medical training do not automatically install the base rate. Prevalence 1 in 1,000. False-positive rate 5 percent. No other information. Modal answer 95 percent (27 of 60). Correct answer about 2 percent, given by 11 of 60.[450] The fluent computation is: the test is wrong 5 percent of the time, so it is right 95 percent of the time. That computation answers a different question. It answers P(positive | no disease), inverted and subtracted, as if it were P(disease | positive). It is the prosecutor's fallacy in a white coat, and Section 8.6 will name the inversion properly. Here the civic and clinical point is simpler. The maxim "consider the base rate" was available in 1978. Eleven of sixty executed it. The rest executed a fluent wrong answer.

Natural frequencies restore insight. Ulrich Hoffrage and Gigerenzer, in *Academic Medicine* in 1998, showed that physicians' diagnostic inferences improve when the same information is recoded as counts rather than as percentages and conditionals.[451] Gigerenzer and Hoffrage (1995, *Psychological Review*) is the representation-effect origin paper: frequency formats improve Bayesian reasoning without instruction.[452] Gigerenzer's 2011 *BMJ* note records that a Cochrane review found people understood natural frequencies better than probabilities; definitional fights about what counts as a "natural frequency" do not erase the representation effect.[453] The teaching recoding is the one this book will steal, labelled as a representation, not as a skill transplant.

Imagine 1,000 people. One has the disease (the base rate). The test catches that one (assume, for the Casscells item's structure, a sensitivity that does not change the punch: the item as put gives false-positive 5 percent and no other information; the usual teaching completion is that nearly all of the sick person and about 50 of the 999 well people test positive). About 50 false positives. One true positive. Of the people who test positive, about 1 in 51 has the disease — about 2 percent. You can read the posterior off the counts. You do not have to remember Bayes' theorem as a formula. You have to keep the 1,000, apply the base rate, apply the false-positive rate, and not invert. That is named knowledge (what prevalence and false-positive *mean*) plus a format (counts in a population)

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449 AMA *Journal of Ethics*, "Shared Decision Making Requires Statistical Literacy," *Virtual Mentor* 11, no. 4 (April 2009): 301–5, https://journalofethics.ama-assn.org/article/shared-decision-making-requires-statistical-literacy/2009-04. Fetched in full 30 August 2026. Teaching restatement of Gigerenzer et al. 2008 and Casscells 1978. Not new data.

450 Ward Casscells, Arno Schoenberger, and Thomas B. Graboys, "Interpretation by Physicians of Clinical Laboratory Results," *New England Journal of Medicine* 299 (1978): 999–1001, https://doi.org/10.1056/NEJM197811022991808. Harvard Medical School faculty, staff, and students. Prevalence 1/1,000, false-positive 5 percent; 27 of 60 said 95 percent; 11 of 60 said about 2 percent. Cited via the AMA *Journal of Ethics* 2009 restatement (fetched in full; re-read this pass) and the 04-life memo.

451 Ulrich Hoffrage and Gerd Gigerenzer, "Using Natural Frequencies to Improve Diagnostic Inferences," *Academic Medicine* 73, no. 5 (1998): 538–40, https://doi.org/10.1097/00001888-199805000-00024. Physicians' inferences improve when information is recoded as natural frequencies.

452 Gerd Gigerenzer and Ulrich Hoffrage, "How to Improve Bayesian Reasoning without Instruction: Frequency Formats," *Psychological Review* 102, no. 4 (1995): 684–704, https://doi.org/10.1037/0033-295X.102.4.684. Representation-effect origin paper.

453 Gerd Gigerenzer, "What Are Natural Frequencies? Doctors Need to Find Better Ways to Communicate Risk to Patients," *BMJ* 343 (2011): d6386, https://doi.org/10.1136/bmj.d6386. Cochrane 2011 cited; definitional fights do not erase the representation effect.

plus an error-cost (a person about to be told they have a rare disease, or a person about to be told they do not). Willingham's constraint is visible in the arithmetic. The maxim without the format is empty. The format without the knowledge of what the counts represent is a worksheet. Together they make the move executable.

Relative versus absolute risk is the poster version of the same problem, and it is the one patients actually see. "25 percent reduction" sounds like a quarter of the danger is gone. 1 in 1,000 is a count you can hold. Survival rates that rise because diagnosis moved earlier, not because death moved later, are another nontransparent frame. Gigerenzer and colleagues' claim is not that patients are stupid. It is that the information ecology — journals, press offices, clinic pamphlets — prefers the frame that makes a small absolute difference look large, and that physicians are not automatically protected. Casscells is the protection failing at Harvard Medical School. Hoffrage and Gigerenzer is the recoding that restores the inference in lab-like tasks. Whether natural-frequency training reduces *real* diagnostic error, or improves shared-decision-making adherence at scale, is a different question. It is live. It is not filled. This chapter will not pretend a representation effect on a test item is a population-health result.

A patient who cannot recode a PSA number or a mammogram number cannot share the decision. That is critical thinking as a life skill *in a domain*, not a generic "be skeptical" badge. Skepticism without the count is how a person refuses a useful screen or accepts a useless one, depending on which fluent sentence they heard last. Illusory truth from Chapter 7 is waiting in the waiting room: a repeated "25 percent reduction" will feel truer than a single "1 in 1,000," even for a reader who could, in principle, recode. Fluency is not a clinic-only problem. It is why the representation has to be practiced, not merely handed over on a pamphlet the patient sees once.

The Casscells item is also a teaching object about *who* fails. Harvard Medical School faculty, staff, and students — not a convenience sample of people who "don't like math." Twenty-seven of sixty offered the inversion as if it were the answer. Eleven of sixty kept the base rate. Prestige did not protect. Medical training, for most of the room, did not install the recoding. That is the Glaser 1941 finding in a white coat, sixty-six years before Willingham: skill in applying a method is limited by pertinent knowledge, and the attitude of wanting evidence transfers better than the skill. Wanting to be good at tests is not the same as having a format for prevalence. The AMA 2009 restatement is useful because it puts the 1978 item next to shared decision-making and says the quiet part: if the clinician cannot recode, the patient cannot be informed by that clinician's number. Teaching restatement, not new data. Steal the juxtaposition.

Journalists are in Gigerenzer's title for a reason this civic book should not skip. The relative-risk frame is a press-office product as often as it is a clinic product. "25 percent reduction" is a sentence that survives contact with an editor; "1 in 1,000" is a sentence that sounds small. Chapter 7's accuracy nudge was about sharing under inattention. Here the parallel is reporting under incentive: the fluent magnitude travels. A media chapter that only treats fake news, and a health chapter that only treats patients, would miss the shared mechanism. Nontransparent framing creates statistical illiteracy. The remedy is not a disposition. It is a recoding practiced on the actual number.

## Diagnostic error, named as synthesis and as modeling

The National Academies report is a synthesis, not a new RCT, and it should be printed as a synthesis. The committee — John R. Ball, chair; Mark Graber, Patrick Croskerry, and others in the orbit — continued the Quality Chasm work that began with *To Err Is Human*. The definition is from the patient's side: failure to establish an accurate and timely explanation of a health problem *or* to communicate that explanation to the patient. Communication is inside the definition. A correct label left in the chart is still an error if the patient never receives it. About 5 percent of U.S. adults seeking outpatient care each year. Postmortem work: about 10 percent of deaths. Medical-record reviews: 6 to 17 percent of hospital adverse events. Leading type of paid malpractice claim. Most people, at least once.[454] NAM's process picture is iterative information-gathering plus communication. Has enough information been collected? Has it been told? Those are the questions a life of evaluation asks in this domain. They are not a second Croskerry literature review. This book will not dump one.

Newman-Toker and colleagues, in *BMJ Quality & Safety* in 2023, produced a modeling estimate of **795,000** Americans permanently disabled or dead annually from misdiagnosis. Plausible range 598,000 to 1,023,000. Conservative sensitivity analysis: 549,000. Methods: disease-specific error and harm rates multiplied by incidence, then extrapolation. Fifteen diseases account for about 50.7 percent of the harms; the top five — stroke, sepsis, pneumonia, venous thromboembolism, lung cancer — about 38.7 percent.[455] **This is a modeling estimate, not a census.** It is compatible with setting-specific studies. It is not a vital-statistics count. Johns Hopkins Medicine's newsroom, 17 July 2023 — **journalism / institutional news release** summarizing the paper — reports 371,000 deaths plus 424,000 permanent disabilities as the split inside the 795,000.[456] Use the newsroom with the journal article. Do not print the newsroom as the finding. Do not print 795,000 without the range and the word *model*.

Geoffrey Norman and colleagues (2017) already supplied the teaching-stream negative, and it belongs here as a constraint rather than as a sequel. Diagnostic error is knowledge access, not a missing general critical-thinking trait. Lots of cases, because of case specificity. The fluent premature label is the failure mode: a pattern recognized too soon, from too little, in a low-validity

[454] National Academies of Sciences, Engineering, and Medicine, Committee on Diagnostic Error in Health Care, *Improving Diagnosis in Health Care* (Washington, DC: National Academies Press, 2015). Report brief fetched in full 30 August 2026: https://nap.nationalacademies.org/resource/21794/DiagnosticError_ReportBrief.pdf. NCBI Bookshelf edition: https://www.ncbi.nlm.nih.gov/books/NBK338596/. Definition from the patient's side; ~5 percent outpatient; ~10 percent of deaths; 6–17 percent of hospital adverse events; most people at least one diagnostic error. Synthesis report, not an RCT. Committee orbit includes Mark Graber and Patrick Croskerry; this chapter does not dump a second Croskerry review.

[455] David E. Newman-Toker et al., "Burden of Serious Harms from Diagnostic Error in the USA," *BMJ Quality & Safety* (2023), https://pubmed.ncbi.nlm.nih.gov/37460118/. Modeling estimate 795,000 serious harms per year (plausible range 598,000–1,023,000; conservative sensitivity 549,000). Fifteen diseases ~50.7 percent of harms; top five ~38.7 percent. **Print as a modeling estimate, not a census.** PubMed abstract fetched.

[456] Johns Hopkins Medicine newsroom, "Report Highlights Public Health Impact of Serious Harms From Diagnostic Error in U.S.," 17 July 2023, https://www.hopkinsmedicine.org/news/newsroom/news-releases/2023/07/report-highlights-public-health-impact-of-serious-harms-from-diagnostic-error-in-us. **Journalism / institutional news release** summarizing Newman-Toker et al.: 371,000 deaths + 424,000 permanent disabilities. Use with the journal article.

corner of a generally valid craft.[457] NAM's iterative picture and Norman's knowledge-plus-cases picture are the same architecture this book has been arguing since Chapter 3. You cannot think critically about a presentation you do not have the knowledge to see. You cannot diagnose from a maxim. You can notice uncertainty, ask for the base rate, and refuse a fluent premature label — from a doctor *or* from a symptom-checker model. That noticing is a habit. The content it operates on is cases.

Natural-frequency training helps inferences in lab-like tasks. Whether it reduces real diagnostic error or improves shared-decision-making adherence at scale is, again, not shown in the sources fetched for this book. The honest life claim is smaller and still worth making. A clinic that recodes the number, names the uncertainty, and asks whether enough information has been collected is practicing the structure: named knowledge, a usable representation, real error-costs. A clinic that hands the patient a relative-risk poster and a portal message written in fluent complete sentences is practicing the failure mode Breakstone already documented on the web. Looking finished is not the same as being checked. Chapter 9 will spend the machine that finishes the sentence. Here the body is the stake, and the 2015 synthesis plus the 2023 model are the scale of the stake, labelled as synthesis and as model.

A note on what 5 percent, 10 percent, 6–17 percent, and 795,000 are allowed to do in a sentence. They are allowed to establish that diagnostic error is common enough to be a life topic rather than a malpractice anecdote. They are not allowed to establish that a critical-thinking course would have prevented the error. They are not allowed to establish that shared decision-making, as a three-step talk model, has been shown to cut the Newman-Toker total. They are not allowed to travel into a recruiter brochure as "why CT is the most important skill in health care." NAM did not write that brochure. Newman-Toker did not run that RCT. This book will not write either.

The modeling method deserves a sentence of respect and a sentence of limit. Disease-specific error and harm rates, multiplied by incidence, summed and extrapolated, with a published range and a conservative sensitivity analysis, is how you estimate a burden you cannot census. It is not how you count death certificates. Stroke, sepsis, pneumonia, venous thromboembolism, and lung cancer dominating the top of the list is a clinical fact about where missed or delayed diagnosis is both common and costly, not a ranking of which specialties lack critical thinking. A chapter that turned those five into a character test of physicians would have left the paper. A chapter that refused to print 795,000 because it is a model would have left the scale of the stake. Print the model. Print the range. Print "not a census." Then return to the structure: knowledge plus cases, a count when a test is in play, a communication duty NAM already wrote into the definition.

Communication inside the definition is the piece a skills-poster misses. A correct diagnosis that never reaches the patient is, on NAM's wording, still error. That is Ennis's *or do* in a chart. Believing the right label is not enough if the doing — telling, recoding, deciding together — does not happen. Chapter 6 argued that the test is not the thing. Here the chart is not the thing. The thing is an accurate, timely explanation *received*. Elwyn's sequence, next, is one way a clinic can make receiving possible. It is still not a trial of a generic skill.

[457] Geoffrey Norman et al., 2017: diagnostic error is knowledge access, not a missing general CT trait; lots of cases because of case specificity. Teaching-stream negative already in Chapter 5; cited here as constraint.

## Shared decision-making is a practice, not a civic RCT

Glyn Elwyn, Dominick Frosch, Richard Thomson, and colleagues, in the *Journal of General Internal Medicine* in 2012, offered a three-step model to make shared decision-making doable in clinic: choice talk, option talk, decision talk.[458] Choice talk: let the patient know that a choice exists, and that their preferences matter. Option talk: name the options, including doing nothing, with benefits and harms in a form a person can use. Decision talk: arrive at a decision that fits the patient's informed preferences. It is a translation of ethical principles into a practice sequence. It is not a demonstration that teaching the sequence produces population-health gains, and it is not a civic-outcome RCT. This book will keep Elwyn as a practice model. It will not upgrade the model to evidence that critical thinking was taught.

Gigerenzer's claim is prior and sharper, and the 2009 AMA restatement is the teaching sentence: shared decision-making requires statistical literacy.[459] Without the recoding, option talk is a fluent brochure. "Twenty-five percent reduction" will travel through the option slot and arrive at decision talk as if it were a count. Choice talk can be performed as theater: "this is your decision," said over a number the patient cannot use. Elwyn's sequence is the right conversational architecture *if* the middle step carries a usable representation. Natural frequencies are one such representation. Absolute risks are another. Relative risks standing alone are not. A clinic that has trained staff in the three talks and not in the recoding has trained a manner. Manners are not consent.

Goals are not given. Stanovich's distinction between epistemic and instrumental rationality is the constraint Chapter 4 already installed: instrumental rationality is relative to *the agent's* goals, not to an exam key.[460] A patient may refuse a screen whose absolute benefit is 1 in 1,000 because 1 in 1,000 is not worth the harm *to them*. That refusal can be informed. A patient may accept a screen because a relative-risk poster frightened them. That acceptance can be fluent and uninformed. Shared decision-making is not a device for producing the clinician's preferred option by nicer means. It is a device for making the patient's goals, and the actual numbers, jointly usable. A critical-thinking book that treats the "right" decision as the one that matches a guideline, regardless of the patient's recoded understanding, has confused clinical quality metrics with evaluation. Guidelines can be knowledge. They are not the patient's goals.

The practice sequence also names a communication duty that NAM already put inside diagnostic error. Failure to communicate the explanation is error. Choice talk that never happens is a communication failure with a decision attached. Option talk that uses a nontransparent frame is a communication failure with a number attached. Decision talk that the patient cannot later explain — to a family member, to themselves, without the portal open — is the clinical version of Bastani's unaided exam, and this book will not sequel the education volume to say so at length. One sen-

[458] Glyn Elwyn, Dominick Frosch, Richard Thomson, et al., "Shared Decision Making: A Model for Clinical Practice," *Journal of General Internal Medicine* 27 (2012): 1361–67, https://doi.org/10.1007/s11606-012-2077-6. Choice talk / option talk / decision talk. Practice framework, not an RCT of civic or population-health outcomes.

[459] AMA *Journal of Ethics*, "Shared Decision Making Requires Statistical Literacy," *Virtual Mentor* 11, no. 4 (April 2009): 301–5, https://journalofethics.ama-assn.org/article/shared-decision-making-requires-statistical-literacy/2009-04. Fetched in full 30 August 2026. Teaching restatement of Gigerenzer et al. 2008 and Casscells 1978. Not new data.

[460] Keith E. Stanovich on epistemic versus instrumental rationality, and instrumental rationality as relative to the agent's goals: constraint from Chapter 4. Do not pretend goals are given as exam keys are given.

tence is enough. If the patient cannot recode the number without the pamphlet, the decision was not shared. It was performed.

Elwyn 2012 will be cited again in Chapter 10 as a deed-adjacent practice, not as a trial. The deed is to notice when you do not know, to ask for the count, and to refuse a fluent premature label. The model is how a clinic can make room for that deed on a Tuesday. Room is not proof. Proof would be a study, which this section does not have, that teaching the three talks plus natural frequencies reduces diagnostic harm or improves adherence at scale. Until that study is fetched, the honest recommendation is architectural: use the sequence; put a count in the middle; do not call the architecture a critical-thinking curriculum.

Option talk is where Gigerenzer and Elwyn either meet or miss. If the options are named with relative risks, the talk has happened and the literacy has not. If they are named with counts — 1 in 1,000, 50 false positives in 1,000, 1 true positive — the patient has something to prefer *with*. Preference without a usable number is taste. Preference with a usable number is a decision. Decision talk that then asks the patient to explain the number back, without the pamphlet, is the oral defense Chapter 6 wanted for judgment and that Chapter 9 will want when a model has drafted the explanation. This chapter does not sequel that argument. It notes the kinship and stays in clinic: the unaided recoding is the check that the sharing was real.

## Workplace: they rate it; they ask a story

Employers say they want critical thinking. What they test, when they test, is a story in an interview. “Critical thinking is the number-one skill employers want” is folklore if unsourced. If sourced to NACE or to the World Economic Forum, it is still a recruiter survey, not a skill assessment, and not evidence that teaching isolated critical-thinking skills produces job performance. Chapter 1 already vetoed the unsourced sentence. This section restates the veto with the interview-story detail, so the sentence cannot re-enter through a career-services door.

The National Association of Colleges and Employers, *Job Outlook 2025* — data collected 5 August to 16 September 2024; 237 respondents; 162 members equal to 19.2 percent of eligible members, plus 75 nonmembers; revised January 2025 — is the document. On a five-point importance scale, communication is 4.57, **critical thinking is 4.49**, teamwork is 4.43, professionalism is 4.31. Recent-graduate *proficiency* in critical thinking: 3.67 (communication 3.62, teamwork 4.00, professionalism 3.50). Share rating critical thinking very or extremely important: 96.1 percent. Share rating graduates very or extremely proficient: 53.5 percent. Resume attribute “problem-solving skills”: 88.3 percent. Skills-based hiring: 64.8 percent, used most at interviewing (89.7 percent of those who use it) and screening (67.5 percent). GPA screening: 46.4 percent, up from a 2023 trough, still well below 73.3 percent in 2019. Practices among the skills-based: competency-based job descriptions 73.7 percent, interview rubrics 52.6 percent, internally created assessments 38.6 percent, game-based assessments 2.6 percent, digital badges 3.5 percent.[461]

[461] National Association of Colleges and Employers, *Job Outlook 2025* (revised January 2025), https://www.naceweb.org/docs/default-source/default-document-library/2025/publication/research-report/2025-nace-job-outlook-jan-2025.pdf. Fetched 30 August 2026. $n$ = 237 (162 members = 19.2 percent of eligible, plus 75 nonmembers); data 5 August–16 September 2024. Communication 4.57, critical thinking 4.49, teamwork 4.43 (importance, 1–5); CT proficiency 3.67; 96.1 percent very/extremely important versus 53.5 percent very/extremely

NACE's own definition of the competency, from the 2025 Career Readiness PDF, is not a Facione skill list and not a Watson–Glaser. "Identify and respond to needs based upon an understanding of situational context and logical analysis of relevant information." Sample behaviors: gather diverse sources, anticipate needs, summarize data with bias awareness, communicate rationale. Sample interview item: "Can you describe a time when you anticipated a problem in advance…?"[462] That is a situated story. STAR, behavioral interview, work sample when skills-based hiring is real. It is closer to Willingham — knowledge plus practice in a domain — than to a generic critical-thinking course. The gap between 4.49 importance and 3.67 proficiency is a recruiter perception gap, not a CCTST finding, and not a license to conclude that colleges failed to teach a transferable skill. NACE did not administer a transferable-skill test. A couple of hundred recruiters, 19.2 percent of eligible members, rated importance and proficiency on Likert scales. Convenience and member-survey, not a probability sample of all employers.

NACE's 2026 skills-based hiring page, reporting on *Job Outlook 2026*, puts skills-based hiring at 70 percent (up from 65 percent) and GPA screening at 42 percent "this year," with fewer than 40 percent of graduating seniors familiar with the term.[463] Association journalism about its own survey, useful as a trendlet, still a survey. What gets tested, when the practice is real, remains a story or a work sample, not a Delphi construct.

The World Economic Forum's *Future of Jobs Report 2025* is the other document in the folklore sentence, and it does not say "critical thinking." **Analytical thinking** remains the top core skill, "seven out of ten companies considering it as essential." Then resilience, flexibility, and agility; leadership and social influence. AI and big data are the *fastest-growing* skills, not the current number one. Reading, writing, and mathematics show a small net *decline* in expected importance — a survey forecast, not a finding that literacy stopped mattering.[464] WEF's sample is WEF respondents. The phrase is analytical thinking. Treating that row as proof that generic critical-thinking courses produce job performance is a category error this book will not make.

Beware the folklore in both directions. Do not write that critical thinking is *the* number-one skill. NACE 2025 ranks communication slightly above it; WEF's word is different. Do not write that employers have stopped caring. 96.1 percent rating CT very or extremely important is a demand

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proficient graduates; problem-solving skills 88.3 percent on resumes; skills-based hiring 64.8 percent; GPA screening 46.4 percent. Survey of recruiters, not a skill assessment, not a probability sample of all employers.

[462] NACE, "Career Readiness Competencies: Critical Thinking" (December 2025), https://www.naceweb.org/docs/default-source/default-document-library/2025/career-readiness/competencies/nace-career-readiness-competencies-critical-thinking-december-2025.pdf. Definition and sample behaviors; sample interview item ("describe a time when you anticipated a problem"). Framework/rubric, not a validity study. Fetched 30 August 2026.

[463] NACE, "What Students Need to Know About the Skills-Based Hiring Process," 2026, https://www.naceweb.org/job-market/trends-and-predictions/what-students-need-to-know-about-the-skills-based-hiring-process. *Job Outlook 2026*: 70 percent skills-based hiring (up from 65 percent); GPA screening 42 percent; fewer than 40 percent of graduating seniors familiar with the term. Association journalism about its own survey.

[464] World Economic Forum, *The Future of Jobs Report 2025*, skills-outlook chapter, https://www.weforum.org/publications/the-future-of-jobs-report-2025/in-full/3-skills-outlook/, and PDF https://reports.weforum.org/docs/WEF_Future_of_Jobs_Report_2025.pdf. Fetched 30 August 2026. Analytical thinking top core skill, seven of ten companies; AI and big data fastest-growing, not current number one; reading, writing, and mathematics a small net decline in expected importance — survey forecast, not a finding that literacy stopped mattering. Employer survey of WEF respondents, not a critical-thinking assessment. Phrase is *analytical thinking*, not "critical thinking."

signal, and demand signals are allowed to be demand signals. Do not write that the proficiency gap proves a crisis of thinking. It proves that a couple of hundred recruiters, on a Likert scale, rate graduates lower than the competency's importance. Do not write that skills-based hiring has replaced degrees. GPA screening ticked back up; internally created assessments are 38.6 percent; game-based assessments are 2.6 percent; badges are 3.5 percent. The interview story is still the dominant instrument among those who claim to hire for skills.

Superforecasting hygiene is closer to thinking-for-life than a Wason card, and it still does not rescue a transferable faculty. Mellers and colleagues (2014): base rates, updating, outside view, scorekeeping.[465] Those are practices in a domain (geopolitical forecasting) with a scoring rule (Brier) and a tournament. Steal the hygiene — keep score, start from a base rate, update — when the domain has a score. Do not pretend a forecasting tournament is a hiring RCT. Almost no workplace critical-thinking RCTs were fetched for this book. A validated CT instrument that predicts job performance after controlling for general cognitive ability and domain knowledge was not in the NACE or WEF pages fetched. Kuncel's incremental-validity warning from Chapter 6 still stands. Open question 14 remains open.

What a workplace chapter can honestly recommend is therefore the structure, not a badge. If you are hiring, ask for a situated story or a work sample in the domain of the job, and do not launder a Watson–Glaser into "we tested critical thinking." If you are being hired, prepare the story NACE's sample item already named: a time you anticipated a problem, in a context, with information you can still recode. If you are designing a curriculum to satisfy recruiters, teach the domain and the evaluation habits the domain requires, and do not stamp "critical thinking" on every syllabus in the hope that Likert ratings will move. Implicit expectation fails in college (Huber and Kuncel; Chapter 5) as it fails in civic life (Chapter 7). A recruiter survey is not a reason to reverse those findings.

Read the NACE table as a table, not as a ranking of human worth. Communication 4.57, critical thinking 4.49, teamwork 4.43, professionalism 4.31 — importance. Proficiency: teamwork 4.00, critical thinking 3.67, communication 3.62, professionalism 3.50. The competency recruiters think graduates have is teamwork. The competencies they think graduates lack relative to importance are communication, critical thinking, and professionalism. None of those rows is a Watson–Glaser. None is a work sample. Skills-based hiring, at 64.8 percent in 2025 and 70 percent in the 2026 association write-up, still lives mostly at the interview (89.7 percent of those who use it). Internally created assessments 38.6 percent; game-based 2.6 percent; badges 3.5 percent. The instrument, overwhelmingly, is a conversation about a time. Willingham would recognize that conversation as a request for domain practice. Facione would not recognize it as a CCTST. Both can be true. Only one of them licenses "we test critical thinking."

The WEF row needs the same hygiene in a second paragraph so it cannot re-enter as "the World Economic Forum says critical thinking is number one." Analytical thinking, seven of ten companies, top *core* skill in a survey of WEF respondents. Fastest-growing: AI and big data. Small net decline expected in reading, writing, and mathematics — a forecast, not a finding that literacy

[465] Barbara Mellers, Lyle Ungar, Jonathan Baron, et al., "Psychological Strategies for Winning a Geopolitical Forecasting Tournament," *Psychological Science* 25 (2014): 1106–15, https://doi.org/10.1177/0956797614524255. Base rates, updating, outside view, scorekeeping. Superforecasting hygiene, not a workplace CT RCT.

stopped mattering, and a forecast this book will not launder into a reason to starve knowledge-rich curriculum. If analytical thinking depends on content, as Willingham says thinking does, then a survey that ranks the thinking and downgrades the reading is describing a wish, not a cognitive architecture. Steal the demand signal. Do not steal the wish.

## Court: designed error-costs, prosecutor's fallacy, contested cab

Law belongs in this book as established teaching objects with sources. It does not belong as a courtroom RCT, and no clean large trial of "legal reasoning course → generic CT gain → better later briefs" was located in the teaching-stream memo.[466] Print the objects. Do not imply a transferable legal-reasoning skill. The maxim "consider the base rate" is empty here too, without the domain content.

Burden of proof is a designed error-cost asymmetry. In criminal law the state bears the burden; the default is not-guilty. You do not prove innocence. You fail to prove guilt beyond a standard. Blackstone's ratio — better that guilty persons walk than that an innocent person be convicted — is the usual classroom gloss of the value choice. It is the same structure as a scientific null: the cost of a false positive (convicting the innocent) has been set higher than the cost of a false negative (acquitting the guilty), by design, not by computational necessity. A critical-thinking course that treats "don't jump to conclusions" as a generic maxim has not yet said *which error is worse, in this institution, and who decided*. The burden is the decision. The maxim is a poster.

The prosecutor's fallacy is the named inversion. Thompson and Schumann, in *Law and Human Behavior* in 1987, described treating P(evidence | innocence) as P(innocence | evidence).[467] A DNA profile that would match 1 in 10,000 innocent people is not a 1-in-10,000 chance of innocence, not without the base rate of how many people were in the relevant population and how the match was found. Casscells is the same inversion with a test: P(positive | no disease) of 5 percent is not a 95 percent chance of disease. The fallacy is small when the prior is high and large when the prior is tiny, which is why rare-disease screening and cold-hit DNA are the dangerous cases. A high-prevalence condition plus a decent test does not produce a courtroom disaster. A one-in-a-million match drawn from a database of millions can.

Oxford's Centre for Evidence-Based Medicine, in a 2020 teaching essay by Kathy Taylor, walks the inversion through named miscarriages and a medical false-positive — **journalism / teaching exposition**, not new data, citing Thompson and Schumann 1987 and Tversky and Kahneman 1974.[468] Sally Clark (1999) and Lucia de Berk (2003) are the named legal cases in that essay. Leonard

[466] Geoffrey Samuel, 2015: legal reasoners construct a virtual fact pattern to fit or not fit a precedent — not the same as medical explanation. No clean large RCT of "legal reasoning course → generic CT gain → better later briefs" was located in the 03-teaching memo.

[467] William C. Thompson and Edward L. Schumann, "Interpretation of Statistical Evidence in Criminal Trials: The Prosecutor's Fallacy and the Defense Attorney's Fallacy," *Law and Human Behavior* 11 (1987): 167–87, https://doi.org/10.1007/BF01044641. P(evidence | innocence) confused with P(innocence | evidence). Teaching and legal object, not a courtroom RCT.

[468] Kathy Taylor, "Getting the Wrong End of the Stick: The Prosecutor's Fallacy," Centre for Evidence-Based Medicine, University of Oxford, 2020, https://www.cebm.ox.ac.uk/news/views/the-prosecutors-fallacy. **Teaching essay**, fetched 30 August 2026. Walks Sally Clark, Lucia de Berk, and Mlodinow's HIV false-positive. Not new data; cites Thompson and Schumann 1987 and Tversky and Kahneman 1974.

Mlodinow's HIV example, as Taylor presents it: a false-positive rate of 1 in 1,000 misread as the chance of being infected given a positive result; in a 1-in-10,000 base-rate group the odds were about 10 to 1 *against*. This book will not retry those cases. It will not claim that a critical-thinking module would have prevented them. It will print them as Taylor prints them: teaching exposition of a published fallacy. The object of practice is the inversion. The error-cost is a person's liberty or a person's diagnosis. The representation that makes the inversion visible is, again, a natural-frequency tree.

Base-rate neglect has a psychology-class object: the cab problem, in Kahneman and Tversky's line (1973, 1974, "Judgment under Uncertainty"). A witness identifies a cab as a color; the witness's accuracy is given; the base rate of that color in the city is given; the modal answer tracks the witness and drops the base rate.[469] The "correct" Bayesian number often taught is about 41 percent. Birnbaum (1983) and later philosophers dispute whether the modal 80 percent answer is even the wrong Bayesian number given how the witness criterion is specified — **a live academic dispute**.[470] Print the cab problem as the *teaching* example of neglecting base rates. Do not pretend the 41 percent is uncontested. A book that needs an uncontested number has Casscells, where about 2 percent is not in dispute as the item was put. A book that needs the psychology-class object can keep the cab, with the dispute on the page, as a reminder that even the "correct" teaching answer can be a live fight about the problem's interpretation. That fight is not a reason to skip base rates. It is a reason not to treat a textbook posterior as a sacrament.

Geoffrey Samuel (2015) is the reminder that legal reasoning is not medical explanation under a wig. Legal reasoners *construct* a virtual fact pattern to fit or not fit a precedent.[471] That is a domain practice: analogizing, distinguishing, building a story the doctrine can receive. It is not diagnosis, and it is not a Wason card. No clean large RCT of a legal-reasoning course producing generic CT gains that then produce better later briefs was located. This section will not invent one. Steal the teaching objects — burden as designed asymmetry, prosecutor's fallacy as inversion, cab as contested base-rate item, natural frequencies as the recoding. Leave the franchise of "think like a lawyer" as a transferable skill where McPeck left "I teach thinking" simpliciter: empty until you say *about what*.

Tversky and Kahneman (1974) remain the established teaching source for the heuristic family — representativeness, availability, anchoring — without this chapter becoming a second dual-process tour.[472] The courtroom is a hostile environment for those heuristics when the prior is tiny and the

[469] Amos Tversky and Daniel Kahneman, "Judgment under Uncertainty: Heuristics and Biases," *Science* 185, no. 4157 (1974): 1124–31, https://doi.org/10.1126/science.185.4157.1124. Established teaching source for base-rate neglect. Daniel Kahneman and Amos Tversky, "On the Psychology of Prediction," *Psychological Review* 80, no. 4 (1973): 237–51. Cab problem associated with this program.

[470] Michael H. Birnbaum, 1983, and later philosophers: whether the modal cab-problem answer is even the wrong Bayesian number given witness-criterion specification is a live academic dispute. Print the cab as a teaching example of neglecting base rates; do not print 41 percent as uncontested. Casscells ~2 percent, as the 1978 item was put, is the uncontested teaching posterior in this chapter.

[471] Geoffrey Samuel, 2015: legal reasoners construct a virtual fact pattern to fit or not fit a precedent — not the same as medical explanation. No clean large RCT of "legal reasoning course → generic CT gain → better later briefs" was located in the 03-teaching memo.

[472] Amos Tversky and Daniel Kahneman, "Judgment under Uncertainty: Heuristics and Biases," *Science* 185, no. 4157 (1974): 1124–31, https://doi.org/10.1126/science.185.4157.1124. Established teaching source for base-rate neglect. Daniel Kahneman and Amos Tversky, "On the Psychology of Prediction," *Psychological Review* 80, no. 4

evidence is fluent. A match statistic spoken as a probability of guilt is fluency doing the inversion. A natural-frequency tree on a whiteboard is the interruption. The interruption is domain content plus a format. It is not a general power the law student will carry, unprompted, into a clinic or a feed a year later. Chapter 4 already spent that transfer default. Open question 2 remains open: year-scale unaided far transfer of the law of large numbers and confound into workplace and civic decisions, scored independently of CT tests, is not in the fetched literature.

Thompson and Schumann's naming is the load-bearing citation; Taylor is the walkthrough, labelled journalism and teaching exposition so it cannot be mistaken for a trial of those cases.[473] [474] The inversion is easy to state and hard to hear in a room that wants a number. P(evidence | innocence) is a laboratory-looking quantity: how often would this trace appear if the person were innocent. P(innocence | evidence) is the question the room is actually asking, and it is not answerable from the first quantity alone. You need the base rate — how many people could have left this trace, how the person was selected into the test. Cold-hit DNA from a large database is the dangerous design because the search *creates* opportunities for false matches. Rare-disease screening is the medical twin. Casscells hid the same structure in one sentence. Eleven of sixty unpacked it. A jury is not a Harvard Medical School sample with more time. A jury is a group under social pressure, with a fluent statistic, without a tree unless someone draws one.

Burden of proof, restated without Blackstone as folklore: the institution chose the costlier error. Science, at its best, chooses a similar asymmetry — fail to reject the null rather than claim a discovery on a fluke. A generic "be careful" maxim does not tell you which way to be careful. The criminal burden tells you. Shared decision-making, oddly, is the place where the asymmetry is *not* pre-chosen: the patient's goals decide which error is worse, a missed cancer or an unnecessary biopsy, and that is why the count has to be usable by the person who will live with it. Court and clinic share the inversion and do not share the value. Teaching them as one "critical thinking skill called base rates" would erase the design. Teaching them as one representation serving different error-costs is the structure.

## Natural-frequency trees as the bridge — and the limit

One representation has now appeared in a clinic, in a courtroom teaching essay, and, by extension, in any civic claim that comes as a percentage. Gigerenzer and Hoffrage (1995) and Hoffrage and Gigerenzer (1998) made Bayes tractable for doctors by recoding conditionals as counts in a popula-

(1973): 237–51. Cab problem associated with this program.

[473] William C. Thompson and Edward L. Schumann, "Interpretation of Statistical Evidence in Criminal Trials: The Prosecutor's Fallacy and the Defense Attorney's Fallacy," *Law and Human Behavior* 11 (1987): 167–87, https://doi.org/10.1007/BF01044641. P(evidence | innocence) confused with P(innocence | evidence). Teaching and legal object, not a courtroom RCT.

[474] Kathy Taylor, "Getting the Wrong End of the Stick: The Prosecutor's Fallacy," Centre for Evidence-Based Medicine, University of Oxford, 2020, https://www.cebm.ox.ac.uk/news/views/the-prosecutors-fallacy. **Teaching essay**, fetched 30 August 2026. Walks Sally Clark, Lucia de Berk, and Mlodinow's HIV false-positive. Not new data; cites Thompson and Schumann 1987 and Tversky and Kahneman 1974.

tion.[475] [476] Kathy Taylor's CEBM essay uses the same recoding for an HIV false-positive and, by structural kinship, for a DNA match.[477] A patient, a juror, and a reader of a relative-risk headline can, in principle, use the same tree: start with 1,000 (or 10,000) people; apply the base rate; apply the hit rate; apply the false-positive rate; read the posterior off the counts. That is domain knowledge about what the counts *mean* in this case, plus a format that makes the move executable, plus an error-cost that makes the move worth the trouble.

Do not claim the tree *is* critical thinking. Do not claim it transfers unprompted a year later at work with identity on the line — that test is not in the fetched literature. Do not claim natural-frequency training reduces real diagnostic error or improves shared decision-making at scale. Do not claim that teaching the prosecutor's fallacy produces better briefs. Do not claim that a recruiter who asks for a story has measured the tree. The tree is this book's thesis in one object: not a transferable badge, not a checklist, not a Likert rating, not a law that says the words. Named knowledge, a usable representation, real error-costs.

Three domains, restated, so the structure cannot be missed.

In health, the named knowledge is prevalence, sensitivity, false-positive rate, absolute versus relative risk, and the difference between a survival rate and a longer life. The representation is the count. The error-cost is a body — NAM's "most people, at least once," and Newman-Toker's modeling estimate of 795,000 serious harms a year, range attached, not a census.[478] [479] The practice model is Elwyn's three talks, empty without the count.[480] The teaching item is Casscells, 11 of

[475] Ulrich Hoffrage and Gerd Gigerenzer, "Using Natural Frequencies to Improve Diagnostic Inferences," *Academic Medicine* 73, no. 5 (1998): 538–40, https://doi.org/10.1097/00001888-199805000-00024. Physicians' inferences improve when information is recoded as natural frequencies.

[476] Gerd Gigerenzer and Ulrich Hoffrage, "How to Improve Bayesian Reasoning without Instruction: Frequency Formats," *Psychological Review* 102, no. 4 (1995): 684–704, https://doi.org/10.1037/0033-295X.102.4.684. Representation-effect origin paper.

[477] Kathy Taylor, "Getting the Wrong End of the Stick: The Prosecutor's Fallacy," Centre for Evidence-Based Medicine, University of Oxford, 2020, https://www.cebm.ox.ac.uk/news/views/the-prosecutors-fallacy. **Teaching essay**, fetched 30 August 2026. Walks Sally Clark, Lucia de Berk, and Mlodinow's HIV false-positive. Not new data; cites Thompson and Schumann 1987 and Tversky and Kahneman 1974.

[478] National Academies of Sciences, Engineering, and Medicine, Committee on Diagnostic Error in Health Care, *Improving Diagnosis in Health Care* (Washington, DC: National Academies Press, 2015). Report brief fetched in full 30 August 2026: https://nap.nationalacademies.org/resource/21794/DiagnosticError_ReportBrief.pdf. NCBI Bookshelf edition: https://www.ncbi.nlm.nih.gov/books/NBK338596/. Definition from the patient's side; ~5 percent outpatient; ~10 percent of deaths; 6–17 percent of hospital adverse events; most people at least one diagnostic error. Synthesis report, not an RCT. Committee orbit includes Mark Graber and Patrick Croskerry; this chapter does not dump a second Croskerry review.

[479] David E. Newman-Toker et al., "Burden of Serious Harms from Diagnostic Error in the USA," *BMJ Quality & Safety* (2023), https://pubmed.ncbi.nlm.nih.gov/37460118/. Modeling estimate 795,000 serious harms per year (plausible range 598,000–1,023,000; conservative sensitivity 549,000). Fifteen diseases ~50.7 percent of harms; top five ~38.7 percent. **Print as a modeling estimate, not a census.** PubMed abstract fetched.

[480] Glyn Elwyn, Dominick Frosch, Richard Thomson, et al., "Shared Decision Making: A Model for Clinical Practice," *Journal of General Internal Medicine* 27 (2012): 1361–67, https://doi.org/10.1007/s11606-012-2077-6. Choice talk / option talk / decision talk. Practice framework, not an RCT of civic or population-health outcomes.

60.[481] The negative is Norman: diagnosis is knowledge plus cases, not a missing trait.[482]

At work, the named knowledge is the domain of the job. The representation, when hiring is real, is a situated story or a work sample, not a Facione subscale. The error-cost is a hire, a promotion, a missed problem a behavioral item asked you to anticipate. The demand signal is NACE 4.49 and WEF seven of ten, surveys, communication slightly above CT on NACE 2025, phrase *analytical thinking* on WEF 2025.[483] [484] The folklore is “number one skill” without a source, or with a source treated as a skill test. Superforecasting hygiene is stealable; a workplace CT RCT after *g* and domain knowledge was not fetched.

In court, the named knowledge is the burden, the inversion, the base rate, and — if you are actually a lawyer — the doctrine into which a virtual fact pattern must fit. The representation is again the tree, plus the designed asymmetry of the burden. The error-cost is liberty, and, in the teaching cases, a name. Thompson and Schumann 1987 is the fallacy’s paper. Taylor 2020 is the teaching essay, labelled as such.[485] [486] The cab problem is the psychology-class object, 41 percent flagged as dispute.[487] No courtroom RCT, no generic-CT gain from a legal-reasoning course located.

Life versus lab was the condition. Recognition-primed expertise is trustworthy where Kahneman

[481] Ward Casscells, Arno Schoenberger, and Thomas B. Graboys, “Interpretation by Physicians of Clinical Laboratory Results,” *New England Journal of Medicine* 299 (1978): 999–1001, https://doi.org/10.1056/NEJM197811022991808. Harvard Medical School faculty, staff, and students. Prevalence 1/1,000, false-positive 5 percent; 27 of 60 said 95 percent; 11 of 60 said about 2 percent. Cited via the AMA *Journal of Ethics* 2009 restatement (fetched in full; re-read this pass) and the 04-life memo.

[482] Geoffrey Norman et al., 2017: diagnostic error is knowledge access, not a missing general CT trait; lots of cases because of case specificity. Teaching-stream negative already in Chapter 5; cited here as constraint.

[483] National Association of Colleges and Employers, *Job Outlook 2025* (revised January 2025), https://www.naceweb.org/docs/default-source/default-document-library/2025/publication/research-report/2025-nace-job-outlook-jan-2025.pdf. Fetched 30 August 2026. $n$ = 237 (162 members = 19.2 percent of eligible, plus 75 nonmembers); data 5 August–16 September 2024. Communication 4.57, critical thinking 4.49, teamwork 4.43 (importance, 1–5); CT proficiency 3.67; 96.1 percent very/extremely important versus 53.5 percent very/extremely proficient graduates; problem-solving skills 88.3 percent on resumes; skills-based hiring 64.8 percent; GPA screening 46.4 percent. Survey of recruiters, not a skill assessment, not a probability sample of all employers.

[484] World Economic Forum, *The Future of Jobs Report 2025*, skills-outlook chapter, https://www.weforum.org/publications/the-future-of-jobs-report-2025/in-full/3-skills-outlook/, and PDF https://reports.weforum.org/docs/WEF_Future_of_Jobs_Report_2025.pdf. Fetched 30 August 2026. Analytical thinking top core skill, seven of ten companies; AI and big data fastest-growing, not current number one; reading, writing, and mathematics a small net decline in expected importance — survey forecast, not a finding that literacy stopped mattering. Employer survey of WEF respondents, not a critical-thinking assessment. Phrase is *analytical thinking*, not “critical thinking.”

[485] William C. Thompson and Edward L. Schumann, “Interpretation of Statistical Evidence in Criminal Trials: The Prosecutor’s Fallacy and the Defense Attorney’s Fallacy,” *Law and Human Behavior* 11 (1987): 167–87, https://doi.org/10.1007/BF01044641. P(evidence | innocence) confused with P(innocence | evidence). Teaching and legal object, not a courtroom RCT.

[486] Kathy Taylor, “Getting the Wrong End of the Stick: The Prosecutor’s Fallacy,” Centre for Evidence-Based Medicine, University of Oxford, 2020, https://www.cebm.ox.ac.uk/news/views/the-prosecutors-fallacy. **Teaching essay**, fetched 30 August 2026. Walks Sally Clark, Lucia de Berk, and Mlodinow’s HIV false-positive. Not new data; cites Thompson and Schumann 1987 and Tversky and Kahneman 1974.

[487] Michael H. Birnbaum, 1983, and later philosophers: whether the modal cab-problem answer is even the wrong Bayesian number given witness-criterion specification is a live academic dispute. Print the cab as a teaching example of neglecting base rates; do not print 41 percent as uncontested. Casscells ~2 percent, as the 1978 item was put, is the uncontested teaching posterior in this chapter.

and Klein said it was: valid environments, learned cues, feedback. Where those fail, fluency — 95 percent, "25 percent reduction," a match statistic spoken as guilt, a recruiter's "we hire for critical thinking" — will fill the slot. The tree, the leave-the-page habit from Chapter 7, and the situated story are interruptions. They are not one skill. They are what evaluation looks like when the stake is real and the bicycle does not exist.

The default document of the 2020s is a fluent paragraph a model will finish. A cloaked `.org` fooled students because it looked finished. A relative-risk poster fools a waiting room because it looks like a number. A GPT solution, in the education book this volume is not sequeling, fooled students because it looked like math. One chapter, not five, is what that condition earns. Evaluation is the remaining human work. Clinic, workplace, and court were the demonstration that the work is domain-bound. The machine does not relax the bound. It makes the fluent wrong answer cheaper.

What this chapter did not show, because the sources did not show it: natural-frequency training cutting real diagnostic error or lifting shared-decision-making adherence at scale; a validated CT instrument predicting job performance after *g* and domain knowledge; a legal-reasoning course producing generic CT gains that then produce better briefs; recognition-primed experts trained to notice invalid environments without losing speed; year-scale far transfer of the law of large numbers into workplace and civic decisions. Those remain open. Filling them with a tree drawn on a blackboard in prose would be the folk collapse in professional dress.

Named knowledge. A representation that makes a move executable. Real error-costs. That is what critical thinking for life looks like when the stake is a body, a hire, or a verdict. It is not a badge, not a NACE competency PDF, not a recruiter's number-one, not a courtroom legend, and not 795,000 printed as a census. The next chapter is the fluent sentence. This one was the stake.

# Chapter 9

## Machines That Finish the Sentence

Figure 10: A closed laptop, a handwritten sheet, a fountain pen, and a brass desk lamp.

A machine will finish the sentence. That is the life condition this chapter is about, and it is almost the whole of what this book still owes after *Autonomous AI and Education*. The default document of the 2020s arrives already fluent. A cloaked `.org` fooled students because it *looked* finished. A generated solution fooled students because it *looked* like math. A generated essay fooled graders because it *looked* like thought. The surface form of completed judgment is now cheap. Evaluation is not.

Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and Rei Mariman gave nearly a thousand students in a large Turkish high school four ninety-minute sessions of mathematics practice under one of three conditions: no AI; unguarded GPT-4 in a ChatGPT-style interface; or the same interface with guardrails that withheld full solutions and used teacher-written correct answers and common-mistake hints. Unguarded GPT raised practice scores about 48 percent. On the closed-book isomorphic exam that followed, those same students scored about 17 percent *worse* than classmates who never had AI. The hint tutor raised practice about 127 percent and left the unaided exam indistinguishable from control. Students in the unguarded arm did not perceive that they had learned less.[488] That dissociation — performance up, unaided judgment down, the loss invisible to the person who suffered it — is already treated at length in the education book. It is cited here, not retold as pedagogy. What it exhibits for a *life* book is simpler. When a machine finishes the sentence, the remaining human work is evaluation. If that work is also offloaded, nothing that this book has been calling critical thinking has occurred.

## 9.1 One chapter, not five: the division of labor

Daniel Willingham's 2007 argument, Bastani's field trial, and Yizhou Fan and colleagues' 2025 study of ChatGPT and essay writing already live in *Autonomous AI and Education*. That book had to take classroom AI design seriously: what a constrained tutor does that an unguarded chat window does not; why a polished take-home is the wrong measure of a formed skill; how detectors, companion products, and unattended runtimes fail different tests. This book does not. The division of labor is not a courtesy to a sibling manuscript. It is a constraint on what a life book is allowed to become. A chapter that re-litigates tutor architecture, product classes, or age-band standard operating procedure is no longer a chapter about critical thinking for life. It is a sequel.

The job here is the condition under which the rest of a life now proceeds. Civic pages, clinic numbers, workplace drafts, and the sentence you are about to share all arrive, increasingly, as fluent paragraphs a model will complete. UNESCO connectivity figures, Khanmigo, Kestin, Hermes, Grok Bot, and the education book's age-band design live there, not here.[489] Willingham's bicycle passage is already on the page in Chapter 3; it is not quoted again. Bastani and Fan are cited at the grain the record already used. A reader who wants the classroom design can open the other book. A reader who wants to know what happens to judgment when fluency is free is in the right chapter.

That division also keeps a coalition veto from going missing. Ennis's public sentence — reasonable reflective thinking focused on deciding what to believe or do — still names the work, including

[488]Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and Rei Mariman, "Generative AI without Guardrails Can Harm Learning: Evidence from High School Mathematics," *Proceedings of the National Academy of Sciences* 122, no. 26 (2025): e2422633122, https://doi.org/10.1073/pnas.2422633122. Field RCT, large Turkish high school, nearly 1,000 students, grades 9–11, four 90-minute sessions. Assisted practice: GPT Base about +48 percent; GPT Tutor about +127 percent. Unaided closed-book isomorphic exam: GPT Base about −17 percent versus never-AI; GPT Tutor indistinguishable from control. Students in the unguarded arm did not perceive the loss. Already treated at length in *Autonomous AI and Education*; cited here as the exhibit, not retold as classroom design. Education-book ITT grain: GPT Base practice +0.137 (control mean 0.284) ≈ 48 percent; exam −0.054 (control mean 0.321) ≈ 17 percent; GPT Tutor practice +0.361 ≈ 127 percent; exam −0.004 ≈ 0.

[489]Division of labor as stated in the research brief for this book, 30 August 2026. UNESCO connectivity figures, Khanmigo, Kestin, Hermes, Grok Bot, and age-band SOP are not re-fetched or reused here as this project's findings.

under machine conditions, because the work is still a decision about belief or action.[490] Willingham still vetoes a content-free "AI literacy" module as the engine of that work. The processes of thinking remain intertwined with the content of thought. A maxim that says "evaluate the chatbot" is empty in a domain the evaluator does not know, for the same reason a maxim that says "consider both sides" is empty in a domain the student does not know. The machine does not lift that constraint. It makes the constraint more expensive to ignore, because the fluent output *looks* like the constraint has already been met.

The temptation this chapter has to refuse is the temptation to treat "machines" as a fifth conception of critical thinking. It is not. Hitchcock's shared concept — careful, goal-directed thinking — does not acquire a new rival because the document was generated. Ennis's aim (belief or action) does not split into a special AI-aim. What changes is the *object* on which the work is practiced. Chapter 7's object was a live page. Chapter 8's objects were a clinic number, a hiring story, a courtroom inversion. This chapter's object is a fluent paragraph that did not have to be written by the person who will act on it. Same work. New object. One chapter.

A second temptation is to treat the education book's causal record as if it had already answered the life question. It has not. Bastani measured the next unaided exam, not a year of civic sharing. Fan measured an essay task and a knowledge test, not a consent form. Stadler measured justification quality after twenty minutes on sunscreen nanoparticles, not calibration at work. Those are the right studies to cite. They are the wrong studies to stretch into a life-span policy. The stretch is how sequels get written. This book will cite, not stretch.

## 9.2 Fluency fools: look finished, look like math, look like thought

Three findings, already in this book under other headings, join under one civic mechanism.

Joel Breakstone, Mark Smith, Sam Wineburg, and colleagues gave 3,446 high-school students a live internet connection and six constructed-response tasks. Asked to investigate a site claiming to "disseminate factual reports" on climate science, 96 percent never learned the organization's fossil-fuel ties. Two thirds could not distinguish news stories from ads on a popular homepage. More than half treated an anonymously posted Facebook video, shot in Russia, as "strong evidence" of U.S. voter fraud. The signals that fooled them were look, top-level domain, About pages, and quantity of information — the surface form of a finished document.[491] Chapter 7 spent that paper as a civic baseline and as the case against checklist literacy. Those are exactly the signals CRAAP-style checklists train students to trust: an author, a reference list, a `.org`, an About page free of typos. Wineburg, Breakstone, Nadav Ziv, and Smith's working paper *Educating for Misunderstanding* already named the failure mode — vertical, feature-based evaluation of a page that adversaries can fake.[492] The machine-age restatement is one sentence: a cloaked `.org` fooled students because

[490] Robert H. Ennis, working definition as restated in Ennis, "The Nature of Critical Thinking: An Outline of Critical Thinking Dispositions and Abilities," criticalthinking.net (2011; already the 1985 *Educational Leadership* working definition): "Critical thinking is reasonable reflective thinking focused on deciding what to believe or do."

[491] Joel Breakstone, Mark Smith, Sam Wineburg, Amie Rapaport, Jill Carle, Marshall Garland, and Anna Saavedra, "Students' Civic Online Reasoning: A National Portrait," *Educational Researcher* 50, no. 8 (2021): 505–515. n = 3,446 high-school students, live internet, six constructed-response tasks. Spent at length in Chapter 7; restated here as the look-finished mechanism.

[492] Sam Wineburg, Joel Breakstone, Nadav Ziv, and Mark Smith, *Educating for Misunderstanding: How Approaches*

it *looked* finished. A model that can generate a cloaked `.org`'s prose on demand does not invent that failure. It industrializes the look. Quantity of information, the fourth of Breakstone's fooling signals, is now free. Fluency plus volume is not a new species of error. It is the old error with the production costs removed.

Bastani's unguarded arm is the school version of the same look. The generated solution *looked* like math. Practice scores rose. The unaided exam, on isomorphic items, fell. Students used the unguarded model as a crutch and did not perceive the loss. The hint tutor — GPT Tutor, which withheld solutions and used teacher-authored hints — raised practice more and left the unaided exam approximately at control. The difference is not "AI versus no AI." It is whether the interface finished the sentence or forced the student to keep evaluating. Collapsing that difference into a single story about "AI in schools" is how a life book would sequel the education book. Keeping the difference is how this chapter stays on its title. The unperceived loss is the life detail. A person who knows they used a crutch can, at least in principle, put it down on the next stake. A person who did not perceive the loss has a fluent document and a false sense that evaluation occurred. Chapter 4 already refused to treat performance as later unaided judgment. Bastani is that refusal with a model in the loop. The practice score was the artefact of the session. The exam was the judgment. They came apart, and the students could not see the seam.

Yizhou Fan, Luzhen Tang, Huixiao Le, and colleagues, including Dragan Gašević, assigned 117 university students to four supports — ChatGPT, a human expert, writing-analytics tools, or none — on an essay task. The ChatGPT group improved essay scores more. There were no significant differences in knowledge gain or transfer. Process mining showed fewer metacognitive processes — evaluation and orientation among them — than in the human-expert group. The authors' term is "metacognitive laziness."[493] The essay *looked* like thought. The knowledge did not move. The evaluation moves dropped. That is not a finding that writing with a model is always shallow. It is a finding that a better artefact is not evidence of better judgment, and that the missing work is exactly the work this book has been calling critical thinking. Fan's "metacognitive laziness" is an authors' term, not a diagnosis of character. Process mining showed fewer evaluation and orientation moves than the human-expert arm. The students were not scored as lazy people. They were scored as people whose process skipped the steps this book's coalition treats as the activity — Ennis's deciding, Dewey's suspended judgment, Willingham's implementing of a maxim in a domain. A life book that treated a higher essay score as a critical-thinking gain would have confused Chapter 6's warning (the test is not the thing) with a new, worse version: the *artefact* is not the thing, and now the artefact can be generated.

The civic mechanism that joins the three is older than the models. Lisa Fazio, Nadia Brashier,

*to Teaching Digital Literacy Make Students Susceptible to Scammers, Rogues, Bad Actors, and Hate Mongers,* SHEG working paper A-21322 (2020). CRAAP / vertical-checklist critique. PDF fetch timed out in the research for this book; abstract and stacks snippet used. Do not invent extra percentages from that paper. Working paper, not a journal article.

[493] Yizhou Fan, Luzhen Tang, Huixiao Le, Kejie Shen, Shufang Tan, Yueying Zhao, Yuan Shen, Xinyu Li, and Dragan Gašević, "Beware of Metacognitive Laziness: Effects of Generative Artificial Intelligence on Learning Motivation, Processes, and Performance," *British Journal of Educational Technology* 56, no. 2 (2025): 489–530, https://doi.org/10.1111/bjet.13544. N = 117 university students; four supports (ChatGPT, human expert, writing-analytics tools, none). ChatGPT group improved essay scores more; no significant differences in knowledge gain or transfer; fewer metacognitive processes (evaluation, orientation) than the human-expert group. Authors' term: "metacognitive laziness." Already treated at length in *Autonomous AI and Education*.

B. Keith Payne, and Elizabeth Marsh showed that repetition increases judged truth even when you know better: knowledge does not protect against illusory truth.[494] A fluent sentence, repeated, feels true. A machine that can emit fluent sentences at the scale of a feed is a fluency engine. Combined with Breakstone's look, Bastani's copied arithmetic, and Fan's evaluation drop, the mechanism is not mysterious. People use the feeling of a finished product as a cue that evaluation occurred. When the product is generated, that cue is a lie about the human's work, not about the model's syntax. Hasher's original illusory-truth finding, and Fazio's demonstration that prior knowledge does not protect, were not about machines. They were about repetition and fluency as a truth-cue in human prose. Machine fluency does not replace that literature. It scales the cue. A generated sentence can be repeated across a feed, a slide deck, a parent-teacher email, and a clinic portal in the time it used to take to type one of them. The Debunking Handbook's advice — do not skip correction for fear of backfire; continued influence is the real problem — still applies.[495] What it cannot do by itself is restore the evaluation that was never done because the sentence arrived finished.

None of this requires a claim that models cannot be useful. Bastani's hint tutor is useful in the narrow sense that it did not harm the unaided exam. Fan's essays were better as essays. The life problem is not usefulness. It is the substitution of fluency for evaluation, and the fact that the substitution is pleasant. Ease is the feature. Depth is the cost. The next section is the older psychology of that trade.

## 9.3 Offloading is old; the new risk is offloading evaluation

Evan Risko and Sam Gilbert, writing in *Trends in Cognitive Sciences* in 2016, defined cognitive offloading as the use of physical action to alter the information-processing requirements of a task so as to reduce cognitive demand: tilting the head at a rotated image, setting a reminder, writing it down.[496] Their framework is metacognitive. People choose internal versus external strategies based on beliefs and feelings about their own capacities. Those evaluations can be wrong, producing suboptimal offloading. Intention offloading — creating an external cue for a delayed intention — is a central real-world case that laboratory prospective-memory studies used to treat as noise. Offloading is old. It is often adaptive. A calendar is not a moral failure. A shopping list is not a decline of civilization. The critical-thinking problem does not begin when information is stored outside the head. It begins when the thing offloaded is the *evaluation* itself, and the metacognitive cue that evaluation occurred is the fluency of the external product.

Betsy Sparrow, Jenny Liu, and Daniel Wegner's 2011 paper on Google effects is the pre-LLM cousin, not a sequel. When people expect information to remain available, they remember where,

[494] Lisa K. Fazio, Nadia M. Brashier, B. Keith Payne, and Elizabeth J. Marsh, "Knowledge Does Not Protect against Illusory Truth," *Journal of Experimental Psychology: General* 144, no. 5 (2015): 993–1002. Also the *Debunking Handbook 2020* consensus restatement: repetition increases judged truth even when you know better.

[495] Stephan Lewandowsky, John Cook, Ullrich K. H. Ecker, Dolores Albarracín, Michelle A. Amazeen, Panayiota Kendeou, et al., *The Debunking Handbook 2020* (2020), https://doi.org/10.17910/b7.1182. Twenty-two-author consensus. Do not skip correction for fear of backfire; continued influence is the surviving problem. Spent at length in Chapter 7.

[496] Evan F. Risko and Sam J. Gilbert, "Cognitive Offloading," *Trends in Cognitive Sciences* 20, no. 9 (2016): 676–688. Offloading = physical action to alter information-processing requirements so as to reduce cognitive demand; metacognitively governed; evaluations of one's own capacities can be wrong.

not what.[497] That is a memory finding, not a critical-thinking finding, and it should not be inflated into a claim that search destroyed thought. It is the right ancestor because it already showed the substitution: an external store changes what the person keeps, and the person may not notice. A language model is a more aggressive external store. It does not only hold the fact. It holds a finished-looking argument about the fact.

Matthias Stadler, Maria Bannert, and Michael Sailer ran the comparison that makes the cost visible. Ninety-one university students had twenty minutes to research nanoparticles in sunscreen and write a recommendation with justifications; notes were then removed. ChatGPT-3.5 versus Google. After adjusting for prior knowledge, justification quality averaged 1.87 in the search condition and 1.20 in the LLM condition on a 0–7 scheme whose observed maximum was 4 ($\eta^2 = 0.11$). Germane cognitive load — the load associated with learning-relevant processing — was substantially lower with the LLM ($\eta^2 = 0.27$) and fully mediated the quality difference.[498] Extraneous and intrinsic load also fell. Homogeneity of recommendations did not differ. The model made the task easier and shallower. Fluent answers arrived. The justifications were thinner. Ease is not depth, and the missing depth is exactly the germane work evaluation is supposed to be. Prior-knowledge-adjusted figures, for the record: extraneous load 3.96 versus 3.00 ($\eta^2 = 0.09$); intrinsic load 4.43 versus 3.16 ($\eta^2 = 0.15$); germane load 4.79 versus 3.14 ($\eta^2 = 0.27$). The germane-load drop fully mediated the quality difference (indirect $\beta = 0.15$, $p = 0.020$; the direct path was not significant). That mediation is the chapter's mechanism in a statistic. The model did not merely correlate with thinner justifications. The thinning ran through the reduction in learning-relevant processing. Students were not "using AI wrong." They were using a tool that made the evaluative work optional, and optional work is the work people skip.

Stadler measured product quality, not calibration. That limit belongs on the page. We do not learn from this study whether students *knew* their justifications were thinner, or whether the fluency of the generated draft fooled them into thinking they had evaluated. Fan's process-mining result — fewer evaluation and orientation moves — is the closer measure of the missing work, and it is still a process measure on an essay task, not a life-span finding about judgment. The open question is the metacognitive one Risko and Gilbert already named: when do people correctly treat fluent external output as unevaluated? The studies we have are consistent with a wrong answer to that question. They are not a fitted theory of adult calibration in a workplace that keeps the model in the room.

The moralism that sometimes attaches to offloading is a distraction. No serious account in this book requires a person to compute a posterior in their head when a natural-frequency tree would make the same number usable, or to reconstruct a climate organization's funding from memory when a new tab would show it. Chapter 7's lateral reading *is* offloading of a kind: leave the page, use other pages. Chapter 8's natural-frequency tree is offloading of a kind: recode the problem onto paper. Those are evaluation tools. The new risk is the tool that performs the evaluation's *appearance* and

[497] Betsy Sparrow, Jenny Liu, and Daniel M. Wegner, "Google Effects on Memory: Cognitive Consequences of Having Information at Our Fingertips," *Science* 333, no. 6043 (2011): 776–778. Pre-LLM cousin, not a sequel: remember where, not what.

[498] Matthias Stadler, Maria Bannert, and Michael Sailer, "Cognitive Ease at a Cost: LLMs Reduce Mental Effort but Compromise Depth in Student Scientific Inquiry," *Computers in Human Behavior* 160 (2024): 108386. N = 91; ChatGPT-3.5 vs. Google; 20 minutes on nanoparticles in sunscreen; notes removed. Justification quality 1.87 vs. 1.20, $\eta^2 = 0.11$; germane load $\eta^2 = 0.27$, fully mediating the quality difference (indirect $\beta = 0.15$, $p = 0.020$; direct ns). Stadler measured product quality, not calibration.

thereby removes the occasion to evaluate. A search result still looks like a source you might check. A generated paragraph looks like a conclusion you might adopt. That is the difference that makes this a critical-thinking chapter rather than a technology chapter.

## 9.4 Kosmyna is a preprint

Nataliya Kosmyna, Eugene Hauptmann, Ye Tong Yuan, and colleagues posted "Your Brain on ChatGPT" to arXiv in 2025 (arXiv:2506.08872). It is a preprint. It was not peer-reviewed as of 30 August 2026. Three groups — LLM, search engine, brain-only — wrote essays across three sessions; a fourth session crossed LLM users to brain-only and brain-only users to LLM. The sample is 54 for sessions 1–3 and 18 for session 4. EEG connectivity (dDTF) was strongest and most distributed in the brain-only group, moderate in search, weakest in the LLM group. The LLM group reported the lowest ownership of the text and was impaired at quoting their own writing. Session-4 LLM-to-brain showed reduced alpha/beta connectivity, which the authors read as under-engagement. They call the accumulation "cognitive debt."[499]

That paper will be cited as proof that AI is rotting brains. It does not prove that. Connectivity during a lab essay is not critical-thinking ability. A sample of 54, shrinking to 18, is not a population. There is no delayed knowledge test of the Willingham kind, no school RCT, no workplace outcome, no civic measure. The finding is consistent with Stadler's ease-without-depth and with Risko and Gilbert's offloading-plus-metacognitive-error. Consistency is not a license to extrapolate EEG onto a life of judgment. Viral summaries of this preprint are not this preprint. An honest book plants the citation as a caution people will use, and then refuses to let it carry the chapter. Session 4's n = 18 is small enough that a single later trial could move the EEG pattern. Even if the pattern replicates, the construct still would not be this book's construct. Reduced alpha/beta connectivity after LLM use is a neural correlate of a writing session. Critical thinking, in Ennis's sentence, is reasonable reflective thinking focused on deciding what to believe or do. No bridge from dDTF to that sentence has been built. Do not build it in a footnote.

The refusal is not a defense of unguarded models. Bastani's −17 percent and Fan's evaluation drop are peer-reviewed reasons to worry about offloaded evaluation. They are enough. Inflating a preprint into a civilizational diagnosis is how a life book would acquire a morality play it has not earned. The remaining human work is still evaluation. Kosmyna does not measure that work. It measures connectivity while people write essays with and without a model. Keep the label. Keep the n. Keep moving.

[499] Nataliya Kosmyna, Eugene Hauptmann, Ye Tong Yuan, Jessica Situ, Xian-Hao Liao, Ashly Vivian Beresnitzky, Iris Braunstein, and Pattie Maes, "Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task," arXiv:2506.08872 (2025). **Preprint; not peer-reviewed as of 30 August 2026** (still marked preprint / under review on the arXiv PDF this pass; MIT Media Lab page: peer review started, not completed). n = 54 sessions 1–3; n = 18 session 4. EEG (dDTF) connectivity strongest in brain-only, weakest in LLM. Authors' term: "cognitive debt." Connectivity is not critical-thinking ability; no delayed Willingham-style knowledge test; not a population finding.

## 9.5 Detectors fail and punish the wrong writers

If the life problem is fluent output that looks like thought, a tempting policy is to detect the fluency and punish it. Weixin Liang, Mert Yuksekgonul, Yining Mao, Eric Wu, and James Zou tested that policy as a measurement claim. GPT detectors misclassified 61.22 percent of human TOEFL essays as AI-generated, while remaining near-perfect on U.S. eighth-grade essays. Eighty-nine of ninety-one TOEFL essays (97.8 percent) were flagged by at least one of seven detectors.[500] The bias is against non-native English writers. The instrument does not catch cheaters. It catches a writing profile. Near-perfect performance on U.S. eighth-grade essays is not a success. It is a demonstration that the classifier is keyed to a register — shorter sentences, a certain lexical simplicity — that TOEFL writers, writing honestly in a second language, do not occupy. A life policy built on that classifier would not police “AI use.” It would police whose English looks like a child’s, and whose looks like a model’s, and it would get the second group wrong more than half the time on the TOEFL set.

Debora Weber-Wulff, Alla Anohina-Naumeca, Sonja Bjelobaba, and colleagues tested fourteen detectors — twelve public tools plus the commercial systems Turnitin and PlagiarismCheck. The tools were neither accurate nor reliable, and they were worse after paraphrase and other obfuscation.[501] A student who can paraphrase can evade. A student whose English is a second language, writing honestly, cannot evade a false positive rate north of 60 percent on TOEFL-like prose. Detector scores are not critical thinking. They are not evidence of cheating. They punish the wrong writers.

Chapter 6 already treated assessment redesign as the surviving response when a take-home essay is no longer, by itself, evidence of the student’s judgment. That sentence is older than generative models; the models made it unskippable. The life-policy restatement is the same redesign, pointed at adults. Process evidence, oral defense, unaided performance on the stakes that are the person’s own — consent forms, votes, diagnoses they must understand — are evaluations. A detector score is not. Collapsing performance into a detector is Liang’s false positives with a workplace or a civic office attached. It is a way of pretending that the fluency problem has an instrument, when the instrument is a bias against the writers this book’s equity chapter will not be able to ignore. Chapter 10 will print Breakstone’s SES, race, maternal-education, and free/reduced-price-lunch gradient on civic online reasoning — an equity finding from *before* ChatGPT. A detector that flags non-native writers at 61.22 percent is a second gradient, now on language status, laid on top of the first. Combining them in a school or a workplace that already under-teaches lateral reading is not a critical-thinking reform. It is a way of calling the wrong people untrustworthy.

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[500]Weixin Liang, Mert Yuksekgonul, Yining Mao, Eric Wu, and James Zou, “GPT Detectors Are Biased against Non-Native English Writers,” *Patterns* 4, no. 7 (2023): 100779, https://doi.org/10.1016/j.patter.2023.100779. **61.22 percent** is the authors’ average false-positive rate on 91 human TOEFL essays (arXiv HTML). *Patterns* HTML rounds the same average to **61.3 percent**. Body keeps 61.22, matching the education book’s note apparatus. Near-perfect on 88 U.S. 8th-grade essays. 18/91 flagged by all seven detectors; **89/91 (97.8 percent)** flagged by at least one.

[501]Debora Weber-Wulff, Alla Anohina-Naumeca, Sonja Bjelobaba, Tomáš Foltýnek, Jean Guerrero-Dib, Olumide Popoola, Petr Šigut, and Lorna Waddington, “Testing of Detection Tools for AI-Generated Text,” *International Journal for Educational Integrity* 19, art. 26 (2023), https://doi.org/10.1007/s40979-023-00146-z. Preprint arXiv:2306.15666. Fourteen tools: twelve public plus Turnitin and PlagiarismCheck. Authors’ verdict: neither accurate nor reliable; bias toward classifying output as human-written; worse after paraphrase, machine translation, and other obfuscation. Not a finding that detectors catch cheaters.

Do not claim detectors catch cheaters. The papers do not say that, and the numbers they do report run the other way. A school, a firm, or a platform that treats a detector dashboard as a critical-thinking policy has confused a failed classifier with a judgment.

## 9.6 Two lanes are a design hypothesis, not a fitted schedule

The education book used two lanes as a theoretically motivated assessment design: *formation*, in which the skill being built is practiced without the model, or with a hint-only tutor; and *performance*, in which the model is allowed, logged, and judged as a tool. This chapter restates the same distinction as a life-design, and labels it as what it is. It is not existing district policy. It is not a fitted life-span schedule. It is not a finding that any workplace already runs it. It is a design hypothesis motivated by Bastani, Fan, Stadler, and Liang, and constrained by Willingham.

Formation, in Bastani's terms, is the unaided exam, or the hint tutor that withholds solutions. GPT Tutor's unaided exam was approximately at control. The skill being formed was still the student's. Performance is the adult workplace and the civic feed: the model is in the room, the output is a draft or a summary or a number, and the human is supposed to be the evaluator. Collapsing formation into ChatGPT is the −17 percent result. Students practiced with a machine that finished the sentence and then could not do the isomorphic work alone. Collapsing performance into a detector is Liang: the institution pretends to evaluate the human by classifying the prose, and classifies the wrong humans.

Adults will keep the model in the room. Bastani's unaided exam is a school. A consent form, a hiring recommendation, a share button, and a generated brief about a medical number are not closed-book tests. When refusal of fluent output is costly — when the draft is due, when the meeting is now, when the alternative is to look slow — the life problem is harder than the exam problem. The two-lane rule does not dissolve that cost. It names it. Some stakes have to remain formation: you must be able to understand the number, the source, the diagnosis, without the window. Some stakes are performance: you may use the model, but then the work is logged and the judgment is yours. Mixing the lanes without saying which stake you are on is how a person spends a decade with a fluent crutch and calls it expertise.

The adult version of formation is not a nostalgia for closed-book school. It is the set of stakes on which the person must still be able to notice that they do not know. A generated relative-risk sentence about a screening test is not a draft to be edited for tone. It is a number that will be used on a body. Chapter 8 already showed that physicians, not only patients, invert those numbers; Casscells, Schoenberger, and Graboys found 11 of 60 in a Harvard Medical School sample giving the correct ~2 percent.[502] A model that emits the 95 percent answer fluently has not made the inversion rarer. It has made the inversion look like expertise. Formation, on that stake, means being able to recode the number with the window closed. Performance, on a neighboring stake, might mean using the model to draft a letter to a specialist, logged, and then defending the ask. The lanes are not moral grades. They are different objects of practice.

[502] Casscells, Schoenberger, and Graboys, "Interpretation by Physicians of Clinical Laboratory Results," *New England Journal of Medicine* 299 (1978): 999–1001. Harvard Medical School faculty, staff, and students; prevalence 1/1,000, false-positive 5 percent; 11 of 60 ~2 percent; 27 of 60 said 95 percent. Spent in Chapter 8; restated here as the stake on which a fluent inversion is costly. Original NEJM paywalled; numbers via AMA 2009 restatement.

The hypothesis can be wrong in several ways the record has not settled. There is no fitted schedule that says formation until 22 and performance thereafter, or formation for health and performance for email. There is no RCT that a two-lane workplace produces better later judgment than an always-on model or a ban. Metacognitive monitoring of offloaded work — when people correctly treat fluent output as unevaluated — is the measurement Stadler did not take. This book will not invent the schedule to make the hypothesis look like a curriculum. Chapter 10's deed is the honest remainder: refuse a fluent wrong answer, including from a model, including when refusal is costly. The deed is not a lane. It is what the lanes are for.

## 9.7 The evaluative loop *is* CT — and it has no transfer RCT

The remaining human work, named as a loop rather than as a skill, is this: interpret the output; check it laterally, numerically, or against a source you actually have; refuse a fluent wrong answer; orally defend the conclusion with the model closed. That loop is Ennis (deciding what to believe or do) plus Willingham (you need knowledge to check) plus Wineburg (leave the page) plus Gigerenzer (recode the number). It is not a new conception of critical thinking. It is the book's conception under the life condition that a machine will finish the sentence.

Interpret is not optional. A generated paragraph about a climate organization, a generated proof, a generated relative-risk sentence, and a generated recommendation about nanoparticles in sunscreen are all objects that look finished. Interpretation means noticing what claim is being made and what would have to be true for it to be usable. Without that, there is nothing to check.

Check is the domain move. Laterally: who's behind this, what's the evidence, what do other sources say — Civic Online Reasoning's three questions, already practiced in Chapter 7, now pointed at a model that can fake the look of a source. Numerically: recode the percentage as a natural frequency, as Chapter 8 required of a mammogram poster and a prosecutor's statistic. Against a source: a worked example, a primary document, a number you can locate. The check is empty without the knowledge the check requires. Willingham's constraint does not relax because the sentence was generated. If anything it tightens. The generated sentence is more likely to be locally fluent in a domain you do not know, which is exactly when you cannot see that it is wrong.

Refuse is the deed Chapter 10 will print. Bastani's students copied arithmetic errors they could have caught. Fan's essays got better while evaluation dropped. Liang's detectors were fooled by fluency in the other direction, flagging human TOEFL writers. The adult habit is not handing the conclusion to whoever sounds finished — a `.org`, a relative-risk poster, a colleague's story, or a model. Refusal is costly. The loop that omits it is not a loop. It is adoption.

Orally defend with the model closed is the life version of Bastani's unaided exam and of Chapter 6's oral-defense redesign. If you cannot say the conclusion, the check, and the remaining uncertainty without the window, you have a document, not a judgment. AIAS, the AI Assessment Scale, is language for that redesign, not a trial that it improves critical thinking.[503] The point is construct validity for judgment, not a new badge.

[503] Mike Perkins, Jasper Roe, Leon Furze, and Jason MacVaugh's AI Assessment Scale (AIAS, 2024) is language for assessment redesign, not an RCT that oral defense improves critical thinking. Mike Perkins, not David Perkins. Dissemination claims ("used in more than 350 institutions") are not effect sizes. See Chapter 6.

No RCT shows that practice evaluating chatbot output transfers to non-AI source evaluation, or the reverse.[504] Willingham predicts it will not, without shared domain knowledge. The prediction follows from everything Chapter 3 installed. Surface structure captures attention; deep structure is what would transfer; novices do not see the analog. A unit called "evaluate the chatbot" is a surface. The deep structure — leave the page, recode the number, notice when you do not know — transfers, when it transfers, because the person has practiced those moves on enough real objects in a domain they know. This chapter will not fill the gap with analogy. The transfer question is Open question 12 of the book, and it stands. A unit that practiced "spot the AI tell" would, on Willingham's account, train a surface. Tells change. Fluency is the durable property. The durable move is not a tell. It is the same move Chapter 7 already had evidence for: leave the page. A generated paragraph is a page. Who's behind it, in this case, may be a model plus a prompt plus whatever the prompt retrieved. What's the evidence, in this case, is still a number or a document that has to be located. What do other sources say, in this case, is still a new tab. If those moves transfer, they transfer because they were practiced on real objects in a domain, not because they were branded as AI literacy. That is a prediction. It is not a trial.

The coalition veto, restated for machines, is the same veto the rest of the book has been carrying. Ennis's sentence still names the work: decide what to believe or do, including when the sentence arrives finished. Willingham still vetoes a content-free general course, now in the form of an "AI literacy" module that promises to make a person a critical user of models as such. Dewey still vetoes fallacy-spotting as the whole, which includes spotting "AI tells" as if those were the inquiry. McPeck's *about what?* still applies: thinking about a generated medical number is thinking about medicine, not about machines. Delphi still cannot be the meaning, and a detector score still cannot be the test. One AI chapter was enough.

What this chapter has shown is narrower than a sequel and more usable than a slogan. Fluency fools because it looks like finished judgment. Offloading is old; offloading evaluation is the new risk. A preprint about EEG is not population decline. Detectors fail and fail unfairly. Two lanes are a design hypothesis, not a schedule. The evaluative loop is this book's title under machine conditions, and it has no transfer RCT. The next chapter does not award a badge for having read that. It names the deeds a life of this work actually contains — pause, check laterally, notice when you do not know, change a mind including your own, refuse a fluent wrong answer — and the stages and inequities in which those deeds either get practiced or become a luxury.

[504] No RCT showing that "evaluate the chatbot" transfers to non-AI source evaluation, or the reverse, was found in the research for this book (access date 30 August 2026). Willingham's prediction follows from the knowledge constraint in Chapter 3. Open question 12; not filled by analogy.

# Chapter 10

## A Life of Deeds, Not a Badge

Figure 11: A metronome, folded newspapers, a notebook, a card marked X, and a slim closed device in a row.

A life of critical thinking is not a certificate, not a NACE competency PDF, not a NAMLE principle, not a completed Philosophy for Children unit, and not a CCTST score. It is five deeds the literatures in this book actually describe, each with a source already on the page.

Pause — before sharing, ask whether it is true (Pennycook and Rand); slow down and consider plausibility against alternatives (*Debunking Handbook 2020*). Check a source laterally — leave the page; who's behind it, what's the evidence, what do other sources say (Civic Online Reasoning); Wikipedia as a first hop, not a banned last hop (McGrew and colleagues). Notice when you don't know — you cannot think critically about a domain you lack (Willingham); if you cannot recode the number, you are not informed (Gigerenzer); has enough information been collected? (National Academies, 2015). Change a mind in public, including your own — corrections usually move factual belief a little (Nyhan 2020; the handbook); they often fail to move votes in polarized settings (Swire-Thompson and colleagues); the deed is still to say "I was wrong about the number" rather than to wear a critical-thinking identity. Refuse a fluent wrong answer — from a `.org`, a relative-risk poster, a colleague's story, or a model (Bastani's copied arithmetic; Fan's evaluation drop; Liang's fluency fooling detectors the other way).[505]

Those are not a bicycle. Each deed is empty without the domain knowledge Chapter 3 installed. The rest of this chapter is what "for life" can mean once that is admitted: not a unit, not a developmental RCT this book does not have, not a decline story about older adults, not an equity sermon that invents a trial, and not a political morality smuggled into the meaning of the phrase. Knowledge plus evaluation habits, restated at each stake, including your own mind.

The title's payload is the word *life*, and the word is doing two jobs that a curriculum poster collapses. One job is duration: not a semester, not a course, not a unit test, but the decades in which a person will meet new pages, new numbers, new fluent sentences, and old identities. The other job is stakes: not a scored item, but a vote, a share, a test result, a hiring story, a sentence a machine will finish. Wineburg's program is the closest the civic literature comes to a bicycle that *does* exist — a short set of evaluation habits practiced on real pages in a civic domain — and even then six hours doubles a still-failing score. Inoculation and accuracy nudges are narrower still: technique recognition and sharing attention, perishable. NACE and WEF want a word that sounds like critical thinking and then test a story in context. Gigerenzer shows the same structure in health: a representation plus a decision, not a disposition. Bastani and Fan show what happens when fluency is outsourced. Edwards names the political risk: the evaluative adult becomes private. The deeds are what remains when those findings are not allowed to become a slogan.

## 10.1 "For life" is not a unit

Self-help closes reinstall the folk collapse. They take a list of habits, print them as a poster, and promise that practice will make a person a critical thinker *everywhere*. That sentence is the conjunction Chapter 1 refused, now in the second person. This chapter will not write it.

Philip Abrami and colleagues' 2008 headline, spent in Chapter 5 as a teaching result, restates here as a life rule: improvement "cannot be a matter of implicit expectation."[506] Objectives have to be explicit. Immersion — critical thinking as a hoped-for by-product of a decent life, a decent school, a decent feed — was the weakest design in that meta-analysis. A life of this work is therefore

[505] The five deeds as specified in the research brief for this book, 30 August 2026, from sources already spent in Chapters 3, 7, 8, and 9. Individual citations follow in §10.5.

[506] Philip C. Abrami, Robert M. Bernard, Evgueni Borokhovski, Anne Wade, Michael A. Surkes, Rana Tamim, and Dai Zhang, "Instructional Interventions Affecting Critical Thinking Skills and Dispositions: A Stage 1 Meta-Analysis," *Review of Educational Research* 78, no. 4 (2008): 1102–1134. 117 studies, 20,698 participants, 161 effects; g+ = 0.341 (SD 0.610). Headline: improvement "cannot be a matter of implicit expectation." Type of intervention plus pedagogical grounding accounted for 32 percent of variance; immersion weakest.

practiced on named stakes, with feedback, over years. Willingham's 2019 planning note, that skills meant to stick need three to five years of revisiting, now applies to a life's domains rather than to a school subject.[507] Money, health, news, work, friendship, citizenship are not "everything in general." They are the subjects. The knowledge and the practiced recognitions have to be specified per domain, or the deeds above are maxims again.

What this is not, item by item. It is not a K–12 unit you can schedule on Tuesdays and stamp as complete. Wineburg's six Civic Online Reasoning lessons moved credibility-task scores and still left students at about half the possible points; that is a real, modest, incomplete gain after six hours, not a life.[508] It is not a NACE Career Readiness Competency. Recruiters rate critical thinking 4.49 on a five-point importance scale, behind communication at 4.57, and then ask a situated interview story.[509] A story about anticipating a problem is closer to Willingham than to a badge. It is not a NAMLE Core Principle. The 2023 revision asks learners to "practice active inquiry, reflection, and critical thinking"; that is a professional association's position, not a trial.[510] It is not a completed P4C unit. Gorard, Siddiqui, and See's 2015 efficacy trial found small attainment effects; the Education Endowment Foundation's 2021 effectiveness trial, 198 schools, five padlocks, found **0 months**.[511] Lipman still earns the childhood page later in this chapter as pedagogy. He does not earn a life-certificate. It is not a CCTST badge. Chapter 6 already refused to treat the test as the thing. A score is an operationalization of Delphi, not a life of judgment.

The design rule that survives is Abrami's explicitness plus Willingham's revisiting plus Ennis's aim. Name the stake. Practice the move on that stake. Come back in years, because even well-learned content is about half forgotten in three (Pawl and colleagues, as Willingham cites).[512] Do not write "practice these twelve habits and you will be a critical thinker everywhere." There is no twelve. There are five deeds, and they are domain-bound, and they do not transfer as a bicycle transfers. Chapter 4 already spent that constraint. This chapter inherits it.

Huber and Kuncel's college result is the adult version of the same warning. Ordinary college already moves generic critical-thinking tests about 0.5 to 0.6 standard deviations over four years. Curriculum-wide "critical thinking" campaigns, and nursing's mandated critical-thinking curricu-

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[507] Daniel T. Willingham, *How to Teach Critical Thinking*, Education: Future Frontiers Occasional Paper Series (NSW Department of Education, 2019). Four-step plan; skills meant to stick: plan on three to five years of practice (citing Bahrick 1984; Bahrick and Hall 1991). Even well-learned content: about half forgotten in three years (Pawl et al. 2012, as cited by Willingham).

[508] Sam Wineburg, Joel Breakstone, Sarah McGrew, Mark D. Smith, and Teresa Ortega, "Lateral Reading on the Open Internet: A District-Wide Field Study in High School Government Classes," *Journal of Educational Psychology* 114, no. 5 (2022): 893–909. Treatment n = 271, matched control n = 228; six ~50-minute COR lessons. Doubling / half-points from Stanford GSE news, 19 April 2022 — journalism/PR summarizing the paper, not the PDF.

[509] National Association of Colleges and Employers, *Job Outlook 2025*. 237 respondents; 162 members = 19.2 percent of eligible. Communication 4.57, critical thinking 4.49, teamwork 4.43. Sample interview: "describe a time when you anticipated a problem." Recruiter survey, not a skill assessment; not a probability sample of all employers.

[510] National Association for Media Literacy Education, Core Principles (2023 revision). Framework, not a trial. Principle 4: "practice active inquiry, reflection, and critical thinking."

[511] Stephen Gorard, Nadia Siddiqui, and Beng Huat See, "Philosophy for Children: Evaluation Report and Executive Summary," Education Endowment Foundation (2015); EEF, "Philosophy for Children: Evaluation Report" (2021). 2015: 48 English primary schools; KS2 n = 1,529; reading +0.12. 2021: 198 schools, 3,601 pupils, 5/5 padlock, **0 months**. Do not hide the 2021 result. Do not say Gorard "showed P4C works."

[512] Willingham 2019, citing Pawl et al. 2012; Bahrick 1984; Bahrick and Hall 1991. See n. 452.

lum, did not add a reliable increment on those same tests.[513] Stamping the phrase on every syllabus is immersion-by-committee. Stamping it on a life — a podcast, a poster, a LinkedIn badge, a completed MOOC — is the same design with worse feedback. Abrami 2015's overall g+ of 0.30 on standardized critical-thinking tests is real and small and not far transfer. The combination that actually showed up, dialogue plus authentic problems plus mentoring, g = 0.57 from 19 studies, is a teaching design, not a life-hack.[514] A life that wants that combination has to build it: a real problem, a person who will talk back, a mentor who knows the domain. That is expensive. The badge is cheap. This chapter is about the expensive thing.

The implementation unit for a non-school life is the architectural decision the definitions memo already named. Willingham's four steps assume subjects, years, and a curriculum. A book for adults needs an analogue: domains of a life as the "subjects," with knowledge and practice specified per domain. Money is not health is not news is not work. The maxim "consider the base rate" is empty until the base rate is a number you have. The maxim "leave the page" is empty until there is a page, and a second page, and enough knowledge to notice that they disagree. "For life" is that analogue. It is not a unit because a life is not a semester.

## 10.2 Stages as what the literatures imply, not as a developmental RCT

No developmental civic-outcome RCT was fetched for this book. Stage, in what follows, is what the existing literatures imply. The implication is labeled as implication. It is not a fitted curriculum. It is not a finding that critical thinking unfolds on a birthday schedule.

Early childhood. Willingham notes that even three-year-olds can think critically in domains they know, and that even trained scientists fail outside theirs.[515] Stage is domain knowledge, not a birthday. What "for life" can mean at three is: build knowledge, and build the habit of asking "how do we know?" in concrete domains. There is no Civic Online Reasoning RCT for K–2 in the sources this book stands on. There is no inoculation RCT for that age. Do not invent one. Lipman earns the childhood page here as *pedagogy*, not as EEF months. His 1988 formula — skillful, responsible thinking that facilitates good judgment because it relies upon criteria, is self-correcting, and is sensitive to context — is a reason to start before college, a reason to treat the outcome as *judgment* rather than as argument analysis, and a reason to keep community of inquiry in view.[516] Chapter 5's veto still applies: Gorard's small attainment effects and EEF 2021's zero

[513] Christopher R. Huber and Nathan R. Kuncel, "Does College Teach Critical Thinking? A Meta-Analysis," *Review of Educational Research* 86, no. 2 (2016): 431–468. Mixed-design 4-year CT-skill gain 0.59 SD; purely longitudinal 0.46 SD; disposition (CCTDI) 0.55 SD over 4 years. Curriculum-wide efforts not necessarily incremental. Nursing mandate did not outperform other majors on domain-general tests.

[514] Philip C. Abrami, Robert M. Bernard, Evgueni Borokhovski, David I. Waddington, C. Anne Wade, and Tonje Persson, "Strategies for Teaching Students to Think Critically: A Meta-Analysis," *Review of Educational Research* 85, no. 2 (2015): 275–314. 341 *effect sizes*, not 341 studies; g+ = 0.30; dialogue + authentic problems + mentoring combined g = 0.57 (19 studies). Ennis-type mixed/infusion/general/immersion differences **not statistically significant**. Do not cite 2015 as proof that mixed won a horse race.

[515] Daniel T. Willingham, "Critical Thinking: Why Is It So Hard to Teach?," *American Educator* 31, no. 2 (Summer 2007). "It is a type of thought that even 3-year-olds can engage in — and even trained scientists can fail in." Full bicycle passage quoted in Chapter 3, not here.

[516] Matthew Lipman, "Critical Thinking — What Can It Be?," *Educational Leadership* 46, no. 1 (September 1988): 38–43. "Skillful, responsible thinking that facilitates good judgment because it (1) relies upon criteria, (2) is self-

months are what happened when P4C was scaled as an attainment intervention. They do not erase Lipman's pedagogical claim. They forbid treating a completed P4C unit as the life. Trickey and Topping's 2004 mean effect size of 0.43, and Topping and Trickey's 2007 Dundee CAT effect size of 0.7 on a complete n of 115, are the smaller, developer-adjacent layer underneath Gorard.[517] They are not hidden. They are also not the 198-school result. A life book that used the Dundee number and skipped EEF 2021 would be doing vendor work. A life book that used EEF 2021 to erase Lipman's formula would be doing the opposite overclaim. Keep both. Childhood judgment, community of inquiry, criteria, self-correction, context-sensitivity: those are pedagogical reasons to start before a Watson–Glaser. Zero months on KS2 reading at scale: that is what happened when the pedagogy was asked to be an attainment program. The deeds in this chapter, for a child, are closer to Lipman's "how do we know?" in a known domain than they are to a media-literacy law.

Elementary. Knowledge-rich curriculum enters here as E. D. Hirsch's *equity argument*, not as a media-literacy trial. Hirsch, as printed by the Core Knowledge Foundation: "Only a well-rounded, knowledge-specific curriculum can impart needed knowledge to all children and overcome inequality of opportunity."[518] That is an argument. It is aligned with Willingham's constraint. Outcome evidence that a named U.S. knowledge-rich curriculum closed Civic-Online-Reasoning-like gaps was not fetched. Do not invent the RCT. Media-literacy *laws* sometimes start later; Illinois's high-school unit (2021) and New Jersey's K–12 requirement (2022) are named in secondary roundups, with statutory details unverified pending bill text.[519] They are laws, not elementary lateral-reading field experiments of Wineburg-2022 grade.

Secondary. The Civic Online Reasoning trials live here. Wineburg, Breakstone, McGrew, Smith, and Ortega (2022): six ~50-minute lessons in a district-mandated high-school government course, treatment n = 271, matched control n = 228; experimental classrooms grew significantly on digital-credibility tasks; Stanford GSE news, journalism summarizing the paper, reports they roughly doubled pre-test scores and still earned only about half the possible points.[520] Government class is a natural home. No turnout RCT. No polarization RCT. The remaining miss rate is the story, not the doubling. McGrew's 2020 eight-lesson intervention improved three of four online tasks. Smaller n; still a trial, not a slogan.

College and early career. Accuracy nudges and inoculation videos are tested on adults, often on on-

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correcting, and (3) is sensitive to context." Pedagogy, not EEF months.

[517] Steve Trickey and Keith J. Topping, "Philosophy for Children: A Systematic Review," *Research Papers in Education* 19, no. 3 (2004): 365–380, mean ES 0.43 on a mixed bag; Keith J. Topping and Steve Trickey, "Collaborative Philosophical Inquiry for Schoolchildren," *British Journal of Educational Psychology* 77 (2007), Dundee CAT ES 0.7, complete n = 115. Developer-adjacent layer; not the EEF 2021 effectiveness result. Colom et al. 2014 Madrid study: Gorard et al. 2015 already treated it as high-caution (two private schools, not randomised); not added here as a confirming RCT.

[518] E. D. Hirsch, Jr., *Why Knowledge Matters: Rescuing Our Children from Failed Educational Theories* (Cambridge, MA: Harvard Education Press, 2016), as printed on the Core Knowledge Foundation homepage, fetched 30 August 2026. Argument, not a fetched RCT that gaps closed.

[519] Illinois high-school unit (2021), New Jersey K–12 (2022), Delaware Digital Citizenship Education Act (2022) are named in secondary roundups. Statutory details **UNVERIFIED** pending bill text. Media Literacy Now, January 2026: 25 states have media-literacy laws; MLN has not evaluated classroom implementation.

[520] Wineburg et al. 2022; Stanford Graduate School of Education news, "It doesn't take long to learn how to spot misinformation online, Stanford study finds," 19 April 2022. Journalism/university communications. Remaining miss rate is the story.

line convenience samples. Pennycook, Epstein, Mosleh, Arechar, Eckles, and Rand (2021): veracity of headlines has little effect on sharing intentions despite a large effect on accuracy judgments; a subtle accuracy prompt raises the quality of subsequent shares.[521] Roozenbeek, van der Linden, Goldberg, Rathje, and Lewandowsky (2022): lab $d$ 0.28 to 0.68 on technique recognition; YouTube field $h = 0.09$; Maertens and colleagues (2021): Bad News effects decay to nonsignificance over two months without regular testing.[522] NACE interviews reward a situated story. Bastani and Fan are high-school and university learning studies, already spent in Chapter 9. Transfer of "evaluate the chatbot" to civic evaluation remains unshown.

Midlife work and health. Shared decision-making plus numeracy at the clinic; sharing discernment on the feed; two-lane AI at work. Elwyn and colleagues' 2012 model — choice talk, option talk, decision talk — is a practice sequence, not a civic RCT; Gigerenzer's prior claim is that shared decision-making *requires* statistical literacy or the "informed" in informed consent is empty.[523] Almost no workplace critical-thinking RCTs were fetched. Chapter 8 already refused to treat recruiter ratings as a skill assessment. Superforecasting hygiene — base rates, update, outside view, scorekeeping (Mellers and colleagues, 2014) — is closer to thinking-for-life than Wason-card performance, and it is still not a workplace RCT of "critical thinking."[524] The two-lane design from Chapter 9 is a hypothesis here, not a midlife curriculum.

Midlife is also where the error-costs Chapter 8 named stop being academic. The National Academies' synthesis: most people will experience at least one diagnostic error in their lifetime; about 5 percent of U.S. adults seeking outpatient care each year; postmortem work about 10 percent of deaths. Newman-Toker's 2023 modeling estimate — 795,000 Americans permanently disabled or dead annually from misdiagnosis, plausible range 598,000 to 1,023,000 — is a modeling estimate, not a census, and it still names a stake.[525] A midlife person who cannot recode a PSA or a mammogram number cannot share the decision. That is critical thinking as a life skill in a domain, not a generic "be skeptical" badge. The workplace analog is the interview story NACE actually asks, not a Facione skill. Almost no workplace RCT fetched means this paragraph is implication, labeled as such, not a finding that two-lane AI at work has been tried.

The whole sequence is implication from existing literatures. A writer who converted it into a

[521] Gordon Pennycook, Ziv Epstein, Mohsen Mosleh, Antonio A. Arechar, Dean Eckles, and David G. Rand, "Shifting Attention to Accuracy Can Reduce Misinformation Online," *Nature* 592 (2021): 590–595. Sharing intentions ≠ accuracy judgments. Not a belief-change paper.

[522] Jon Roozenbeek, Sander van der Linden, Beth Goldberg, Steve Rathje, and Stephan Lewandowsky, "Psychological Inoculation Improves Resilience against Misinformation on Social Media," *Science Advances* 8, no. 34 (2022): eabo6254. Lab $d$ 0.28–0.68; YouTube field $h = 0.09$. Rakoen Maertens, Jon Roozenbeek, Melisa Basol, and Sander van der Linden, "Long-Term Effectiveness of Inoculation against Misinformation," *Journal of Experimental Psychology: Applied* 27, no. 1 (2021): 1–23. Decay to nonsignificance over two months without regular testing.

[523] Glyn Elwyn et al., "Shared Decision Making: A Model for Clinical Practice," *Journal of General Internal Medicine* 27, no. 10 (2012): 1361–1367. Practice model, not a civic RCT. Gerd Gigerenzer, Wolfgang Gaissmaier, Elke Kurz-Milcke, Lisa M. Schwartz, and Steven Woloshin, "Helping Doctors and Patients Make Sense of Health Statistics," *Psychological Science in the Public Interest* 8, no. 2 (2008): 53–96.

[524] Barbara Mellers et al., "Psychological Strategies for Winning a Geopolitical Forecasting Tournament," *Psychological Science* 25, no. 5 (2014): 1106–1115. Superforecasting hygiene, not a workplace CT RCT.

[525] National Academies 2015, as n. 485. David E. Newman-Toker et al., "Burden of Serious Harms from Diagnostic Error in the USA," *BMJ Quality & Safety* 33, no. 2 (2023): 109–120. Modeling estimate 795,000 (range 598,000–1,023,000). Not a census. Johns Hopkins Medicine newsroom 17 July 2023 is journalism summarizing the paper.

K–through-retirement scope-and-sequence would have invented the developmental civic-outcome RCT this section opened by refusing. Stages are where the deeds get their objects. They are not a theory of unfolding.

## 10.3 Older adults: sharing is not detection, and decline is not enough

Andrew Guess, Jonathan Nagler, and Joshua Tucker linked a survey to Facebook profile data from the 2016 election. Sharing articles from fake-news domains was rare overall. Users over 65 shared nearly seven times as many such articles as the youngest age group, and the association held after partisanship and ideology.[526] That is a sharing finding, on Facebook, in 2016. It is not a 2026 detection RCT. It is not a finding that older adults *believe* more false news. Detection is not sharing. Print the number. Print the platform. Print the year.

Nadia Brashier and Daniel Schacter's 2020 review is the constraint that has to travel with Guess. Do **not** reduce the sharing pattern to cognitive decline. Fluency is intact. Accumulated knowledge helps evaluation. Social goals, trust, lie-detection, and digital-newcomer status are candidate mechanisms.[527] Brashier cites 2016 Census figures: older adults vote at high rates — 70.9 percent versus 46.1 percent ages 18–29. A group that shares more and votes more is a civic fact, not a neuropsychological cartoon. Nir Grinberg, Kenneth Joseph, Lisa Friedland, Briony Swire-Thompson, and David Lazer, on Twitter in the same election, found users over 50 overrepresented among "supersharers."[528] Same caveat: 2016, sharing, not detection.

The popular story that will attach to these papers is that aging brains cannot do critical thinking. The papers do not say that. Willingham's three-year-olds-can / scientists-fail already vetoed birthday as the stage variable. An older adult who has spent decades in a domain can evaluate in that domain; an older adult who is new to a platform can be fooled by the platform's fluency the way Breakstone's high-school students were fooled by a `.org`. Digital-newcomer status is a knowledge-and-practice gap, which is this book's constraint, not a decline story. Social goals — sharing to connect, to warn, to belong — are a different mechanism again, and they are not cured by a maxim. Accuracy prompts, which redirect attention toward veracity, are tested mostly on mixed-age online samples; a 2026 older-adult Civic Online Reasoning RCT was not fetched. Open question 15 stands: 2016 Facebook and Twitter are not today's short-video ecology. Targeted digital-literacy trials for 65+ are thinner than the sharing descriptive papers. This chapter will not fill that gap with a cognitive-decline analogy.

Sharing rare overall is the other half of Guess that slogan-versions drop. Most people, including

[526] Andrew Guess, Jonathan Nagler, and Joshua Tucker, "Less than You Think: Prevalence and Predictors of Fake News Dissemination on Facebook," *Science Advances* 5, no. 1 (2019): eaau4586. Survey linked to Facebook profile data, 2016. Sharing rare overall; over-65s ~7× the youngest age group after partisanship and ideology. Sharing, not detection; 2016, not 2026.

[527] Nadia M. Brashier and Daniel L. Schacter, "Aging in an Era of Fake News," *Current Directions in Psychological Science* 29, no. 3 (2020): 316–323. Do not reduce older-adult fake-news sharing to cognitive decline. Fluency intact; accumulated knowledge helps evaluation; social goals, trust, lie-detection, digital-newcomer status as candidate mechanisms. 2016 Census voting rates as cited by Brashier: 70.9 percent vs. 46.1 percent ages 18–29.

[528] Nir Grinberg, Kenneth Joseph, Lisa Friedland, Briony Swire-Thompson, and David Lazer, "Fake News on Twitter during the 2016 U.S. Presidential Election," *Science* 363, no. 6425 (2019): 374–378. Over-50s overrepresented among supersharers. Sharing/exposure study, not a detection RCT.

most older adults, were not supersharers. A life book that treated "the elderly share fake news" as the finding would have done to Guess what recruiter folklore does to NACE: strip the number from its design and print a moral. The number is ~7×, after partisanship, on Facebook in 2016, against a low base rate. Hold all four pieces.

The mechanism list Brashier and Schacter offer is itself a Willingham list. Fluency intact: older adults are not, on this review, missing the feeling of a finished sentence; they may be *more* susceptible to fluency as a truth-cue because they have a lifetime of treating fluent prose as a signal of care. Accumulated knowledge helps evaluation: that is Chapter 3, now as a protective factor, which is why reducing the sharing pattern to decline gets the cognitive science backwards. Digital-newcomer status is a practice gap on a new surface — the feed, the share button, the look of a source — not a missing faculty. Social goals and trust are not bugs in a critical-thinking program; they are why a prompt that puts accuracy in working memory can move sharing at the margin, and why a partisan identity can put it back. Lie-detection is a different skill again, and this book has no RCT that older adults are worse at it in 2026. Candidate mechanisms are not a diagnosis. They are a list of things a decline story erases.

Grinberg's supersharers are the concentrated tail, not the mean. A small number of accounts produced a large share of the fake-news circulation on Twitter in 2016; older users were overrepresented in that tail. A life of deeds for an older adult is not a population intervention aimed at "the elderly." It is the same five deeds, on a platform that may be new, in domains that may be old. Pause before the share. Leave the page. Notice whether you know the organization. Change a forwarded claim if the number is wrong. Refuse a fluent post that looks finished. None of that requires a theory of aging. All of it requires not confusing 2016 Facebook with 2026 short video, and not confusing sharing with belief.

## 10.4 Equity: the gradient is older than the chatbot

Breakstone and colleagues' 2021 national portrait already found that civic-online-reasoning scores varied by socioeconomic status, race, maternal education, and free/reduced-price lunch.[529] That is the equity finding for civic reasoning, and it is *before* ChatGPT. The students who most need the public skill are the ones the assessment says lack it. Checklist literacy that trains `.org`-trust will not close that gradient. *Educating for Misunderstanding* argued it may widen it, because the signals checklists train are the signals adversaries fake, and the students with less background knowledge have less with which to notice the fake. A machine that industrializes the look, Chapter 9's point, does not create the gradient. It lands on it.

Hirsch's equity argument is the knowledge-side of the same gradient. Comprehension and critical thought require word-and-world knowledge that affluent children get at home. A specific, communal curriculum is, on Hirsch's account, the instrument that imparts that knowledge to children who will not get it elsewhere. Willingham is the cognitive-science partner to the claim: you cannot think critically about a domain you lack; choosing content expresses values; not choosing is still a

[529] Joel Breakstone, Mark Smith, Sam Wineburg, Amie Rapaport, Jill Carle, Marshall Garland, and Anna Saavedra, "Students' Civic Online Reasoning: A National Portrait," *Educational Researcher* 50, no. 8 (2021): 505–515. Scores varied by SES, race, maternal education, free/reduced-price lunch. Equity gradient pre-ChatGPT.

choice.[530] England's "knowledge-rich" turn is a policy experiment; a 2026 New Zealand commentary arguing that gaps persisted under a Hirsch-influenced English reform is advocacy/secondary, not a trial this book will launder.[531] Do not claim that the U.S. Core Knowledge sequence has closed racial or SES gaps on the basis of the research this book stands on. Do claim the *logic*: if critical thinking is domain knowledge plus evaluation habits, then starving some children of domain knowledge and giving them a "critical thinking" standard is how the thing becomes a luxury good.

David Edwards, general secretary of Education International, in a World Economic Forum interview dated 2 July 2024 — journalism, not a trial — said AI could make human teachers "the reserve of the privileged," while children who cannot afford them "sit in rooms with a chatbot."[532] That is a union leader's argument. It is consistent with the education book's connectivity figures, which this book does not re-fetch. For *this* book the implication is local. If evaluation habits require a knowledgeable adult to model lateral reading, numeracy, and "I don't know," then chatbot-for-the-poor is the opposite of public critical thinking. Fan's metacognitive laziness and Bastani's unperceived −17 percent are what happens when the machine finishes the sentence and the adult is not there to keep evaluation in the room. Edwards is not a randomized trial of that sentence. He is journalism that names the political risk the trials already made empirical.

The luxury-good mechanism is not a metaphor. Breakstone's gradient says who already scores lower on the civic habit with trial evidence. Hirsch's argument says why: the knowledge that makes lateral reading and numeracy executable is unequally supplied at home, and a skills standard without a communal curriculum leaves that inequality in place. Edwards's journalism says what a plausible next inequality looks like: the knowledgeable adult who would model "I don't know," recode a number, and leave a page becomes private, and the public remainder is a chatbot. Fan and Bastani say what the chatbot does when it is the formation environment: better artefacts, less evaluation, an unperceived loss on the unaided test. Liang says what happens if the institution then polices the artefacts with a detector: it flags the wrong writers. Put those four in one paragraph and the equity chapter writes itself. The book still cannot claim that a named U.S. knowledge-rich curriculum closed the COR-like gap, or that Edwards measured classroom substitution. Open question 16 and the journalism label stay on the page. The logic does not require those missing trials to be sayable. It requires them not to be invented.

Standpoint is the last equity constraint, and it is not solved in this chapter. Facione's Delphi panel was North American, college-experienced, philosophy-heavy, 1988–89; McPeck was absent; Siegel was not listed. P21 is employer-facing. UNESCO's transversal-competencies work is Asia-Pacific ministerial. A global *life* book that treats these as "what critical thinking is" has a standpoint

[530] Willingham 2019: choosing content expresses values; "not choosing is still making a choice." Cognitive-science partner to Hirsch's argument, not a factional banner in the English/Australian knowledge-rich curriculum wars.

[531] A 2026 New Zealand commentary (Bay Science — advocacy/secondary) arguing gaps persisted under a Hirsch-influenced English reform is not a trial and is not laundered here as outcome evidence.

[532] David Elliott, "AI Won't Replace Teachers, Says This Global Union," World Economic Forum, 2 July 2024, https://www.weforum.org/stories/2024/07/artificial-intelligence-education-teachers-union/. Interview with David Edwards (Education International). **Journalism.** Human teachers as "the reserve of the privileged"; children who cannot afford them "sit in rooms with a chatbot." Edwards also: "Education is the place where we build societies and we build democracy." Not a study.

problem.[533] Who is left out of expert panels remains Open question 21. Naming it is not a solution. Pretending Delphi, P21, or UNESCO TVC were representative of the lives this book addresses would be a worse one. The five deeds do not escape the problem — they were drawn from U.S. and European literatures, SHEG tasks, Pennycook samples, Bastani's Turkish high school, Fan's university students. They are the deeds the fetched record describes. They are not a claim that the record was complete.

## 10.5 Five deeds, each with a source

The deeds are the close of the argument, not a new argument. Print them as a list a practitioner can steal, with the source on the row, and then say what each is not.

| Deed | What it is | Source already in this book | What it is not |
|---|---|---|---|
| Pause | Before sharing, ask whether it is true; slow down; consider plausibility against alternatives | Pennycook and Rand accuracy prompt; *Debunking Handbook 2020* | A belief vaccine; a finding that the prompt works for every partisan group |
| Check a source laterally | Leave the page. Who's behind it? What's the evidence? What do other sources say? Wikipedia as a first hop | COR three questions; Wineburg and McGrew 2019; McGrew et al. 2017 | CRAAP; a six-hour civic transformation; a turnout RCT |
| Notice when you don't know | You cannot think critically about a domain you lack; if you cannot recode the number, you are not informed; has enough information been collected? | Willingham 2007; Gigerenzer et al. 2008; NAM 2015 | A personality trait; a CCTST subscore; a reason to skip correction |
| Change a mind in public, including your own | Say "I was wrong about the number." Corrections usually move facts a little; they often fail to move votes | Nyhan 2020; *Debunking Handbook 2020*; Swire-Thompson et al. | A CT identity; a backfire-avoidance rule; a franchise virtue |

[533] Standpoint as named in the 01-definitions memo and the brief's Open question 21. Delphi: 46 panelists, six rounds, 11 February 1988–25 September 1989; philosophy-heavy; McPeck absent; Siegel not listed. P21: advocacy coalition, business plus education. UNESCO Bangkok / ERI-Net transversal competencies: Asia-Pacific ministerial.

| Deed | What it is | Source already in this book | What it is not |
|---|---|---|---|
| Refuse a fluent wrong answer | From a `.org`, a relative-risk poster, a colleague's story, or a model | Breakstone 2021; Bastani 2025; Fan 2025; Liang 2023 | A detector policy; an "AI literacy" module; a twelve-habit bicycle |

**Pause.** Gordon Pennycook and David Rand's limited-attention account is the mechanism: people *prefer* accuracy; the feed distracts them; a prompt that puts accuracy in working memory raises sharing quality, mostly by cutting false shares.[534] The 2021 *Nature* paper combined survey experiments with a Twitter field experiment. The 2020 *Psychological Science* paper was the COVID-era origin. The 2022 *Nature Communications* paper argues the prompt is replicable and generalizable. Limits, from the same literature and its critics: the prompt changes *sharing discernment* under inattention, not belief. Persistence and habituation are open. Rathje, Roozenbeek, Traberg, Van Bavel, and van der Linden's letter claims little or no effect for U.S. conservatives in a meta-analysis — a published dispute, not settled fact. A 2024 adversarial collaboration treats partisan moderation as unresolved; this book does not print its coefficients.[535] Guess, Lerner, Lyons, Montgomery, Nyhan, Reifler, and Sircar (2020) is the neighboring "tips" paper, not a Pennycook nudge: Facebook-style digital media literacy tips improved discernment 26.5 percent in a U.S. national sample and 17.5 percent in a highly educated Indian online sample, decayed, and had **no effect** in a largely rural northern Indian sample with low social-media use.[536] Checklist tips help a little, fade, and fail where the information ecology is different. The *Debunking Handbook* adds the slower cousin of the same deed: slow down; consider plausibility against alternatives. Pause is attention, not a vaccine.

**Check a source laterally.** Wineburg and McGrew watched professional fact checkers leave an unfamiliar page and open new tabs; Stanford undergraduates and historians stayed on the page and read vertically — slower *and* less accurate.[537] Civic Online Reasoning distills three questions in the curriculum's own words. McGrew, Ortega, Breakstone, and Wineburg, in *American Educator* (Fall 2017), named the other two fact-checker moves: click restraint (not taking Google's ranking as a trust ranking) and wise Wikipedia use — harvest references, read Talk pages — as a *first* hop,

[534] Pennycook et al. 2021; Gordon Pennycook, Jonathon McPhetres, Yunhao Zhang, Jackson G. Lu, and David G. Rand, "Fighting COVID-19 Misinformation on Social Media," *Psychological Science* 31, no. 7 (2020): 770–780; Gordon Pennycook and David G. Rand, "Accuracy Prompts Are a Replicable and Generalizable Approach for Reducing the Spread of Misinformation," *Nature Communications* 13 (2022): 2333. Sharing discernment, not belief.

[535] Rathje et al. letter to *Psychological Science*: meta-analysis claiming little or no effect for U.S. conservatives — a published dispute. 2024 adversarial collaboration (*Psychological Science*) treats partisan moderation as unresolved; coefficients **UNVERIFIED** pending a full-PDF fetch this book did not perform. Do not mash Mosleh, Martel, Eckles, and Rand (CHI 2021) public social correction — perverse effects of being publicly corrected on Twitter — with a private accuracy prompt.

[536] Andrew M. Guess, Michael Lerner, Benjamin Lyons, Jacob M. Montgomery, Brendan Nyhan, Jason Reifler, and Neelanjan Sircar, "A Digital Media Literacy Intervention Increases Discernment between Mainstream and False News in the United States and India," *Proceedings of the National Academy of Sciences* 117, no. 27 (2020): 15536–15545. +26.5 percent U.S.; +17.5 percent educated India online; decay; null in rural northern India.

[537] Sam Wineburg and Sarah McGrew, "Lateral Reading and the Nature of Expertise: Reading Less and Learning More when Evaluating Digital Information," *Teachers College Record* 121 (2019). Fact checkers vs. historians vs. Stanford undergraduates.

not a banned last hop.[538] Six COR lessons moved a test of the same skill and left a high remaining miss rate. That is the trial evidence. CRAAP is the contrast case, not a sibling intervention. The deed is a habit of leaving, practiced on real pages. A generated paragraph, Chapter 9's object, is a page. The three questions still apply. No RCT shows that practicing them on a chatbot transfers to practicing them on a cloaked `.org`, or the reverse. Willingham predicts little transfer without shared domain knowledge. The deed does not become a bicycle because the page was generated.

**Notice when you don't know.** Willingham: you cannot think critically about a domain you lack. Gerd Gigerenzer, Wolfgang Gaissmaier, Elke Kurz-Milcke, Lisa Schwartz, and Steven Woloshin: statistical illiteracy is common in patients, journalists, *and physicians*; a "25 percent reduction" for mammography is 1 in 1,000 absolute; if you cannot recode the number, you are not informed.[539] The National Academies' 2015 diagnostic-error report asks, in the process picture, whether enough information has been collected; diagnostic error is failure to establish an accurate and timely explanation *or* to communicate it to the patient.[540] Noticing not-knowing is the opposite of a critical-thinking identity. The identity says you have the skill. The deed says you may not have the knowledge, and in that case the honest next move is to get it or to defer, not to emit a fluent opinion. Kruger and Dunning's 1999 finding, spent in Chapter 4, is the monitoring half: the bottom quartile, actually at about the 12th percentile, estimated themselves at about the 62nd; improving the skill improved the monitoring.[541] "Rate your confidence" without teaching the skill is theater. Notice-when-you-don't-know without domain practice is the same theater.

**Change a mind in public, including your own.** Nyhan and Reifler (2010) found, in limited conditions, that corrections could strengthen misperceptions among ideological subgroups; the Iraq WMD item is the famous one. Wood and Porter (2019), five experiments, more than 10,000 participants, 52 issues: **no item triggered a worldview backfire.** Nyhan (2020, *PNAS*) later wrote that interpretations of the 2010 paper outstripped the findings and that backfire effects are "extremely rare in practice"; the 2020 full PDF was not independently opened for this book, and the phrasing is from snippets plus the handbook.[542] The *Debunking Handbook 2020*: "Do not refrain from attempt-

[538] Sarah McGrew, Teresa Ortega, Joel Breakstone, and Sam Wineburg, "The Challenge That's Bigger Than Fake News: Civic Reasoning in a Social Media Environment," *American Educator* (Fall 2017). Click restraint; wise Wikipedia use; 10-to-30-item checklists fail. Classroom translation, not an RCT.

[539] Gigerenzer et al. 2008. Relative vs. absolute risk; survival rates are not longer life; conditional probabilities vs. natural frequencies. Casscells, Schoenberger, and Graboys, "Interpretation by Physicians" (1978). Harvard Medical School faculty, staff, and students; 11 of 60 ~2 percent; 27 of 60 said 95 percent.

[540] National Academies of Sciences, Engineering, and Medicine, *Improving Diagnosis in Health Care* (Washington, DC: National Academies Press, 2015). Diagnostic error from the patient's side; ~5 percent of U.S. adults seeking outpatient care each year; postmortem work ~10 percent of deaths; medical-record reviews 6–17 percent of hospital adverse events; "most people will experience at least one diagnostic error in their lifetime." Synthesis report, not a new RCT. Newman-Toker et al. 2023 modeling estimate of 795,000 serious harms annually (range 598,000–1,023,000) is a modeling estimate, not a census; spent in Chapter 8.

[541] Justin Kruger and David Dunning, "Unskilled and Unaware of It," *Journal of Personality and Social Psychology* 77, no. 6 (1999): 1121–1134. Bottom quartile ~12th percentile actual, ~62nd estimated. Spent in Chapter 4.

[542] Brendan Nyhan and Jason Reifler, "When Corrections Fail: The Persistence of Political Misperceptions," *Political Behavior* 32, no. 2 (2010): 303–330. Thomas Wood and Ethan Porter, "The Elusive Backfire Effect: Mass Attitudes' Steadfast Factual Adherence," *Political Behavior* 41 (2019): 135–163. Five experiments, >10,000 participants, 52 issues; no worldview backfire. Brendan Nyhan, "Facts and Myths about Misperceptions," *Proceedings of the National Academy of Sciences* 117, no. 13 (2020). Full PDF **UNVERIFIED**; phrasing from snippets plus the handbook. *The Debunking Handbook 2020*, fetched in full 30 August 2026.

ing to debunk or correct misinformation out of fear that doing so will backfire." Swire-Thompson, DeGutis, and Lazer (2020) conclude backfire is not a robust empirical phenomenon. Continued influence — the sticky myth after accepted correction — is the real problem. Corrections usually move factual belief a little. They often fail to move votes in polarized settings.[543] The deed is still to say "I was wrong about the number." Wearing a critical-thinking identity is the folk use Chapter 1 named as a political compliment. Changing a number in public is the opposite of that use. It is also harder, which is why it is a deed and not a trait.

**Refuse a fluent wrong answer.** Chapter 7's `.org`, Chapter 8's relative-risk poster, a colleague's confident story, Chapter 9's model. Bastani: students copied arithmetic errors they could have caught. Fan: the essay got better while evaluation dropped. Liang: fluency fooled detectors in the other direction. The adult habit is *not handing the conclusion to whoever sounds finished*. Refusal is costly. Chapter 9 already said so. Adults keep the model in the room; the draft is due; the meeting is now. The deed is not a detector policy. It is not an AI-literacy module. It is the evaluation that remains when fluency is free. Without the domain knowledge that makes the wrongness visible, refusal has nothing to refuse. The five deeds are therefore not a set of context-free skills. They are five names for evaluation, each empty until Chapter 3's constraint is met.

Two further limits on the table. First, the deeds are not a sequence you must always run in order. Pause without a later check is still a sharing improvement at the margin; Pennycook and Rand's prompt does not require a COR lesson. A lateral check without a public mind-change is still Wineburg's move. The table is a set, not an algorithm. Second, the deeds are not scored by a generic test. A person can pause, leave the page, notice a gap, correct a number, and refuse a fluent draft, and still not move a CCTST subscore, a CLA constructed-response, or a Watson–Glaser. Chapter 6's rule holds: the test is not the thing. If what would count as improvement in a life is these deeds on named stakes, then a semester of practice-item gains is the wrong evidence. Unseen cases, process logs, oral defense with the model closed, a share you didn't make — those are closer. They have rater problems, halo problems, and no HALO test. They are still better matched to the construct than a badge.

Inoculation belongs next to pause, not as a sixth deed. Roozenbeek's short videos train technique recognition — emotionally manipulative language, incoherence, false dichotomies, scapegoating, ad hominem — with lab effects that are medium and a YouTube field effect that is small and that missed the authors' own smallest effect size of interest. Maertens says the effect decays without boosters. Technique recognition is not the same object as lateral reading and not the same object as an accuracy prompt. A life book that mashed the three into "media literacy" would have reinstalled the slogan-machine. Keep them adjacent. Do not average 0.30 with 0.57 with Gorard +0.12 with Roozenbeek's field $h$ of 0.09. Heterogeneity is the story.

[543] Briony Swire-Thompson, Joseph DeGutis, and David Lazer, "Searching for the Backfire Effect: Measurement and Design Considerations," *Journal of Applied Research in Memory and Cognition* 9, no. 3 (2020): 286–299. Backfire not a robust empirical phenomenon. Vote-movement limit as stated in the brief: corrections often fail to move votes in polarized settings (Swire-Thompson et al.). Continued influence (Johnson and Seifert 1994 line; handbook 2020) is the surviving problem.

## 10.6 Correlative care, strong-sense warning, not a franchise

The dispositions debate from Chapter 2.8 returns as an ethics close, not as a new definition. Ennis, in 1996, treated care for persons' dignity as a *correlative* disposition: a critical-thinking program without it "would be deficient and perhaps dangerous."[544] Correlative, not constitutive. The meaning of the activity remains reasonable reflective thinking focused on deciding what to believe or do. The warning is that a program which trains the activity without care for persons can do harm. That is the honest way to keep danger in view without building a political morality into the *meaning* of the phrase and then pretending Delphi, Ennis, or Willingham did that.

Richard Paul's 1981 distinction belongs here as the moral-psychological close, not as a Foundation franchise. Atomistic fallacy-spotting can make students "more sophistic rather than less so." Weak-sense critical thinking is skilled defense of one's side. Strong-sense is the dialectical, self-correcting use that includes one's own view as an object.[545] Steal the warning. Leave the franchise. Do not outsource this book's theory to criticalthinking.org. Do not pretend there is one canonical list of nine intellectual standards; Foundation lists vary. Do not treat Moseley and colleagues 2005 or Payette and Ross 2016 as RCTs of Paul–Elder. The warning is older and better than the brand: a life of deeds that is only attack on the other side is the political compliment from Chapter 1, now with a skills certificate attached.

Delphi's majority refused to build ethics into the meaning. Facione's 1990 report: 52 percent rejected an ethical component in the meaning of "critical thinking"; the panel split about 61/30 on whether dispositions are *in* the meaning versus a strict proceduralism.[546] Folk use does the opposite. "We are the critical thinkers; they are the sheeple" is ethics-in-the-meaning as a compliment and an insult. Freire's "critical" as conscientization, which Hitchcock's encyclopedia entry treats as a non-example of the mainstream concept, lives in that folk use.[547] The book's position, restated from Chapter 2.8: meaning = Ennis's sentence; correlative care named; constitutive partisan ethics refused. A person can use the five deeds in a weak sense. Pause, check, notice, change, refuse — pointed only at the other side — is Paul's sophistry with better citations. The correlative warning is that this book's title does not license that use. The constitutive refusal is that this book's title also does not license a political program as the definition.

Willingham 2019, on the nearby question of need-for-cognition, treats that tendency as possibly

[544] Robert H. Ennis, "Critical Thinking Dispositions: Their Nature and Assessability," *Informal Logic* 18, nos. 2–3 (1996): 165–182. Care for persons' dignity as a *correlative* disposition; without it a CT program "would be deficient and perhaps dangerous" — not constitutive of the meaning.

[545] Paul, "Teaching Critical Thinking in the 'Strong' Sense," *Informal Logic* 4, no. 2 (1981). Exact 1981 wording: students trained on " 'egocentrically' neutral" cases "become more sophistic rather than less so." Hitchcock's "more able sophists" is a paraphrase. Strong sense vs. weak sense. Steal the warning; leave the Foundation franchise. Scriven and Paul 1987 draft still posted as "Defining Critical Thinking" on the Foundation page, fetched 30 August 2026, is not this chapter's definition.

[546] Peter A. Facione, *Critical Thinking: A Statement of Expert Consensus for Purposes of Educational Assessment and Instruction* (The Delphi Report), ERIC ED315423 (1990). 46 panelists; six rounds, 11 February 1988–25 September 1989. ~61 percent dispositions-in-the-meaning vs. ~30 percent strict proceduralism; 52 percent reject building ethics into the meaning of "CT." Expert consensus for assessment and instruction, not "the APA's official definition."

[547] David Hitchcock, "Critical Thinking," *Stanford Encyclopedia of Philosophy*, first published 21 July 2018, substantive revision 12 October 2022. Freire's *Pedagogy of the Oppressed* (1970 trans.) treated as a non-example of the mainstream concept.

personality, with limited educational evidence.[548] Do not convert a dispositional hope into a curriculum. Care for persons is a constraint on a program, not a module that produces fairmindedness as a transferable trait. The five deeds can be practiced in a way that respects persons, or not. Ennis's "deficient and perhaps dangerous" is the sentence this close needs. It is not a seventh deed.

Siegel's *Educating Reason* (1988) constrains the *value* of the program more than the definition of the activity: respect for persons, self-sufficiency, initiation into rational traditions, democratic citizenship.[549] Those justifications can support a life book's claim that the work is worth doing. They cannot be loaded into Ennis's sentence as extra conjuncts and then treated as what "critical thinking" means. Adjacent accounts — Bailin against generic discrete skills, Halpern's teach-for-transfer, Fisher and Scriven's interpretation-and-evaluation formula, Stanovich's override of the autonomous mind, Kuhn's dialogic practice — exist, and Chapter 2 named them as adjacent. They do not get a victory lap here. The ethics close is correlative care plus the sophistry warning. Anything more is a different book's constitution.

## 10.7 Coalition contradictions, still standing

Chapter 2 chose in public. The coalition this manuscript can honestly run is Ennis (believe *or do*) plus Dewey (inquiry, not only appraisal) plus Willingham (knowledge and practice) plus a Paul-ish warning about weak-sense sophistry plus Lipman for childhood judgment, with McPeck's *about what?* kept as a constraint. That choice is still a coalition. Coalitions hide contradictions. The close of the book does not get to dissolve them.

Ennis's sentence still names the work, including the five deeds, because each deed is a decision about belief or action. Willingham still vetoes a content-free general course and a twelve-habit bicycle, including a five-deed bicycle. The deeds are names for evaluation in a domain. They are not a skill-set that, once practiced on news, deploys on mammograms. Dewey still vetoes fallacy-spotting as the whole, including a life of spotting other people's biases while one's own inquiry never leaves perplexity for a tested hypothesis. Paul's warning still vetoes weak-sense sophistry as "critical thinking," including the folk compliment. Lipman still earns childhood judgment and community of inquiry, and still does not earn EEF months. McPeck's *about what?* still vetoes "everything in general," including "for life" as a mass noun. Delphi still cannot be the meaning; the test is still not the thing; CCTST is still an operationalization of one consensus statement. One AI chapter was enough. Two lanes are still a design hypothesis. Kosmyna is still a preprint.

The live debate from Chapter 1 — whether "critical thinking" is still the right title-phrase — is still live. Dewey preferred reflective thinking. Some cognitive scientists prefer scientific thinking, historical thinking, rationality, judgment. The market will punish a title change; the argument might not.[550] This book holds the title. No better public sentence than Ennis's has appeared in the drafting. Hold unless one does. That is Open question 18, still standing, not filled by the five deeds.

[548] Willingham 2019: a second informal sense of critical thinking — thinking when others might not; need for cognition (Cacioppo et al. 1996) — set aside as under-researched for education.

[549] Harvey Siegel, *Educating Reason: Rationality, Critical Thinking, and Education* (New York: Routledge, 1988), via Hitchcock. Four justifications of CT as an educational ideal. Constrains the value chapter more than the definition. Adjacent, not one of this book's six accounts.

[550] Open question 18 of the brief. Hold the title-phrase unless a later draft finds a better public sentence than Ennis's. No such sentence appeared in this drafting.

Open question 1 is the coalition itself. Where Willingham vetoes Ennis-style general courses, this book has sided with Willingham on pedagogy and kept Ennis on the public definition. Where Ennis's long outline of dispositions and abilities still earns pages, it earns them as a specification of the *aim*, not as a reason to ignore the knowledge constraint. That is a choice, not a finding those authors signed. A later writer who wants a different coalition — Halpern's teach-for-transfer optimism, Siegel's critical spirit as the educational ideal, Stanovich's rationality, Freire's politically engaged "critical" — will have to design a different book. This one named its coalition and its vetoes. It does not get to pretend, on the last chapter's last pages, that they were the same view.

Hitchcock's verdict on McPeck travels here as the last coalition note. McPeck's central argument "needs elaboration"; *nevertheless* the claim that critical-thinking dispositions and abilities are best developed in subject-matter instruction is shared by Dewey, Glaser, Passmore, Weinstein, Bailin and colleagues, and Willingham 2019.[551] This book rejected McPeck's ban on general courses — writing and speaking analogies, and Ennis's conceptual/epistemological/empirical subject-specificity papers, weaken a claim that there are no general abilities of any kind. It kept the *about what?* demand. The five deeds are the test of that keeping. Pause is general in aim and empty without a headline. Lateral check is general in form and empty without a page and a domain. Noticing not-knowing is general as humility and empty without a domain to lack. Changing a mind is general as a social act and empty without a number. Refusal is general as a habit and empty without a fluent wrongness you can see. McPeck still vetoes the sentence "I teach thinking." He does not veto this chapter. He constrains it.

## 10.8 Not a badge

A person who can pause, leave the page, notice when they don't know, change a number in public, and refuse a fluent wrong answer has not "acquired critical thinking." They have practiced evaluation on named stakes. The next stake will require knowledge they may not have. A three-year-old can do this in a domain she knows. A scientist can fail at it outside his. An over-65 Facebook user in 2016 could share a fake-news-domain article for reasons that are not cognitive decline. A high-school student facing a cloaked climate `.org` could miss the fossil-fuel ties 96 percent of the time before anyone had heard of ChatGPT. A university student could hand in a better essay and have done less evaluation. A TOEFL writer could be flagged as a machine for writing like herself. A child who cannot afford a knowledgeable adult could sit in a room with a chatbot and be told that this is twenty-first-century skill.

None of those people is described by a badge. The badge is the folk collapse's last product: a trait you possess, a course you completed, a score you can print, a compliment you can pay yourself. This book's title is a life of work on stakes that do not grade you. The work is domain knowledge plus evaluation habits. The evaluation is the remaining human work when a machine will finish the sentence. The habits are deeds, not a transferable faculty.

[551] Hitchcock, SEP entry, on McPeck: the central argument "needs elaboration"; the subject-matter-instruction claim is shared by Dewey, Glaser, Passmore, Weinstein, Bailin et al., and Willingham 2019. Ennis 1989 on conceptual, epistemological, and empirical subject-specificity; Paul 1985 and Siegel 1985 on the writing/speaking analogy. Reject the ban; keep the knowledge constraint. McPeck 1981, p. 8: "the propensity and skill to engage in an activity with reflective scepticism"; thinking is always about X; X is never "everything in general."

A badge has a completion date. A life of this work does not. Willingham's three-to-five-year revisiting, applied to a life's domains, means the health number you recoded at forty is not the health number you will need at sixty, and the civic page you learned to leave in a government class is not the generated paragraph you will be asked to share tonight. Abrami's explicitness means the revisiting cannot be a hoped-for by-product of being a serious person. Ennis's sentence means the work includes *doing*, not only appraising: the share you don't make, the consent you don't give, the fluent draft you send back. Paul's warning means you can do all of that in a weak sense, as a more able combatant, and still have failed the correlative care Ennis named. Lipman's childhood page means the work starts before the first test, in a community that asks "how do we know?" about something the children actually know. McPeck's question means you still have to say *about what*.

The honest remainder is the coda: what this book can promise, what it cannot, and the open questions it will not fill with analogy.

# Coda

## What an Honest Book Can Promise versus What It Cannot

Critical thinking is not a transferable bicycle-skill. It is domain knowledge plus evaluation habits practiced on real stakes, for life, including when a machine will finish the sentence. Robert Ennis's public sentence — reasonable reflective thinking focused on deciding what to believe or do — is the definition this life book can use, because it includes action.[552] Daniel Willingham's 2007/2019 constraint is how you teach toward that definition without lying: thinking processes are intertwined with the content of thought; maxims without knowledge and practice will not implement; domain-specific critical thinking *can* be taught; there is no proven way to teach a content-free general skill directly.[553] The coalition this manuscript ran — Ennis plus Dewey plus Willingham plus a Paul-ish sophistry warning plus Lipman for childhood, with McPeck's *about what?* as a constraint — held only because the vetoes were named. Willingham still vetoes a general course as the engine. Ennis's sentence still names the work. Dewey still vetoes fallacy-spotting as the whole. Paul still vetoes weak-sense sophistry as the title. Lipman still earns childhood judgment, not EEF months. McPeck still asks *about what?* Delphi still cannot be the meaning. A close that dissolved those vetoes would be a different, less honest book.

Promise, here, does not mean a prediction. It means a claim the chapters already stood on, with named documents, that survives a second reading after the open questions have been left standing. What this book cannot promise is longer. Leaving that list standing is the honest close. A hedge keeps the claim and adds a qualifier. An open question is a claim the book refuses to make. The popular story fills every item below with an analogy, a vendor page, or a 2030 classroom. This coda will not. It will not say the future is bright if we are thoughtful. The future is not a finding in the record. Brightness is a tone the folk conjunction uses when it has run out of documents. An honest book stops.

The can-list is not a victory lap for Paul–Elder, CCTST, P21, or a sequel to *Autonomous AI and Education*. Those were refused in the chapters that had to refuse them. The cannot-list is not modesty as a style. It is the set of claims a writer could make in the last five pages by filling a gap

[552] Robert H. Ennis, working definition as restated in Ennis, "The Nature of Critical Thinking: An Outline of Critical Thinking Dispositions and Abilities," criticalthinking.net (2011; already the 1985 *Educational Leadership* working definition).

[553] Daniel T. Willingham, "Critical Thinking: Why Is It So Hard to Teach?," *American Educator* 31, no. 2 (Summer 2007); Willingham, *How to Teach Critical Thinking* (NSW Department of Education, 2019). Bicycle passage quoted in Chapter 3. Domain-specific CT can be taught; no proven way to teach general skills directly.

with a likeness. Chapter 9 did not fill "evaluate the chatbot" with an analogy to Wineburg. Chapter 10 did not fill older-adult sharing with cognitive decline, or Hirsch with a fetched RCT, or Edwards with a trial. The coda's job is to copy those refusals into one place so a later reader cannot treat a named open question as a solved chapter.

## What an honest book can promise

It can promise a public definition that includes action, and a pedagogical constraint that includes knowledge. Ennis's sentence is the first. Willingham 2007/2019, and Glaser 1941 at page 175 already, are the second: the attitude of wanting evidence transfers better than skill in applying logical methods, which is limited by pertinent knowledge.[554]

It can promise a family of six non-equivalent accounts, shown before one coalition was chosen, with vetoes on the page. Hitchcock's shared *concept* — careful, goal-directed thinking — is not a license to collapse Ennis, Delphi/Facione, Paul–Elder, Willingham, McPeck, and Lipman. Chapter 2 printed the table. Chapter 10 restated where each still vetoes.

It can promise that transfer is rare and similarity-bound, and that implicit expectation is a failed design. Thorndike, Detterman, Gick and Holyoak (~10/30/75). Abrami 2008: cannot be a hoped-for by-product. Abrami 2015: g+ = 0.30 on standardized critical-thinking tests — real, small, not far; Ennis-type differences not statistically significant; dialogue plus authentic problems plus mentoring g = 0.57 from 19 studies.[555] Those numbers are not interchangeable with Gorard +0.12, EEF 2021's 0 months, van Gelder's pre–post pool, Roozenbeek's field *h* of 0.09, or Schwichow's 0.61. Heterogeneity is the story. Do not average them.

It can promise domain-embedded named moves as the load-bearing beam. Control-of-variables strategy, meta-analytic g = 0.61 (Schwichow and colleagues, 2016). Reading Like a Historian, Reisman 2012, a quasi-experiment, significant MANCOVA on four outcomes, not an RCT, not a generic-CT result, effect sizes not invented from an abstract. Norman and colleagues 2017: diagnosis is knowledge-plus-cases, not a missing critical-thinking trait.[556]

It can promise that the test is not the thing. Delphi is expert consensus for assessment, not a discovery of nature. Independent CCTST subscale alphas as low as 0.21. CLA+ constructed-response reliability 0.43. HCTA retired. No HALO test. Kuncel: little discriminant validity versus *g*. Arum and Roksa's 45 percent is a CLA finding, partly motivation, not "critical thinking cannot be taught."[557] CCTST is an operationalization of Delphi, not the meaning.

It can promise lateral reading as the civic habit with trial evidence, CRAAP as the contrast case,

[554] Edward M. Glaser, *An Experiment in the Development of Critical Thinking* (1941), p. 175, via Hitchcock.

[555] Philip C. Abrami et al., *Review of Educational Research* 78, no. 4 (2008) and 85, no. 2 (2015). g+ = 0.30; combination g = 0.57, 19 studies; Ennis-type NS. Do not mash with Gorard, EEF, van Gelder, Roozenbeek, or Schwichow.

[556] Schwichow et al. 2016, CVS g = 0.61 (95% CI 0.53–0.69). Avishag Reisman, "Reading Like a Historian," *Cognition and Instruction* 30, no. 1 (2012): QES, 236 eleventh-graders; MANCOVA significant on four outcomes from the abstract; full PDF HTTP 500 this fetch; not an RCT; effect sizes not invented. Geoffrey R. Norman, Sandra D. Monteiro, Jonathan Sherbino, et al., on diagnostic error as knowledge access, 2017.

[557] Liu, Frankel, and Roohr, ETS RR-14-10 (2014); Bernard et al. 2008; Zahner 2013 CLA+ CR 0.43; HCTA retired 2024; Kuncel 2011 via Huber and Kuncel 2016; Arum and Roksa 2011; Liu, Bridgeman, and Adler 2012. No HALO test found.

backfire rare and continued influence real. Breakstone 2021: 96 percent never uncovered a climate site's fossil-fuel ties. Wineburg 2022: six COR lessons moved credibility-task scores; remaining miss rate is the story. Wood and Porter 2019; Nyhan 2020; *Debunking Handbook 2020*: do not skip correction.[558]

It can promise clinic, work, and court as domains with named knowledge and error-costs, not as proof of a transferable faculty. Casscells: 11 of 60. Gigerenzer: natural frequencies restore insight. NAM 2015: most people, at least one diagnostic error. Newman-Toker 795,000 is a modeling estimate, range 598,000–1,023,000. NACE and WEF are recruiter surveys. Legal examples are teaching objects.[559]

It can promise one AI chapter: when a machine finishes the sentence, evaluation is the remaining human work. Bastani: practice +48 percent, unaided exam −17 percent. Fan: better essays, no extra knowledge, metacognitive laziness. Stadler: ease without depth. Liang: 61.22 percent of human TOEFL essays flagged. Kosmyna is a preprint, n = 54 / 18. Two lanes labelled as a design hypothesis, not a fitted schedule, not existing policy.[560]

It can promise five deeds with sources, not a badge. Pause. Check a source laterally. Notice when you don't know. Change a mind in public, including your own. Refuse a fluent wrong answer. Each empty without domain knowledge. Correlative care, not constitutive politics. Guess 2019: over-65s ~7× on 2016 Facebook sharing, not cognitive decline. Breakstone's SES/race/maternal-education/FRPL gradient is older than ChatGPT. Hirsch is an argument. Edwards is journalism.[561]

What the can-list adds up to is not a program you could hand a ministry. It is a set of constraints a program would have to survive. If it promises a bicycle, it fails Willingham. If it promises a short test as the meaning, it fails Chapter 6. If it promises civic transformation from six lessons, it fails Wineburg's remaining miss rate. If it promises that employers test the skill, it fails NACE's interview story. If it promises that unguarded fluency is learning, it fails Bastani. If it promises a badge, it fails the deeds. A ministry that wanted a poster would find this list unsatisfying. That is a feature of an honest book, not a defect to be repaired in a coda. The poster is Chapter 1's slogan-machine. The honest remainder is a definition, a constraint, a family of accounts, a transfer default, a teaching combination that is not a horse race, a civic habit with a remaining miss rate, domains rather than a faculty, one AI chapter, and five deeds. That is enough to work with. It is

[558] Breakstone et al. 2021, *Educational Researcher*; Wineburg et al. 2022, *Journal of Educational Psychology*; Wood and Porter 2019; Nyhan 2020 *PNAS* (full PDF UNVERIFIED; phrasing from snippets plus handbook); *Debunking Handbook 2020*, fetched in full 30 August 2026.

[559] Casscells, Schoenberger, and Graboys 1978, 11 of 60; Gigerenzer et al. 2008; NAM 2015; Newman-Toker et al. 2023, modeling estimate; NACE *Job Outlook 2025*; WEF *Future of Jobs 2025*; Thompson and Schumann 1987. Cab-problem "correct" 41 percent is a live academic dispute (Birnbaum 1983 and later); print as teaching example, not uncontested Bayesian number.

[560] Bastani et al. 2025, *PNAS* 122(26) e2422633122; Fan et al. 2025, *BJET* (title: "Learning Motivation, Processes, and Performance"; N = 117; four supports); Stadler, Bannert, and Sailer 2024; Liang et al. 2023, *Patterns*, 61.22 percent (arXiv HTML; *Patterns* HTML rounds to 61.3 percent; 89/91 = 97.8 percent flagged by at least one detector); Kosmyna et al. 2025, arXiv:2506.08872, **preprint**, n = 54 / 18. Two lanes: design hypothesis.

[561] Five deeds, Chapter 10. Guess, Nagler, and Tucker 2019, ~7× (abstract: "nearly seven times"; column-2 IRR e^1.9 ≈ 6.69); Brashier and Schacter 2020; Breakstone 2021 gradient; Hirsch 2016 via Core Knowledge, argument not RCT; Edwards / WEF, David Elliott, "AI Won't Replace Teachers, Says This Global Union," 2 July 2024, journalism. WWC intervention-reports index, 30 August 2026: no named-CT report. Open questions as numbered in the research brief.

not enough to congratulate ourselves, and it is not a reason to invent the rest. An honest close stops on the documents it had, and on the questions it did not fill.

## What it cannot promise

It cannot promise year-scale unaided far transfer of specific reasoning principles into workplace and civic decisions, scored independently of critical-thinking tests.

It cannot promise a small set of generic skills with a proven direct teaching method. Willingham 2019: we are not even sure they exist.

It cannot promise that Abrami 2015 proved mixed greater than infusion greater than general. The differences were not statistically significant.

It cannot promise that Philosophy for Children raises attainment at scale. EEF 2021: **0 months**, 198 schools, five padlocks. Gorard 2015's small attainment effects are not hidden and are not the 2021 result.

It cannot promise a gold-standard RCT of argument mapping. Van Gelder 2015 said none existed; none was found since.

It cannot promise incremental validity of CCTST, Watson–Glaser, or CLA+ after *g*.

It cannot promise civic outcomes from Civic Online Reasoning or inoculation — turnout, vote quality, scam victimization, vaccine uptake, wild sharing over a year.

It cannot promise that twenty-five state media-literacy laws changed classrooms. Media Literacy Now has not evaluated implementation. Named-state bill text remains unverified.

It cannot promise that employers test critical thinking, or that a validated instrument predicts job performance after *g* and domain knowledge.

It cannot promise that natural-frequency training reduces real diagnostic error at scale.

It cannot promise that "evaluate the chatbot" transfers to non-AI thinking. Willingham predicts it will not, without domain knowledge. No such RCT.

It cannot promise that Kosmyna proves population cognitive decline. Preprint; n = 54 / 18; connectivity is not critical-thinking ability.

It cannot promise that a named U.S. knowledge-rich curriculum closed COR-like or racial/SES gaps. Hirsch's argument; RCT not fetched.

It cannot promise a What Works Clearinghouse intervention report for P4C, argument mapping, Paul–Elder, or named critical-thinking programs. None as of 30 August 2026.

It cannot promise a fitted life-span two-lane AI schedule, or a twelve-habit bicycle.

It cannot promise that the coalition's contradictions have been dissolved, that dispositions are in or out of the meaning, that "critical thinking" is the right title-phrase, or that 1988–89 expert panels spoke for the lives this book addressed. Open questions 1, 3, 18, and 21 stand.

It cannot promise dose of domain knowledge versus maxims versus incentives, fitted. It cannot promise debiasing under identity load; successful training is mostly non-tribal, and Kahan's paradigm is the transfer test that has not been passed. It cannot promise what P4C actually does when attainment is zero at scale, on better measures than two survey items. It cannot promise why nursing did not beat other majors on generic tests. It cannot promise how much of the college-gain literature is effort. It cannot promise a 2026 older-adult Civic Online Reasoning RCT, or that 2016 Facebook is today's short-video ecology. It cannot promise Dual Process 2.0 logical intuitions as trainable mindware. It cannot print *Nature Communications* 2025 booster coefficients; they remain unverified; Maertens 2021 is the decay finding this book used. Dewey 1910 pages 74 and 82 have now been opened against Gutenberg and the Mead Project; they use the English phrase for suspended judgment and for "verified critical thinking." Hitchcock's claim that those pages name an educational goal remains his framing, not a sentence on those pages.

The cannot-list is not a confession that the book found nothing. It is a specification of what the findings are not. Abrami's 0.30 is a finding. It is not far transfer. Wineburg's growth is a finding. It is not turnout. Bastani's −17 percent is a finding. It is not a life-span schedule. Guess's ~7× is a finding. It is not decline. A coda that converted those "nots" into a program for 2030 would have rejoined the popular story on the last page.

Do not close with "practice these habits and you will be a critical thinker everywhere." That sentence reinstalls the bicycle, the badge, and the already-promised in one instruction. Close with Ennis's sentence and Willingham's constraint held in one hand: decide what to believe or do, in a domain you actually know, on a stake that can hurt, including when the sentence arrives already finished. The next stake will require knowledge you may not have. A three-year-old can do this in a domain she knows. A scientist can fail at it outside his. A person who paused, left the page, noticed a gap, changed a number, and refused a fluent draft has practiced evaluation. They have not acquired a faculty. The work remains. The open questions remain. The future, bright or otherwise, was not in the documents.

That is not a failure of the book. It is the activity the book named.

# A Note on Sources

This book is sourced to documents opened on 30 August 2026, or to the intermediary that was actually read when a full text could not be opened. It does not invent titles, publishers, years, page numbers, author names, coefficients, or quotations. Where a bibliographic element could not be verified from a catalog, a journal landing page, a fetched PDF, or a manuscript note, the element is omitted. An UNVERIFIED flag in a note or in the bibliography means the full text was not read. Unfetched is not a finding.

A first fact-check pass corrected identities and wording that an earlier draft had wrong. Those corrections are locked and are not to be reverted. Dewey 1910, pages 74 and 82, have been opened against Project Gutenberg and the Mead Project transcription of chapters 6–7; the sentences use the English phrase "critical thinking" for suspended judgment and for verified critical thinking. They do not say that the phrase is the name of an educational goal; that framing remains David Hitchcock's. Richard Paul's 1981 *Informal Logic* article says that students trained on " 'egocentrically' neutral" cases "become more sophistic rather than less so." Hitchcock's later gloss "more able sophists" is a paraphrase, not the 1981 sentence. The three-feature core Hitchcock uses is Bailin, Case, Coombs, and Daniels 1999b, "Conceptualizing Critical Thinking," not the companion 1999a paper "Common Misconceptions of Critical Thinking" in the same issue. *Journal of Environmental Psychology* 101865 is Pröpper, Geiger, Blanken, and Brick 2022, a weak cultural-cognition replication; Kácha, Vintr, and Brick 2022 is a different paper (101815, "Four Europes"). Fan, Luzhen Tang, Le, and colleagues 2025 is the metacognitive-laziness paper (N = 117; four supports); the second author is not Linquan Tang. Liang, Yuksekgonul, Mao, Wu, and Zou 2023: 61.22 percent is the authors' average false-positive rate on 91 human TOEFL essays (arXiv HTML); *Patterns* HTML rounds the same average to 61.3 percent; 18 of 91 were flagged by all seven detectors; 89 of 91 (97.8 percent) were flagged by at least one. The book keeps 61.22 in the body and reports the rounding and the 89/91 count in the notes. Kosmyna and colleagues, arXiv:2506.08872: the author line includes Ye Tong Yuan and Ashly Vivian Beresnitzky, not Yeo Wei Yuan or Ava Vivian Beresnitzky. CCTST is about 34 items in about 45 minutes (Liu, Frankel, and Roohr 2014), not "34–45 minutes." Wineburg and McGrew 2019 compared 10 fact checkers, 10 Ph.D. historians, and 25 Stanford undergraduates. Casscells, Schoenberger, and Graboys 1978 asked Harvard Medical School faculty, staff, and students, not "Harvard" simpliciter. Chi, Feltovich, and Glaser 1981 is Michelene T. H. Chi.

Full texts that remain unread are flagged, not filled. Among them: Dewey 1933; Ennis 1985 *Educational Leadership* (paywalled; later restatement used as the public sentence; an ERIC reprint records a 1985 p. 45 variant); Ennis 1962, 1989, and 1991; Glaser 1941; the Eight-Year Study primary reports; Bloom 1956 p. 38; McPeck 1981; Siegel 1985 and 1988; Bailin 1999a/1999b; Halpern 1998; Kahane 1971; Nyhan 2020 *PNAS* (Cloudflare-blocked; "extremely rare in practice" from snippets plus *The Debunking Handbook 2020* — the UNVERIFIED flag stays); Fan et al. 2025 *BJET* (publisher blocked; ERIC abstract used); Reisman 2012 (HTTP 500; no effect size invented from the abstract); Murphy et al. 2009; Abrami 2008 and 2015 *Review of Educational Research* (paywalled; combination g = 0.57, k = 19, and the 0.57/0.23/0.33 splits remain TUM-mediated); Wineburg et al. 2016 *Evaluating Information*; *Educating for Misunderstanding* (2020); Maertens 2021; McGrew 2020; Guess 2020 *PNAS*; Gigerenzer et al. 2008 *Psychological Science in the Public Interest*; Casscells 1978 *New England Journal of Medicine* (numbers via AMA 2009); the 2007 AFT compiled-issue PDF (body verified via Reading Rockets; prefer the body); blicket-detector primary papers (follow Willingham 2007); Kant as a critical-thinking source (*sapere aude* is atmosphere; no primary Kant text opened that uses the English educational slogan); *Nature Communications* 2025 booster-shot coefficients; 2024 adversarial-collaboration coefficients on partisan moderation of accuracy prompts; and several stream-memo coefficients (Niu 2013, Sala and Gobet 2016/2017, Morewedge 2015, Kahan 2012, van Gelder 2004, a 2025 *Higher Education Quarterly* argument-mapping review) whose primary PDFs were not re-opened on the mid-pass. Kuncel 2011 is via Huber and Kuncel 2016. A live re-crawl of the What Works Clearinghouse intervention-reports index was not repeated on the end pass; the negative finding (no named-critical-thinking report as of 30 August 2026) is from the teaching-stream access date.

Author lines and citations that could not be locked are omitted rather than guessed. That includes remaining authors after the named first authors of several Quality Talk, Butler 2024, Newman-Toker 2023, Elwyn 2012, Mellers 2014, DeAngelo 2009, and Swan 2023 papers; the 2024 adversarial collaboration's full author line; a 2026 four-pillar AI-resilient assessment paper cited as language, not as an RCT; Permzadian and Credé 2016's journal path (δ = 0.02 GPA / 0.11 retention left untouched); Ortiz 2007 (unpublished master's; not a third Abrami); Colom et al. 2014 Madrid Philosophy for Children (not added as a confirming RCT); and a handful of volume/page omissions that still have DOIs. Tsui 1998 versus 1999 in the *Journal of Higher Education* was not re-confirmed as a second object. WEF 2025 "seven of ten" is WEF's wording; an underlying 69 percent from secondary roundups was not read off the PDF table this pass. NACE 2026 skills-based hiring journalism (70 percent / GPA 42 percent) was not re-fetched on the mid-pass.

Illinois 2021, New Jersey 2022, and Delaware's Digital Citizenship Education Act 2022 are named in secondary roundups only. Statutory details remain UNVERIFIED pending bill text. Media Literacy Now's 25-state count, and its statement that it has not evaluated classrooms, are held. No systematic census of U.S. state standards' use of "critical thinking" is asserted.

Instrument and vendor holes are left open. No instrument named HALO was found on 30 August 2026; HCTSR, the halo *effect*, and the retired Halpern Critical Thinking Assessment are distinct. CART is a research prototype (Stanovich 2016), not a fielded school test. Watson–Glaser and CCTST are vendor products; their subscores are unusable for steering. Rationale, bCisive, Austhink, and Kritisch Denken BV are labelled vendor; van Gelder 2015 is pre–post, not an RCT. The Foundation for Critical Thinking is a curriculum brand, not a meta-analysis; Paul–Elder QEPs are acceptance, not trials. AIAS dissemination ">350 institutions" is a claim, not an effect size; Mike Perkins, not David Perkins.

The book does not revive coinage myths, Dewey 74/82 as Hitchcock's educational-goal-name *sen-*

*tence*, Paul 1981 quoted as “more able sophists,” Bailin 1999a as the three-feature core, Kácha as author of 101865, Fan’s second author as Linquan Tang, Liang 61.22 as the only number, Kosmyna as population decline or as peer-reviewed, CCTST “34–45 minutes,” Wineburg and McGrew without n = 10/10/25, Casscells as “Harvard” simpliciter, Klein *Sources of Power* as 1989, Abrami 2015 as 341 *studies*, Ennis-type mixed/infusion/general/immersion as a statistically significant horse race, Willingham as “critical thinking cannot be taught,” EEF 2015 as 40 schools, van Gelder as an RCT, Reisman as an RCT with invented effect sizes, six Civic Online Reasoning lessons as producing informed citizens, detector scores as catching cheaters, Newman-Toker 795,000 as a census, NACE 4.49 as “#1 skill,” WEF analytical thinking as “critical thinking,” backfire as robust, two lanes as existing policy, a HALO test, a WWC named-critical-thinking intervention report, an invented Pew poll, an invented classroom, or an invented dedication. Effect sizes are not averaged. Gaps are not filled by analogy.

# Bibliography

Chicago Manual of Style, 17th edition, notes-bibliography system

This list is a working apparatus for a university-press or serious trade book. It is compiled from the four research source files of 30 August 2026 (`01-sources.json` through `04-sources.json`) and from every work actually named in the manuscript notes (`manuscript/00-preface.md`, `00-introduction.md`, `ch01.md` through `ch10.md`, `coda.md`). It does not invent titles, publishers, years, DOIs, URLs, or page numbers. Where a bibliographic element could not be verified from a catalog, landing page, fetched PDF, or manuscript note, it is omitted and recorded in `GAPS.md`. Errata (`factcheck/front-errata.md`, `mid1-errata.md`, `mid2-errata.md`, `end-errata.md`) were read so that cut claims are not revived and so that authorship and wording corrections (Dewey 1910: 74 and 82 opened; Paul 1981 actual sentence; Bailin 1999b; Pröpper not Kácha for *JEP* 101865; Fan second author Luzhen Tang; Liang 61.22 / *Patterns* 61.3 / 89/91; Kosmyna author line; CCTST ~34 items / ~45 minutes; Wineburg and McGrew *n* = 10/10/25; Casscells at Harvard Medical School) are the versions listed here.

**Status tags** (after each entry, in square brackets):

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Digital sources: access date **30 August 2026**, unless a fact-check fetch the same day is noted.

Hanging-indent ready: first line flush; subsequent lines indented three spaces.

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Elwyn, Glyn, Dominick Frosch, Richard Thomson, et al. "Shared Decision Making: A Model for Clinical Practice." *Journal of General Internal Medicine* 27 (2012): 1361–1367. https://doi.org/10.1007/s11606-012-2077-6. Choice / option / decision talk. Practice framework, not an RCT of civic outcomes. Remaining authors after Thomson not locked. [Cited]

Murphy, P. Karen, and colleagues. "Quality Talk: Developing Students' Discourse to Promote High-level Comprehension." *American Educational Research Journal* (2018). https://doi.org/10.3102/0002831218771303. Year-long grade-4 study, two classrooms. Small, not a national RCT. Remaining authors not locked. [Cited]

Murphy, P. Karen, Jeffrey A. Greene, and colleagues. "Examining the Effects of Quality Talk Discussions on 4th- and 5th-Grade Students' High-level Comprehension of Text." QES, QT n = 133 vs comparison n = 155. Abstract via ASU record. Not an RCT. Remaining authors and full journal citation not locked this pass. [Cited; UNVERIFIED full citation]

"The Use of Quality Talk to Increase Critical Analytical Speaking and Writing of Students in Three Secondary Schools." *British Educational Research Journal* (2016). https://doi.org/10.1002/berj.3210. Seven intervention classes vs one comparison; d = 0.92 on critical-analytic writing. Large ES, small non-randomized design. Authors unlisted in the JSON. [Cited; UNVERIFIED authors]

van der Linden, Sander, Edward Maibach, John Cook, Anthony Leiserowitz, and Stephan

Lewandowsky. "Inoculating against Misinformation." *Science* 358, no. 6367 (2017): 1141–1142. https://doi.org/10.1126/science.aar4533. Letter/comment, not a full RCT report. [Cited]

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## 3. Policy, frameworks, and union / association documents

College Board. *Academic Preparation for College: What Students Need to Know and Be Able to Do*. 1983. ERIC ED232517. Names reasoning as one of six basic academic competencies. Cited via Hitchcock History. [Cited]

Core Knowledge Foundation / E. D. Hirsch, Jr. Homepage. https://www.coreknowledge.org/. Accessed 30 August 2026. Prints the Hirsch 2016 quote: "Only a well-rounded, knowledge-specific curriculum can impart needed knowledge to all children and overcome inequality of opportunity." Advocacy / curriculum organization. [Cited]

DeAngelo, Linda, Sylvia Hurtado, John H. Pryor, and colleagues. *The American College Teacher: National Norms for the 2007–2008 HERI Faculty Survey*. Los Angeles: Higher Education Research Institute, UCLA, 2009. https://heri.ucla.edu/. >99 percent of faculty rate developing CT very important or essential. Mission language, not evidence. Remaining authors not locked. [Cited]

Digital Inquiry Group (formerly Stanford History Education Group). History Lessons / Reading Like a Historian. https://www.inquirygroup.org/history-lessons. Accessed 30 August 2026. Curriculum dissemination, not a trial. AIR study of new RLH+digital-literacy lessons under US ED grant S422B2300049 is ongoing, not a completed RCT as of the access date. [Cited]

Dumke, Glenn S. / California State University Trustees. Chancellor's Executive Order on Critical Thinking. 1980. System-level operational definition quoted in full by Hitchcock History. Primary order **not independently fetched.** [Cited; UNVERIFIED primary]

Education Endowment Foundation. See Gorard, Siddiqui, and See 2015 and the 2021 effectiveness-trial project page, both in §2. SAPERE "Going for Gold" labelled vendor/charity package there.

Institute of Education Sciences, What Works Clearinghouse. "Find What Works / All Intervention Reports" index. Searched 30 August 2026. https://ies.ed.gov/ncee/wwc/AllInterventionReports. **No intervention report located** for Philosophy for Children, argument mapping, Thinking Maps, Building Thinking Classrooms, Paul–Elder, or "critical thinking" as a named product. Negative search is the finding. End-errata did not re-crawl this index; bibliography follows the 03-stream / ch. 5 access-date search. [Cited]

Institute of Education Sciences, What Works Clearinghouse. Intervention Report: *Leadership and Assistance for Science Education Reform (LASER)*. 2019. https://ies.ed.gov/ncee/WWC/Docs/InterventionReports report.pdf. Inquiry science + leadership PD; science achievement, **not** CT tests. Named so WWC is not over-invoked as a CT clearinghouse. [Cited]

Institute of Education Sciences, What Works Clearinghouse. Intervention Report: *Science Teachers Learning through Lesson Analysis (STeLLA)*. May 2021. https://ies.ed.gov/ncee/WWC/Docs/InterventionRepo Teacher PD; science achievement. Exists; not a CT-inventory finding. [Cited]

Institute of Education Sciences. "Improving Comprehension and Writing Through Reasoned Argumentation." Award page (Collaborative Reasoning units). https://ies.ed.gov/use-work/awards/improving-comprehension-and-writing-through-reasoned-argumentation. Funded program, not a completed WWC intervention report for CR. [Cited]

Koretz, Daniel, and colleagues / CRESST. *Alignment, High Stakes, and the Inflation of Test Scores*. CRESST report, 2005. https://cresst.org/wp-content/uploads/R655.pdf. Teaching-to-the-test / score inflation framework. Applied to coachable CT inventories. Not a CT-specific RCT. [Cited]

Media Literacy Now. "Current Policy" (state media literacy laws map). https://medialiteracynow.org/impact/curren policy/. Accessed 30 August 2026. Last updated January 2026. **25 states** with laws; eleven new steps since January 2024. MLN states it has **not evaluated** implementation. Illinois 2021, New Jersey 2022, Delaware 2022 statutory details **UNVERIFIED** pending bill text. [Cited]

Media Literacy Now. *U.S. Media Literacy Policy & Impact Report*. Through 30 October 2025; released January 2026. https://medialiteracynow.org/wp-content/uploads/2026/01/MLN_Policy_Impact_Report-1.2026.pdf. Legislative tracking, not classroom outcomes. PDF snippets via search. [Cited]

Michaels, Sarah, Catherine O'Connor, and Lauren B. Resnick. "Deliberative Discourse Idealized and Realized: Accountable Talk in the Classroom and in Civic Life." *Studies in Philosophy and Education* 27 (2008): 283–297. https://doi.org/10.1007/s11217-007-9071-1. ERIC EJ797081. Three accountabilities: community, reasoning, knowledge. Accountable Talk is trademarked. Research-and-PD program, not a WWC CT intervention report. [Cited]

National Academies of Sciences, Engineering, and Medicine. Committee on Diagnostic Error in Health Care. *Improving Diagnosis in Health Care*. Washington, DC: National Academies Press, 2015. Report brief fetched in full 30 August 2026: https://nap.nationalacademies.org/resource/21794/DiagnosticEr NCBI Bookshelf: https://www.ncbi.nlm.nih.gov/books/NBK338596/. Diagnostic error = failure to establish an accurate and timely explanation **or** communicate it to the patient. ~5 percent of U.S. adults who seek outpatient care each year; postmortem ~10 percent of deaths; 6–17 percent of hospital adverse events; "most people will experience at least one diagnostic error in their lifetime." John R. Ball chair; Graber and Croskerry on the committee. Synthesis report, not an RCT. [Cited]

National Association for Media Literacy Education. "Core Principles of Media Literacy Education." 2023 revision. https://namle.org/resources/core-principles/. Accessed 30 August 2026. Fetched in full. Professional-association framework, **not an RCT**. SHEG listed among sources drawn on. Principle 4: "practice active inquiry, reflection, and critical thinking." [Cited]

National Association of Colleges and Employers. *Job Outlook 2025*. Revised January 2025. https://www.naceweb.org/docs/default-source/default-document-library/2025/publication/research-report/2025-nace-job-outlook-jan-2025.pdf. Accessed 30 August 2026. PDF fetched. n = 237 (162 members = **19.2 percent** of eligible, plus 75 nonmembers); data 5 August–16 September 2024. Communication **4.57**, critical thinking **4.49**, teamwork 4.43 (importance, 1–5). CT proficiency 3.67; 96.1 percent very/extremely important vs 53.5 percent very/extremely proficient. Skills-based hiring 64.8 percent; GPA screening 46.4 percent. Recruiter survey, **not a skill assessment**, not a probability sample of all employers. Recruiter "#1 skill" treated as folklore unless sourced. [Cited]

National Association of Colleges and Employers. "Career Readiness Competencies: Critical Thinking." December 2025. https://www.naceweb.org/docs/default-source/default-document-library/2025/career-readiness/competencies/nace-career-readiness-competencies-critical-thinking-december-2025.pdf?sfvrsn=e24130d6_3. Accessed 30 August 2026. Definition and sample behaviors (sample interview: "describe a time when you anticipated a problem"). Framework/rubric, not a validity study. [Cited]

National Association of Colleges and Employers. "What Students Need to Know About the Skills-Based Hiring Process." 2026. https://www.naceweb.org/job-market/trends-and-predictions/what-students-need-to-know-about-the-skills-based-hiring-process. *Job Outlook 2026*: 70 percent skills-based hiring; GPA screening 42 percent; <40 percent of seniors familiar with the term. Association journalism about its own survey. Not re-fetched on the mid2 pass. [Cited]

National Association of Colleges and Employers. "Recruiters and Students Have Differing Perceptions of New Grads' Proficiency in Competencies." 2024. https://www.naceweb.org/career-readiness/competencies/31ad4621-2693-4815-bf78-8d0667ca3e46. Comparison of 2023 Student Survey vs *Job Outlook 2024* (18,966 students / 255 employers). In the JSON; not a load-bearing manuscript citation. [Consulted]

National Commission on Excellence in Education. *A Nation at Risk: The Imperative for Educational Reform*. Washington, DC: U.S. Government Printing Office, 1983. Willingham 2007 cites its lament about missing "higher-order" intellectual skills. Demand-signal, not a CT definition. [Cited]

National Council for the Social Studies. *The College, Career, and Civic Life (C3) Framework for Social Studies State Standards*. 2013. https://www.socialstudies.org/sites/default/files/c3/c3-framework-for-social-studies-rev0617.pdf. Four-dimension Inquiry Arc. Overview language includes "critical thinking"; operational content is inquiry, sources, evidence, informed action. Not a CT course and not a trial. [Cited]

National Governors Association Center for Best Practices and Council of Chief State School Officers. *Common Core State Standards for Mathematics*: Standards for Mathematical Practice. 2010. http://www.corestandards.org/Math/Practice/. Practice 3: construct viable arguments and critique the reasoning of others. Naming is not teaching. [Cited]

National Research Council. *A Framework for K–12 Science Education*. Washington, DC: National Academies Press, 2012. NGSS Lead States, *Next Generation Science Standards*, 2013. Appendix F, Science and Engineering Practices: https://neacadsci.org/docs/Appendix_F_Science_and_Engineering_Practices_FINAL_060513.pdf. Practice 7: engaging in argument from evidence. Standards, not an intervention trial. [Cited]

OECD. *Future of Education and Skills 2030: Critical Thinking* (Learning Compass construct note). 2019. https://www.oecd.org/content/dam/oecd/en/topics/policy-issues/future-of-education-and-skills/learning-compass-constructs/Critical%20Thinking.pdf. Defines CT as "questioning and evaluating ideas and solutions (OECD, 2016)." Policy construct note, not an RCT. [Cited]

Partnership for 21st Century Skills. *P21 Framework Definitions*. December 2009. ERIC ED519462, https://files.eric.ed.gov/fulltext/ED519462.pdf. Accessed 30 August 2026. Advocacy coalition, not an RCT. 4Cs unpacking of critical thinking and problem solving; also the often-dropped clause that CT must build on core academic knowledge. [Cited]

Partnership for 21st Century Skills. *21st Century Skills, Education and Competitiveness*. 2008. ERIC ED519337, https://files.eric.ed.gov/fulltext/ED519337.pdf. Advocacy report; employer-survey atmosphere. Framework, not evidence of teachability. Not named separately from the 2009 *Framework Definitions* on the manuscript page. [Consulted]

Resnick, Lauren B., Christa S. C. Asterhan, and Sherice N. Clarke. *Accountable Talk: Instructional Dialogue That Builds the Mind*. UNESCO IBE Educational Practices Series 29, 2018. ERIC ED612793. Eight principles. Practice guide, not an RCT. [Cited]

Stanford History Education Group. Civic Online Reasoning curriculum site. https://cor.stanford.edu/. Accessed 30 August 2026. Three questions in the project's own words. Free lessons; not an outcome trial. [Cited]

Stanford History Education Group / Civic Online Reasoning. "Lateral Reading on the Open Internet" (project summary). 2022. https://cor.stanford.edu/research/lateral-reading-on-the-open-internet/. Accessed 30 August 2026. Use with Wineburg et al. 2022, not instead of it. Names the cluster-randomized, repeated-measures analysis. [Cited]

Swan, Kathy, S. G. Grant, John Lee, and colleagues. "The State of the C3 Framework." *Social Education* (November/December 2023). https://www.socialstudies.org/system/files/2023-12/se-870623361.pdf. State-standards uptake survey. Implementation variation; not outcome evidence. Remaining authors not locked. [Cited]

Tertiary Education Quality and Standards Agency (TEQSA, Australia). Noted AIAS in 2024 as an option. An option is not a trial. Named in ch. 6 n. 35. [Cited]

UNESCO Bangkok / NEQMAP / ERI-Net. *Assessment of Transversal Competencies: Policy and Practice in the Asia-Pacific Region*. 2016. https://neqmap.bangkok.unesco.org/wp-content/uploads/2019/09/246590eng.pdf. TVC domain 1 is "critical and innovative thinking" (creativity, entrepreneurship, resourcefulness, application skills, reflective thinking, reasoned decision-making). Policy vocabulary, not a validated construct. Diagnostic that CT sits in a bin with entrepreneurship. Named in introduction n. 2 and ch. 1 n. 23. [Cited]

World Economic Forum. *The Future of Jobs Report 2025*, skills-outlook chapter. https://www.weforum.org/publica future-of-jobs-report-2025/in-full/3-skills-outlook/. PDF: https://reports.weforum.org/docs/WEF_Future_of_Jobs Accessed 30 August 2026. **Analytical thinking** top core skill, "seven out of 10 companies considering it as essential" (WEF's own wording). Employer survey of WEF respondents, **not a CT assessment**. Phrase is *analytical thinking*, not "critical thinking." Skills-outlook HTML timed out; "seven of ten" confirmed from WEF digest/PDF snippet. Underlying 69 percent from secondary roundups, not read off PDF figure 3.3 this pass. [Cited]

# 4. Product, vendor, and test publishers

Council for Aid to Education. Collegiate Learning Assessment / CLA+. **Vendor.** Constructed-response performance task plus selected-response. Blend of critical thinking, problem solving, and writing. Huber and Kuncel: criterion contamination — not a pure CT test. [Cited; vendor]

Ennis, Robert H., Jason Millman, and Thomas N. Tomko. Cornell Critical Thinking Test, Levels X and Z. Commercially distributed. Independent psychometric work thinner than for Watson–Glaser and CCTST (Liu et al. 2014 inventory table). [Cited; vendor]

Ennis, Robert H., and Eric Weir. *The Ennis–Weir Critical Thinking Essay Test*. 1985. Letter-to-the-editor scored on argument moves. Closer to Ennis's conception than Facione's Delphi list. [Cited]

Facione, Peter A., and Noreen C. Facione / Insight Assessment. The Holistic Critical Thinking Scoring Rubric (HCTSR). https://insightassessment.com/iaresource/the-holistic-critical-thinking-scoring-rubric/. Accessed 30 August 2026. **Vendor.** Four-level holistic rating. Publisher states rating tools weaker than standardized instruments; rater training required; Kappa is the reliability statistic. Not a HALO test. [Cited; vendor]

Facione, Noreen C., and Peter A. Facione. "Critical Thinking: The Statement of Expert Consensus and the CCTST" (nursing accreditation context). ERIC ED380509, 1994. https://files.eric.ed.gov/fulltext/ED380509.pdf. CCTST development for NLN Criterion 20. Pilot n = 1,196 college students. Vendor-origin instrument history. Not a load-bearing manuscript citation beyond the instrument map. [Consulted; vendor]

Foundation for Critical Thinking. "Defining Critical Thinking." https://www.criticalthinking.org/pages/defining-critical-thinking/766. Accessed 30 August 2026. Curriculum-brand site, not a meta-analysis. Hosts the 1987 Scriven–Paul draft, Glaser 1941 excerpt, and Elder 2007 one-liner. History blurb should not be followed as historiography. [Cited; vendor]

Foundation for Critical Thinking. "The Elements of Reasoning and the Intellectual Standards." https://www.criticalthinking.org/pages/the-elements-of-reasoning-and-the-intellectual-standards/480. Accessed 30 August 2026. This page lists seven "virtually universal" standards; other Foundation publications add significance, fairness, completeness. Internal list-variation is a fact. [Cited; vendor]

Halpern, Diane F. Halpern Critical Thinking Assessment (HCTA). Schuhfried — **vendor**. Everyday scenarios; open-ended plus forced-choice. Described as **retired** in a 2024 *Journal of Intelligence* review (PMC10890380). [Cited; vendor]

Insight Assessment. California Critical Thinking Skills Test (CCTST) and California Critical Thinking Disposition Inventory (CCTDI). **Vendor.** CCTST: about **34 items in about 45 minutes** (Liu, Frankel, and Roohr 2014 table), not "34–45 minutes." CCTST is not the meaning of critical thinking. Independent subscale alphas 0.21–0.51 (Leppa 1997, via Liu). [Cited; vendor]

Pearson / TalentLens. Watson–Glaser Critical Thinking Appraisal (WGCTA, Watson–Glaser II, short form). **Vendor.** Five advertised buckets. Employment screen, especially law-firm hiring — a different validity argument from college learning. Independent subscale alphas 0.17–0.74 (Loo and Thorpe 1999, via Liu); often one factor (Bernard et al. 2008, 60 published studies). [Cited;

vendor]

Perkins, Mike, Jasper Roe, Leon Furze, and Jason MacVaugh / Learning Innovation Practice Ltd. The AI Assessment Scale (project site). https://aiassessmentscale.com/. Accessed 30 August 2026. HTML fetched. AIAS 2.1 wording; dissemination claim (>350 institutions, 30+ languages). Vendor/consultancy-maintained but free non-commercial use. Not outcome evidence. [Cited; vendor]

Rationale / bCisive / Austhink / Kritisch Denken BV. Argument-mapping software identified in van Gelder 2015. **Vendor.** Reason! Project Studies, 1999–2002, project report PDF (search snippet 30 August 2026): https://www.reasoninglab.com/wp-content/uploads/2025/10/ReasonProjectReport2002_1.pdf. Developer document, not a refereed multi-lab RCT. Used only as corroboration of the 2015 chapter's description. [Cited; vendor]

Zahner, D. (Council for Aid to Education). "Reliability and Validity – CLA+." 2013. http://cae.org/images/uploads/pdf/Reliability_and_Validity_of_CLA_Plus.pdf. **Vendor** technical note, cited by Liu et al. 2014 for CR reliability 0.43 and test-level 0.87. Direct CAE PDF not re-fetched this pass; number taken from the ETS report. [Cited; vendor]

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# 5. Journalism and teaching restatements

American Medical Association. "Shared Decision Making Requires Statistical Literacy." *Virtual Mentor* (*AMA Journal of Ethics*) 11, no. 4 (April 2009): 301–305. https://journalofethics.ama-assn.org/article/shared-decision-making-requires-statistical-literacy/2009-04. Accessed 30 August 2026. Fetched in full. Teaching restatement of Gigerenzer et al. 2008 and Casscells 1978 (11 of 60 correct; **Harvard Medical School**). Not new data. [Cited]

Elliott, David. "AI Won't Replace Teachers, Says This Global Union." World Economic Forum, 2 July 2024. Interview with David Edwards (Education International). https://www.weforum.org/stories/2024/07/artificial-intelligence-education-teachers-union/. Accessed 30 August 2026. Fetched in full. **Live title** — do not use the invented title " 'Education is a place where we build democracy.' Why a teacher's union isn't afraid AI will replace teachers." Journalism, **not a trial.** Quoted claims confirmed on the page: human teachers as "the reserve of the privileged"; children who cannot afford them "sit in rooms with a chatbot." Adjacent sentence: "Education is the place where we build societies and we build democracy." [Cited]

Johns Hopkins Medicine newsroom. "Report Highlights Public Health Impact of Serious Harms From Diagnostic Error in U.S." 17 July 2023. https://www.hopkinsmedicine.org/news/newsroom/news-releases/2023/07/report-highlights-public-health-impact-of-serious-harms-from-diagnostic-error-in-us. **Journalism / institutional news release** summarizing Newman-Toker et al.: 371,000 deaths + 424,000 permanent disabilities. Use with the journal article. [Cited]

PR Newswire / Media Literacy Now. "New U.S. Media Literacy Report Finds State Policy Accelerating as Lawmakers Respond to Phones, Social Media, and AI." 15 January 2026. https://www.prnewswire.com/news-releases/new-us-media-literacy-report-finds-state-policy-accelerating-as-lawmakers-respond-to-phones-social-media-and-ai-302662137.html. Journalism / press release. Use only as a pointer to the MLN report. [Cited]

Stanford Graduate School of Education news. “It doesn’t take long to learn how to spot misinformation online, Stanford study finds.” 19 April 2022. https://ed.stanford.edu/news/it-doesn-t-take-long-learn-how-spot-misinformation-online-stanford-study-finds. Accessed 30 August 2026. **Journalism / university communications** summarizing Wineburg et al. 2022. Source of “twice as well” / “twice as likely” and “earned only half the total number of possible points.” Funded by Google.org. Not the paper. Do not upgrade GSE language into a civic-outcome RCT. [Cited]

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*Compiled 30 August 2026. Unfetched = UNVERIFIED. Vendor counts and recruiter surveys remain labelled as such. Cut claims in the errata are not revived here. Dewey 1910: 74 and 82 are opened; Paul 1981 uses the 1981 sentence; Bailin 1999b is the three-feature core; JEP 101865 is Pröpper et al.; Fan’s second author is Luzhen Tang; Liang reports 61.22 / 61.3 / 89/91; Kosmyna names follow the arXiv PDF; CCTST is ~34 items / ~45 minutes; Wineburg and McGrew n = 10/10/25; Casscells is Harvard Medical School.*